

A Compact Paravector Form of Space-Time Algebra

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Abstract—Space-time geometry, electromagnetism, relativistic wave mechanics, and spin dynamics are shown to fit inside one associative product. A Pauli paravector contains one complex scalar and one complex three-vector; its multiplication rule packages dot and cross products, the inverse divisor, the space-time interval, rotations, boosts, Maxwell’s equations, and the Klein–Gordon and Dirac systems. Spin-axis projectors, ladder elements, and the quaternion rotation basis arise from the same product, and the leading Pauli equation produces the expected magnetic spin separation. Events, fields, and wave functions retain the different transformation laws required by their physical roles. The main construction is intrinsic to the paravector algebra; optional matrix applications are identified where they are used, and the standard two-by-two coordinate realization is isolated in Appendix A. No new physical law is proposed. A notation-light, self-contained reorganization is supplied so that connections usually divided among vector, indexed-array, and matrix descriptions remain visible.

Index Terms—space-time algebra, geometric algebra, Lorentz transformations, Maxwell electrodynamics, Dirac equation, Pauli equation, spin dynamics.

Introduction

Shared geometry is often hidden when space-time, electromagnetic fields, and quantum states are assigned different mathematical languages. Their rotations, boosts, intervals, and order-sensitive differential products are placed here in one small algebra. Notation light enough for plain-text transcription is retained, while scalar/vector and real/imaginary structure remains visible.

Three generators σ_n are used. A generator is an element from which other elements are built by addition and multiplication. Each generator squares to 1, while the product of two different generators changes sign when its order is exchanged. Their full product $i = \sigma_1\sigma_2\sigma_3$ commutes with them and satisfies $i^2 = -1$. It is conventionally called a pseudoscalar [12, 15]. Every algebra element is written uniquely as

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad S \in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3.$$

Elements of this form are called Pauli paravectors and are referred to below simply as paravectors. Their product is supplied by the sigma-basis rule, not componentwise multiplication. An optional one-to-one product-preserving translation into two-by-two complex matrices is supplied in Appendix A; it is conventionally called a faithful representation [12, 14, 15]. No intrinsic result depends on that translation; optional matrix applications are identified explicitly where they are used.

The product yields dot and cross products, while multiplication by the adjugate supplies the determinant used in an inverse. For real events that determinant is the space-time interval. Exponentials generate rotations and boosts; one equation contains all four Maxwell equations; and ordered first-order products produce the Dirac and leading Pauli equations [2, 4, 9, 10]. Events are transformed by a two-sided Hermitian law, fields by similarity, and one Dirac field by a fixed algebraic projection. Spin-axis projectors, ladder elements, and the magnetic separation of the two Pauli spin parts are obtained from the same product. These distinctions and spin structures are derived rather than suppressed.

Established flat-space physics is reorganized rather than altered. Here flat means that coordinates exist in which the interval coefficients are constant. Position-dependent coefficients produced only by a coordinate relabeling do not change that geometry, and gravity is absent from the model. The construction is grounded in the space-time-algebra and paravector literature [12, 14, 15], with roots in quaternions and geometric algebra [1, 3]. Natural units set the reduced Planck constant and the speed of light to 1. Rationalized Heaviside–Lorentz units absorb the vacuum constants and factors of 4π [16]. The signature $(+, -, -, -)$ gives $t^2 - \mathbf{r}^2$. Ordinary three-vectors are bold and their components are displayed in braces:

$$\mathbf{A} := \{A_1, A_2, A_3\}. \quad (1)$$

Operator order and the derivative product rule are retained explicitly. The notation $O(x^n)$ collects terms of order x^n and beyond. The prime mark is defined as a value label:

$$x' := \text{a distinct value carrying the same declared type and role as } x. \quad (2)$$

When a transformation has been specified, the prime labels the transformed value. In other uses it only distinguishes two values that share the same meaning and algebraic properties. It never denotes a derivative; derivatives are written explicitly with ∂ or d .

Wave-function roles are identified by upright text subscripts, as in ψ_{KG} , $\psi_{\text{KG_split}}(\pm)$, and $\Psi_{\text{Dirac_with_potential}}$. Parentheses attached directly to a wave function contain only actual inputs t , $\mathbf{r}$, and $+$ or $-$; unneeded inputs are omitted.

The next page collects the principal formulas. Algebra, space-time, transforms, differential calculus, electromagnetism, wave mechanics, probability, and spin are developed in Sections 1–6. Conclusions, open work, and symbolic checks are separated in Sections 7–9. Quaternion relabeling accompanies rotations; Appendix A isolates the two-way translation between paravectors and ordinary two-by-two complex matrices.

Equation Reference Sheet (Summary)

Natural units: $\hbar = 1$ (reduced Planck constant), $c = 1$ (speed of light).

Conventions: signature $(+, -, -, -)$; unmarked spatial integrals cover all space at fixed time with measure $dr_1 dr_2 dr_3$.

Name	Formula	Ref.
Custom functions, maps, and identities		
Pauli-paravector spaces	$a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{R}}, \quad a \in \mathbb{R}, \quad \mathbf{A} \in \mathbb{R}^3,$	(13)
Complex paravector	$S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{C}}, \quad S \in \mathbb{C}, \quad \mathbf{V} \in \mathbb{C}^3, \quad \mathbb{P} \text{ when the coefficient type is immaterial or fixed by context}$	(19)–(20)
Identity elements	$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma} = S + \mathbf{V} \cdot \boldsymbol{\sigma} = X + iY \in \mathbb{P}_{\mathbb{C}}$ $\mathbf{0} = \{0, 0, 0\}, \quad 0 = 0 + \mathbf{0} \cdot \boldsymbol{\sigma}, \quad 1 = 1 + \mathbf{0} \cdot \boldsymbol{\sigma},$	(26)–(28)
Prime value label	$Z + 0 = Z, \quad Z1 = 1Z = Z$	(2)
Real and img functions	$x' = \text{a distinct value with the same declared type and role as } x; \quad ' \neq \text{derivative}$	(21), (22)
Scalar and vector functions	$\text{real}(Z) = X, \quad \text{img}(Z) = Y$	(23), (24)
Axis functions	$\text{scalar}(Z) = S, \quad \text{vector}(Z) = \mathbf{V}$	(39), (40)
Order functions	$\text{par}(\mathbf{A}) = (\mathbf{A} \cdot \mathbf{u})\mathbf{u}, \quad \text{prp}(\mathbf{A}) = \mathbf{A} - \text{par}(\mathbf{A}), \quad \mathbf{u}^2 = 1$	(43), (44)
Adjugate function	$\text{com}(Z, Z') = ZZ' - Z'Z, \quad \text{acom}(Z, Z') = ZZ' + Z'Z$	(54), (55)
Determinant function	$\text{adj}(Z) = Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}, \quad Z^{\text{H}} = \text{adj}(Z)^* = \text{adj}(Z^*)$	(57), (58)
Inverse	$\det(Z) = \text{adj}(Z)Z = Z\text{adj}(Z) = S^2 - \mathbf{V}^2$	(65)
Matrix-map functions	$Z^{-1} = \text{adj}(Z) / \det(Z), \quad \det(Z) \neq 0$	(290)–(292)
Algebraic progression		
Pseudoscalar	$\text{mat}(Z) = \mathbf{M} \iff Z = \text{mat}^{-1}(\mathbf{M})$	(3)
Sigma product	$i = \sigma_1 \sigma_2 \sigma_3, \quad i^2 = -1, \quad i\sigma_n = \sigma_n i$	(4)
Sigma/H operations	$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k$	(7)
Star and H actions	$XY^* = X^* Y^*, \quad XY^{\text{H}} = Y^{\text{H}} X^{\text{H}}, \quad XY^{*\text{H}} = Y^{*\text{H}} X^{*\text{H}}$	(29)–(31)
Vector product	$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad Z^{\text{H}} = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad Z^{*\text{H}} = S - \mathbf{V} \cdot \boldsymbol{\sigma}$	(41)
Paravector product	$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(42)
Positive norms	$ZZ' = SS' + \mathbf{V} \cdot \mathbf{V}' + (S\mathbf{V}' + S'\mathbf{V} + i\mathbf{V} \times \mathbf{V}') \cdot \boldsymbol{\sigma}$	(50)–(52)
Determinant products	$ S ^2 = SS^*, \quad \ \mathbf{V}\ ^2 = \mathbf{V} \cdot \mathbf{V}^*, \quad \ Z\ ^2 = \text{scalar}(ZZ^{\text{H}}) = S ^2 + \ \mathbf{V}\ ^2$	(62), (61)
Axis projectors	$\det(ZZ') = \det(Z)\det(Z'), \quad \det(\text{adj}(Z)) = \det(Z)$	(71), (72)
Local eigenvalues	$\mathbf{u}^2 = 1, \quad \Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad Z(\pm) = \Pi(\pm)Z, \quad Z = Z(+) + Z(-)$ $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}, \quad \mathbf{r} \neq \mathbf{0}, \quad \mathbf{u} := \mathbf{r}/\ \mathbf{r}\ : \quad \lambda(\pm) = t \pm \ \mathbf{r}\ ,$	(77), (81), (82)
Ladder function	$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}, \quad z^2 = \mathbf{V}^2 \neq 0: \quad \mathbf{u} := \mathbf{V}/z, \quad \mathbf{u}^2 = 1, \quad \lambda(\pm) = S \pm z$	(75)
Functions of an axis element	$a(\pm) = \frac{1}{2}(\sigma_1 \pm i\sigma_2), \quad a(\pm)^2 = 0, \quad \text{com}(\sigma_3, a(\pm)) = \pm 2a(\pm)$	(73)
Vector exponential	$\mathbf{u}^2 = 1: \quad f(\mathbf{u} \cdot \boldsymbol{\sigma}) = f(+1)\Pi(+) + f(-1)\Pi(-)$	(83)
Two-generator exponential	$\exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) = \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}$	(89)
Paravector logarithm	$\exp(\theta \mathbf{u} \cdot i\boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot i\boldsymbol{\sigma}$	(87), (88)
Transforms and kinematics		
Inverse-pair transform	$\mathbf{B} \neq \mathbf{0}, \quad \mathbf{u} := \mathbf{B}/\ \mathbf{B}\ , \quad \lambda(\pm) = b \pm \ \mathbf{B}\ , \quad \log(Y) = \log(\lambda(+))\Pi(+) + \log(\lambda(-))\Pi(-)$	(93), (94)
Event two-sided law	$TZ' = ZT \text{ implies } Z' = T^{-1}ZT$	(97)
Active rotation	$X' = T^{-1}X(T^{-1})^{\text{H}} = T^{*\text{H}}XT^*, \quad \det(T) = 1$	(92), (101)
Passive event boost	$R = \exp(\theta \mathbf{u} \cdot i\boldsymbol{\sigma}/2), \quad X' = R^{\text{H}}XR$	(84), (102)
Passive field boost	$L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2), \quad X' = L^{-1}X(L^{-1})^{\text{H}} = L^*XL^*$	(149)
Space-time interval	$F' = L^{-1}FL = L^*FL$	(116)
Velocity and momentum	$ds^2 = \det(dX) = dt^2 - d\mathbf{r}^2$	(117), (121)
Acceleration	$U = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \quad mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$	(122), (123)
Differential and electromagnetic fields		
Spatial derivatives	$A = dU/ds, \quad \text{scalar}(U^*A) = 0$	
Paravector derivatives	$\partial_{\mathbf{r}} = \{\partial_1, \partial_2, \partial_3\}, \quad \partial_n = \partial/\partial r_n, \quad \partial_{\mathbf{r}}^2 = \partial_1^2 + \partial_2^2 + \partial_3^2$	(133), (134)
Wave operator	$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \quad \partial^* = \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}$	(135), (136)
Ordered product rule	$\square = \partial\partial^* = \partial^*\partial = \partial^2 - \partial^2$	(137), (138)
Field and potential	$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z'), \quad \sum_n \text{com}(\sigma_n, Z) \partial_n Z'$	(139)
Potential gradient	$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad \Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$	(141), (142)
Maxwell equation	$\partial\Phi = S + F^*, \quad S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$	(143), (144)
Maxwell components	$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$	(146)
Gauge functions	$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad \partial_t \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}$	(147), (148)
Lorentz force	$\Phi(\lambda) = \Phi + \partial^* \lambda, \quad F(\lambda) = F$	(153), (158)
Energy and flux	$m dU/ds = q \text{ real}(FU)$	(159)
Full energy-momentum function	$T = \frac{1}{2}FF^{\text{H}} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$	(161)
Relativistic wave equations		
Frequency split	$T(a) = \frac{1}{2}FaF^{\text{H}}, \quad a \in \mathbb{P}_{\mathbb{R}}, \quad T = T(1)$	(165)
Potential energy/momentum	$\Omega = \sqrt{m^2 - \partial_{\mathbf{r}}^2}, \quad (\partial_t \pm i\Omega)\psi_{\text{KG_split}}(\pm) = 0$	(169), (171)
Klein–Gordon	$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}$	(172), (173)
Ordered factors	$(-E^2 + \mathbf{P}^2 + m^2)\psi_{\text{KG_with_potential}} = 0$	(174)
Dirac system	$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$	(181)
Fixed right-projector	$(i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2} = m\Psi_{\text{Dirac_with_potential},1},$	(194)
Dirac block transform	$(i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} = m\Psi_{\text{Dirac_with_potential},2}$	(198), (199)
Squared blocks	$\mathbf{u} := \mathbf{e}_3, \quad \Pi(+) = \frac{1}{2}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(1 + \sigma_3),$	(188), (189)
Dirac current	$\Psi_{\text{Dirac},n} = \Psi_{\text{Dirac},n}\Pi(+)$	
Klein–Gordon pairing	$\Psi'_{\text{Dirac},1}(t', \mathbf{r}') = T^{\text{H}}\Psi_{\text{Dirac},1}(t, \mathbf{r}),$	
Projected-wave pairings	$\Psi'_{\text{Dirac},2}(t', \mathbf{r}') = T^{-1}\Psi_{\text{Dirac},2}(t, \mathbf{r})$	
Leading Pauli limit	$(-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_{\text{Dirac_with_potential},1} = 0,$	(202)
Pauli spin split	$(-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_{\text{Dirac_with_potential},2} = 0$	(207)
Free energy/axis split	$J(\Psi_{\text{Dirac_with_potential}}) = \rho(\Psi_{\text{Dirac_with_potential}}) + \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) \cdot \boldsymbol{\sigma}, \quad \partial_t \rho(\Psi_{\text{Dirac_with_potential}}) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) = 0$	(183), (186)
	$\langle \phi_{\text{KG_with_potential}} \psi_{\text{KG_with_potential}} \rangle = \iiint \rho(\phi_{\text{KG_with_potential}}, \psi_{\text{KG_with_potential}}) dr_1 dr_2 dr_3$	(212), (214)
	$\langle \phi_{\text{block}} \psi_{\text{block}} \rangle = 2 \iiint \text{scalar}(\phi_{\text{block}}^{\text{H}} \psi_{\text{block}}) dr_1 dr_2 dr_3,$	
	$\langle \Psi_{\text{Dirac_left}} \Psi_{\text{Dirac_right}} \rangle = 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_{\text{Dirac_left},n}^{\text{H}} \Psi_{\text{Dirac_right},n}) dr_1 dr_2 dr_3$	(225)
	$E\psi_{\text{Pauli_large}} = (\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}}/(2m) + O(m^{-2})$	(230), (231), (233)
	$\mathbf{u} := \mathbf{B}/\ \mathbf{B}\ , \quad \psi_{\text{Pauli_split}}(\pm) = \Pi(\pm)\psi_{\text{Pauli_large}},$	
	$i\partial_t \psi_{\text{Pauli_split}}(\pm) = (\mathbf{P}^2 \mp q\ \mathbf{B}\) \psi_{\text{Pauli_split}}(\pm)/(2m) + qV\psi_{\text{Pauli_split}}(\pm) + O(m^{-2})$	
	$\Psi_{\text{Dirac_energy}}(\pm) = \frac{1}{2}(\Psi_{\text{Dirac}} \pm \mathcal{H}(\Psi_{\text{Dirac}})/E(\mathbf{p})),$	(234)–(242)
	$\Psi_{\text{Dirac_positive_energy_axis}}(\pm), \quad \Psi_{\text{Dirac_negative_energy_axis}}(\pm)$	

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1 Algebraic Foundations

The algebra is fixed before physical meanings are assigned to particular elements. Product grouping does not matter: $(XY)Z = X(YZ)$. This property is called associativity. Multiplication by 1 leaves every element unchanged. Every complex coefficient used in the master product commutes with every algebra element; such a coefficient is called central. Only later calculations containing differential operators require the written order to be retained. These standard algebra terms are used in precisely these senses [12, 15].

1.1 Basis and pseudoscalar

For $n \in \{1, 2, 3\}$, every algebra element is built from real-number multiples of products of the three basis elements σ_n . Products containing zero, one, two, or three different generators provide the four basis types. The product containing all three generators is called the pseudoscalar, following the standard Pauli-algebra terminology [12, 14, 15]. Altogether, eight independent real coefficients are required. The resulting algebra is commonly called the real Pauli algebra or the three-dimensional real Clifford algebra. The basis relations are

$$\begin{aligned} \sigma_n^2 &= 1, & \sigma_n \sigma_m &= -\sigma_m \sigma_n \quad (n \neq m), \\ i &:= \sigma_1 \sigma_2 \sigma_3, & i^2 &= -1. \end{aligned} \quad (3)$$

The full three-generator product commutes with each generator, $i\sigma_n = \sigma_n i$, and its square is the scalar -1 :

$$\begin{aligned} (\sigma_1 \sigma_2 \sigma_3)^2 &= \sigma_1 \sigma_2 (\sigma_3 \sigma_1) \sigma_2 \sigma_3, \\ &= -\sigma_1 (\sigma_2 \sigma_1) (\sigma_3 \sigma_2) \sigma_3, \\ &= -\sigma_1^2 \sigma_2^2 \sigma_3^2, \\ &= -1. \end{aligned}$$

With the Kronecker delta and Levi-Civita symbol [12, 15],

$$\sigma_n \sigma_m = \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k, \quad (4)$$

where

$$\begin{aligned} \delta_{nm} &:= \begin{cases} 1, & n = m, \\ 0, & n \neq m, \end{cases} \\ \epsilon_{nmk} &:= \begin{cases} +1, & (n, m, k) \text{ even permutation of } (1, 2, 3), \\ -1, & (n, m, k) \text{ odd permutation of } (1, 2, 3), \\ 0, & \text{otherwise.} \end{cases} \end{aligned}$$

The cyclic orderings $(1, 2, 3)$, $(2, 3, 1)$, and $(3, 1, 2)$ are the even permutations. Exchanging any two entries gives an odd permutation. The display is therefore also a complete definition of the signs used for ϵ_{nmk} .

1.2 The Conjugate and Hermitian Operators

The sigma-conjugate operator is defined by

$$(\cdot)^* := \text{flip the signs of } \sigma_n. \quad (5)$$

The Hermitian operator is defined by

$$(\cdot)^H := \text{reverse the order of multiplied } \sigma_n. \quad (6)$$

Within the present algebra, the Hermitian operation is defined solely by reversing sigma-product order. Appendix A shows that this intrinsic operation becomes the familiar Hermitian operation after the optional matrix translation is made. Both intrinsic operations distribute over addition and multiplication by real numbers. Star preserves product order, whereas H reverses it. Because both send i to $-i$, each also conjugates a complex coefficient: applying either operation to cZ produces c^* times the operated element. This property is conventionally called conjugate linearity. Their combined operation preserves complex coefficients and reverses product order [12, 15]. These statements are expressed directly by

$$\begin{aligned} (XY)^* &= X^* Y^*, & (XY)^H &= Y^H X^H, \\ X^{*H} &= (X^H)^* = (X^*)^H, & (X^*)^* &= (X^H)^H = X. \end{aligned} \quad (7)$$

When applied to the generators and the pseudoscalar, these rules are reduced to

$$\sigma_n^* = -\sigma_n, \quad \sigma_n^H = \sigma_n, \quad \sigma_n^{*H} = -\sigma_n, \quad (8)$$

$$i^* = -i, \quad i^H = -i, \quad i^{*H} = i. \quad (9)$$

For a product of two different generators, with $n \neq m$,

$$\begin{aligned} (\sigma_n \sigma_m)^* &= \sigma_n^* \sigma_m^* = (-\sigma_n)(-\sigma_m) = \sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^H &= \sigma_m^H \sigma_n^H = \sigma_m \sigma_n = -\sigma_n \sigma_m, \\ (\sigma_n \sigma_m)^{*H} &= (\sigma_m \sigma_n)^* = \sigma_m^* \sigma_n^* \\ &= (-\sigma_m)(-\sigma_n) = \sigma_m \sigma_n = -\sigma_n \sigma_m. \end{aligned} \quad (10)$$

The same sign pattern on the sigma-number-two elements is recorded in Eq. (10). Each basic operation returns the original element when applied twice, as shown in Eq. (7); an operation with this property is called an involution. The word ‘‘Hermitian’’ names the operation; it does not assert that every operated element is unchanged. An element is self-Hermitian precisely when $Z^H = Z$. When this algebra is compared with products containing an even number of basis vectors in the usual space-time algebra, reversal of a space-time product corresponds to the combined operation $(\cdot)^{*H}$, not to $(\cdot)^H$ alone [12, 15]. This comparison introduces no additional rule.

The sigma number of a reduced basis product is defined as the number of sigma generators that remain after repeated use of $\sigma_n^2 = 1$ and $\sigma_n \sigma_m = -\sigma_m \sigma_n$. Sigma numbers 0, 1, 2, and 3 are assigned respectively to 1, σ_n , $\sigma_n \sigma_m$ with $n \neq m$, and i . A product not yet simplified by these rules is called unreduced. Every such product is reduced to one of the four cases. The basis elements and operator signs are summarized by sigma number in Tables 1 and 2.

Table 1. Sigma basis elements by sigma number.

sigma number	representative basis elements
0 (scalar)	1
1 (vector)	$\sigma_1, \sigma_2, \sigma_3$
2 (bivector)	$\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2$
3 (pseudoscalar)	$\sigma_1\sigma_2\sigma_3$

Table 2. Operator signs by sigma number.

sigma number	$(\cdot)^*$	$(\cdot)^H$	$(\cdot)^{*H}$
0 (scalar)	+	+	+
1 (vector)	−	+	−
2 (bivector)	+	−	−
3 (pseudoscalar)	−	−	+

1.3 Generic Complex Paravector

The three generators are collected into the sigma-vector

$$\boldsymbol{\sigma} := \{\sigma_1, \sigma_2, \sigma_3\}, \quad (11)$$

which is a vector of algebraic objects rather than a vector of scalar components. The associated three products of two different generators are obtained by multiplication by the pseudoscalar:

$$i\boldsymbol{\sigma} := \{i\sigma_1, i\sigma_2, i\sigma_3\} = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \quad (12)$$

because $i = \sigma_1\sigma_2\sigma_3$. For each n , the two indices other than n remain in the displayed order after reduction. Equation (12) therefore fixes the chosen ordered basis for the sigma-number-two products. The real and complex Pauli-paravector spaces are defined directly by the allowed scalar and vector coefficients. The unqualified symbol $\mathbb{P}$ denotes whichever of the two spaces is applicable when the coefficient type is immaterial or already fixed by context:

$$\begin{aligned} \mathbb{P}_{\mathbb{R}} &:= \{a + \mathbf{A} \cdot \boldsymbol{\sigma} \mid a \in \mathbb{R}, \mathbf{A} \in \mathbb{R}^3\}, \\ \mathbb{P}_{\mathbb{C}} &:= \{S + \mathbf{V} \cdot \boldsymbol{\sigma} \mid S \in \mathbb{C}, \mathbf{V} \in \mathbb{C}^3\}, \\ \mathbb{P} &:= \text{either } \mathbb{P}_{\mathbb{R}} \text{ or } \mathbb{P}_{\mathbb{C}}, \text{ as fixed by context.} \end{aligned} \quad (13)$$

Thus every member of $\mathbb{P}_{\mathbb{R}}$ is constructed from one real scalar and one real three-vector. Every member of $\mathbb{P}_{\mathbb{C}}$ is constructed in the same way with a complex scalar and a complex three-vector. The $\mathbb{R}$ or $\mathbb{C}$ subscript identifies the coefficient type. An element with a real or zero scalar and a complex vector is therefore still in $\mathbb{P}_{\mathbb{C}}$. Every member of $\mathbb{P}_{\mathbb{R}}$ is also a member of $\mathbb{P}_{\mathbb{C}}$. A product of two members of $\mathbb{P}_{\mathbb{R}}$ can nevertheless leave that set because of the imaginary cross-product term in Eq. (41). In standard algebra language, $\mathbb{P}_{\mathbb{R}}$ is therefore not closed under the product, whereas $\mathbb{P}_{\mathbb{C}}$ is. Equivalently, $X \in \mathbb{P}_{\mathbb{R}}$ exactly when $X^H = X$.

With this basis fixed, for real scalars $a, b \in \mathbb{R}$ and real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, with $\mathbf{A} \cdot \boldsymbol{\sigma} = A_1\sigma_1 + A_2\sigma_2 + A_3\sigma_3$, the real paravectors $X, Y \in \mathbb{P}_{\mathbb{R}}$ are defined by

$$X := a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (14)$$

$$Y := b + \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (15)$$

The complex scalar and vector components are defined by

$$S := a + ib \in \mathbb{C}, \quad (16)$$

$$\mathbf{V} := \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3. \quad (17)$$

Every algebra element can then be defined as a complex paravector $Z \in \mathbb{P}_{\mathbb{C}}$, written as a real paravector plus an imaginary paravector:

$$Z := X + iY. \quad (18)$$

The equivalent real-component and scalar–vector forms are

$$Z = (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (19)$$

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}. \quad (20)$$

On coefficients alone, the actions inherited from either basic operation are $S^* = S^H = a - ib$ and $\mathbf{V}^* = \mathbf{V}^H = \mathbf{A} - i\mathbf{B}$. When either operation acts on the full algebra term $\mathbf{V} \cdot \boldsymbol{\sigma}$, its action on the sigma basis is included as well, according to Eqs. (7) and (8). The real function is defined by

$$\text{real}(Z) := X = \frac{Z + Z^H}{2}. \quad (21)$$

The imaginary function is defined by

$$\text{img}(Z) := Y = \frac{Z - Z^H}{2i}. \quad (22)$$

The scalar function is defined by

$$\text{scalar}(Z) := S = \frac{Z + Z^{*H}}{2}. \quad (23)$$

The vector function is defined by

$$\text{vector}(Z) := \mathbf{V}. \quad (24)$$

The algebra element carried by that vector is recovered as

$$\text{vector}(Z) \cdot \boldsymbol{\sigma} = \frac{Z - Z^{*H}}{2}. \quad (25)$$

The zero three-vector and the two paravector identity elements can now be defined without introducing another kind of object:

$$\mathbf{0} := \{0, 0, 0\}, \quad 0 := 0 + \mathbf{0} \cdot \boldsymbol{\sigma}, \quad 1 := 1 + \mathbf{0} \cdot \boldsymbol{\sigma}. \quad (26)$$

Their scalar and vector parts are

$$\begin{aligned} \text{scalar}(0) &= 0, & \text{vector}(0) &= \mathbf{0}, \\ \text{scalar}(1) &= 1, & \text{vector}(1) &= \mathbf{0}. \end{aligned} \quad (27)$$

Their defining actions on every paravector Z are

$$Z + 0 = 0 + Z = Z, \quad Z1 = 1Z = Z. \quad (28)$$

Thus 0 is the additive identity and 1 is the multiplicative identity. An inverse, when it exists, is defined by a product equal to 1. The

actions of the conjugate and Hermitian operators on the generic complex paravector are

$$Z^* = (a - ib) + (-\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (29)$$

$$Z^H = X - iY = (a - ib) + (\mathbf{A} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad (30)$$

$$Z^{*H} = (a + ib) - (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (31)$$

For the real paravector X from Eq. (14), these are reduced to

$$X^* = a - \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (32)$$

$$X^H = X = a + \mathbf{A} \cdot \boldsymbol{\sigma}, \quad (33)$$

$$X^{*H} = a - \mathbf{A} \cdot \boldsymbol{\sigma} = X^*. \quad (34)$$

Equations (29)–(31) are the generic formulas used below. In particular, conjugation of the complex coefficients is already encoded by H ; no overbar or additional intrinsic operation is needed. Equations (32)–(34) are the real specialization. The real and complex paravector pieces and the functions that select them are fixed by Eqs. (14)–(24). The type assignments are recorded in Table 3, the recovery formulas for the selection functions are recorded in Table 4, and the same split is recorded in components in Table 5.

Table 3. Type assignments for the generic complex paravector notation.

Type name	Underlying set	Example
real scalar	$\mathbb{R}$	a, b
real vector	$\mathbb{R}^3$	$\mathbf{A}, \mathbf{B}$
complex scalar	$\mathbb{C}$	S
complex vector	$\mathbb{C}^3$	$\mathbf{V}$
real paravector	$\mathbb{P}_{\mathbb{R}}$	X, Y
complex paravector	$\mathbb{P}_{\mathbb{C}}$	Z

Table 4. Components of Z .

scalar:	$S = (Z + Z^{*H})/2$
vector:	$\mathbf{V} \cdot \boldsymbol{\sigma} = (Z - Z^{*H})/2$
real:	$X = (Z + Z^H)/2$
img:	$Y = (Z - Z^H)/(2i)$

Table 5. Real and imaginary scalar/vector parts of Z .

real scalar:	$a = (S + S^H)/2$ $= (X + X^*)/2$
img scalar:	$b = (S - S^H)/(2i)$ $= (Y + Y^*)/2$
real vector:	$\mathbf{A} \cdot \boldsymbol{\sigma} = (\mathbf{V} + \mathbf{V}^H) \cdot \boldsymbol{\sigma}/2$ $= (X - X^*)/2$
img vector:	$\mathbf{B} \cdot \boldsymbol{\sigma} = (\mathbf{V} - \mathbf{V}^H) \cdot \boldsymbol{\sigma}/(2i)$ $= (Y - Y^*)/2$

1.4 Dot products and cross products

For real vectors $\mathbf{A}, \mathbf{B} \in \mathbb{R}^3$, the dot product is defined by

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &:= \sum_{n=1}^3 \sum_{m=1}^3 \delta_{nm} A_n B_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3. \end{aligned} \quad (35)$$

The square of a vector is defined by

$$\mathbf{A}^2 := \mathbf{A} \cdot \mathbf{A}. \quad (36)$$

For the cross product, the Levi-Civita form is

$$(\mathbf{A} \times \mathbf{B})_k := \sum_{n=1}^3 \sum_{m=1}^3 \epsilon_{nmk} A_n B_m, \quad (37)$$

from which the vector is obtained:

$$\mathbf{A} \times \mathbf{B} = \begin{Bmatrix} A_2 B_3 - A_3 B_2, \\ A_3 B_1 - A_1 B_3, \\ A_1 B_2 - A_2 B_1 \end{Bmatrix}. \quad (38)$$

Dot products are commutative, whereas cross products are anti-symmetric:

$$\begin{aligned} \mathbf{A} \cdot \mathbf{B} &= +\mathbf{B} \cdot \mathbf{A}, \\ \mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}. \end{aligned}$$

For a fixed unit vector $\mathbf{u}$, where $\mathbf{u}^2 = 1$, the parallel component function is defined by

$$\text{par}(\mathbf{A}) := (\mathbf{A} \cdot \mathbf{u})\mathbf{u}. \quad (39)$$

The perpendicular component function is defined by

$$\text{prp}(\mathbf{A}) := \mathbf{A} - \text{par}(\mathbf{A}). \quad (40)$$

With Eqs. (3) and (4),

$$\begin{aligned} (\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) &= \sum_{n=1}^3 \sum_{m=1}^3 A_n B_m \sigma_n \sigma_m \\ &= \sum_{n=1}^3 A_n B_n \\ &\quad + \sum_{n < m} (A_n B_m - A_m B_n) \sigma_n \sigma_m \\ &= A_1 B_1 + A_2 B_2 + A_3 B_3 \\ &\quad + (A_2 B_3 - A_3 B_2) i \sigma_1 \\ &\quad - (A_1 B_3 - A_3 B_1) i \sigma_2 \\ &\quad + (A_1 B_2 - A_2 B_1) i \sigma_3. \end{aligned}$$

This can be efficiently expressed in terms of the dot product and cross product as

$$(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) = \mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (41)$$

The same product is recorded in component form in Table 6. For vectors in $\mathbb{C}^3$, the dot and cross products are extended linearly in each input, without complex conjugation. Formula (41) may also be used for commuting scalar-valued functions. A separate ordered calculation is required when a coefficient contains a derivative, because the ordinary derivative product rule acts on everything to its right.

Only the antisymmetric term is flipped by the reversed order:

$$\begin{aligned}(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \mathbf{A} \cdot \mathbf{B} - i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) + (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2\mathbf{A} \cdot \mathbf{B}, \\(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma}) - (\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{A} \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The following identities are obtained immediately from the same product:

$$\begin{aligned}(\mathbf{A} \cdot \boldsymbol{\sigma})^2 &= \mathbf{A}^2, \\(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{A}^2)^n, \\(\mathbf{A} \cdot \boldsymbol{\sigma})^{2n+1} &= (\mathbf{A}^2)^n (\mathbf{A} \cdot \boldsymbol{\sigma}), \\(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} &= \mathbf{A} - i\mathbf{A} \times \boldsymbol{\sigma}.\end{aligned}$$

These are the basic sigma-product identities from which the later paravector formulas are built.

1.5 Products

With the scalar/vector split for real vectors fixed by Eq. (41), the fully complex product is obtained as the master formula from which the simpler cases are recovered by restriction.

For Z from Eq. (20) and $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$ defined analogously to Eqs. (20) and (17),

$$ZZ' = (SS' + \mathbf{V} \cdot \mathbf{V}') + (S\mathbf{V}' + S'\mathbf{V} + i(\mathbf{V} \times \mathbf{V}')) \cdot \boldsymbol{\sigma}. \quad (42)$$

The generic complex paravector and its master multiplication rule are fixed by Eqs. (20) and (42): pure scalar, real vector, real paravector, and mixed products are all obtained by setting the unused parts to zero.

1.5.1 Order functions

For arbitrary paravectors $A, B \in \mathbb{P}$, the commutator function is defined by

$$\text{com}(A, B) := AB - BA. \quad (43)$$

The anticommutator function is defined independently by

$$\text{acom}(A, B) := AB + BA. \quad (44)$$

1.5.2 Products of real scalars, vectors, and paravectors

For the real paravectors X and Y from Eqs. (14) and (15), Eq. (41) gives

$$\begin{aligned}XY &= (a + \mathbf{A} \cdot \boldsymbol{\sigma})(b + \mathbf{B} \cdot \boldsymbol{\sigma}) \\&= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} + i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma}, \\YX &= (ab + \mathbf{A} \cdot \mathbf{B}) + (a\mathbf{B} + b\mathbf{A} - i(\mathbf{A} \times \mathbf{B})) \cdot \boldsymbol{\sigma} \\&= (XY)^{\text{H}}.\end{aligned}$$

The same product is given in component form in Table 7. The scalar part is symmetric, while the sign of the imaginary-vector part is changed by the order.

Two immediate consequences are

$$\begin{aligned}\text{com}(X, Y) &= 2i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\XY \in \mathbb{P}_{\mathbb{R}} &\text{ if and only if } \mathbf{A} \times \mathbf{B} = 0.\end{aligned}$$

Thus exact commutation of two real paravectors is obtained precisely when $\mathbf{A} \times \mathbf{B} = 0$, including either zero-vector case. For two nonzero vector parts, this means that they are parallel or antiparallel.

The antisymmetric term is removed automatically by setting $Y = X$:

$$X^2 = a^2 + \mathbf{A}^2 + 2a\mathbf{A} \cdot \boldsymbol{\sigma}.$$

If $X^2 = b + \mathbf{B} \cdot \boldsymbol{\sigma}$, the following relations are obtained by matching scalar and vector parts:

$$\begin{aligned}b &= a^2 + \mathbf{A}^2, \\ \mathbf{B} &= 2a\mathbf{A},\end{aligned}$$

hence

$$\begin{aligned}a^2 &= \frac{1}{2} \left(b \pm \sqrt{b^2 - \mathbf{B}^2} \right), \\ \mathbf{A}^2 &= b - a^2.\end{aligned}$$

These equations are necessary, but the direction equation $\mathbf{B} = 2a\mathbf{A}$ must also be retained. When $\mathbf{B} \neq \mathbf{0}$, real paravector roots exist precisely when $b \geq \|\mathbf{B}\|$. For either choice of the inner sign, either sign of the resulting nonzero a may be chosen, with

$$\mathbf{A} = \frac{\mathbf{B}}{2a}.$$

When $\mathbf{B} = \mathbf{0}$, a real root requires $b \geq 0$; the scalar roots are $X = \pm\sqrt{b}$, and the exceptional pure-vector roots satisfy $a = 0$ and $\mathbf{A}^2 = b$. Thus the displayed quadratic relation selects among possible square-root values but is not, by itself, a complete square-root formula. Such a selection among the possible values of a multivalued function is called a branch choice.

1.5.3 Products of complex paravectors

The master product law (42) is the complex extension of the real formulas. When Z is written as in Eqs. (20)–(19) and $Z' = X' + iY' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, every special case is obtained by setting the unused scalar or vector parts to zero. The quantities S, S' and the components of $\mathbf{V}, \mathbf{V}'$ may be passed through one another and through every sigma generator. Complex scalars with this property are called central. The full components of ZZ' are given in Table 8, those of Z^2 are given in Table 9, and the order-sensitive products built from star and H are recorded in Tables 10–15.

For products with ordinary scalar coefficients, exchanging the order of two factors does not change the extracted scalar:

$$\text{scalar}(ZZ') = \text{scalar}(Z'Z).$$

This order-independence of the scalar part is conventionally called cyclicity. In particular, if $Z' = \mathcal{O}Z^{*\text{H}}$, then

$$\begin{aligned}\text{scalar}(ZOZ^{*\text{H}}) &= \text{scalar}(\mathcal{O}Z^{*\text{H}}Z) \\ &= \text{scalar}(\mathcal{O})(Z^{*\text{H}}Z),\end{aligned}$$

because $Z^{*\text{H}}Z$ is scalar.

The order-sensitive variants are divided into three algebraic types:

ZZ^* is a real scalar plus a complex vector in $\mathbb{P}_{\mathbb{C}}$,

ZZ^{H} is a real scalar plus a real vector in $\mathbb{P}_{\mathbb{R}}$,

$ZZ^{*\text{H}}$ is a complex scalar in $\mathbb{P}_{\mathbb{C}}$.

The positive norms, determinant forms, inverses, and transformation formulas are supplied by those three cases.

If a component is instead a derivative operation or another quantity whose order cannot be exchanged, the ordered expansion must be used before any simplification: in general $\mathbf{V}S' \neq S'\mathbf{V}$, terms from the ordinary derivative product rule may occur, and cyclicity of the algebraic scalar projection alone does not justify cycling the operator coefficients. The displayed ordering is therefore required when the factors act on functions. Such results cannot be obtained by automatic substitution into the central-coefficient formulas.

The basic product decompositions and order-dependent component expansions are collected in the following tables.

Table 6. Components of $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$.

real scalar:	$\mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$\mathbf{0}$
img vector:	$\mathbf{A} \times \mathbf{B}$

Table 7. Components of XY .

real scalar:	$ab + \mathbf{A} \cdot \mathbf{B}$
img scalar:	0
real vector:	$a\mathbf{B} + b\mathbf{A}$
img vector:	$\mathbf{A} \times \mathbf{B}$

The full component expansions for the generic product ZZ' and the special case Z^2 are collected in Tables 8 and 9.

Table 8. Components of ZZ' .

real scalar:	$aa' - bb' + \mathbf{A} \cdot \mathbf{A}' - \mathbf{B} \cdot \mathbf{B}'$
img scalar:	$ab' + ba' + \mathbf{A} \cdot \mathbf{B}' + \mathbf{B} \cdot \mathbf{A}'$
real vector:	$a\mathbf{A}' - b\mathbf{B}' + a'\mathbf{A} - b'\mathbf{B} - \mathbf{A} \times \mathbf{B}' - \mathbf{B} \times \mathbf{A}'$
img vector:	$a\mathbf{B}' + b\mathbf{A}' + a'\mathbf{B} + b'\mathbf{A} + \mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}'$

Table 9. Components of Z^2 .

real scalar:	$a^2 - b^2 + \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	$2ab + 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$2(a\mathbf{A} - b\mathbf{B})$
img vector:	$2(a\mathbf{B} + b\mathbf{A})$

The order dependence of ZZ^* and Z^*Z is made explicit in Tables 10 and 11.

Table 10. Components of ZZ^* .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$-2\mathbf{A} \times \mathbf{B}$ ← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Table 11. Components of Z^*Z .

real scalar:	$a^2 + b^2 - \mathbf{A}^2 - \mathbf{B}^2$
img scalar:	0
real vector:	$2\mathbf{A} \times \mathbf{B}$ ← differs
img vector:	$2(a\mathbf{B} - b\mathbf{A})$

Products involving the Hermitian operation are treated analogously in Tables 12 and 13.

Table 12. Components of ZZ^{H} .

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} + \mathbf{A} \times \mathbf{B})$ ← differs
img vector:	$\mathbf{0}$

Table 13. Components of $Z^{\text{H}}Z$.

real scalar:	$a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	0
real vector:	$2(a\mathbf{A} + b\mathbf{B} - \mathbf{A} \times \mathbf{B})$ ← differs
img vector:	$\mathbf{0}$

The collapse of products formed with the star-H operation to purely scalar and imaginary-scalar parts is shown in Tables 14 and 15.

Table 14. Components of $ZZ^{*\text{H}}$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 15. Components of $Z^{*\text{H}}Z$.

real scalar:	$a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2$
img scalar:	$2ab - 2\mathbf{A} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

1.5.4 Order dependence and commutators

The order-dependent part of a product is represented by the commutator function defined in Eq. (43). For generic complex paravectors $Z, Z' \in \mathbb{P}_{\mathbb{C}}$,

$$\text{com}(Z, Z') = 2i(\mathbf{V} \times \mathbf{V}') \cdot \sigma \quad (45)$$

Equation (45) is the compact commutator law. In real and imaginary vector components, the following expansion is obtained:

$$\begin{aligned} \text{com}(Z, Z') &= -2(\mathbf{A} \times \mathbf{B}' + \mathbf{B} \times \mathbf{A}') \cdot \sigma \\ &\quad + 2i(\mathbf{A} \times \mathbf{A}' - \mathbf{B} \times \mathbf{B}') \cdot \sigma. \end{aligned}$$

The real-paravector commutator

$$\text{com}(X, Y) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \sigma$$

is the corresponding special case. In every case the scalar terms cancel, so the cross-product part of the multiplication law is isolated by the commutator. The following identity is obtained by multiplying the n th basis commutator by B_n and summing:

$$\sum_{n=1}^3 B_n \text{com}(\mathbf{A} \cdot \sigma, \sigma_n) = \text{com}(\mathbf{A} \cdot \sigma, \mathbf{B} \cdot \sigma) = 2i(\mathbf{A} \times \mathbf{B}) \cdot \sigma. \quad (46)$$

This is the component bridge between the vector cross-product law and the commutator correction when the ordinary derivative product rule is applied to a paravector product.

The symmetric companion is the anticommutator from Eq. (44). For pure vectors,

$$\begin{aligned} \text{acom}(\mathbf{A} \cdot \sigma, \mathbf{B} \cdot \sigma) &= (\mathbf{A} \cdot \sigma)(\mathbf{B} \cdot \sigma) \\ &\quad + (\mathbf{B} \cdot \sigma)(\mathbf{A} \cdot \sigma) \\ &= 2\mathbf{A} \cdot \mathbf{B}. \end{aligned} \quad (47)$$

Thus the symmetric dot-product part is isolated by $\text{acom}(\cdot, \cdot)$, just as the antisymmetric cross-product part is isolated by $\text{com}(\cdot, \cdot)$.

More generally, for generic complex paravectors $Z, Z' \in \mathbb{P}_{\mathbb{C}}$, Eq. (44) gives

$$\text{acom}(Z, Z') = ZZ' + Z'Z. \quad (48)$$

The anticommutator is therefore

$$\text{acom}(Z, Z') = 2(SS' + \mathbf{V} \cdot \mathbf{V}') + 2(S\mathbf{V}' + S'\mathbf{V}) \cdot \sigma. \quad (49)$$

The cross-product contribution to the anticommutator cancels identically. Only the symmetric scalar and vector couplings are retained.

The same order dependence is already present in the sigma basis itself. Commutation of the pseudoscalar with the generators is expressed by

$$\begin{aligned} \text{com}(i, \sigma_n) &= 0, \\ i\sigma &= \sigma i = \{\sigma_2\sigma_3, \sigma_3\sigma_1, \sigma_1\sigma_2\}, \end{aligned}$$

while the following relation is satisfied by distinct sigma generators:

$$\text{com}(\sigma_n, \sigma_m) = 2\sigma_n\sigma_m = 2 \sum_{k=1}^3 \epsilon_{nmk} i\sigma_k \quad (n \neq m).$$

Thus commutators of the sigma-number-one generators reproduce the entire oriented sigma-number-two basis.

The same product types are displayed in full component form in Tables 12–15. Their positive-norm and determinant-form consequences are fixed by Eqs. (52), (60), and (58); no additional quadratic definitions are required.

1.6 Positive Norm, Adjugate, and Determinant

Two distinct expressions formed from two copies of a paravector are used: a positive coefficient norm and a generally complex determinant. For a complex scalar $S = a + ib \in \mathbb{C}$, the scalar norm is defined by

$$|S| := \sqrt{SS^*}. \quad (50)$$

Its squared value is

$$\begin{aligned} |S|^2 &= SS^* \\ &= a^2 + b^2. \end{aligned}$$

If $S \in \mathbb{R}$, then the ordinary absolute value $|S| = \sqrt{S^2}$ is obtained from Eq. (50). For a complex vector $\mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3$, the vector norm is defined by

$$\|\mathbf{V}\| := \sqrt{\mathbf{V} \cdot \mathbf{V}^*}. \quad (51)$$

The corresponding vector-norm square is

$$\begin{aligned} \|\mathbf{V}\|^2 &= \mathbf{V} \cdot \mathbf{V}^* \\ &= \mathbf{A}^2 + \mathbf{B}^2. \end{aligned}$$

The quadratic product $\mathbf{V}^2 = \mathbf{V} \cdot \mathbf{V}$ is not the same object as the positive norm square $\mathbf{V} \cdot \mathbf{V}^*$. For the complex paravector Z from Eqs. (20)–(19), the paravector norm is defined by

$$\|Z\| := \sqrt{|S|^2 + \|\mathbf{V}\|^2}. \quad (52)$$

Its square is reduced directly to the coefficient norm:

$$\begin{aligned} \|Z\|^2 &= |S|^2 + \|\mathbf{V}\|^2 \\ &= a^2 + b^2 + \mathbf{A}^2 + \mathbf{B}^2 \\ &= \text{scalar}(ZZ^H) = \text{scalar}(Z^H Z). \end{aligned} \quad (53)$$

This is a nonnegative real norm: it is the sum of the squared magnitudes of the scalar and vector coefficients.

The determinant is obtained through construction of the multiplicative inverse. For $Z = S + \mathbf{V} \cdot \sigma$, the adjugate is obtained by reversing the sign of the vector part:

$$\text{adj}(Z) := Z^{*H} = S - \mathbf{V} \cdot \sigma. \quad (54)$$

Because the two basic operations commute and each returns the original element when applied twice, the following useful consistency chain is also obtained from Eq. (54):

$$Z^H = (Z^{*H})^* = (Z^*)^{*H} = \text{adj}(Z)^* = \text{adj}(Z^*). \quad (55)$$

In the compact superscript notation, this is $Z^H = (Z^{*H})^* = (Z^*)^{*H} = \text{adj}(Z)^* = \text{adj}(Z^*)$. Because S is central and $(\mathbf{V} \cdot \boldsymbol{\sigma})^2 = \mathbf{V}^2$, the vector terms are canceled by multiplication by the adjugate:

$$\begin{aligned} Z \text{adj}(Z) &= (S + \mathbf{V} \cdot \boldsymbol{\sigma})(S - \mathbf{V} \cdot \boldsymbol{\sigma}) \\ &= S^2 - \mathbf{V}^2 \\ &= \text{adj}(Z) Z. \end{aligned} \quad (56)$$

The remaining central scalar is exactly the divisor required to normalize the product to the multiplicative identity 1. The determinant function is therefore defined by

$$\det(Z) := \text{adj}(Z) Z. \quad (57)$$

Equation (56) shows that the order of the two factors does not matter. The determinant is reduced to

$$\det(Z) = Z \text{adj}(Z) = S^2 - \mathbf{V}^2. \quad (58)$$

In real and imaginary components, the same result is

$$\det(Z) = (a^2 - b^2 - \mathbf{A}^2 + \mathbf{B}^2) + 2i(ab - \mathbf{A} \cdot \mathbf{B}). \quad (59)$$

The name determinant is assigned because this central scalar is exactly the divisor required by the inverse in Eq. (65). Its agreement with the ordinary two-by-two determinant is verified independently in Appendix A. No square root is included in the definition.

For the real paravector X from Eq. (14), $X^{*H} = X^*$, and the determinant is restricted to the Minkowski quadratic form

$$\begin{aligned} \det(X) &= a^2 - \mathbf{A}^2 \\ &= XX^* = X^* X. \end{aligned} \quad (60)$$

Product order is reversed by the adjugate, so for complex paravectors Z and Z' ,

$$\begin{aligned} \text{adj}(\text{adj}(Z)) &= Z, \\ \text{adj}(ZZ') &= \text{adj}(Z') \text{adj}(Z), \\ \det(\text{adj}(Z)) &= \det(Z). \end{aligned} \quad (61)$$

Consequently, the determinant of a product is the product of the two determinants, even when the two paravectors do not commute:

$$\begin{aligned} \det(ZZ') &= ZZ' \text{adj}(Z') \text{adj}(Z) \\ &= Z \det(Z') \text{adj}(Z) \\ &= \det(Z') \det(Z) \\ &= \det(Z) \det(Z') = \det(Z'Z). \end{aligned} \quad (62)$$

For a real θ , the factor $\exp(i\theta)$ has magnitude one and is called a phase factor. Under multiplication by this factor, the positive norm and determinant are transformed as

$$\begin{aligned} \|\exp(i\theta)Z\|^2 &= \|Z\|^2, \\ \det(\exp(i\theta)Z) &= \exp(2i\theta) \det(Z). \end{aligned}$$

More generally, under complex scalar multiplication by $r \exp(i\theta)$ with $r \in \mathbb{R}$, the two quadratic constructions are scaled differently:

$$\|r \exp(i\theta)Z\|^2 = r^2 \|Z\|^2, \quad (63)$$

$$\det(r \exp(i\theta)Z) = r^2 \exp(2i\theta) \det(Z). \quad (64)$$

The positive norm is unchanged by a phase factor, while the determinant retains that phase factor squared. The term ‘‘Minkowski metric’’ below is reserved for the real-paravector expression in Eq. (60) [6].

1.7 Inverse formulas

The inverse construction is completed by Eqs. (56) and (58). For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, when $\det(Z) \neq 0$, the inverse is defined by

$$Z^{-1} := \frac{\text{adj}(Z)}{\det(Z)}. \quad (65)$$

The equivalent explicit forms are

$$Z^{-1} = \frac{Z^{*H}}{ZZ^{*H}} = \frac{Z^{*H}}{Z^{*H}Z} = \frac{S - \mathbf{V} \cdot \boldsymbol{\sigma}}{S^2 - \mathbf{V}^2}. \quad (66)$$

If $\det(Z) = 0$, the inverse does not exist.

By differentiation of the identity $Z^{-1}Z = 1$, the following result is obtained:

$$d(Z^{-1}) = -Z^{-1}(dZ)Z^{-1}. \quad (67)$$

For $Z' = S' + \mathbf{V}' \cdot \boldsymbol{\sigma}$, if $\mathbf{V} \times \mathbf{V}' = \mathbf{0}$, then

$$Z^{-1}Z' = Z'Z^{-1}. \quad (68)$$

1.8 Supporting Algebraic Derivations

The explicit sigma-basis and operator calculations, together with one generic map decomposition, are recorded in the supporting derivations without repetition of the product development already collected above.

1.8.1 Basis-Element Classification

The four sigma numbers and corresponding basis-element types are listed in Table 1. The signs acquired under the conjugate, Hermitian, and combined star-H operations are listed in Table 2.

1.8.2 Complex Paravector-Valued Maps

A map is a rule that assigns an output to each allowed input. A generic map whose output is a complex paravector is formed from real scalar and vector functions without imposing any physical interpretation:

$$\begin{aligned} Z(t, \mathbf{r}) &:= (a(t, \mathbf{r}) + i b(t, \mathbf{r})) \\ &\quad + (\mathbf{A}(t, \mathbf{r}) + i \mathbf{B}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (69)$$

Its allowed inputs and coefficient-function outputs are

$$\begin{aligned} a, b &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{A}, \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \\ Z &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{P}_{\mathbb{C}}. \end{aligned}$$

1.9 Projector Algebra

A projector is an element that is unchanged when squared. Projectors are introduced next so that the later function formulas and the elements that exchange their two parts can be organized. Once Eqs. (4) and (41) are fixed, simple unit-direction elements can be used to split a paravector into two parts.

An algebraic unit direction is any $\mathbf{u} \in \mathbb{C}^3$ satisfying $\mathbf{u}^2 = 1$. For a real physical axis obtained from a nonzero $\mathbf{A} \in \mathbb{R}^3$, the direction is defined before the projector by

$$\mathbf{u} := \frac{\mathbf{A}}{\|\mathbf{A}\|}, \quad \mathbf{u}^2 = 1. \quad (70)$$

The projector function for the chosen direction is then defined by

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (71)$$

The corresponding left-projected part of any algebra element Z is defined by

$$Z(\pm) := \Pi(\pm)Z. \quad (72)$$

The identities $\Pi(\pm)^2 = \Pi(\pm)$, $\Pi(+)\Pi(-) = 0$, $\Pi(+)+\Pi(-) = 1$, $Z = Z(+) + Z(-)$, and $(\mathbf{u} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\Pi(\pm)$ are satisfied. Thus each projector is unchanged when squared, the two projectors annihilate one another, and their sum is the identity. These properties are conventionally called idempotence, orthogonality, and completeness, respectively [19]. Algebraically, the parts aligned and anti-aligned with the direction $\mathbf{u}$ are isolated by $\Pi(+)$ and $\Pi(-)$. Their role is purely structural: an object is split into two parts adapted to a chosen axis.

For a scalar-valued function $f : \{+1, -1\} \rightarrow \mathbb{C}$,

$$f(\mathbf{u} \cdot \boldsymbol{\sigma}) := f(+1)\Pi(+) + f(-1)\Pi(-). \quad (73)$$

A compact projector-based formula for roots, inverses, exponentials, logarithms, and related functions is thereby obtained. If an element satisfies a polynomial equation, that equation can be used to replace its higher powers by lower powers. The lowest-degree such equation is called its minimal polynomial. When its two scalar roots are distinct, the function is obtained by evaluating it at those roots. When the roots coincide, the limiting formula also contains the derivative of the function. These statements are derived explicitly below rather than assumed.

For the ladder construction, the chosen direction is specialized to the third axis:

$$\begin{aligned} \mathbf{u} &:= \mathbf{e}_3 = \{0, 0, 1\}, \quad \mathbf{u}^2 = 1, \\ \mathbf{u} \cdot \boldsymbol{\sigma} &= \sigma_3, \quad \Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(1 \pm \sigma_3). \end{aligned} \quad (74)$$

The ladder elements about this axis are defined by

$$a(\pm) := \frac{1}{2}(\sigma_1 \pm i\sigma_2). \quad (75)$$

The associated ordered-difference, zero-square, and product identities are

$$\begin{aligned} \text{com}(\sigma_3, a(\pm)) &= \pm 2a(\pm), \quad a(\pm)^2 = 0, \\ a(+)a(-) &= \Pi(+), \quad a(-)a(+) = \Pi(-). \end{aligned} \quad (76)$$

Equation (76) shows that each ladder element has square zero; such an element is conventionally called nilpotent. These combinations of σ_1 and σ_2 exchange the two projector parts selected by Eq. (74). This third-axis specialization applies only to the ladder identities; the supporting function formulas are restored to an arbitrary chosen unit direction. The corresponding projector values, mixed products, and ladder identities built from Eqs. (71), (73), and (75) are collected next.

1.9.1 Functions of $\mathbf{u} \cdot \boldsymbol{\sigma}$

For this subsection, a new algebraic unit direction $\mathbf{u}$, satisfying $\mathbf{u}^2 = 1$, is chosen. For a nonzero real vector $\mathbf{A}$, it is supplied by $\mathbf{u} := \mathbf{A}/\|\mathbf{A}\|$ in Eq. (70). Multiplication by $\mathbf{u} \cdot \boldsymbol{\sigma}$ gives $+1$ on one projector part and -1 on the other, as shown in Eq. (71). Consequently, a scalar function is applied by assigning its value at $+1$ to the first part and its value at -1 to the second. This is the projector-based function rule defined in Eq. (73):

$$\begin{aligned} f(\mathbf{u} \cdot \boldsymbol{\sigma}) &= f(+1)\Pi(+) + f(-1)\Pi(-) \\ &= \frac{1}{2}(f(+1) + f(-1)) + \frac{1}{2}(f(+1) - f(-1))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are the basic projector formulas for a unit-direction element. Square roots and logarithms can have more than one value. A consistent choice of one such value is called a branch. With the conventional scalar choices $\sqrt{+1} = 1$ and $\sqrt{-1} = i$, the resulting identities are

$$\begin{aligned} \sqrt{\mathbf{u} \cdot \boldsymbol{\sigma}} &= \sqrt{+1}\Pi(+) + \sqrt{-1}\Pi(-) \\ &= \Pi(+) + i\Pi(-), \\ \log(\mathbf{u} \cdot \boldsymbol{\sigma}) &= 2i\pi(n\Pi(+) + (\tfrac{1}{2} + m)\Pi(-)), \\ (z + \mathbf{u} \cdot \boldsymbol{\sigma})^{-1} &= \frac{1}{z+1}\Pi(+) + \frac{1}{z-1}\Pi(-) \\ &= \frac{z - \mathbf{u} \cdot \boldsymbol{\sigma}}{z^2 - 1}, \\ \frac{i - \mathbf{u} \cdot \boldsymbol{\sigma}}{i + \mathbf{u} \cdot \boldsymbol{\sigma}} &= i\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cosh(\theta) + \sinh(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}, \\ \exp(i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) &= \cos(\theta) + i\sin(\theta)\mathbf{u} \cdot \boldsymbol{\sigma}, \end{aligned}$$

where m and n are integers selecting the independent logarithm values, and the inverse formula requires $z \neq \pm 1$. Other square roots are obtained by independently changing the sign of either coefficient in the displayed square-root formula. The selected square root is verified by

$$\begin{aligned} (\Pi(+) + i\Pi(-))^2 &= \Pi(+) - \Pi(-) \\ &= \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

These identities are obtained by applying the chosen scalar value separately on the two projector-selected parts. The corresponding domains for real and complex paravectors are stated with the exponential and logarithm formulas.

1.9.2 Functions of X

The scalar, vector, and unit-direction data are specified by

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ t &= \text{scalar}(X) \in \mathbb{R}, \\ \mathbf{r} &= \text{vector}(X) \in \mathbb{R}^3, \\ \mathbf{u} &:= \mathbf{r}/\|\mathbf{r}\| \quad \text{when } \mathbf{r} \neq 0. \end{aligned}$$

If $\mathbf{r} \neq 0$, its decomposition is $\mathbf{r} = \|\mathbf{r}\|\mathbf{u}$, with $\mathbf{u}^2 = 1$. The two scalar values associated with the two projector parts are defined by

$$\lambda(\pm) := t \pm \|\mathbf{r}\|. \quad (77)$$

They are called eigenvalues because multiplication by X gives each projector back multiplied by the corresponding scalar, as verified below. For $\mathbf{r} \neq 0$, the quadratic relation is

$$\begin{aligned} X^2 - 2tX + (t^2 - \mathbf{r}^2) &= 0, \\ 0 &= (X - \lambda(-))(X - \lambda(+)) \\ &= X^2 - \lambda(-)X \\ &\quad - X\lambda(+) + \lambda(-)\lambda(+). \end{aligned}$$

No degree-one polynomial with scalar coefficients can vanish on this non-scalar X . The displayed quadratic is therefore the lowest-degree scalar-coefficient polynomial satisfied by X , which is its minimal polynomial. If $\mathbf{r} = \mathbf{0}$, then $X = t$, the minimal polynomial is instead $x - t$, and every function reduces directly to $f(X) = f(t)$. Here x is only the placeholder input of the polynomial. The projector formulas below apply to $\mathbf{r} \neq \mathbf{0}$. The projector function from Eq. (71) gives the corresponding decomposition directly. Its local direction is restated immediately before the projector:

$$\begin{aligned} \mathbf{u} &:= \mathbf{r}/\|\mathbf{r}\|, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ X &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm). \end{aligned}$$

If f is defined at the two eigenvalues, then

$$\begin{aligned} f(X) &= f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-) \\ &= \frac{1}{2}(f(\lambda(+)) + f(\lambda(-))) \\ &\quad + \frac{1}{2}(f(\lambda(+)) - f(\lambda(-)))\mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Its scalar and vector parts are therefore

$$\begin{aligned} \text{scalar}(f(X)) &= \frac{1}{2}(f(\lambda(+)) + f(\lambda(-))), \\ \text{vector}(f(X)) &= \frac{1}{2}(f(\lambda(+)) - f(\lambda(-)))\mathbf{u}. \end{aligned}$$

For complex-valued f , both parts may be complex. The projector summary obtained from Eqs. (77) and (71) may therefore be

recorded as

$$\begin{aligned} \lambda(\pm) &= t \pm \|\mathbf{r}\|, \\ \mathbf{u} &= \mathbf{r}/\|\mathbf{r}\|, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \\ \Pi(\pm)X &= \lambda(\pm)\Pi(\pm), \\ X &= \lambda(+)\Pi(+) + \lambda(-)\Pi(-), \\ f(X) &= f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-). \end{aligned}$$

The standard examples are

$$\begin{aligned} \exp(X) &= \exp(t) \left(\cosh(\|\mathbf{r}\|) + \sinh(\|\mathbf{r}\|) \mathbf{u} \cdot \boldsymbol{\sigma} \right), \\ \sqrt{X} &= \sqrt{\lambda(+)} \Pi(+) + \sqrt{\lambda(-)} \Pi(-), \\ \log(X) &= \log(\lambda(+)) \Pi(+) + \log(\lambda(-)) \Pi(-), \\ X^{-1} &= \frac{\Pi(+)}{\lambda(+)} + \frac{\Pi(-)}{\lambda(-)} \\ &= \frac{\text{adj}(X)}{\det(X)} = \frac{X^*}{t^2 - \mathbf{r}^2}, \\ X^n &= \lambda(+)^n \Pi(+) + \lambda(-)^n \Pi(-) \\ &= \frac{1}{2}(\lambda(+)^n + \lambda(-)^n) \\ &\quad + \frac{1}{2}(\lambda(+)^n - \lambda(-)^n) \mathbf{u} \cdot \boldsymbol{\sigma}. \end{aligned}$$

One scalar square root of each $\lambda(\pm)$ is selected in the displayed square-root formula. The other square roots are obtained by independently changing the sign of either selected scalar root. The scalar logarithms must likewise be taken with specified choices of their possible values. A square root that is itself a real paravector exists when $\lambda(\pm) \geq 0$, equivalently $t \geq \|\mathbf{r}\|$; otherwise the displayed roots are complex. The displayed logarithm is defined only when $\lambda(\pm) \neq 0$. When both values are positive, the ordinary single-valued real logarithm is used; this choice is conventionally called the real principal logarithm. The condition $\lambda(+)\lambda(-) \neq 0$ is required for the inverse. The power formula is valid for $n \geq 0$, and also for negative integers when $\lambda(+)\lambda(-) \neq 0$. For two paravectors written with projector pairs along possibly different axes, the second real paravector and its local eigenvalues are defined by

$$X' = t' + \mathbf{r}' \cdot \boldsymbol{\sigma}, \quad \lambda'(\pm) := t' \pm \|\mathbf{r}'\|. \quad (78)$$

When $\mathbf{r}' \neq \mathbf{0}$, its unit axis is $\mathbf{u}' := \mathbf{r}'/\|\mathbf{r}'\|$, with $(\mathbf{u}')^2 = 1$. The second projector pair is the full specialization of Eq. (71),

$$\Pi'(\pm) := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}). \quad (79)$$

Thus $\lambda(\pm)$ belongs to X , while the prime on $\lambda'(\pm)$ identifies the corresponding distinct values for X' , in accordance with Eq. (2). Applying the function rule separately to the two paravectors gives

$$\begin{aligned} f(X)g(X') &= \left(f(\lambda(+))\Pi(+) + f(\lambda(-))\Pi(-) \right) \\ &\quad \times \left(g(\lambda'(+))\Pi'(+) + g(\lambda'(-))\Pi'(-) \right), \end{aligned}$$

so

$$\begin{aligned} f(X)g(X') &= f(\lambda(+))g(\lambda'(+))\Pi(+)\Pi'(+)) \\ &\quad + f(\lambda(+))g(\lambda'(-))\Pi(+)\Pi'(-) \\ &\quad + f(\lambda(-))g(\lambda'(+))\Pi(-)\Pi'(+)) \\ &\quad + f(\lambda(-))g(\lambda'(-))\Pi(-)\Pi'(-). \end{aligned}$$

The mixed projector products are

$$\begin{aligned} \Pi(+)\Pi'(+)) &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' + \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(+)\Pi'(-) &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' + \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-)\Pi'(+)) &= \frac{1}{4}(1 - \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(\mathbf{u}' - \mathbf{u} - i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}, \\ \Pi(-)\Pi'(-) &= \frac{1}{4}(1 + \mathbf{u} \cdot \mathbf{u}') \\ &\quad + \frac{1}{4}(-\mathbf{u}' - \mathbf{u} + i(\mathbf{u} \times \mathbf{u}')) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The hyperbolic tangent $\tanh(x)$ is the ratio $\sinh(x)/\cosh(x)$. The function $\operatorname{arctanh}(x)$ is its inverse on the real interval $-1 < x < 1$. As a useful example, the following rational expression is itself a projector exponential:

$$\begin{aligned} \frac{1 - \mathbf{r} \cdot \boldsymbol{\sigma}}{1 + \mathbf{r} \cdot \boldsymbol{\sigma}} &= \exp(-2 \operatorname{arctanh}(\mathbf{r} \cdot \boldsymbol{\sigma})) \\ &= \exp(-2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(+)) \\ &\quad + \exp(+2 \operatorname{arctanh}(\|\mathbf{r}\|))\Pi(-). \end{aligned}$$

The rational expression is defined only when $\|\mathbf{r}\| \neq 1$. For a real-valued $\operatorname{arctanh}$ and the displayed real exponentials, the additional condition $\|\mathbf{r}\| < 1$ is required; otherwise a complex branch must be selected from the possible complex values. When $\mathbf{r} = \mathbf{0}$, the rational expression and exponential are both 1, and no axis is needed.

1.9.3 Quadratic Relation and Projector Decomposition

The function formulas, projector identities, and exponential identities are determined by the following quadratic relation for a complex paravector, as shown by Eqs. (42) and (73):

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{C}}, \\ Z^2 - 2SZ + (S^2 - \mathbf{V}^2) &= 0. \end{aligned}$$

When $\mathbf{V}^2 \neq 0$, a value z with $z^2 = \mathbf{V}^2$ is chosen. The local algebraic unit direction is then defined by

$$\mathbf{u} := \frac{\mathbf{V}}{z}, \quad \mathbf{u}^2 = 1. \quad (80)$$

For a real $\mathbf{V}$, the choice $z = \|\mathbf{V}\|$ reduces this definition to $\mathbf{u} = \mathbf{V}/\|\mathbf{V}\|$. For a general complex vector, z is required because the positive norm $\|\mathbf{V}\|^2 = \mathbf{V} \cdot \mathbf{V}^*$ need not equal $\mathbf{V}^2$. The two scalar values associated with the two projector parts are defined by

$$\lambda(\pm) := S \pm z. \quad (81)$$

The projector function from Eq. (71) is thereby specialized without adding the paravector as an input. The local direction is restated immediately before the projector:

$$\begin{aligned} \mathbf{u} &:= \frac{\mathbf{V}}{z}, \\ \Pi(\pm) &:= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned} \quad (82)$$

Then

$$\begin{aligned} (Z - \lambda(-))(Z - \lambda(+)) &= 0, \\ \Pi(\pm)Z &= \lambda(\pm)\Pi(\pm), \\ Z &= \lambda(+)\Pi(+)) \\ &\quad + \lambda(-)\Pi(-), \\ f(Z) &= f(\lambda(+))\Pi(+)) \\ &\quad + f(\lambda(-))\Pi(-) \end{aligned}$$

for any scalar function defined on the two values. Because multiplication by Z gives each projector back multiplied by its associated scalar, these scalars are the eigenvalues of Z . The two labels are merely exchanged when the sign of z is changed. Since $\mathbf{V}^2 \neq 0$ makes Z non-scalar, the displayed quadratic is its minimal polynomial.

The case $\mathbf{V}^2 = 0$ must be treated separately. If $\mathbf{V} = \mathbf{0}$, then $Z = S$ is scalar, its minimal polynomial is $x - S$, and $f(Z) = f(S)$. If $\mathbf{V} \neq \mathbf{0}$, define $N := \mathbf{V} \cdot \boldsymbol{\sigma}$. Then $N^2 = 0$. A nonzero element whose positive power is zero is called nilpotent. In this case $(Z - S)^2 = 0$, so the minimal polynomial is $(x - S)^2$. The two distinct projector parts have then merged into one repeated value, so the two-projector formula cannot be used. For a function with a convergent power series about S ,

$$f(Z) = f(S) + \left. \frac{df(x)}{dx} \right|_{x=S} N.$$

Every term containing N^2 or a higher power vanishes, which proves the formula directly. It is the exact replacement for the two-projector formula when the two roots become equal.

1.10 Exponential and Logarithmic Algebra

With the projector formulas fixed, exponential and logarithmic formulas can be developed in one place. Exponentials of elements whose square is $+1$ produce the $\cosh(\theta/2)$ and $\sinh(\theta/2)$ factors used for boosts. Exponentials of elements whose square is -1 produce the $\cos(\theta/2)$ and $\sin(\theta/2)$ factors used for rotations and for the quaternion correspondence.

1.10.1 Real Exponential and Half-Angle Elements

The sigma-number-one exponential is evaluated by applying the scalar exponential separately to the two projector parts defined in Eq. (73). For the real paravector

$$X = a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{R}},$$

the vector part is decomposed as

$$\begin{aligned}\mathbf{A} &= \theta \mathbf{u}, & \theta &= \|\mathbf{A}\|, \\ \mathbf{u} &= \frac{\mathbf{A}}{\|\mathbf{A}\|} \quad (\mathbf{A} \neq 0), & \mathbf{u} &= \{u_1, u_2, u_3\}, \\ \mathbf{u}^2 &= 1.\end{aligned}$$

When $\mathbf{A} = \mathbf{0}$, the vector exponential is 1 and no direction is needed. The directional formulas below apply to $\mathbf{A} \neq \mathbf{0}$. The exponential is therefore given by the projector-based function rule of Eq. (73):

$$\begin{aligned}\exp(\pm \mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm \theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\pm \theta) \Pi(+) + \exp(\mp \theta) \Pi(-) \\ &= \cosh(\theta) \pm \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}\quad (83)$$

An equivalent derivation is obtained from the power series by separating the even and odd powers:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \sum_{n \geq 0} \frac{\theta^n (\mathbf{u} \cdot \boldsymbol{\sigma})^n}{n!} \\ &= \sum_{n \geq 0} \frac{\theta^{2n}}{(2n)!} + \sum_{n \geq 0} \frac{\theta^{2n+1}}{(2n+1)!} \mathbf{u} \cdot \boldsymbol{\sigma} \\ &= \cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}$$

With

$$\begin{aligned}\gamma &:= \cosh(\theta), \\ \mathbf{v} &:= \tanh(\theta) \mathbf{u},\end{aligned}$$

Eq. (83) gives the same formula in terms of $\cosh(\theta)$, $\sinh(\theta)$, and their ratio $\tanh(\theta)$:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\ \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}).\end{aligned}$$

For the real paravector X from Eq. (14), the scalar part commutes with the vector part, so

$$\begin{aligned}\exp(X) &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}), \\ \exp(X^*) &= \exp(a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^*, \\ \exp(-X) &= \exp(-a) \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) = \exp(X)^{-1}.\end{aligned}$$

The corresponding half-angle element is

$$L(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot \boldsymbol{\sigma}\right). \quad (84)$$

Its conjugation properties are

$$L^H = L, \quad L^* = L^{-1}.$$

The action of this element on real events is derived from the two-sided event law in Eq. (97); it is not obtained from the inverse-pair

product in Eq. (94). The exponential and its inverse are specified by Eq. (84). The two projector weights are exchanged by the conjugate operator:

$$\begin{aligned}\exp(\mathbf{A} \cdot \boldsymbol{\sigma})^* &= \exp(-\theta) \Pi(+) + \exp(\theta) \Pi(-) \\ &= \gamma(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \exp(-\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp((\mathbf{A} \cdot \boldsymbol{\sigma})^*) \\ &= \exp(\mathbf{A} \cdot \boldsymbol{\sigma})^{-1}.\end{aligned}$$

1.10.2 Real Logarithm of a Paravector

The logarithm is the inverse problem for the same projector formulas. The real paravector Y from Eq. (15) is assumed to satisfy $b > \|\mathbf{B}\|$, so that the two real values in Eq. (77) are positive. The unique ordinary real logarithm of each positive value is used; this choice is called the principal logarithm. For $\mathbf{B} \neq \mathbf{0}$, the local direction is defined by

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|}, \quad \mathbf{u}^2 = 1. \quad (85)$$

Thus $\mathbf{B} = \|\mathbf{B}\| \mathbf{u}$. The projector function from Eq. (71) is then specialized immediately after its direction:

$$\Pi(\pm) := \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}). \quad (86)$$

The eigenvalue construction from Eq. (77) is then specialized to Y . Within this subsection, the two eigenvalues of Y are denoted locally by $\lambda(\pm)$:

$$\begin{aligned}\lambda(\pm) &:= b \pm \|\mathbf{B}\|, \\ Y &= \lambda(+) \Pi(+) + \lambda(-) \Pi(-).\end{aligned}\quad (87)$$

The logarithm is therefore applied separately to the two eigenvalues:

$$\begin{aligned}\log(Y) &= \log(\lambda(+)) \Pi(+) \\ &\quad + \log(\lambda(-)) \Pi(-) \\ &= \frac{1}{2} \log(\lambda(+)\lambda(-)) \\ &\quad + \frac{1}{2} \log(\lambda(+)/\lambda(-)) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}\quad (88)$$

When $\mathbf{B} = \mathbf{0}$, the axis is unnecessary and this formula is reduced directly to $\log(Y) = \log(b)$. For $\mathbf{B} \neq \mathbf{0}$, if $Y = \exp(X)$ with X from Eq. (14) and $\mathbf{A} = \theta \mathbf{u}$, then $\lambda(\pm) = \exp(a \pm \theta)$. The relation is expanded as

$$\begin{aligned}Y &= \exp(X) \\ &= \exp(a + \mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) \exp(\mathbf{A} \cdot \boldsymbol{\sigma}) \\ &= \exp(a) (\cosh(\theta) + \sinh(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= b + \|\mathbf{B}\| \mathbf{u} \cdot \boldsymbol{\sigma},\end{aligned}$$

and the recovered parameters are

$$\begin{aligned}\theta &= \operatorname{arctanh}(\|\mathbf{B}\|/b), \\ \exp(a) &= \sqrt{b^2 - \mathbf{B}^2}, \\ a &= \frac{1}{2} \log(b^2 - \mathbf{B}^2), \\ \mathbf{A} = \theta \mathbf{u} &= \operatorname{arctanh}(\|\mathbf{B}\|/b) \frac{\mathbf{B}}{\|\mathbf{B}\|}.\end{aligned}$$

The condition $b > \|\mathbf{B}\|$ gives a positive scalar part and $\det(Y) = b^2 - \mathbf{B}^2 > 0$. In special-relativity language this is the future-directed timelike region [5, 6]. Thus its principal logarithm is again a real paravector,

$$\log(Y) = a + \mathbf{A} \cdot \boldsymbol{\sigma}.$$

1.10.3 Circular Exponentials and Half-Angle Elements

For this subsection, a nonzero real vector $\mathbf{A}$ is written with the local values

$$\theta := \|\mathbf{A}\|, \quad \mathbf{u} := \frac{\mathbf{A}}{\|\mathbf{A}\|}, \quad \mathbf{A} = \theta \mathbf{u}, \quad \mathbf{u}^2 = 1.$$

Replacing θ with $i\theta$ changes the hyperbolic functions into the ordinary sine and cosine functions. The resulting sigma-number-two exponential is therefore called circular:

$$\begin{aligned}\exp(\pm i\mathbf{A} \cdot \boldsymbol{\sigma}) &= \exp(\pm i\theta \mathbf{u} \cdot \boldsymbol{\sigma}) \\ &= \exp(\pm i\theta) \Pi(+) + \exp(\mp i\theta) \Pi(-) \\ &= \cos(\theta) \pm i \sin(\theta) \mathbf{u} \cdot \boldsymbol{\sigma}.\end{aligned}\quad (89)$$

Hamilton's quaternion units are obtained by relabeling the same sigma-number-two basis, without introducing new generators or a new product [1]:

$$\begin{aligned}q_1 &:= -\sigma_2 \sigma_3 = -i\sigma_1, & q_2 &:= -\sigma_3 \sigma_1 = -i\sigma_2, \\ q_3 &:= -\sigma_1 \sigma_2 = -i\sigma_3, & \mathbf{q} &:= \{q_1, q_2, q_3\} = -i\boldsymbol{\sigma}.\end{aligned}\quad (90)$$

The multiplication and intrinsic-operation rules are inherited directly from Eqs. (4) and (10):

$$\begin{aligned}q_n q_m &= -\delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} q_k, \\ q_n^* &= q_n, & q_n^H &= -q_n.\end{aligned}\quad (91)$$

In particular, $q_1^2 = q_2^2 = q_3^2 = q_1 q_2 q_3 = -1$. The negative-sign exponential may therefore be written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot (-i\boldsymbol{\sigma}).$$

In the quaternion basis from Eq. (90), the same formula is written as

$$\exp(-i\mathbf{A} \cdot \boldsymbol{\sigma}) = \cos(\theta) + \sin(\theta) \mathbf{u} \cdot \mathbf{q}.$$

For a real angle $\theta \in \mathbb{R}$ and a unit real vector $\mathbf{u} \in \mathbb{R}^3$, the corresponding circular half-angle exponential is

$$R(\theta, \mathbf{u}) := \exp\left(\frac{\theta}{2} \mathbf{u} \cdot i\boldsymbol{\sigma}\right). \quad (92)$$

A real scalar coefficient and a purely imaginary vector coefficient are carried, so $R \in \mathbb{P}_{\mathbb{C}}$. Its conjugation properties are

$$R^H R = 1, \quad R^* = R.$$

Equation (92) is the circular half-angle element built from Eq. (89). This element is called a rotor because its two-sided action produces a spatial rotation. That action is derived in the Rotation subsection from Eqs. (101) and (95), where the result is reduced to Rodrigues's rotation formula [15]. The equivalent quaternion form is obtained from the same sigma-number-two algebra.

1.10.4 Exponentials of Complex Paravectors

The real paravectors are written as

$$\begin{aligned}X &= a + \mathbf{A} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{R}}, \\ Y &= b + \mathbf{B} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{R}}, \\ Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma} \in \mathbb{P}_{\mathbb{C}}, \\ &= X + iY,\end{aligned}$$

where

$$S = a + ib \in \mathbb{C}, \quad \mathbf{V} = \mathbf{A} + i\mathbf{B} \in \mathbb{C}^3.$$

Because the scalar part S commutes with $\mathbf{V} \cdot \boldsymbol{\sigma}$, the exponential is separated into scalar and directional factors:

$$\begin{aligned}\exp(Z) &= \exp(S + \mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \exp(S) \exp(\mathbf{V} \cdot \boldsymbol{\sigma}).\end{aligned}$$

The associated star and inverse identities are

$$\begin{aligned}\exp(Z^*) &= \exp(S^*) \exp(-\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ \exp(Z)^{-1} &= \exp(-S) \exp(-\mathbf{V} \cdot \boldsymbol{\sigma}), \\ \exp(Z^*)^{-1} &= \exp(-S^*) \exp(\mathbf{V}^* \cdot \boldsymbol{\sigma}), \\ (\mathbf{V} \cdot \boldsymbol{\sigma})^2 &= \mathbf{V} \cdot \mathbf{V} = \mathbf{V}^2, \\ &= \mathbf{A}^2 - \mathbf{B}^2 + 2i\mathbf{A} \cdot \mathbf{B}.\end{aligned}$$

The key simplification is that all higher powers are reduced to the two basis elements 1 and $\mathbf{V} \cdot \boldsymbol{\sigma}$:

$$\begin{aligned}(\mathbf{V} \cdot \boldsymbol{\sigma})^{2n} &= (\mathbf{V}^2)^n, \\ z &= \sqrt{\mathbf{V}^2} \quad \text{complex scalar,} \\ \exp(\mathbf{V} \cdot \boldsymbol{\sigma}) &= \sum_{n \geq 0} \frac{(\mathbf{V} \cdot \boldsymbol{\sigma})^n}{n!}, \\ &= \sum_{n \geq 0} \frac{(\mathbf{V}^2)^n}{(2n)!} + \sum_{n \geq 0} \frac{(\mathbf{V}^2)^n}{(2n+1)!} (\mathbf{V} \cdot \boldsymbol{\sigma}), \\ &= \cosh(z) + \sinh(z) \frac{\mathbf{V} \cdot \boldsymbol{\sigma}}{z}.\end{aligned}$$

The final expression is independent of the sign chosen for z . When $\mathbf{V}^2 = 0$, the quotient $\sinh(z)/z$ is understood by its continuous value 1, and the formula is reduced to

$$\exp(Z) = \exp(S)(1 + \mathbf{V} \cdot \boldsymbol{\sigma}).$$

1.11 Generalized Transform Algebra

With the projector and exponential formulas fixed, the remaining purely mathematical transform material can be collected in one place. The general law is given first, followed by basis changes and then by the rotation and boost elements built from the earlier exponentials:

$$TZ' = ZT, \quad (93)$$

$$Z' = T^{-1}ZT. \quad (94)$$

Here Z is a generic complex paravector and T is an algebra element with an inverse. Equation (93) requires that applying the transformed element and then T give the same result as applying T and then the original element. This compatibility condition is conventionally called an intertwining relation. Multiplication by T^{-1} gives the unique result (94). A transform of the form $T^{-1}ZT$ is called a similarity transform. It is used below for basis changes, rotations, and electromagnetic fields [12, 14, 15]. A different two-sided law is needed for a real event paravector; its two factors are related by H , and that law is derived and named when it is introduced.

1.11.1 Intertwining and the General Similarity Transform

Compatibility between two descriptions of the same algebraic action is expressed by the starting relation (93). An arbitrary algebra input $\psi'_{\text{algebra_input}}$ in the primed description is represented by $T\psi'_{\text{algebra_input}}$ in the unprimed description. Acting first with Z' and then changing description must give the same result as changing description first and then acting with Z :

$$T(Z'\psi'_{\text{algebra_input}}) = Z(T\psi'_{\text{algebra_input}}).$$

This equality holds for every $\psi'_{\text{algebra_input}}$. Setting the arbitrary right factor to 1 gives $TZ' = ZT$. This compatibility equation is conventionally called an intertwining relation. Because T is invertible, its unique solution is

$$Z' = T^{-1}ZT.$$

Thus Eq. (94) adds no law of motion. It is the change-of-description rule forced by the displayed compatibility requirement, and its inverse-pair form is called a similarity transform. When the direction of the identification is reversed, T is replaced everywhere by T^{-1} , and the equivalent convention $Z' = T^{-1}ZT$ is obtained. Under the inverse-first convention, the displayed positive sign in Rodrigues's formula is produced by the positive exponent in Eq. (92) [15].

The fully general complex paravectors $Z, T \in \mathbb{P}_{\mathbb{C}}$ are written as

$$\begin{aligned} Z &= S + \mathbf{V} \cdot \boldsymbol{\sigma}, & S &\in \mathbb{C}, & \mathbf{V} &\in \mathbb{C}^3, \\ T &= f + \mathbf{g} \cdot \boldsymbol{\sigma}, & f &\in \mathbb{C}, & \mathbf{g} &\in \mathbb{C}^3, \end{aligned}$$

where the ordinary dot and cross products are expanded linearly in each input, with no complex conjugation of either input. This

is the meaning of their extension to complex vectors. The determinant $\det(T) = f^2 - \mathbf{g}^2$ is a complex scalar and therefore commutes with every algebra element. When $\det(T) \neq 0$, the inverse is given by the general formula (65):

$$T^{-1} = \frac{T^{*\text{H}}}{\det(T)} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}.$$

Because the scalar part S is central,

$$Z' = S + \frac{(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma})(f + \mathbf{g} \cdot \boldsymbol{\sigma})}{\det(T)}.$$

The first multiplication is

$$(f - \mathbf{g} \cdot \boldsymbol{\sigma})(\mathbf{V} \cdot \boldsymbol{\sigma}) = -\mathbf{g} \cdot \mathbf{V} + (f\mathbf{V} - i\mathbf{g} \times \mathbf{V}) \cdot \boldsymbol{\sigma}.$$

After multiplication by the final factor, the scalar terms cancel and the remaining vector terms are

$$\begin{aligned} Z' &= S + \mathbf{V}' \cdot \boldsymbol{\sigma}, \\ \mathbf{V}' &= \frac{(f^2 + \mathbf{g}^2)\mathbf{V} - 2(\mathbf{g} \cdot \mathbf{V})\mathbf{g} - 2if(\mathbf{g} \times \mathbf{V})}{f^2 - \mathbf{g}^2}. \end{aligned} \quad (95)$$

This is the similarity law for every complex paravector Z and every invertible complex paravector T . The complex scalar part is fixed. Neither Eq. (94) nor Eq. (95) is changed when T is multiplied by a nonzero central complex scalar.

Products, inverses, and functions are governed by the same intertwining. For a second algebra element W ,

$$\begin{aligned} (ZW)' &= T^{-1}ZWT = T^{-1}ZT T^{-1}WT = Z'W', \\ (Z')^{-1} &= T^{-1}Z^{-1}T \quad \text{when } Z^{-1} \text{ exists,} \\ (Z')^n &= T^{-1}Z^nT, \\ h(Z') &= T^{-1}h(Z)T. \end{aligned} \quad (96)$$

The last line is valid when h is given by a convergent power series, and therefore in particular

$$\exp(Z') = T^{-1}\exp(Z)T.$$

The same relation is obeyed by logarithms, roots, and ratios of polynomials wherever the same choice among their possible values is made and every required inverse exists.

Applying the adjugate before the similarity gives the same result as applying it afterward. Since $T^{*\text{H}} = \det(T)T^{-1}$ and $(T^{-1})^{*\text{H}} = T/\det(T)$,

$$\begin{aligned} Z'^{*\text{H}} &= T^{*\text{H}}Z^{*\text{H}}(T^{-1})^{*\text{H}} = T^{-1}Z^{*\text{H}}T, \\ Z'Z'^{*\text{H}} &= T^{-1}(ZZ^{*\text{H}})T = ZZ^{*\text{H}}. \end{aligned}$$

Thus the determinant form $S^2 - \mathbf{V}^2$, the two scalar values defined in Eq. (81), and the inverse formula are preserved by similarity for every invertible T . The positive coefficient norm is a different construction. It is not fixed by the similarity equations and need not be preserved.

Algebra elements whose action is represented by multiplication are transported by the similarity law; this law is not automatically the Lorentz law for every physical use of the algebra. A real event paravector is characterized by $X^H = X$. Pairing the chosen left factor T^{-1} with its Hermitian image on the right preserves that defining property for every real event and reduces to the identity map when $T = 1$. Direct application of H gives

$$(T^{-1}X(T^{-1})^H)^H = T^{-1}X(T^{-1})^H.$$

The same law is exposed by any product $\psi_{\text{algebra_input}}\psi_{\text{algebra_input}}^H$, which is self-Hermitian. If $\psi'_{\text{algebra_input}} = T^{-1}\psi_{\text{algebra_input}}$, then

$$\psi'_{\text{algebra_input}}\psi_{\text{algebra_input}}^H = T^{-1}(\psi_{\text{algebra_input}}\psi_{\text{algebra_input}}^H)(T^{-1})^H.$$

For an element that satisfies the determinant-one condition $TT^{*H} = 1$, as required for the rotations and boosts considered here, it is

$$\begin{aligned} X' &= T^{-1}X(T^{-1})^H \\ &= T^{*H}XT^*. \end{aligned} \quad (97)$$

The right factor is the Hermitian image of the left factor. A two-sided transformation with this relation between its factors is called a Hermitian congruence [14, 15]. It is not a similarity, whose two factors are an inverse pair. The second factor occurs because an event can be built from products $\psi_{\text{algebra_input}}\psi_{\text{algebra_input}}^H$. The complex scalar part of every paravector is preserved by a similarity according to Eq. (95); the time coordinate therefore cannot be mixed with a spatial coordinate by that law. Distinct roles are assigned to star and H . Their combination *H is the adjugate, which equals T^{-1} only after the determinant-one normalization $TT^{*H} = 1$ has been imposed. Appendix A gives the optional translation of these operations. The set $\mathbb{P}_{\mathbb{R}}$ is preserved because $(X')^H = X'$, and the event determinant is preserved because T^{-1} has unit determinant. When X is invertible, the transformed inverse is obtained without a new definition:

$$(X')^{-1} = T^H X^{-1} T, \quad (98)$$

because the identity is obtained by direct multiplication with $X' = T^{-1}X(T^{-1})^H$. The same inverse pattern is used later when the derivative paravector is transformed. An equivalent event compatibility equation is obtained by multiplication of Eq. (97) on the left by T and on the right by T^H :

$$\begin{aligned} TX'T^H &= X, \\ TX' &= XT^*. \end{aligned} \quad (99)$$

The determinant-one identity $(T^{-1})^H = T^*$ is used in the second line. The form $TX' = XT$ is obtained for a circular rotor because $R^* = R$, but not for a boost. The similarity and event compatibility equations are therefore identical for circular rotors but remain distinct for boosts.

In contrast, the electromagnetic field paravector is defined later by $F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}$ in Eq. (141), where $\mathbf{E}$ and $\mathbf{B}$ are the real

electric and magnetic field vectors. Its action is represented by left multiplication on an algebra element. The following similarity is then required by the original compatibility argument:

$$F' = T^{-1}FT. \quad (100)$$

For a circular rotor, $R^* = R$ and $R^{-1} = R^H$, so Eq. (97) rotates the represented spatial vector while the coordinate axes are held fixed. This description is called active, and the event law is reduced to

$$X' = R^H X R. \quad (101)$$

For the hyperbolic element in Eq. (84), $L^H = L$ and $L^{-1} = L^* = L^{*H}$. In this case the coordinates assigned to the same event are changed to those of the boosted frame. This description is called passive [4–6], and the same general event law is reduced to

$$\begin{aligned} X' &= L^{-1}X(L^{-1})^H \\ &= L^*XL^*. \end{aligned} \quad (102)$$

The equal factors in the final line are obtained from $(L^{-1})^H = L^{-1}$. They are not an inverse pair. For comparison, the following results are obtained from the two possible similarity orderings:

$$\begin{aligned} L^{-1}XL &= t + (\text{par}(\mathbf{r}) + \cosh(\theta) \text{prp}(\mathbf{r}) \\ &\quad - i \sinh(\theta) \mathbf{u} \times \mathbf{r}) \cdot \boldsymbol{\sigma}, \\ LXL^{-1} &= t + (\text{par}(\mathbf{r}) + \cosh(\theta) \text{prp}(\mathbf{r}) \\ &\quad + i \sinh(\theta) \mathbf{u} \times \mathbf{r}) \cdot \boldsymbol{\sigma}. \end{aligned} \quad (103)$$

Under both similarities, t and $\text{par}(\mathbf{r})$ are left unchanged and a generic real perpendicular component is made complex. A Lorentz boost of the real event X is therefore not obtained. In the simple case $\mathbf{r} = a\mathbf{u}$, where the position vector is parallel to the boost axis, $\text{com}(X, L) = 0$ is obtained, so the entire event is left unchanged by both similarities instead of t and a being mixed. By contrast, Eq. (100) is reduced to $F' = L^{-1}FL = L^*FL$. The unequal right factors are therefore consequences of the event and field having different algebraic actions, not of an inconsistent choice of inverse. Under the definition of $L(\theta, \mathbf{u})$, the two-sided product $L^HXL = LXL$ is the inverse-direction event boost, obtained by replacing θ with $-\theta$. It is physically legitimate, but it is not the passive $+\theta$ convention used in Eq. (102).

The hyperbolic reduction of the general complex formula is obtained directly with

$$f = \cosh(\theta/2), \quad \mathbf{g} = \sinh(\theta/2)\mathbf{u}, \quad \mathbf{u}^2 = 1.$$

Substitution into Eq. (95) gives

$$\mathbf{V}' = (\mathbf{u} \cdot \mathbf{V})\mathbf{u} + \cosh(\theta)(\mathbf{V} - (\mathbf{u} \cdot \mathbf{V})\mathbf{u}) - i \sinh(\theta) \mathbf{u} \times \mathbf{V}. \quad (104)$$

For $\mathbf{V} = \mathbf{E} + i\mathbf{B}$, its real and imaginary parts are exactly the standard electric–magnetic boost formulas [16]. For a real $\mathbf{V}$,

however, the cross term is generally imaginary; hyperbolic similarity is therefore a field or basis transformation rather than an event boost.

A transform element $T(\lambda)$ is taken to depend on one real parameter λ . All displayed derivatives are ordinary derivatives with respect to that parameter. The quantity that measures the parameter-rate of change of T , expressed on its inverse-carried side, is defined by

$$\Omega := T^{-1} \frac{dT}{d\lambda}. \quad (105)$$

It is called the similarity generator because it determines the first-order change of every similarity-transformed element. For a small parameter change, the local factor is

$$T(\lambda)^{-1} T(\lambda + d\lambda) = 1 + \Omega d\lambda, \\ \text{with terms containing } (d\lambda)^2 \text{ discarded.}$$

This first-order form is conventionally called infinitesimal. The similarity and event actions of this local factor are

$$Z(\lambda + d\lambda) = Z(\lambda) + \text{com}(Z(\lambda), \Omega) d\lambda, \\ X(\lambda + d\lambda) = X(\lambda) - \Omega X(\lambda) d\lambda - X(\lambda) \Omega^H d\lambda.$$

The two Hermitian cases reduce the added event term to

$$\Omega^H = -\Omega : \quad \text{com}(X(\lambda), \Omega) d\lambda, \\ \Omega^H = \Omega : \quad -\text{acom}(\Omega, X(\lambda)) d\lambda.$$

The first term generates a rotation. The second mixes the scalar time component with the vector component along the boost direction. Differentiation of the inverse and of the transported paravector then gives

$$\begin{aligned} \frac{d(T^{-1})}{d\lambda} &= -T^{-1} \frac{dT}{d\lambda} T^{-1}, \\ \frac{dZ'}{d\lambda} &= T^{-1} \frac{dZ}{d\lambda} T + \text{com}(Z', \Omega), \\ \frac{d^2 Z'}{d\lambda^2} &= T^{-1} \frac{d^2 Z}{d\lambda^2} T \\ &\quad + 2 \text{com}\left(\frac{dZ'}{d\lambda}, \Omega\right), \\ &\quad + \text{com}\left(Z', \frac{d\Omega}{d\lambda}\right), \\ &\quad + \text{com}(\Omega, \text{com}(Z', \Omega)). \end{aligned}$$

They are not space-time derivatives.

1.11.2 Transformed Basis

The parameter-dependent element is written as

$$T(\lambda) = f(\lambda) + \mathbf{g}(\lambda) \cdot \boldsymbol{\sigma}, \quad f \in \mathbb{C}, \quad \mathbf{g} \in \mathbb{C}^3, \\ T^{-1} = \frac{f - \mathbf{g} \cdot \boldsymbol{\sigma}}{f^2 - \mathbf{g}^2}, \quad f^2 - \mathbf{g}^2 \neq 0.$$

The transformed basis element is defined by

$$\sigma'_n := T^{-1} \sigma_n T. \quad (106)$$

Its inverse relation is

$$\sigma_n = T \sigma'_n T^{-1}.$$

Both real hyperbolic elements and imaginary-vector rotors are included by allowing complex $\mathbf{g}$. The following relations are obtained immediately by similarity:

$$\begin{aligned} (\sigma'_n)^2 &= 1, \\ \sigma'_n \sigma'_m &= \delta_{nm} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma'_k, \\ \sigma'_1 \sigma'_2 \sigma'_3 &= i. \end{aligned}$$

Thus the same sigma multiplication table is satisfied by the transformed generators for every invertible T .

The separate actions of star and H on the transformed generators are represented by

$$\begin{aligned} \sigma'^{*}_n &= (T^{-1})^* \sigma_n^* T^* = -(T^*)^{-1} \sigma_n T^*, \\ \sigma'^H_n &= T^H \sigma_n^H (T^{-1})^H = T^H \sigma_n (T^H)^{-1}. \end{aligned}$$

The scalar part is also preserved by similarity, so $\text{scalar}(\sigma'_n) = \text{scalar}(\sigma_n) = 0$. The individual star and H actions are recorded separately by these formulas and are combined by the adjugate relation.

For every invertible T , applying the adjugate before the similarity gives the same result as applying it afterward. This follows from $T^{*H} = (f^2 - \mathbf{g}^2) T^{-1}$ and $(T^{-1})^{*H} = T / (f^2 - \mathbf{g}^2)$:

$$\begin{aligned} (\sigma'_n)^{*H} &= (f^2 - \mathbf{g}^2) T^{-1} (-\sigma_n) \frac{T}{f^2 - \mathbf{g}^2} \\ &= -T^{-1} \sigma_n T = -\sigma'_n. \end{aligned}$$

Consequently a paravector written in the transformed basis has the form

$$Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}'.$$

Here S and $\mathbf{V}$ denote the coefficients in the displayed primed basis; no fixed-basis vector extraction is being applied. The same determinant expression is then assigned:

$$ZZ^{*H} = S^2 - \mathbf{V}^2.$$

For differentiation with respect to λ , the generator from Eq. (105) gives

$$\frac{d\sigma'_n}{d\lambda} = \text{com}(\sigma'_n, \Omega), \quad \frac{d\boldsymbol{\sigma}'}{d\lambda} = \text{com}(\boldsymbol{\sigma}', \Omega).$$

The coefficient form is obtained directly from

$$\begin{aligned} \Omega &= \frac{f \frac{df}{d\lambda} - \mathbf{g} \cdot \frac{d\mathbf{g}}{d\lambda}}{f^2 - \mathbf{g}^2} \\ &\quad + \frac{f \frac{d\mathbf{g}}{d\lambda} - \frac{df}{d\lambda} \mathbf{g} - i \mathbf{g} \times \frac{d\mathbf{g}}{d\lambda}}{f^2 - \mathbf{g}^2} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Relative to the primed basis, its vector coefficient is written uniquely as

$$\Omega = \text{scalar}(\Omega) + \frac{1}{2}(\mathbf{k} + i\boldsymbol{\omega}) \cdot \boldsymbol{\sigma}', \quad \mathbf{k}, \boldsymbol{\omega} \in \mathbb{R}^3.$$

The same generator carried into the unprimed basis is defined by

$$\Omega' := T\Omega T^{-1}. \quad (107)$$

The definition in Eq. (105) then gives

$$\Omega' = \frac{dT}{d\lambda} T^{-1}.$$

The coefficients are recovered through scalar extraction. For $n \in \{1, 2, 3\}$, the coefficients are given by cyclicity of the algebraic scalar part:

$$k_n + i\omega_n = 2 \text{ scalar}(\Omega\sigma'_n) = 2 \text{ scalar}(\Omega'\sigma_n).$$

The scalar part is central with respect to the basis, while

$$\begin{aligned} \frac{d\boldsymbol{\sigma}'}{d\lambda} &= (-\boldsymbol{\omega} + i\mathbf{k}) \times \boldsymbol{\sigma}', \\ \frac{dZ}{d\lambda} &= \frac{dS}{d\lambda} + \frac{d\mathbf{V}}{d\lambda} \cdot \boldsymbol{\sigma}' \\ &\quad - (\mathbf{V} \times \boldsymbol{\omega}) \cdot \boldsymbol{\sigma}' \\ &\quad + i(\mathbf{V} \times \mathbf{k}) \cdot \boldsymbol{\sigma}'. \end{aligned}$$

These identities describe a basis that depends on λ , and are therefore called moving-basis identities. Similarity alone does not define λ as space-time, energy, or proper time. Such an interpretation requires a separately specified event law of the form Eq. (97) and a definition relating the changing basis to a physical reference frame.

1.11.3 Rotation

Equation (101) is obtained as the rotation case of the general transform law by specializing the transform element to $T = R$. Sigma number two is assigned to the generator, so the exponential is circular rather than hyperbolic. The rotor function defined in Eq. (92) has the expanded form

$$\begin{aligned} R(\theta, \mathbf{u}) &= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})), \\ &= \cos(\tfrac{1}{2}\theta) + \sin(\tfrac{1}{2}\theta)(\mathbf{u} \cdot i\boldsymbol{\sigma}). \end{aligned}$$

The inverse is obtained as

$$R^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot i\boldsymbol{\sigma})).$$

$\theta \in \mathbb{R}$, and $\mathbf{u} \in \mathbb{R}^3$ is a unit real vector. The element R is generated by the sigma-number-two element $\mathbf{u} \cdot i\boldsymbol{\sigma}$. The inverse relation is $R^{-1} = R^H = R^{*H}$. The sign of the $\mathbf{u} \times \mathbf{r}$ term, and therefore the positive rotation direction, is fixed by the substitution

$$f = \cos(\theta/2), \quad \mathbf{g} = i \sin(\theta/2)\mathbf{u}$$

into Eq. (95). Then

$$\begin{aligned} f^2 - \mathbf{g}^2 &= 1, \\ f^2 + \mathbf{g}^2 &= \cos(\theta), \\ -2(\mathbf{g} \cdot \mathbf{r})\mathbf{g} &= (1 - \cos(\theta))(\mathbf{u} \cdot \mathbf{r})\mathbf{u}, \\ -2i f(\mathbf{g} \times \mathbf{r}) &= \sin(\theta)(\mathbf{u} \times \mathbf{r}). \end{aligned}$$

Therefore the transformed paravector is

$$\begin{aligned} X' &= R^H X R, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

The scalar and vector parts are therefore obtained as

$$\begin{aligned} t' &= t, \\ \mathbf{r}' &= \mathbf{r} \cos(\theta) + (\mathbf{u} \times \mathbf{r}) \sin(\theta), \\ &\quad + (\mathbf{u} \cdot \mathbf{r})(1 - \cos(\theta))\mathbf{u}. \end{aligned}$$

The vector formula is the standard axis-angle expression conventionally called Rodrigues's formula [15]. For a parameter-dependent rotor, the specialization of the general generator is $R^H(dR/d\lambda)$. Its form carried to the other side is $(dR/d\lambda)R^H$. The following identities are obtained by differentiating $R^H R = 1$:

$$\left(R^H \frac{dR}{d\lambda}\right)^H = -R^H \frac{dR}{d\lambda}, \quad \left(\frac{dR}{d\lambda} R^H\right)^H = -\frac{dR}{d\lambda} R^H.$$

The rotation transport law is the specialization of the general one-parameter similarity formula:

$$\begin{aligned} \frac{dX'}{d\lambda} &= R^H \frac{dX}{d\lambda} R + \text{com}\left(X', R^H \frac{dR}{d\lambda}\right) \\ &= R^H \left(\frac{dX}{d\lambda} + \text{com}\left(X, \frac{dR}{d\lambda} R^H\right)\right) R. \end{aligned}$$

1.11.4 Quaternion Products

For $a \in \mathbb{R}$ and $\mathbf{B} \in \mathbb{R}^3$, a quaternion is the complex paravector

$$Q := a + \mathbf{B} \cdot \mathbf{q} = a - i \mathbf{B} \cdot \boldsymbol{\sigma}. \quad (108)$$

The quaternion basis is fixed by star, so $Q^* = Q$. The ordinary algebra square is therefore

$$\begin{aligned} QQ^* &= Q^2 \\ &= (a + \mathbf{B} \cdot \mathbf{q})^2 \\ &= a^2 - \mathbf{B}^2 + 2a\mathbf{B} \cdot \mathbf{q} \\ &= a^2 - \mathbf{B}^2 - 2ia\mathbf{B} \cdot \boldsymbol{\sigma}. \end{aligned} \quad (109)$$

Thus star is not the operation conventionally called quaternion conjugation. In the present notation that operation is H :

$$Q^H = \text{adj}(Q) = a - \mathbf{B} \cdot \mathbf{q} = a + i \mathbf{B} \cdot \boldsymbol{\sigma}.$$

The determinant and the positive quaternion norm square consequently agree:

$$\det(Q) = QQ^H = Q^H Q = a^2 + \mathbf{B}^2. \quad (110)$$

Here the norm square is the nonnegative real sum of the squared coefficients, whereas Eq. (109) is generally another quaternion. A quaternion satisfying $\det(Q) = 1$ is called a unit quaternion.

1.11.5 Quaternion Form of the Rotor Law

For the rotor $R(\theta, \mathbf{u})$ used in the preceding Rotation subsection, the corresponding unit quaternion is

$$Q := R^H = \cos\left(\frac{\theta}{2}\right) + \sin\left(\frac{\theta}{2}\right) \mathbf{u} \cdot \mathbf{q}.$$

Its Hermitian counterpart is $Q^H = R$. Multiplication of the sigma-vector rotor law by the central scalar $-i$ gives

$$\begin{aligned} R^H (\mathbf{r} \cdot \mathbf{q}) R &= -i R^H (\mathbf{r} \cdot \boldsymbol{\sigma}) R \\ &= -i \mathbf{r}' \cdot \boldsymbol{\sigma} = \mathbf{r}' \cdot \mathbf{q}. \end{aligned}$$

The quaternion rotation law is therefore

$$\mathbf{r}' \cdot \mathbf{q} = Q (\mathbf{r} \cdot \mathbf{q}) Q^H. \quad (111)$$

This has the same positive rotation direction as the Rodrigues formula above. If the alternative identification $Q = R$ is made, the two quaternion factors must instead be written in the opposite order. No second rotation algebra is introduced by either convention.

1.11.6 Spin Rotors

Each oriented product of two different sigma generators is identified by the pseudoscalar without an index-order ambiguity:

$$i\sigma_1 = \sigma_2\sigma_3, \quad i\sigma_2 = \sigma_3\sigma_1, \quad i\sigma_3 = \sigma_1\sigma_2.$$

Such a two-generator product is the sigma-number-two element already defined above; it is conventionally called a bivector. For unit $\mathbf{n}, \mathbf{u} \in \mathbb{R}^3$, the rotor function defined in Eq. (92) is evaluated as

$$\begin{aligned} R(\phi, \mathbf{n}) &= \exp\left(\frac{\phi}{2} \mathbf{n} \cdot i\boldsymbol{\sigma}\right) \\ &= \cos(\phi/2) + \sin(\phi/2) \mathbf{n} \cdot i\boldsymbol{\sigma}. \end{aligned}$$

Then

$$R^{-1} = R^H = R^{*H}, \quad R^H R = 1,$$

and

$$R^H (\mathbf{u} \cdot \boldsymbol{\sigma}) R = \mathbf{u}' \cdot \boldsymbol{\sigma}$$

with $\mathbf{u}'$ defined by Rodrigues's formula. The rotor itself is single-valued and 4π -periodic:

$$R(\phi + 2\pi, \mathbf{n}) = -R(\phi, \mathbf{n}), \quad R(\phi + 4\pi, \mathbf{n}) = R(\phi, \mathbf{n}).$$

The induced spatial rotation is two-to-one because the same conjugation is produced by R and $-R$.

Under the active convention associated with this vector action, an algebra input and its opposite-order product transform as

$$\begin{aligned} \psi'_{\text{algebra_input}} &= R^H \psi_{\text{algebra_input}}, \\ \psi'_{\text{algebra_input}} \psi'^H_{\text{algebra_input}} &= R^H (\psi_{\text{algebra_input}} \psi^H_{\text{algebra_input}}) R. \end{aligned}$$

A physical spin state additionally requires a scalar rule for pairing two states, a normalization, one fixed right-projector restriction, a Hermitian element representing the measurement, and a rule for its predicted average. Those choices are defined in Eqs. (212) and (249). The algebra action is thereby separated from the choice of a single physical spin state.

1.11.7 Spin-Axis Projectors

A nonzero real axis vector $\mathbf{A}$ is chosen first, and its unit direction is

$$\mathbf{u} := \frac{\mathbf{A}}{\|\mathbf{A}\|} \in \mathbb{R}^3, \quad \mathbf{u}^2 = 1,$$

and its left action is resolved by the projector family from Eq. (71). For clarity, that definition is restated here as an ordinary equality:

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}), \quad \text{from Eq. (71).}$$

Equation (71) also shows that each projector is unchanged when squared, their mixed products are zero, their sum is 1, and multiplication by $\mathbf{u} \cdot \boldsymbol{\sigma}$ returns each projector multiplied by its sign. The following star and H actions are obtained:

$$\begin{aligned} \Pi(\pm)^H &= \Pi(\pm), \\ \Pi(+)^* &= \Pi(-), \\ \Pi(-)^* &= \Pi(+). \end{aligned}$$

Accordingly,

$$\mathbf{u} \cdot \boldsymbol{\sigma} = (+1)\Pi(+) + (-1)\Pi(-),$$

so the eigenvalues $+1$ and -1 of the axis element are carried by the two projector parts. Here an eigenvalue is the scalar returned when the action leaves a nonzero projector part otherwise unchanged.

These left projectors split any algebra element into the two axis parts defined in Eq. (72). A later physical two-amplitude wave field is additionally restricted by one fixed projector on the right. The left axis split and the fixed right restriction have different roles and must not be identified.

If the axis is rotated by the rotor R from Eq. (101), the transformed axis and its second projector pair satisfy

$$\begin{aligned} \mathbf{u}' \cdot \boldsymbol{\sigma} &= R^H (\mathbf{u} \cdot \boldsymbol{\sigma}) R, \quad \mathbf{u}'^2 = 1, \\ \Pi'(\pm) &= \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}) = R^H \Pi(\pm) R. \end{aligned} \quad (112)$$

Then

$$\mathbf{u}' \cdot \boldsymbol{\sigma} = (+1)\Pi'(+) + (-1)\Pi'(-).$$

For an axis state $\psi_{\text{axis_state}} \in \mathbb{P}_{\mathbb{C}}$, the left-projected parts from Eq. (71) are

$$\psi_{\text{axis_state}}(\pm) = \Pi(\pm) \psi_{\text{axis_state}}.$$

The following identities are obtained:

$$\begin{aligned} (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}(\pm) &= \pm \psi_{\text{axis_state}}(\pm), \\ \psi_{\text{axis_state}} &= \psi_{\text{axis_state}}(+) + \psi_{\text{axis_state}}(-). \end{aligned}$$

Thus each Pauli state is split into two parts on which the chosen axis acts by multiplication by $+1$ or -1 . The sets of states with these fixed multipliers are conventionally called eigenspaces [9, 19]. The Hermitian operation is represented by

$$\begin{aligned} \psi_{\text{axis_state}}(\pm)^H &= \psi_{\text{axis_state}}^H \Pi(\pm)^H \\ &= \psi_{\text{axis_state}}^H \Pi(\pm), \end{aligned}$$

and therefore

$$\begin{aligned}\psi_{\text{axis_state}}(\pm)^H \psi_{\text{axis_state}}(\pm) &= \psi_{\text{axis_state}}^H \\ \Pi(\pm) \psi_{\text{axis_state}}, \\ \psi_{\text{axis_state}}(+)^H \psi_{\text{axis_state}}(-) &= \psi_{\text{axis_state}}^H \\ \Pi(+)\Pi(-) \psi_{\text{axis_state}} &= 0, \\ \psi_{\text{axis_state}}(-)^H \psi_{\text{axis_state}}(+) &= \psi_{\text{axis_state}}^H \\ \Pi(-)\Pi(+)\psi_{\text{axis_state}} &= 0.\end{aligned}$$

For the axis states $\phi_{\text{axis_state}}$ and $\psi_{\text{axis_state}}$, and a specified measurement element $\mathcal{O}$, the algebraic pairing at one position is $\text{scalar}(\phi_{\text{axis_state}}^H \mathcal{O} \psi_{\text{axis_state}})$. For a nonzero $\psi_{\text{axis_state}}$, the identity pairing $\text{scalar}(\psi_{\text{axis_state}}^H \psi_{\text{axis_state}})$ is positive by Eq. (52). Division by this positive value gives

$$\frac{\text{scalar}(\psi_{\text{axis_state}}^H \mathcal{O} \psi_{\text{axis_state}})}{\text{scalar}(\psi_{\text{axis_state}}^H \psi_{\text{axis_state}})}.$$

This quotient is a normalized operator pairing because the same state occurs on both sides of $\mathcal{O}$. When $\mathcal{O}^H = \mathcal{O}$, the numerator is real. If $\mathcal{O}$ represents a measurement in quantum mechanics, the quotient is its predicted average, conventionally called its expectation value [19]. For a general $\mathcal{O}$, it is only a normalized pairing and may be complex. The common scalar-extraction factor cancels in the normalized pairing. The corresponding spatially integrated pairing is defined separately in Eq. (212).

An active rotation means that the state and measurement element are rotated while the coordinate axes are held fixed. Under this convention,

$$\psi'_{\text{axis_state}} = R^H \psi_{\text{axis_state}}, \quad \mathcal{O}' = R^H \mathcal{O} R, \quad R R^H = 1,$$

both the numerator and denominator of the normalized pairing are unchanged:

$$\frac{\text{scalar}(\psi_{\text{axis_state}}'^H \mathcal{O}' \psi'_{\text{axis_state}})}{\text{scalar}(\psi_{\text{axis_state}}'^H \psi'_{\text{axis_state}})} = \frac{\text{scalar}(\psi_{\text{axis_state}}^H \mathcal{O} \psi_{\text{axis_state}})}{\text{scalar}(\psi_{\text{axis_state}}^H \psi_{\text{axis_state}})}. \quad (113)$$

The unchanged form of this value under a common rotation is called rotational covariance. Equivalently, if the Hermitian measurement element $\mathcal{O}$, conventionally called an observable, is held fixed while only the state is actively rotated, its value in the original state is computed with the rotated-back observable $R \mathcal{O} R^H$. Describing the same change by rotating the axes while holding the state fixed is called the passive description. This active-versus-passive equivalence is expressed in Eq. (112).

The Hermitian element $\mathbf{u} \cdot \boldsymbol{\sigma}$ represents measurement along the chosen axis and is therefore an axis observable. The following identity is obtained by cyclic scalar extraction:

$$\begin{aligned}\text{scalar}(\psi_{\text{axis_state}}^H (\mathbf{u} \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}) \\ = \mathbf{u} \cdot \text{vector}(\psi_{\text{axis_state}} \psi_{\text{axis_state}}^H).\end{aligned} \quad (114)$$

Consequently its normalized value at one position is

$$\frac{\mathbf{u} \cdot \text{vector}(\psi_{\text{axis_state}} \psi_{\text{axis_state}}^H)}{\text{scalar}(\psi_{\text{axis_state}}^H \psi_{\text{axis_state}})}.$$

For a spatial wave field, the same statement is valid with both quantities integrated as in Eqs. (212) and (249).

The unnormalized numerator at one position for the rotated axis is expanded as

$$\begin{aligned}\text{scalar}\left(\psi_{\text{axis_state}}^H (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}\right) \\ = \text{scalar}\left(\psi_{\text{axis_state}}^H (+) (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}(+)\right) \\ + \text{scalar}\left(\psi_{\text{axis_state}}^H (+) (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}(-)\right) \\ + \text{scalar}\left(\psi_{\text{axis_state}}^H (-) (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}(+)\right) \\ + \text{scalar}\left(\psi_{\text{axis_state}}^H (-) (\mathbf{u}' \cdot \boldsymbol{\sigma}) \psi_{\text{axis_state}}(-)\right).\end{aligned}$$

The algebra governing these same-side and cross-side terms is

$$\begin{aligned}(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} + i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma}) &= \mathbf{u}' \cdot \mathbf{u} - i(\mathbf{u}' \times \mathbf{u}) \cdot \boldsymbol{\sigma}, \\ (\mathbf{u} \cdot \boldsymbol{\sigma})(\mathbf{u}' \cdot \boldsymbol{\sigma})(\mathbf{u} \cdot \boldsymbol{\sigma}) &= (2(\mathbf{u}' \cdot \mathbf{u})\mathbf{u} - \mathbf{u}') \cdot \boldsymbol{\sigma}.\end{aligned}$$

The second axis is decomposed into its par and prp parts by the axis functions from Eqs. (39) and (40):

$$\mathbf{u}' = \text{par}(\mathbf{u}') + \text{prp}(\mathbf{u}'), \quad \text{par}(\mathbf{u}') = (\mathbf{u} \cdot \mathbf{u}')\mathbf{u}.$$

Then the projector products are

$$\Pi(\pm)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm(\mathbf{u} \cdot \mathbf{u}')\Pi(\pm).$$

The off-diagonal products are obtained as

$$\begin{aligned}\Pi(+)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(-) &= \frac{1}{2}(\text{prp}(\mathbf{u}') + i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(-), \\ \Pi(-)(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi(+)&= \frac{1}{2}(\text{prp}(\mathbf{u}') - i\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma} \Pi(+).\end{aligned}$$

The diagonal terms are therefore weighted by the axis overlap $\mathbf{u} \cdot \mathbf{u}'$, while the off-diagonal terms are controlled by the prp and commutator parts of the rotated axis.

If θ is the angle between $\mathbf{u}$ and $\mathbf{u}'$, then

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\theta).$$

The normalized projector overlap is

$$\begin{aligned}2 \text{ scalar}(\Pi(\pm)\Pi'(\pm)) &= \frac{1}{2}(1 + \mathbf{u} \cdot \mathbf{u}'), \\ 2 \text{ scalar}(\Pi(+)\Pi'(-)) &= 2 \text{ scalar}(\Pi(-)\Pi'(+)) \\ &= \frac{1}{2}(1 - \mathbf{u} \cdot \mathbf{u}').\end{aligned}$$

When a normalized two-amplitude state lies entirely in one projector part, these overlaps give the probabilities for the same-sign and opposite-sign parts along the second axis. This predicted

change from one selected axis part to another is called a transition probability [19]. In particular, the same-side and opposite-side probabilities are

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\theta/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\theta/2).$$

If $\mathbf{u}'$ is obtained by rotating $\mathbf{u}$ through angle ϕ about a unit axis $\mathbf{n}$, the overlap is given by the standard axis-angle formula commonly called Rodrigues's formula [15]:

$$\begin{aligned} \mathbf{u}' \cdot \mathbf{u} &= (\mathbf{u} \cdot \mathbf{u}) \cos(\phi) + (\mathbf{u} \cdot (\mathbf{n} \times \mathbf{u})) \sin(\phi) \\ &\quad + (\mathbf{n} \cdot \mathbf{u})(1 - \cos(\phi))(\mathbf{n} \cdot \mathbf{u}) \\ &= (\mathbf{n} \cdot \mathbf{u})^2 + (1 - (\mathbf{n} \cdot \mathbf{u})^2) \cos(\phi). \end{aligned}$$

In the perpendicular case $\mathbf{n} \cdot \mathbf{u} = 0$, this is reduced to

$$\mathbf{u}' \cdot \mathbf{u} = \cos(\phi),$$

hence

$$\frac{1}{2}(1 + \mathbf{u}' \cdot \mathbf{u}) = \cos^2(\phi/2), \quad \frac{1}{2}(1 - \mathbf{u}' \cdot \mathbf{u}) = \sin^2(\phi/2).$$

These are the standard transition probabilities for a two-outcome Pauli spin state, conventionally called a spin- $\frac{1}{2}$ state [9, 19]. Without normalization and the later two-amplitude right-projector restriction, only algebraic projector overlaps are represented.

1.11.8 Ladder Operations About the z Axis

For raising and lowering about the z axis, the following axis and projector pair are chosen:

$$\begin{aligned} \mathbf{u} &= \mathbf{e}_3 = \{0, 0, 1\}, \quad \mathbf{u}^2 = 1, \\ \Pi(\pm) &= \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(1 \pm \sigma_3). \end{aligned}$$

The ladder elements already defined in Eq. (75) are used:

$$a(\pm) = \frac{1}{2}(\sigma_1 \pm i\sigma_2).$$

The commutator relation is satisfied:

$$\text{com}(\sigma_3, a(\pm)) = \pm 2a(\pm),$$

so the σ_3 eigenvalue is raised by $a(+)$ and lowered by $a(-)$.

Their projector products are

$$a(+)\Pi(-) = \Pi(+), \quad a(-)\Pi(+)=\Pi(-),$$

and each ladder element has square zero:

$$a(\pm)^2 = 0.$$

An element for which a positive power is zero is conventionally called nilpotent; here that power is 2. The one-sided projector actions are

$$\begin{aligned} a(+)\Pi(+)&=0, & a(+)\Pi(-)&=a(+), \\ \Pi(+)\Pi(+)&=\Pi(+), & \Pi(-)\Pi(+)&=0, \\ a(-)\Pi(-)&=0, & a(-)\Pi(+)&=a(-), \\ \Pi(-)\Pi(-)&=\Pi(-), & \Pi(+)\Pi(-)&=0, \end{aligned}$$

by which the raising/lowering interpretation is made explicit. The original sigma generators are recovered from the same ladder/projector data:

$$\begin{aligned} \sigma_1 &= a(+) + a(-), & \sigma_2 &= \frac{a(+) - a(-)}{i}, \\ \sigma_3 &= \Pi(+)-\Pi(-). \end{aligned}$$

1.11.9 Spin-Axis Noncommutativity

For unit directions $\mathbf{u}, \mathbf{u}' \in \mathbb{R}^3$,

$$\begin{aligned} \text{acom}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2\mathbf{u} \cdot \mathbf{u}', \\ \text{com}(\mathbf{u} \cdot \boldsymbol{\sigma}, \mathbf{u}' \cdot \boldsymbol{\sigma}) &= 2i(\mathbf{u} \times \mathbf{u}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

The axis overlap is recorded by the anticommutator, while the oriented area spanned by the two axes is recorded by the commutator. Thus exact commutation of the axis elements is obtained when the directions are parallel or antiparallel. These identities are purely algebraic. A normalized state, a rule for calculating measurement averages, and each measurement's mean squared deviation from its average are additionally required. The last quantity is called the variance, and its square root is called the uncertainty. Only after these choices are supplied can the commutator be used in the standard inequality that bounds the product of two measurement uncertainties, called the Robertson or Robertson–Schrödinger uncertainty relation. No such inequality is inferred from the commutator alone [11].

2 Space-Time Kinematics

The relations describing motion before forces are introduced are conventionally called kinematics. On real event paravectors, the squared space-time interval is obtained from the determinant, and a four-velocity is represented by a real paravector with determinant 1. Standard special-relativistic kinematics is thereby expressed in paravector form [4–6, 12, 14].

2.1 Differential space-time and four-velocity

The differential event is obtained by specializing the real paravector X from Eq. (14):

$$dX := dt + d\mathbf{r} \cdot \boldsymbol{\sigma} \quad (115)$$

The interval is the differential version of Eq. (58):

$$ds^2 = \det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2. \quad (116)$$

Here $t, s, \gamma \in \mathbb{R}$, $\mathbf{r}, \mathbf{v}, \mathbf{P} \in \mathbb{R}^3$, and X, U are real paravectors. Equation (116) is the real-event restriction of the determinant in Eq. (58). A worldline is the path of an object through space-time. It is called timelike here when $\det(dX) > 0$, and future-directed when $dt > 0$. Along such a path, the positive increment $ds = \sqrt{dt^2 - d\mathbf{r}^2} > 0$ is the time measured by a clock moving with the object and is conventionally called proper time [5, 6].

The coordinate velocity and Lorentz factor are defined by $\mathbf{v} := d\mathbf{r}/dt \in \mathbb{R}^3$ and $\gamma := (1 - \mathbf{v}^2)^{-1/2} \in \mathbb{R}$. The four-velocity is defined by

$$U := \frac{dX}{ds} = \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \quad (117)$$

Its normalization is

$$\det(U) = 1. \quad (118)$$

The energy and spatial momentum of a particle with real rest mass $m > 0$ are defined by

$$E := m\gamma, \quad (119)$$

$$\mathbf{P} := m\gamma\mathbf{v}. \quad (120)$$

The four-momentum is therefore

$$mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}. \quad (121)$$

from which $E^2 - \mathbf{P}^2 = m^2$ is obtained immediately. The normalized tangent U is conventionally called the four-velocity; only its spatial part $\gamma\mathbf{v}$ is often denoted by “proper velocity.” The four-momentum of a massive particle is obtained by multiplication by $m > 0$. A paravector with zero determinant is conventionally called null. For a massless particle $ds = 0$, so U is undefined and a null momentum paravector must instead be introduced directly.

2.2 Four-acceleration and free motion

The four-acceleration is the real paravector

$$A := \frac{dU}{ds} = \frac{d^2 X}{ds^2}. \quad (122)$$

It obeys

$$\text{scalar}(U^* A) = 0. \quad (123)$$

Equation (123) states that the scalar part of the product of velocity and acceleration is zero. This is the standard four-acceleration orthogonality relation. The proper-acceleration magnitude $\sqrt{-\det(A)}$ is unchanged by a Lorentz transform; the paravector A itself is not unchanged.

A Lagrangian is a scalar function whose time integral is made stationary; its derivatives give the equations of motion through the Euler–Lagrange equation [5, 16]. For a free particle, it is defined here by

$$\mathcal{L} := -\frac{m}{\gamma}, \quad (124)$$

from whose Euler–Lagrange equation constant-velocity motion and the usual low-speed expansion are reproduced:

$$\mathcal{L} = -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4). \quad (125)$$

Equation (125) is displayed explicitly to exhibit the reduction of the relativistic action to the classical kinetic expression plus an additive rest-energy constant in the low-speed limit.

2.3 Supporting Kinematic Derivations

Equations (116), (117), and (121) are expanded in the supporting derivations, followed by remarks on coordinates that need not be straight Cartesian coordinates and on the interval coefficients associated with them.

2.3.1 Space-Time Numbers

A spacetime increment is the real paravector

$$dX = dt + dr_1\sigma_1 + dr_2\sigma_2 + dr_3\sigma_3 = dt + \mathbf{dr} \cdot \boldsymbol{\sigma},$$

with conjugate

$$dX^* = dt - dr_1\sigma_1 - dr_2\sigma_2 - dr_3\sigma_3 = dt - \mathbf{dr} \cdot \boldsymbol{\sigma}.$$

Here $dt \in \mathbb{R}$ is the scalar part and $\mathbf{dr} \in \mathbb{R}^3$ is the spatial vector part. Their recovery formulas are

$$\frac{dX + dX^*}{2} = dt, \quad \frac{dX - dX^*}{2} = \mathbf{dr} \cdot \boldsymbol{\sigma},$$

so

$$\text{scalar}(dX) = dt, \quad \text{vector}(dX) = \mathbf{dr}.$$

The determinant of the increment is

$$\det(dX) = dX dX^* = dX^* dX = dt^2 - \mathbf{dr}^2 = ds^2.$$

The following identities are direct consequences of the difference $dt^2 - \mathbf{dr}^2$:

$$\begin{aligned} dt^2 > \mathbf{dr}^2, \quad dX \neq 0 \text{ implies } \det(dX) > 0, \\ \det(dX + dX') - \det(dX) &= 2(dt dt' - \mathbf{dr} \cdot \mathbf{dr}'), \\ \det(a dX) &= a^2 \det(dX), \quad a \in \mathbb{R}. \end{aligned}$$

The commutator of two spacetime increments has zero scalar part and lies entirely in the imaginary-vector part:

$$\begin{aligned} \text{com}(dX, dX') &= \text{com}(\mathbf{dr} \cdot \boldsymbol{\sigma}, \mathbf{dr}' \cdot \boldsymbol{\sigma}) \\ &= 2i(\mathbf{dr} \times \mathbf{dr}') \cdot \boldsymbol{\sigma}. \end{aligned}$$

A path is now taken for which $dt > 0$ and $dt^2 - \mathbf{dr}^2 > 0$. The first condition selects increasing time, and the second makes the interval timelike. Such a path is conventionally called future-directed and timelike [5, 6]. Its positive interval length satisfies

$$\begin{aligned} ds &= \sqrt{dt^2 - \mathbf{dr}^2} \\ &= \sqrt{1 - \mathbf{dr}^2/dt^2} dt \\ &= \sqrt{1 - \mathbf{v}^2} dt, \end{aligned}$$

where

$$\mathbf{v} := \frac{\mathbf{dr}}{dt} = \{v_1, v_2, v_3\} \in \mathbb{R}^3, \quad \gamma := \frac{dt}{ds} = \frac{1}{\sqrt{1 - \mathbf{v}^2}}.$$

The factor γ , which compares coordinate time with proper time, is conventionally called the Lorentz factor. For a real non-null increment, $dX^H = dX$, so the inverse exists and

$$\begin{aligned} (dX)^{-1} &= \frac{\text{adj}(dX)}{\det(dX)} \\ &= \frac{dX^*}{\det(dX)} \\ &= \frac{dX^*}{dX dX^*} \\ &= \frac{dX^*}{ds^2}. \end{aligned}$$

An increment with $ds^2 = 0$ is called null below. Such increments are singled out algebraically because the inverse breaks down at exactly that value.

2.3.2 Interval Classes and Proper Time

For a spacetime increment $dX = dt + d\mathbf{r} \cdot \boldsymbol{\sigma}$, the interval is

$$\det(dX) = dX dX^* = dt^2 - d\mathbf{r}^2 = ds^2.$$

The sign of this displayed quantity defines the three standard interval classes [4–6]:

$$\begin{aligned} ds^2 > 0 & \text{ implies a timelike interval,} \\ ds^2 = 0, dX \neq 0 & \text{ implies a null interval,} \\ ds^2 < 0 & \text{ implies a spacelike interval.} \end{aligned}$$

If $\det(dX) > 0$, the interval is timelike. The time read by a clock moving along the path is called proper time. In the units used here, it is the accumulated positive interval length s :

$$\begin{aligned} s &= \int ds = \int \sqrt{1 - \mathbf{v}^2} dt \\ &= \int \frac{1}{\gamma} dt. \end{aligned}$$

The difference between dt and the moving clock reading ds is the time-dilation relation. In the International System of Units (SI), the interval has length units, and the corresponding clock-time increment is

$$\frac{ds}{c}.$$

If $\det(dX) = 0$ with $dX \neq 0$, the interval is null or lightlike. Then

$$dt^2 = d\mathbf{r}^2 \quad \text{implies} \quad dt = \pm \|d\mathbf{r}\|,$$

and $(dX)^{-1}$ does not exist.

If $\det(dX) < 0$, the interval is spacelike.

Thus the interval classification, the moving-clock formula, and the failure of $(dX)^{-1}$ for a null increment all follow from the one algebraic identity $dX dX^* = dt^2 - d\mathbf{r}^2$.

2.3.3 Four-Velocity

A particle path through space and time is called its worldline. For a timelike worldline, division of its increment by the corresponding proper time gives

$$\begin{aligned} \frac{dX}{ds} &= \frac{dt}{ds} + \frac{d\mathbf{r}}{ds} \cdot \boldsymbol{\sigma} \\ &= \frac{dt + \boldsymbol{\sigma} \cdot d\mathbf{r}}{\sqrt{dt^2 - d\mathbf{r}^2}} \\ &= \frac{dt/dt + \boldsymbol{\sigma} \cdot (d\mathbf{r}/dt)}{\sqrt{(dt/dt)^2 - (d\mathbf{r}/dt)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \boldsymbol{\sigma}}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \boldsymbol{\sigma}). \end{aligned}$$

Its scalar and vector parts are therefore

$$\text{scalar}\left(\frac{dX}{ds}\right) = \gamma, \quad \text{vector}\left(\frac{dX}{ds}\right) = \gamma\mathbf{v}.$$

Its determinant is unity:

$$\begin{aligned} \det\left(\frac{dX}{ds}\right) &= \gamma^2(1 + \mathbf{v} \cdot \boldsymbol{\sigma})(1 - \mathbf{v} \cdot \boldsymbol{\sigma}) \\ &= \gamma^2(1 - \mathbf{v}^2) \\ &= 1. \end{aligned}$$

Also,

$$\frac{dX}{dt} = \frac{dX}{ds} \frac{ds}{dt} = \frac{1}{\gamma} \frac{dX}{ds}.$$

This determinant-one rate of change is called the four-velocity paravector and is defined by

$$\begin{aligned} U &= \frac{dX}{ds} \\ &= \gamma + \gamma\mathbf{v} \cdot \boldsymbol{\sigma}, \\ \text{scalar}(U) &= \gamma, \quad \text{vector}(U) = \gamma\mathbf{v}. \end{aligned}$$

Since $ds^2 = dX dX^*$ with $ds > 0$,

$$\begin{aligned} \frac{dX^*}{ds} &= (dX)^{-1} ds \\ &= \left(\frac{dX}{ds}\right)^{-1}, \end{aligned}$$

so $U^{-1} = U^*$ and

$$\det(U) = UU^* = 1.$$

Differentiation of the determinant-one condition gives

$$\begin{aligned} \frac{d}{ds} \det(U) &= \frac{dU}{ds} U^* + U \left(\frac{dU}{ds}\right)^* \\ &= \left(\frac{dU}{ds} U^*\right) + \left(\frac{dU}{ds} U^*\right)^{*H}, \end{aligned}$$

hence

$$\text{scalar}\left(\frac{dU}{ds} U^*\right) = \text{scalar}\left(U^* \frac{dU}{ds}\right) = 0.$$

The determinant identity is recovered as

$$\begin{aligned} \det(dX) &= dX dX^* \\ &= \left(\frac{dX}{ds} ds\right) \left(\left(\frac{dX}{ds}\right)^* ds\right) \\ &= UU^* ds^2 \\ &= \det(U) ds^2 \\ &= ds^2. \end{aligned}$$

The acceleration formulas below are based on this normalization and on the result that the displayed scalar part vanishes. The

four-velocity paravector is resolved into its scalar and spatial components in the following table.

Table 16. Components of dX/ds .

scalar:	γ
vector:	$\gamma \mathbf{v}$

2.3.4 Four-Acceleration

The rate of change of the three-velocity with respect to the chosen time coordinate is called the coordinate acceleration and is defined by

$$\mathbf{a} := \frac{d\mathbf{v}}{dt} = \{a_1, a_2, a_3\} \in \mathbb{R}^3.$$

With $\gamma = (1 - \mathbf{v}^2)^{-1/2}$, the following relations hold:

$$\begin{aligned} \frac{d\gamma}{dt} &= \gamma^3(\mathbf{v} \cdot \mathbf{a}), \\ \frac{d\gamma}{ds} &= \gamma^4(\mathbf{v} \cdot \mathbf{a}). \end{aligned}$$

The scalar derivative with respect to the three velocity components is

$$\frac{\partial}{\partial \mathbf{v}} \left(-\frac{1}{\gamma} \right) = \gamma \mathbf{v}.$$

The following derivatives are obtained by direct differentiation of the vector product $\gamma \mathbf{v}$:

$$\begin{aligned} \frac{d(\gamma \mathbf{v})}{dt} &= \gamma \mathbf{a} + \frac{d\gamma}{dt} \mathbf{v} \\ &= \gamma \mathbf{a} + \gamma^3(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}, \\ \frac{d(\gamma \mathbf{v})}{ds} &= \frac{d(\gamma \mathbf{v})}{dt} \frac{dt}{ds} \\ &= \gamma^2 \mathbf{a} + \gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}. \end{aligned}$$

The scalar and vector parts of four-velocity are $\text{scalar}(U) = \gamma$ and $\text{vector}(U) = \gamma \mathbf{v}$, as obtained above. The rate of change of four-velocity with respect to proper time is called four-acceleration. It is therefore

$$\begin{aligned} A &= \frac{dU}{ds} \\ &= \frac{d\gamma}{ds} + \frac{d(\gamma \mathbf{v})}{ds} \cdot \sigma \\ &= \gamma^4(\mathbf{v} \cdot \mathbf{a}) + \left(\gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a} \right) \cdot \sigma. \end{aligned}$$

Equivalently,

$$\begin{aligned} \text{scalar}(A) &= \gamma^4(\mathbf{v} \cdot \mathbf{a}), \\ \text{vector}(A) &= \gamma^4(\mathbf{v} \cdot \mathbf{a}) \mathbf{v} + \gamma^2 \mathbf{a}. \end{aligned}$$

For $\mathbf{v} \neq 0$, the unit direction is defined by $\mathbf{u} := \mathbf{v}/\|\mathbf{v}\|$, with $\mathbf{u}^2 = 1$. The following components are then obtained from the axis functions in Eqs. (39) and (40):

$$\begin{aligned} \text{par}(\mathbf{a}) &= (\mathbf{a} \cdot \mathbf{u}) \mathbf{u}, \\ \text{prp}(\mathbf{a}) &= \mathbf{a} - \text{par}(\mathbf{a}). \end{aligned}$$

Then

$$\begin{aligned} \mathbf{a}^2 &= \text{par}(\mathbf{a})^2 + \text{prp}(\mathbf{a})^2, \\ (\mathbf{v} \cdot \mathbf{a})^2 &= \mathbf{v}^2 \text{par}(\mathbf{a})^2, \end{aligned}$$

so the proper-acceleration determinant is

$$\begin{aligned} \det(A) &= -\gamma^4 \left(\mathbf{a}^2 + \gamma^2 (\mathbf{v} \cdot \mathbf{a})^2 \right) \\ &= -\gamma^4 \left(\text{prp}(\mathbf{a})^2 + \gamma^2 \text{par}(\mathbf{a})^2 \right). \end{aligned}$$

The determinant is negative unless the acceleration vanishes, so the four-acceleration is spacelike. Its positive magnitude $\sqrt{-\det(A)}$ is unchanged by a Lorentz transformation. It is called proper acceleration because it equals the acceleration measured at that event in the momentarily comoving frame, where the particle velocity is zero.

The products with U are

$$\begin{aligned} UA &= (\text{scalar}(U) \text{scalar}(A) + \text{vector}(U) \cdot \text{vector}(A)) \\ &\quad + (\text{scalar}(U) \text{vector}(A) + \text{scalar}(A) \text{vector}(U) \\ &\quad + i(\text{vector}(U) \times \text{vector}(A))) \cdot \sigma, \\ U^*A &= (\text{scalar}(U) \text{scalar}(A) - \text{vector}(U) \cdot \text{vector}(A)) \\ &\quad + (\text{scalar}(U) \text{vector}(A) - \text{scalar}(A) \text{vector}(U) \\ &\quad - i(\text{vector}(U) \times \text{vector}(A))) \cdot \sigma. \end{aligned}$$

Their component forms are

$$\begin{aligned} UA &= 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \\ &\quad + (\gamma^3 \mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}) \cdot \sigma \\ &\quad + i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \sigma, \\ U^*A &= \gamma^3 \mathbf{a} \cdot \sigma - i\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot \sigma. \end{aligned}$$

In particular,

$$\text{scalar}(U^*A) = 0.$$

The vanishing scalar part is the precise meaning of orthogonality between four-velocity and four-acceleration in this notation. The scalar and vector parts of U are recorded in Table 16, while the corresponding parts of UA and U^*A are compared in Tables 17 and 18. The products UA and U^*A are compared in the following two tables, and the orthogonality structure of proper acceleration is thereby made explicit.

Table 17. Components of UA .

real scalar:	$2\gamma^5(\mathbf{v} \cdot \mathbf{a})$
img scalar:	0
real vector:	$\gamma^3 \mathbf{a} + 2\gamma^5(\mathbf{v} \cdot \mathbf{a}) \mathbf{v}$
img vector:	$\gamma^3(\mathbf{v} \times \mathbf{a})$

Table 18. Components of U^*A .

real scalar:	0
img scalar:	0
real vector:	$\gamma^3 \mathbf{a}$
img vector:	$-\gamma^3(\mathbf{v} \times \mathbf{a})$

2.3.5 Four-Momentum

For a massive particle with rest mass $m > 0$, the energy–momentum paravector is produced by multiplication of four-velocity by m :

$$mU = E + \mathbf{P} \cdot \sigma.$$

Its scalar part is the relativistic energy,

$$\begin{aligned} E &:= \text{scalar}(mU) \\ &= m\gamma. \end{aligned}$$

If $\mathbf{v} = 0$, then

$$E = m \quad \text{rest energy} \quad (= mc^2 \text{ in SI units}).$$

The kinetic energy is therefore $E - m$.

Its vector part is the relativistic momentum,

$$\begin{aligned} \mathbf{P} &:= \text{vector}(mU) \\ &= m\gamma\mathbf{v}. \end{aligned}$$

Because $\det(U) = 1$, the following restriction on the allowed energy and momentum is obtained immediately:

$$\begin{aligned} \det(mU) &= m^2 \\ &= (E + \mathbf{P} \cdot \sigma)(E - \mathbf{P} \cdot \sigma) \\ &= E^2 - \mathbf{P}^2. \end{aligned}$$

Hence

$$E^2 = \mathbf{P}^2 + m^2.$$

The set of energy–momentum values satisfying this equation is conventionally called the mass shell [10, 17]. For massless motion it reduces to the null branch $E^2 = \mathbf{P}^2$. That branch must be introduced directly as a null energy–momentum paravector: the equation $mU = E + \mathbf{P} \cdot \sigma$ applies only to massive timelike motion because $ds = 0$ and $U = dX/ds$ is undefined on a null worldline.

2.3.6 Flat Paravector Coordinate Map

The same flat event paravector may be described by functions of another time label t and three position labels $\mathbf{r}$. The scalar and vector functions are taken to have the domains

$$\begin{aligned} f &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, \\ \mathbf{g} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3. \end{aligned}$$

Their outputs are used as the scalar and vector components of a real paravector:

$$X := f(t, \mathbf{r}) + \mathbf{g}(t, \mathbf{r}) \cdot \sigma. \quad (126)$$

Thus $(t, \mathbf{r})$ supplies labels, while $(f, \mathbf{g})$ supplies the components inserted into the already defined flat interval. The numerical coefficients of that interval need not remain constant when nonlinear labels are used. The differential and its conjugate are

$$dX = df + d\mathbf{g} \cdot \sigma, \quad dX^* = df - d\mathbf{g} \cdot \sigma.$$

Their product is

$$\begin{aligned} \det(dX) &= dX \, dX^* \\ &= (df + d\mathbf{g} \cdot \sigma)(df - d\mathbf{g} \cdot \sigma) \\ &= df^2 - d\mathbf{g}^2. \end{aligned}$$

Along a future-directed timelike path, the conditions $df > 0$ and $df^2 > d\mathbf{g}^2$ are assumed. Then

$$\begin{aligned} ds &= \sqrt{df^2 - d\mathbf{g}^2} \\ &= \sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2} df. \end{aligned}$$

Within this subsection, the bare symbols $\mathbf{v}$, γ , and $\mathbf{a}$ are defined relative to the components $(f, \mathbf{g})$. They reduce to the earlier definitions when $f = t$ and $\mathbf{g} = \mathbf{r}$. The velocity and Lorentz factor are

$$\begin{aligned} \mathbf{v} &:= \frac{d\mathbf{g}}{df}, \\ \gamma &:= \frac{df}{ds} \\ &= \frac{df}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{1}{\sqrt{1 - \mathbf{v}^2}}. \end{aligned}$$

The corresponding four-velocity is

$$\begin{aligned} U &= \frac{dX}{ds} \\ &= \frac{df + d\mathbf{g} \cdot \sigma}{\sqrt{df^2 - d\mathbf{g}^2}} \\ &= \frac{1 + \left(\frac{d\mathbf{g}}{df}\right) \cdot \sigma}{\sqrt{1 - \left(\frac{d\mathbf{g}}{df}\right)^2}} \\ &= \frac{1 + \mathbf{v} \cdot \sigma}{\sqrt{1 - \mathbf{v}^2}} \\ &= \gamma(1 + \mathbf{v} \cdot \sigma), \end{aligned}$$

where the division by df is valid because $df > 0$. Thus $\mathbf{v}^2 < 1$. For a massive particle, the energy–momentum relation remains unchanged:

$$mU = E + \mathbf{P} \cdot \sigma.$$

The coordinate acceleration relative to f is defined, in direct analogy with $\mathbf{a} = d\mathbf{v}/dt$ above, by

$$\begin{aligned} \mathbf{a} &:= \frac{d\mathbf{v}}{df} \\ &= \frac{d^2\mathbf{g}}{df^2}. \end{aligned}$$

Because $d/ds = \gamma d/df$, the four-acceleration has the same form as before:

$$\frac{dU}{ds} = \gamma^4(\mathbf{v} \cdot \mathbf{a}) + (\gamma^4(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma^2\mathbf{a}) \cdot \sigma.$$

The commutator of four-velocity and four-acceleration is

$$\begin{aligned}\text{com}\left(U, \frac{dU}{ds}\right) &= \text{com}\left(U, \frac{dU}{df}\right) \frac{df}{ds} \\ &= \gamma \text{com}\left(U, \frac{dU}{df}\right) \\ &= 2\gamma^3(\mathbf{v} \times \mathbf{a}) \cdot (i\boldsymbol{\sigma}).\end{aligned}$$

For a massive particle on which no force acts, four-acceleration vanishes in the flat components $(f, \mathbf{g})$:

$$m \frac{dU}{ds} = 0,$$

which is equivalently the straight-worldline equation

$$\frac{d^2 X}{ds^2} = 0.$$

After division by the nonzero factor γ , its scalar and vector parts are

$$\begin{aligned}\gamma^3(\mathbf{v} \cdot \mathbf{a}) &= 0, \\ \gamma^3(\mathbf{v} \cdot \mathbf{a})\mathbf{v} + \gamma\mathbf{a} &= 0.\end{aligned}$$

For $\mathbf{v}^2 < 1$, one has $\gamma > 0$. The relation $\mathbf{v} \cdot \mathbf{a} = 0$ is therefore obtained from the scalar part, and $\mathbf{a} = 0$ is then obtained from the vector part. Hence

$$\frac{d^2 \mathbf{g}}{df^2} = 0,$$

and along such a force-free worldline

$$\mathbf{g} = \mathbf{c}f + \mathbf{c}', \quad \mathbf{v} = \frac{d\mathbf{g}}{df} = \mathbf{c}.$$

Thus motion without force is linear in $(f, \mathbf{g})$, with constant velocity. If nonlinear coordinate labels are used for this same straight path, differentiating the coordinate map twice produces extra terms from the second derivatives of f and $\mathbf{g}$. Such terms are conventionally called connection terms. They can be caused entirely by the coordinate labels and therefore do not by themselves describe curved space-time. Curvature would require a change in the interval geometry that cannot be removed by any coordinate relabeling throughout a region. A path whose force-free equation includes the required connection terms is conventionally called a geodesic [15]. Only a straight path in the fixed flat components has been derived here; no equation for a curved space-time has been assumed or proved.

2.3.7 Lagrangian Mechanics and Conserved Quantities

For a particle trajectory written in terms of time, position, and velocity, the action is defined by

$$\int \mathcal{L}(t, \mathbf{r}, \mathbf{v}) dt =: \text{action}.$$

Its integrand $\mathcal{L}$ is called the Lagrangian. Requiring the first-order change of the action to vanish for fixed endpoints gives the Euler-Lagrange equation [15]:

$$\begin{aligned}\partial_{\mathbf{r}}\mathcal{L} - \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) &= 0, \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) &= \partial_t \mathcal{L}.\end{aligned}$$

Two constant quantities follow directly when the Lagrangian lacks the corresponding dependence. Canonical momentum is defined by

$$\frac{\partial \mathcal{L}}{\partial \mathbf{v}} =: \text{canonical momentum}.$$

If the Lagrangian has no position dependence, so that $\partial_{\mathbf{r}}\mathcal{L} = 0$, then

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) = 0,$$

and the canonical momentum is constant along the path.

The scalar combination is defined by

$$\begin{aligned}\mathcal{H} &:= \mathbf{v} \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{v}} - \mathcal{L} \\ &= \left(\mathbf{v} \cdot \frac{\partial}{\partial \mathbf{v}} - 1 \right) \mathcal{L}\end{aligned}\tag{127}$$

and is called the Hamiltonian. Its time derivative is

$$\begin{aligned}\frac{d\mathcal{H}}{dt} &= \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) \cdot \mathbf{v} + \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt} - \frac{d\mathcal{L}}{dt} \\ &= (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt} \\ &\quad - \left(\partial_t \mathcal{L} + (\partial_{\mathbf{r}}\mathcal{L}) \cdot \mathbf{v} + \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \frac{d\mathbf{v}}{dt} \right) \\ &= -\partial_t \mathcal{L}.\end{aligned}$$

Therefore, if $\partial_t \mathcal{L} = 0$, then

$$\frac{d\mathcal{H}}{dt} = 0,$$

so the Hamiltonian is constant along the path. These identities are applied directly to the free-particle example and to the electromagnetic example in which the potential is added to the action.

2.3.8 Action of a Free Particle

The free-particle action is negative mass times the accumulated proper time along the worldline. With $U = dX/ds$ and $\gamma =$

dt/ds ,

$$\begin{aligned}
\int \mathcal{L} dt &= -m \int \sqrt{\det(dX)} \\
&= -m \int ds \\
&= -m \int \sqrt{\det\left(\frac{dX}{dt}\right)} dt \\
&= -m \int \frac{ds}{dt} dt \\
&= -m \int \sqrt{\det\left(\frac{U}{\gamma}\right)} dt \\
&= -m \int \frac{1}{\gamma} dt \\
&= -m \int \sqrt{1 - \mathbf{v}^2} dt.
\end{aligned}$$

Therefore

$$\mathcal{L} = -\frac{m}{\gamma}.$$

The canonical momentum is obtained by differentiation with respect to velocity:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{v}} &= m\gamma \mathbf{v} \\
&= \mathbf{P}, \\
\left(\frac{\partial}{\partial \mathbf{v}} \cdot \boldsymbol{\sigma}\right) \mathcal{L} &= \mathbf{P} \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The Hamiltonian is

$$\begin{aligned}
\mathcal{H} &= \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \mathbf{v} - \mathcal{L} \\
&= m\gamma \mathbf{v}^2 + \frac{m}{\gamma} \\
&= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) \\
&= m\gamma \\
&= E.
\end{aligned}$$

Because the free Lagrangian has no explicit time or position dependence,

$$\partial_t \mathcal{L} = 0, \quad \partial_{\mathbf{r}} \mathcal{L} = 0,$$

so the conserved quantities are the relativistic energy $E = m\gamma$ and momentum $\mathbf{P} = m\gamma \mathbf{v}$.

2.3.9 Low-Velocity Approximation

When the speed is small compared with $c = 1$, the motion is conventionally called non-relativistic. The free Lagrangian is then expanded directly:

$$\begin{aligned}
\mathcal{L} &= -\frac{m}{\gamma} \\
&= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).
\end{aligned}$$

The canonical momentum is then reduced to

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{v}} &= \frac{\partial}{\partial \mathbf{v}} \left(\frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \right) \\
&= m\mathbf{v} + O(\mathbf{v}^3) \\
&\approx m\mathbf{v}.
\end{aligned}$$

The total energy is the kinetic term plus the rest-mass term:

$$\begin{aligned}
\mathcal{H} &= (m\mathbf{v} + O(\mathbf{v}^3)) \cdot \mathbf{v} + m - \frac{1}{2}m\mathbf{v}^2 - O(\mathbf{v}^4) \\
&= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4) \\
&\approx m + \frac{1}{2}m\mathbf{v}^2.
\end{aligned}$$

Equivalently, the low-velocity series is

$$\begin{aligned}
\mathcal{L} &= -m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4), \\
\mathbf{P} &= m\mathbf{v} + O(\mathbf{v}^3), \\
\mathcal{H} &= m + \frac{1}{2}m\mathbf{v}^2 + O(\mathbf{v}^4).
\end{aligned}$$

The Newtonian free-particle kinetic term together with the additive rest-energy constant is thereby obtained as the low-velocity approximation to the relativistic action.

2.3.10 Matrix Form of the Flat Interval in General Coordinates

As an optional translation into ordinary matrix coordinates, the map $X = f + \mathbf{g} \cdot \boldsymbol{\sigma}$ permits the same flat interval to be written in terms of the original four coordinate labels

$$\mathbf{x} = \begin{bmatrix} t \\ r_1 \\ r_2 \\ r_3 \end{bmatrix}, \quad d\mathbf{x} = \begin{bmatrix} dt \\ dr_1 \\ dr_2 \\ dr_3 \end{bmatrix}.$$

The transpose operation was defined in Eq. (279). After df and $d\mathbf{g}$ are expanded in these four increments, their squared difference can be written using one real symmetric matrix. That matrix is defined by

$$df^2 - d\mathbf{g}^2 =: d\mathbf{x}^T \mathbf{Q}(\mathbf{x}) d\mathbf{x}. \quad (128)$$

It is conventionally called the pullback matrix because it expresses the interval from the $(f, \mathbf{g})$ components in terms of the original $(t, \mathbf{r})$ labels. In this section the name means exactly the displayed equality [15]. Consequently,

$$ds^2 = d\mathbf{x}^T \mathbf{Q}(\mathbf{x}) d\mathbf{x}.$$

The four-by-four matrix of first derivatives of $(f, \mathbf{g})$ with respect to $(t, \mathbf{r})$ is now considered. At a point where this derivative matrix has an inverse, the coordinate map is called locally invertible in the differential sense used here. The matrix $\mathbf{Q}$ is obtained by placing this derivative matrix on both sides of the one-positive, three-negative interval matrix, with its transpose on the left. An invertible real change of coordinates therefore cannot change

the count of one positive and three negative squared directions. That sign count is called the signature $(+, -, -, -)$. The entries of $\mathbf{Q}$ may nevertheless depend on position when nonlinear, or curvilinear, coordinates are used. Such position dependence alone does not imply curvature because $(f, \mathbf{g})$ still supplies components in the fixed flat algebra. If the derivative matrix has no inverse, a nonzero coordinate direction is sent to zero. The same direction is then sent to zero by $\mathbf{Q}$, so $\mathbf{Q}$ necessarily has no inverse. This case is called degenerate.

For a real symmetric matrix, four mutually perpendicular directions can be chosen so that multiplication by $\mathbf{Q}$ only rescales each direction. The rescaling numbers are called eigenvalues, and the corresponding directions are called eigenvectors. At every locally invertible point, the ordered eigenvalue function is defined by

$$\lambda(n) := \text{ordered eigenvalue } n \text{ of } \mathbf{Q}, \quad n \in \{1, 2, 3, 4\}. \quad (129)$$

For the four-entry use required here, the diagonal matrix function is defined by

$$\text{diag}(A, B, C, D) := \begin{bmatrix} A & 0 & 0 & 0 \\ 0 & B & 0 & 0 \\ 0 & 0 & C & 0 \\ 0 & 0 & 0 & D \end{bmatrix}. \quad (130)$$

The normalized eigenvectors are placed in the columns of $\mathbf{N}$, giving

$$\mathbf{Q} = \mathbf{N} \text{diag}(\lambda(1), \lambda(2), \lambda(3), \lambda(4)) \mathbf{N}^T, \quad \mathbf{N}^T \mathbf{N} = \mathbf{I}.$$

The second equality states that the columns have unit length and are mutually perpendicular; $\mathbf{I}$ is the identity matrix of the required four-by-four size. The eigenvalues are ordered numerically so that

$$\lambda(1) > 0 > \lambda(2) \geq \lambda(3) \geq \lambda(4).$$

The corresponding direction function is defined by

$$\mathbf{u}(n) := \text{column } n \text{ of } \mathbf{N}. \quad (131)$$

The defining eigenvalue relation is therefore

$$\mathbf{Q}\mathbf{u}(n) = \lambda(n)\mathbf{u}(n).$$

If two or more eigenvalues are equal, the individual eigenvectors within their shared directions are not unique. Any mutually perpendicular unit vectors spanning those same directions give the same matrix and interval. The local differential directions are defined by

$$w(n) := \sqrt{|\lambda(n)|} \mathbf{u}(n)^T d\mathbf{x}. \quad (132)$$

Each $w(n)$ is the coordinate increment along one eigenvector, rescaled by the square root of the corresponding eigenvalue magnitude. Substitution of the matrix decomposition gives

$$\begin{aligned} ds^2 &= \lambda(1)(\mathbf{u}(1)^T d\mathbf{x})^2 + \lambda(2)(\mathbf{u}(2)^T d\mathbf{x})^2 \\ &\quad + \lambda(3)(\mathbf{u}(3)^T d\mathbf{x})^2 + \lambda(4)(\mathbf{u}(4)^T d\mathbf{x})^2 \\ &= w(1)^2 - w(2)^2 - w(3)^2 - w(4)^2. \end{aligned}$$

An interval with one positive and three negative squared directions is called Lorentzian. A Euclidean squared length would instead contain only plus signs. The three minus signs therefore cannot be dropped by changing matrix directions. When $\mathbf{Q}$ varies with position, the four quantities $w(n)$ describe the interval at one point; a single set of coordinates whose differentials equal all four $w(n)$ need not exist throughout an extended region.

3 Transform Applications

The generic similarity law and the rotation law were established in Section 1.11. The Lorentz action on real event, velocity, and momentum paravectors is separated here from similarity and is applied explicitly.

For $L = \exp(\theta \mathbf{u} \cdot \boldsymbol{\sigma}/2)$, the passive event boost is $X' = L^{-1}X(L^{-1})^H = L^*XL^*$, the $T = L$ specialization of Eq. (97). The Lorentz component formulas are supplied by its scalar and vector parts as derived below, and $\det(X)$ is preserved. It is shown explicitly in Eq. (103) that a Lorentz boost of a real event is not produced by either hyperbolic similarity generated by L . The distinct similarity law (100) is later applied in Section 5 to the electromagnetic field paravector F . In the earlier rotation convention, the object is rotated while the basis is held fixed; this is called an active transform. In the boost convention used here, the same event is described in a changed frame; this is called a passive transform [5, 14].

3.1 Lorentz Boost

A coordinate system in which force-free motion has constant velocity is called an inertial frame. A Lorentz boost relates two inertial frames whose spatial origins move at constant relative velocity [4–6]. The velocity of the primed frame relative to the unprimed frame is written $\beta \mathbf{u}$ in the sign convention used below. Thus $\mathbf{u}$ gives the direction, $\mathbf{u}^2 = 1$, and the signed speed β satisfies $|\beta| < 1$. The parameters and the corresponding algebra element are

$$\begin{aligned} \theta &:= \text{arctanh}(\beta), \\ \beta &= \tanh(\theta), \\ \gamma &:= \cosh(\theta), \\ \gamma\beta &= \sinh(\theta), \\ \mathbf{u} &:= \{u_1, u_2, u_3\}, \quad \mathbf{u}^2 = 1, \\ L(\theta, \mathbf{u}) &:= \exp(\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})), \\ &= \cosh(\tfrac{1}{2}\theta) + \sinh(\tfrac{1}{2}\theta)(\mathbf{u} \cdot \boldsymbol{\sigma}). \end{aligned}$$

The parameter θ , defined by $\beta = \tanh(\theta)$, is conventionally called rapidity, and $\gamma = 1/\sqrt{1 - \beta^2}$ is the Lorentz factor. The functions $\cosh(\theta)$ and $\sinh(\theta)$ replace the cosine and sine of a spatial rotation, which is why L is called a hyperbolic transform element [14, 15]. The element defined in Eq. (84) has the inverse

$$L^{-1} = \exp(-\tfrac{1}{2}\theta(\mathbf{u} \cdot \boldsymbol{\sigma})).$$

Here $\theta, \gamma, \beta \in \mathbb{R}$, $\mathbf{u} \in \mathbb{R}^3$, and $L \in \mathbb{P}_{\mathbb{R}}$. Direct application of the two algebra operations gives $L^{-1} = L^* = L^{*H}$. Elements

with the same direction $\mathbf{u}$, conventionally called collinear boosts, commute:

$$\text{com}(L(\theta, \mathbf{u}), L(\theta', \mathbf{u})) = 0.$$

Consequently, boosts along that same direction are combined by addition of θ . With $\beta' = \tanh(\theta')$ and $\gamma' = \cosh(\theta')$,

$$\begin{aligned} L(\theta, \mathbf{u})L(\theta', \mathbf{u}) &= L(\theta + \theta', \mathbf{u}), \\ \tanh(\theta + \theta') &= \frac{\beta + \beta'}{1 + \beta\beta'}, \\ \cosh(\theta + \theta') &= \gamma\gamma'(1 + \beta\beta'). \end{aligned}$$

The coordinates assigned to one fixed event by the two frames are now compared. This interpretation, in which the event is unchanged while its coordinate description changes, is called a passive boost. With the signs fixed by $\beta\mathbf{u}$ above, the real event paravector $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, with $t \in \mathbb{R}$, $\mathbf{r} \in \mathbb{R}^3$, and $X \in \mathbb{P}_{\mathbb{R}}$, is transformed as

$$\begin{aligned} X &= t + \mathbf{r} \cdot \boldsymbol{\sigma}, \\ X' &= L^{-1}X(L^{-1})^H, \\ &= L^*XL^*, \\ &= t' + \mathbf{r}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

This is the event transformation $X' = T^{-1}X(T^{-1})^H$ from Eq. (97), specialized to $T = L$. A product of the form AXA^H is conventionally called a congruence. It preserves $X^H = X'$ whenever $X^H = X$, which is the Hermitian condition that makes an event paravector real. An active reading would keep the coordinate frame fixed and move the event; the inverse transformation, obtained by replacing θ with $-\theta$, would then describe the corresponding motion. Only the passive reading is used in the component formulas below [5, 14].

$$\begin{aligned} t' &= t \cosh(\theta) - \sinh(\theta)(\mathbf{u} \cdot \mathbf{r}), \\ &= \gamma(t - \beta(\mathbf{u} \cdot \mathbf{r})). \end{aligned}$$

The vector part is made explicit by the axis functions in Eqs. (39) and (40), through which $\mathbf{r}$ is split into components parallel and perpendicular to the chosen direction:

$$\begin{aligned} \text{par}(\mathbf{r}) &= (\mathbf{r} \cdot \mathbf{u})\mathbf{u} && \text{parallel to } \mathbf{u}, \\ \text{prp}(\mathbf{r}) &= \mathbf{r} - \text{par}(\mathbf{r}) && \text{perpendicular to } \mathbf{u}, \\ \mathbf{r} &= \text{par}(\mathbf{r}) + \text{prp}(\mathbf{r}), \\ \text{par}(\mathbf{r}') &= \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ \text{prp}(\mathbf{r}') &= \text{prp}(\mathbf{r}), \\ \mathbf{r}' &= \text{par}(\mathbf{r}') + \text{prp}(\mathbf{r}'), \\ &= \text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u}), \\ &= \mathbf{r} + (\gamma(\mathbf{r} \cdot \mathbf{u} - \beta t) - \mathbf{r} \cdot \mathbf{u})\mathbf{u}. \end{aligned}$$

Equivalently, the full transformed paravector is obtained in one line by direct multiplication:

$$L^*XL^* = \gamma(t - \beta \mathbf{u} \cdot \mathbf{r}) + (\text{prp}(\mathbf{r}) + \gamma(\text{par}(\mathbf{r}) - \beta t\mathbf{u})) \cdot \boldsymbol{\sigma}.$$

Thus the component parallel to $\mathbf{u}$ is mixed with the scalar part while the perpendicular component is left unchanged. In the small-parameter limit $\beta \rightarrow 0$,

$$\gamma \rightarrow 1, t' \rightarrow t, \mathbf{r}' \rightarrow \mathbf{r}.$$

Preservation of the interval is then obtained directly from $L^{-1} = L^*$:

$$\begin{aligned} \det(X') &= X'X'^*, \\ &= (L^*XL^*)(L^*XL^*)^*, \\ &= L^*XL^*LX^*L, \\ &= L^*XX^*L, \\ &= \det(X)L^*L, \\ &= \det(X). \end{aligned}$$

Length contraction is obtained only after the two endpoint events to be compared are specified. A rod is taken to have fixed endpoint positions in the unprimed frame; this frame is therefore the rod's rest frame. The rod is aligned with $\mathbf{u}$. Its two endpoints are denoted by the function inputs (1) and (2), so

$$\text{prp}(\mathbf{r}(2) - \mathbf{r}(1)) = 0, \quad l := \|\text{par}(\mathbf{r}(2) - \mathbf{r}(1))\|.$$

Thus l is the separation of the endpoints in the frame where both positions are fixed. A length measured in the primed frame must instead use two endpoint events occurring at the same primed time, so $t'(2) - t'(1) = 0$. The following relation is then obtained from the inverse boost:

$$\begin{aligned} \text{par}(\mathbf{r}(2) - \mathbf{r}(1)) &= \gamma \text{par}(\mathbf{r}'(2) - \mathbf{r}'(1)) \\ &\quad + \gamma\beta(t'(2) - t'(1))\mathbf{u} \\ &= \gamma \text{par}(\mathbf{r}'(2) - \mathbf{r}'(1)). \end{aligned}$$

and therefore the primed length is

$$l' := \|\text{par}(\mathbf{r}'(2) - \mathbf{r}'(1))\| = \frac{l}{\gamma}.$$

The perpendicular length is unchanged.

The four-momentum of a massive particle is acted on by the same congruence:

$$\begin{aligned} (mU)' &= L^*(mU)L^* \\ &= E' + \mathbf{P}' \cdot \boldsymbol{\sigma}. \end{aligned}$$

Consequently,

$$\begin{aligned} E' &= \gamma(E - \beta \mathbf{u} \cdot \mathbf{P}), \\ \text{par}(\mathbf{P}') &= \gamma(\text{par}(\mathbf{P}) - \beta E\mathbf{u}), \\ \text{prp}(\mathbf{P}') &= \text{prp}(\mathbf{P}). \end{aligned}$$

Because the same real-paravector transformation is used as for X , $E^2 - \mathbf{P}^2 = m^2$ is preserved. As defined in the four-momentum section, this equation is the mass shell condition: it is the restriction that relates the allowed energy and momentum of a particle of mass m .

3.2 Compositions of Boosts and Rotors

The ordered products and their inverses are listed. The order remains significant even when a simple inverse rule is satisfied separately by each factor. These factorizations are purely algebraic. An active or passive physical interpretation is assigned only after the complete product has been inserted into the applicable transformation law.

For one hyperbolic element and one circular element,

$$\begin{aligned} T &= LR, \\ T^{-1} &= T^{*\text{H}} = R^{*\text{H}}L^{*\text{H}} = R^{\text{H}}L^*, \\ T &= RL, \\ T^{-1} &= T^{*\text{H}} = L^{*\text{H}}R^{*\text{H}} = L^*R^{\text{H}}. \end{aligned}$$

For two factors of the same type,

$$\begin{aligned} T &= RR', \\ T^{-1} &= T^{*\text{H}} = R'^{\text{H}}R^{\text{H}}, \\ T &= LL', \\ T^{-1} &= T^{*\text{H}} = L'^*\text{H}L^*. \end{aligned}$$

These formulas are simple order identities, but the inverse of each ordered product is fixed by them without ambiguity.

4 Differential Operators

One fixed set of differential conventions is used in the electromagnetic and wave-equation sections. The conventions are collected here so that the same numbered equations can be referenced later without repeated operator definitions.

For spatial derivatives with respect to $\mathbf{r} = \{r_1, r_2, r_3\}$, the derivatives are defined by $\partial_t := \partial/\partial t$ and $\partial_n := \partial/\partial r_n$ for $n \in \{1, 2, 3\}$. The spatial derivative vector is defined by

$$\partial_{\mathbf{r}} := \left\{ \begin{array}{c} \frac{\partial}{\partial r_1} \\ \frac{\partial}{\partial r_2} \\ \frac{\partial}{\partial r_3} \end{array} \right\} = \{\partial_1, \partial_2, \partial_3\}. \quad (133)$$

The corresponding spatial Laplacian is

$$\partial_{\mathbf{r}}^2 := \partial_1^2 + \partial_2^2 + \partial_3^2. \quad (134)$$

The differential paravector and its conjugate are

$$\partial := \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3, \quad (135)$$

$$\partial^* := \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} = \partial_t - \sigma_1 \partial_1 - \sigma_2 \partial_2 - \sigma_3 \partial_3. \quad (136)$$

Because the coordinate derivatives commute on smooth fields, a scalar second-order wave operator is obtained from their ordered products:

$$\square := \partial \partial^*. \quad (137)$$

The reversed order gives the same scalar operator:

$$\square = \partial^* \partial = \partial_t^2 - \partial_{\mathbf{r}}^2. \quad (138)$$

This operator is conventionally called the d'Alembertian [7, 8]. The ordinary two-term derivative product rule must be corrected when the left field does not commute with the sigma basis. For smooth paravector fields Z and Z' , the following result is obtained by direct expansion:

$$\partial(ZZ') = (\partial Z)Z' + Z(\partial Z') + \sum_{n=1}^3 \text{com}(\sigma_n, Z) \partial_n Z'. \quad (139)$$

The derivatives act on fields written to their right, with parentheses delimiting their scope. The final term is generated when Z is moved through the left-acting basis element σ_n . It is zero for a scalar Z , or whenever $\text{com}(\sigma_n, Z) = 0$ for every contributing direction, and the ordinary product rule is then recovered.

4.1 Complex Fields

The differential identities are first recorded for an arbitrary complex paravector field. The letter Z is used here so that F can be reserved for the electromagnetic field introduced later:

$$\begin{aligned} \partial &= \partial_t + \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma}, \\ Z &:= (a + ib) + (\mathbf{A} + i\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

At each time and position, a and b return real numbers, while $\mathbf{A}$ and $\mathbf{B}$ return real three-component vectors:

$$\begin{aligned} a &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, & \mathbf{A} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \\ b &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}, & \mathbf{B} &: \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{R}^3, \end{aligned}$$

so that

$$Z : \mathbb{R} \times \mathbb{R}^3 \rightarrow \mathbb{P}_{\mathbb{C}}.$$

The letters $\mathbf{A}$ and $\mathbf{B}$ in this subsection are generic placeholders. Their electromagnetic meanings are assigned only in Section 5. Direct multiplication gives the scalar and vector parts of ∂Z and $i\partial Z$:

Table 19. Components of ∂Z .

real scalar:	$\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$
img scalar:	$\partial_t b + \partial_{\mathbf{r}} \cdot \mathbf{B}$
real vector:	$\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$
img vector:	$\partial_{\mathbf{r}} b + \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{A}$

Table 20. Components of $i\partial Z$.

real scalar:	$-\partial_t b - \partial_{\mathbf{r}} \cdot \mathbf{B}$
img scalar:	$\partial_t a + \partial_{\mathbf{r}} \cdot \mathbf{A}$
real vector:	$-\partial_{\mathbf{r}} b - \partial_t \mathbf{B} - \partial_{\mathbf{r}} \times \mathbf{A}$
img vector:	$\partial_{\mathbf{r}} a + \partial_t \mathbf{A} - \partial_{\mathbf{r}} \times \mathbf{B}$

5 Electromagnetic Fields

Electromagnetic fields, their potential freedom, and the Lorentz force can be expressed in compact paravector forms. Different potentials that give the same electric and magnetic fields are said to be related by a gauge change. In physical content, Maxwell theory is recovered in rationalized Heaviside–Lorentz units [2, 16]; the packaging is drawn from later space-time-algebra presentations [12, 14, 15].

5.1 Fields, potentials, and Maxwell equations

The differential conventions are fixed once in Section 4; in particular, the spatial derivative vector, the paravector derivative, its conjugate, and the d'Alembertian are defined by Eqs. (133)–(137). The quantities V, ρ, S, λ are real scalar fields of time and position: V is the scalar potential, ρ is charge density, S is defined below, and λ labels a change of potential. The quantities $\mathbf{E}, \mathbf{B}, \mathbf{A}, \mathbf{J}$ are real three-vector fields: electric field, magnetic field, vector potential, and electric-current density, respectively. The charge q is a real constant. With these meanings fixed, the electromagnetic field vector is defined by

$$\mathbf{F} := \mathbf{E} + i\mathbf{B}. \quad (140)$$

The corresponding pure-vector paravector is defined by

$$F := \mathbf{F} \cdot \boldsymbol{\sigma}. \quad (141)$$

The real potential paravector is defined separately by

$$\Phi := V + \mathbf{A} \cdot \boldsymbol{\sigma}. \quad (142)$$

Thus $\Phi \in \mathbb{P}_{\mathbb{R}}$, while $F \in \mathbb{P}_{\mathbb{C}}$ is restricted to have zero scalar part. Electric and magnetic fields are packaged into F by Eq. (141), and scalar and vector potentials are packaged into Φ .

The scalar part generated by the potential is defined by

$$S := \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \quad (143)$$

The relations in Eq. (145) are then encoded by

$$\partial\Phi = S + F^*. \quad (144)$$

with

$$\mathbf{E} = -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \quad \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{A}. \quad (145)$$

For a vector field, $\partial_{\mathbf{r}} \cdot \mathbf{A}$ measures its local outward flow and is conventionally called the divergence, while $\partial_{\mathbf{r}} \times \mathbf{A}$ measures its local circulation and is conventionally called the curl. The compact Maxwell equation is

$$\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (146)$$

By splitting the real/imaginary scalar/vector parts of (146), the following equations are reproduced:

$$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho, \quad \partial_{\mathbf{r}} \cdot \mathbf{B} = 0, \quad (147)$$

$$\partial_{\mathbf{r}} \times \mathbf{B} = \mathbf{J} + \partial_t \mathbf{E}, \quad \partial_{\mathbf{r}} \times \mathbf{E} = -\partial_t \mathbf{B}. \quad (148)$$

Agreement with standard space-time-algebra treatments is obtained in Eqs. (147) and (148), subject to the sign conventions fixed by Eqs. (135), (136), (137), and (141). Maxwell's field equations are recovered component by component in the present signature and field convention [2, 12, 14, 16]. Separate primitive divergence and curl operators are avoided at this stage in Eq. (146). They are produced automatically when the scalar and vector parts of the single paravector equation are separated.

5.2 Boost transformation of electromagnetic fields

The field paravector is transformed by the same boost element L from Eq. (84):

$$F' = L^{-1} F L = L^* F L. \quad (149)$$

Equation (149) is the similarity form $L^{-1} F L$. It is deliberately different from the Hermitian event congruence $X' = L^{-1} X (L^{-1})^{\dagger}$. The event and field laws were derived from different algebraic actions in Section 1.11; the usual mixing of $\text{par}(\mathbf{E})$, $\text{prp}(\mathbf{E})$, $\text{par}(\mathbf{B})$, and $\text{prp}(\mathbf{B})$ is obtained from the field similarity.

5.3 Wave equations, gauge, and Lorentz force

Two wave-equation consequences of Maxwell's equation are obtained by applying ∂^* to Eq. (146) and ∂ to its star-conjugate equation:

$$\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}), \quad (150)$$

$$\square F^* = \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}). \quad (151)$$

Because $\square$ is scalar and F has zero scalar part, the left side of Eq. (150) has zero scalar part. Its right-side scalar part therefore gives charge conservation:

$$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0. \quad (152)$$

Equation (152) is the continuity equation written in the present notation. It states that a local change of charge density is balanced by charge flowing through the surrounding boundary [16]. Different potentials may describe the same electric and magnetic fields. Such potentials are said to differ by a gauge change [16]. For a real scalar gauge function λ , a function input is used so that confusion with a Lorentz-frame prime is avoided:

$$\Phi(\lambda) := \Phi + \partial^* \lambda. \quad (153)$$

The scalar and vector potential functions contained in this paravector are defined by

$$V(\lambda) := V + \partial_t \lambda, \quad (154)$$

$$\mathbf{A}(\lambda) := \mathbf{A} - \partial_{\mathbf{r}} \lambda. \quad (155)$$

The scalar part generated by the transformed potential is defined by

$$\begin{aligned} S(\lambda) &:= \partial_t V(\lambda) + \partial_{\mathbf{r}} \cdot \mathbf{A}(\lambda) \\ &= S + (\partial_t^2 - \partial_{\mathbf{r}}^2) \lambda \\ &= S + \square \lambda. \end{aligned} \quad (156)$$

The corresponding field function is defined from the transformed potential by the same relation as Eq. (144):

$$\begin{aligned} F(\lambda)^* &:= \partial\Phi(\lambda) - S(\lambda) \\ &= \partial\Phi + \square\lambda - S - \square\lambda \\ &= F^*. \end{aligned} \quad (157)$$

Application of the star operation to the final equality gives

$$F(\lambda) = F. \quad (158)$$

The potential Φ is retained even when the field is physically primary because many potentials related by Eq. (153) give the same F . A quantity unchanged by this replacement is called gauge invariant; the derivation ending at Eq. (158) shows that F has this property [16].

For a particle of real mass $m > 0$, real charge q , and four-velocity U , the four-force formula is

$$m \frac{dU}{ds} = q \operatorname{real}(FU) = q\gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}), \quad (159)$$

in which both relativistic power and the Lorentz three-force are packaged. Since $d/ds = \gamma d/dt$, the equations $dE/dt = q\mathbf{v} \cdot \mathbf{E}$ and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ are obtained by division by γ .

5.4 Lagrangian and energy-momentum remarks

Here potential coupling means adding $-q(V - \mathbf{v} \cdot \mathbf{A})$ to the free-particle Lagrangian. The resulting function is defined by

$$\mathcal{L} := -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}), \quad (160)$$

and $d\mathbf{P}/dt = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ is recovered by substitution into the Euler–Lagrange equation. Thus the potential form gives the same force written directly with $\mathbf{E}$ and $\mathbf{B}$ [16].

The frame-dependent field energy-density and flux paravector is defined by

$$T := \frac{1}{2} F F^H. \quad (161)$$

Its scalar and vector parts are

$$T = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \quad (162)$$

The first term in Eq. (162) is the field energy per unit volume. The vector $\mathbf{E} \times \mathbf{B}$ is the field energy flux and is conventionally called the Poynting vector [16]. These two quantities are packaged into one frame-dependent paravector. The supporting subsection explains how the energy and momentum seen along any chosen observer direction are obtained from a map conventionally called the electromagnetic stress–energy map.

5.5 Supporting Electromagnetic Derivations

The field decomposition, potential reconstruction, source formulas, gauge transformations, force-law expansions, and variational identities underlying Eqs. (133)–(161) are written out in the supporting derivations.

5.5.1 Electricity and Magnetism

The compact field definitions in Eqs. (141) and (146) are expanded into the algebra parts already defined. Rationalized Heaviside–Lorentz units are used, with $c = 1$ [16]. Here $\rho \in \mathbb{R}$, $\mathbf{E}, \mathbf{B}, \mathbf{J} \in \mathbb{R}^3$, $\mathbf{F} = \mathbf{E} + i\mathbf{B} \in \mathbb{C}^3$, and $F = \mathbf{F} \cdot \boldsymbol{\sigma}$ is a pure-vector complex paravector, so $\operatorname{scalar}(F) = 0$. The spatial derivative conventions are

$$\begin{aligned} \partial_{\mathbf{r}} &= \{\partial_1, \partial_2, \partial_3\}, \\ \partial_n &= \partial/\partial r_n, \quad n \in \{1, 2, 3\}, \\ \partial_{\mathbf{r}}^2 &= \partial_1^2 + \partial_2^2 + \partial_3^2, \end{aligned}$$

and the differential paravector is

$$\begin{aligned} \partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}} \\ &= \partial_t + \sigma_1 \partial_1 + \sigma_2 \partial_2 + \sigma_3 \partial_3, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}. \end{aligned}$$

These conventions restate Eqs. (133), (135), and (136). The real/imaginary scalar/vector decomposition of ∂Z and $i\partial Z$ for an arbitrary complex field Z was recorded in Subsection 4.1. The electric and magnetic fields can therefore be inserted into the same multiplication rule.

The usual electric and magnetic fields [2, 16] are collected into the complex three-vector $\mathbf{F}$ and the field paravector F :

$$\begin{aligned} \mathbf{E} &= \{E_1, E_2, E_3\} && \text{real vector electric field,} \\ \mathbf{B} &= \{B_1, B_2, B_3\} && \text{real vector magnetic field,} \\ \mathbf{F} &= \mathbf{E} + i\mathbf{B} && \text{electromagnetic field,} \\ F &= \mathbf{F} \cdot \boldsymbol{\sigma} = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad i = \sigma_1 \sigma_2 \sigma_3. \end{aligned}$$

Its sigma conjugate is obtained as

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &= -\mathbf{F}^* \cdot \boldsymbol{\sigma}. \end{aligned}$$

The two simplest quadratic field combinations are given by

$$\begin{aligned} \frac{1}{2} F^2 &= \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + i\mathbf{E} \cdot \mathbf{B}, \\ \frac{1}{2} F F^H &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The two coefficients in $\frac{1}{2} F^2$ are unchanged by the boosts and rotations generated continuously from the identity. These transformations preserve both spatial orientation and the chosen direction of increasing time; they are the proper, time-orientation-preserving Lorentz transformations used in this document. They do not include a spatial reflection $\mathbf{r} \mapsto -\mathbf{r}$, conventionally called parity. Under parity the electric field changes sign and the magnetic field does not, so $\mathbf{E} \cdot \mathbf{B}$ changes sign [16]. It is therefore the coefficient of the algebra pseudoscalar i , rather than an ordinary scalar under reflection.

The second product separates into $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$, the electromagnetic energy per unit volume, and $\mathbf{E} \times \mathbf{B}$, the energy crossing a unit area per unit time in the chosen frame and units. The latter is conventionally called the Poynting vector [2, 16].

The real source paravector is defined by

$$D := \rho + \mathbf{J} \cdot \boldsymbol{\sigma}. \quad (163)$$

Thus $D \in \mathbb{P}_{\mathbb{R}}$. Its sigma conjugate is

$$D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}.$$

Here $\rho \in \mathbb{R}$ is charge per unit volume, called charge density, and $\mathbf{J} \in \mathbb{R}^3$ is charge flow per unit area per unit time, called current density. Maxwell's equations are then written as

$$\begin{aligned} \partial F &= D^* && \text{Maxwell's equations,} \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The companion equation is obtained by sigma conjugation:

$$\partial^* F^* = D.$$

In this form, the four classical Maxwell equations are reproduced in one step by the scalar, sigma-number-one, and sigma-number-two parts of ∂F . The location of each component is shown by expansion of the product.

The following expression is obtained by expansion of ∂F into the four algebra parts:

$$\begin{aligned} \partial F &= (\partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}), \\ &= (\partial_{\mathbf{r}} \cdot \mathbf{E}), \\ &\quad + (\partial_{\mathbf{r}} \cdot \mathbf{B})i, \\ &\quad + (\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ &\quad + (\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E}) \cdot (i\boldsymbol{\sigma}). \end{aligned}$$

The first two lines are the real-scalar and i -scalar parts. The last two lines are the sigma-number-one and sigma-number-two vector parts. Comparison with $D^* = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ forces the second and fourth parts to vanish and fixes the first and third parts.

Accordingly, the component equations of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$ are

$$\begin{aligned} \partial_{\mathbf{r}} \cdot \mathbf{E} &= \rho, \\ \partial_{\mathbf{r}} \cdot \mathbf{B} &= 0, \\ \partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} &= -\mathbf{J}, \\ \partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} &= 0. \end{aligned}$$

The following relations are then obtained by separation of the vector components:

$$\begin{aligned} \partial_{\mathbf{r}} \times \mathbf{B} &= \mathbf{J} + \partial_t \mathbf{E}, \\ \partial_{\mathbf{r}} \times \mathbf{E} &= -\partial_t \mathbf{B}. \end{aligned}$$

5.5.2 Lorentz Transform of Fields

The boost law is specialized to the electromagnetic field paravector

$$F = (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

For the boost $L = \exp(\frac{1}{2}\theta \mathbf{u} \cdot \boldsymbol{\sigma})$, the unit vector $\mathbf{u}$ gives the boost direction, $\beta = \tanh(\theta)$ is its signed speed, and $\gamma = \cosh(\theta)$ is

the Lorentz factor. The boost is constant here, so none of these quantities depends on position or time. The field law is

$$F' = L^{-1}FL = L^*FL.$$

An inverse on the left and the original element on the right define the similarity transformation of Eq. (100). This differs from the event transformation $T^{-1}(\cdot)(T^{-1})^H$ in Eq. (97). The final field factor must therefore be L , not L^* . A pair of L^* factors would apply the event law to F , and the result would no longer be a pure-vector paravector. For example, if $F = a \mathbf{u} \cdot \boldsymbol{\sigma}$ for a real scalar amplitude a , then

$$L^*FL = F, \quad L^*FL^* = \gamma a \mathbf{u} \cdot \boldsymbol{\sigma} - \gamma \beta a,$$

and a scalar part appears in the second expression. Application of the general complex-vector result (104) to $\mathbf{V} = \mathbf{E} + i\mathbf{B}$, the field and its par and prp parts are obtained:

$$\begin{aligned} F' &= L^*FL, \\ \text{par}(\mathbf{E}') &= \text{par}(\mathbf{E}), \\ \text{par}(\mathbf{B}') &= \text{par}(\mathbf{B}), \\ \text{prp}(\mathbf{E}') &= \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \text{prp}(\mathbf{B}') &= \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}). \end{aligned}$$

Equivalently, the transformed fields can be written in the compact form

$$\begin{aligned} \mathbf{E}' &= \text{par}(\mathbf{E}) + \gamma(\text{prp}(\mathbf{E}) + \beta \mathbf{u} \times \mathbf{B}), \\ \mathbf{B}' &= \text{par}(\mathbf{B}) + \gamma(\text{prp}(\mathbf{B}) - \beta \mathbf{u} \times \mathbf{E}). \end{aligned}$$

The parallel components are unchanged, while the perpendicular components are mixed by the standard electric-magnetic cross terms. Because $LL^* = 1$ and F^2 commutes with every algebra element, the quadratic field combination is preserved:

$$F'^2 = L^*F(LL^*)FL = L^*F^2L = F^2.$$

Thus both coefficients of F^2 are preserved by the proper, time-orientation-preserving Lorentz transformations constructed here. Under the spatial reflection defined above, its real coefficient is unchanged and its i -coefficient changes sign.

For any constant Lorentz element T satisfying $T^{-1} = T^{*H}$, the real potential and source are transformed by the same two-sided rule as event paravectors. The derivative is transformed by the complementary factors required for its product with the field:

$$\Phi' = T^{*H}\Phi T^*, \quad D' = T^{*H}DT^*, \quad \partial' = T^H\partial T.$$

For the boost $T = L$, these expressions are reduced to

$$\Phi' = L^*\Phi L^*, \quad D' = L^*DL^*, \quad \partial' = L\partial L.$$

Consequently,

$$\begin{aligned} \partial' F' &= (T^H\partial T)(T^{*H}FT) \\ &= T^H(\partial F)T \\ &= T^H D^* T \\ &= D'^*. \end{aligned}$$

The first and last lines have the same form, $\partial' F' = D'^*$, as the unprimed equation $\partial F = D^*$. This preservation of the equation's form when every quantity is transformed is what is meant here by covariance. In components, the source transformation is

$$\begin{aligned}\rho' &= \gamma(\rho - \beta \mathbf{u} \cdot \mathbf{J}), \\ \text{par}(\mathbf{J}') &= \gamma(\text{par}(\mathbf{J}) - \beta \rho \mathbf{u}), \\ \text{prp}(\mathbf{J}') &= \text{prp}(\mathbf{J}).\end{aligned}$$

5.5.3 Wave Operator

The operator conventions were fixed in Eqs. (135)–(137). The product of the differential paravector and its conjugate is the scalar operator

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \square &= \partial_t^2 - \partial_{\mathbf{r}}^2 \\ &= \partial \partial^* = \partial^* \partial.\end{aligned}$$

This scalar operator is conventionally called the d'Alembertian [2, 16]. The following relation is also obtained from the sigma algebra:

$$\partial^2 = \partial_t^2 + \partial_{\mathbf{r}}^2 + 2\partial_t \partial_{\mathbf{r}} \cdot \boldsymbol{\sigma},$$

so $\partial^2 \neq \square$. With the conventions of Eqs. (135)–(137), the following result is obtained by applying ∂^* to Maxwell's equation:

$$\begin{aligned}\square F &= \partial^* \partial F \\ &= \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma}) \\ &= (\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J}) \\ &\quad + (-\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho + i \partial_{\mathbf{r}} \times \mathbf{J}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The companion relation is obtained by applying ∂ to the star-conjugate Maxwell equation $\partial^* F^* = D$:

$$\square F^* = \partial D = \partial(\rho + \mathbf{J} \cdot \boldsymbol{\sigma}),$$

which is Eq. (151) in the compact development. Because F is pure vector, the scalar part of the left-hand side vanishes. The scalar and vector parts are therefore given by

$$\begin{aligned}\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} &= 0, \\ \square \mathbf{E} &= -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho, \\ \square \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{J}.\end{aligned}$$

A region with no charge density and no current density, $\rho = 0$ and $\mathbf{J} = 0$, is called vacuum here. In such a region each field component satisfies the ordinary wave equation:

$$\begin{aligned}\square \mathbf{E} = 0 &\quad \text{implies} \quad \partial_t^2 \mathbf{E} = \partial_{\mathbf{r}}^2 \mathbf{E}, \\ \square \mathbf{B} = 0 &\quad \text{implies} \quad \partial_t^2 \mathbf{B} = \partial_{\mathbf{r}}^2 \mathbf{B}.\end{aligned}$$

The scalar and vector parts of Maxwell's equation and its sourced wave consequence are recorded in Tables 21 and 22.

Table 21. Components of $\partial F = \rho - \mathbf{J} \cdot \boldsymbol{\sigma}$.

real scalar:	$\partial_{\mathbf{r}} \cdot \mathbf{E} = \rho$
img scalar:	$\partial_{\mathbf{r}} \cdot \mathbf{B} = 0$
real vector:	$\partial_t \mathbf{E} - \partial_{\mathbf{r}} \times \mathbf{B} = -\mathbf{J}$
img vector:	$\partial_t \mathbf{B} + \partial_{\mathbf{r}} \times \mathbf{E} = 0$

Table 22. Components of $\square F = \partial^*(\rho - \mathbf{J} \cdot \boldsymbol{\sigma})$.

real scalar:	$\partial_t \rho + \partial_{\mathbf{r}} \cdot \mathbf{J} = 0$
img scalar:	$0 = 0$
real vector:	$\square \mathbf{E} = -\partial_t \mathbf{J} - \partial_{\mathbf{r}} \rho$
img vector:	$\square \mathbf{B} = \partial_{\mathbf{r}} \times \mathbf{J}$

5.5.4 Potential Fields

A real scalar field V and a real vector field $\mathbf{A}$ are collected into the real potential paravector $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$. Direct multiplication by the differential paravector gives

$$\begin{aligned}F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ \partial \Phi &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ &\quad + (\partial_t \mathbf{A} + \partial_{\mathbf{r}} V) \cdot \boldsymbol{\sigma}, \\ &\quad + i(\partial_{\mathbf{r}} \times \mathbf{A}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

This expansion contains one real scalar part and one pure-vector complex part. The scalar part is assigned the short name S :

$$\begin{aligned}S &= \text{scalar}(\partial \Phi), \\ &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}.\end{aligned}$$

The component identities extracted from the pure-vector part are

$$\begin{aligned}\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\ \mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}.\end{aligned}$$

The identity is explicit:

$$\begin{aligned}\partial \Phi &= S + F^*, \\ F^* &= \partial \Phi - S, \\ \text{scalar}(F^*) &= \text{scalar}(\partial \Phi - S) = 0, \\ \text{vector}(F^*) &= \text{vector}(\partial \Phi).\end{aligned}$$

The electromagnetic field is therefore exactly the non-scalar part of $\partial \Phi$. A change of potential that alters only S and leaves this field unchanged is derived below. The component content of the potential identity $\partial \Phi = S + F^*$ is recorded in Table 23.

Table 23. Components of $\partial \Phi = S + F^*$.

real scalar:	$\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A} = S$
img scalar:	$0 = 0$
real vector:	$\partial_t \mathbf{A} + \partial_{\mathbf{r}} V = -\mathbf{E}$
img vector:	$\partial_{\mathbf{r}} \times \mathbf{A} = \mathbf{B}$

The differential product $\partial\Phi$ is evaluated first; the involutions are then applied to the resulting field. No right-acting differential operator is implied. Equivalently, without introducing S , the relation can be written as

$$F^* = (\partial\Phi - (\partial\Phi)^{*H})/2.$$

5.5.5 Sources from the Potential

Charge and current can be recovered directly from the potential. From the previously defined potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, scalar part $S = \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}$, and identity $F^* = \partial\Phi - S$, the following relation is obtained:

$$\begin{aligned} F &= \partial^* \Phi^* - S, \\ \partial F &= \partial \partial^* \Phi^* - \partial S, \\ &= \square \Phi^* - \partial S, \\ &= (\square V - \partial_t S) + (-\square \mathbf{A} - \partial_{\mathbf{r}} S) \cdot \boldsymbol{\sigma}, \\ &= \rho - \mathbf{J} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The source components are therefore obtained as

$$\begin{aligned} \rho &= \square V - \partial_t S, \\ \mathbf{J} &= \square \mathbf{A} + \partial_{\mathbf{r}} S. \end{aligned}$$

Thus charge density is determined by the scalar equation and current density by the vector equation.

The condition $S = 0$, or $\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A} = 0$, is conventionally called the Lorenz gauge [16]. Under this condition the relations reduce to

$$\begin{aligned} \partial_t V &= -\partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \rho &= \square V, \\ \mathbf{J} &= \square \mathbf{A}. \end{aligned}$$

5.5.6 Gauge Transform of the Potential

The previously defined scalar potential condition is recalled first:

$$\begin{aligned} \text{scalar}(\partial\Phi) &= S, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}. \end{aligned}$$

An arbitrary real scalar field λ is chosen. A second potential and its components are defined by

$$\begin{aligned} \Phi(\lambda) &= \Phi + \partial^* \lambda, \\ V(\lambda) &= V + \partial_t \lambda, \\ \mathbf{A}(\lambda) &= \mathbf{A} - \partial_{\mathbf{r}} \lambda. \end{aligned}$$

Function inputs are used here so that this change is not confused with the prime used for a change of inertial frame. The potential gradient is defined separately by

$$(\partial\Phi)(\lambda) := \partial\Phi(\lambda). \quad (164)$$

By direct differentiation, the following relation is obtained:

$$(\partial\Phi)(\lambda) = \partial\Phi + \square\lambda.$$

Since $\square\lambda$ is again a scalar field, only the scalar part is changed. Its transformed value was defined in Eq. (156):

$$S(\lambda) = S + \square\lambda.$$

The unchanged field is obtained by substitution:

$$\begin{aligned} F(\lambda)^* &= \partial\Phi(\lambda) - S(\lambda) \\ &= \partial\Phi + \square\lambda - S - \square\lambda \\ &= F^*. \end{aligned}$$

The unchanged field stated in Eq. (158) therefore follows:

$$F(\lambda) = F.$$

Thus different potentials related by the displayed λ change produce exactly the same electric and magnetic fields. Such a change is called a gauge transformation, and the existence of these physically equivalent potentials is called gauge freedom [16]. If $S = 0$ before and after the change, then $\square\lambda = 0$; this is the complete condition for the displayed change to preserve the Lorenz gauge.

5.5.7 Four-Force

The energy-momentum paravector and the relation between coordinate time and proper time are

$$\begin{aligned} mU &= m \frac{dX}{ds}, \\ &= m\gamma + m\gamma \mathbf{v} \cdot \boldsymbol{\sigma}. \end{aligned}$$

The corresponding derivative conversion is

$$\frac{d}{ds} = \gamma \frac{d}{dt}.$$

The scalar part of $d(mU)/dt$ is the rate of change of energy, or power. Its vector part is the rate of change of momentum, or three-force.

The scalar equation is

$$\frac{d(m\gamma)}{dt} = q\mathbf{v} \cdot \mathbf{E}.$$

Here $\mathbf{E}$ denotes the electric field. This equation states that the energy gained per unit time equals charge times electric field dotted with velocity; it is the relativistic work law. No magnetic term appears because $\mathbf{v} \cdot (\mathbf{v} \times \mathbf{B}) = 0$. The vector equation is

$$\begin{aligned} \mathbf{P} &= m\gamma \mathbf{v}, \\ \frac{d\mathbf{P}}{dt} &= q\mathbf{E} + q\mathbf{v} \times \mathbf{B}. \end{aligned}$$

The right side is conventionally called the Lorentz force [4, 16]. Together, the scalar and vector equations are assembled into

$$\frac{d(mU)}{dt} = q\mathbf{v} \cdot \mathbf{E} + q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

The paravector content is exposed explicitly by the expansion

$$\begin{aligned}
FU/\gamma &= ((\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + (\mathbf{E} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\
&\quad + i(\mathbf{B} \cdot \boldsymbol{\sigma})(\mathbf{v} \cdot \boldsymbol{\sigma}), \\
&= \mathbf{E} \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \boldsymbol{\sigma} + \mathbf{E} \cdot \mathbf{v}, \\
&\quad + i(\mathbf{E} \times \mathbf{v}) \cdot \boldsymbol{\sigma} + i\mathbf{B} \cdot \mathbf{v} - (\mathbf{B} \times \mathbf{v}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The product FU separates into real and imaginary parts. Four-force is a real paravector, so the real projection is required. It contains exactly the work and force terms displayed above; the remaining imaginary components are not part of the electromagnetic force law. The component form of the product FU/γ used in the Lorentz-force law is given in Table 24.

Table 24. Components of FU/γ .

real scalar:	$\mathbf{E} \cdot \mathbf{v}$
img scalar:	$\mathbf{B} \cdot \mathbf{v}$
real vector:	$\mathbf{E} + \mathbf{v} \times \mathbf{B}$
img vector:	$\mathbf{B} + \mathbf{E} \times \mathbf{v}$

The required real projection is therefore obtained as

$$\begin{aligned}
\text{real}(FU) &= \frac{1}{2}(FU + UF^H), \\
&= \gamma(\mathbf{v} \cdot \mathbf{E} + (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \boldsymbol{\sigma}).
\end{aligned}$$

This real paravector also has the orthogonality required by Eq. (123):

$$\begin{aligned}
\text{scalar}(U^* \text{real}(FU)) &= \gamma^2(\mathbf{v} \cdot \mathbf{E} - \mathbf{v} \cdot (\mathbf{E} + \mathbf{v} \times \mathbf{B})) \\
&= 0.
\end{aligned}$$

The rate of change of four-momentum with respect to proper time is called four-force. It is therefore

$$\begin{aligned}
m \frac{dU}{ds} &= q \text{real}(FU), \\
&= \frac{1}{2}q(FU + UF^H).
\end{aligned}$$

The equivalent event-path form is

$$m \frac{d^2 X}{ds^2} = q \text{real}\left(F \frac{dX}{ds}\right) \quad \text{four-force.}$$

5.5.8 Action Derivation of the Lorentz Force

The free-particle action is supplemented by a scalar product of charge, potential, and four-velocity. From Eq. (160), together with the potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, four-velocity $U = dX/ds$, and $\gamma = dt/ds$, the action is obtained in the form

$$\begin{aligned}
\text{action} &= -m \int ds - q \int \text{scalar}(\Phi^* U) ds \\
&= \int \left[-\frac{m}{\gamma} - q \text{scalar}((V - \mathbf{A} \cdot \boldsymbol{\sigma})(1 + \mathbf{v} \cdot \boldsymbol{\sigma})) \right] dt \\
&= \int \left(-\frac{m}{\gamma} - qV + q\mathbf{v} \cdot \mathbf{A} \right) dt.
\end{aligned}$$

The coordinate-time Lagrangian is therefore

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

Adding the potential through these terms is conventionally called minimal electromagnetic coupling [16, 17]. In the present notation, it means exactly the displayed addition to the free-particle Lagrangian.

Differentiation with respect to velocity gives the canonical momentum:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial \mathbf{v}} &= m\gamma \mathbf{v} + q\mathbf{A}, \\
&= \mathbf{P} + q\mathbf{A}.
\end{aligned}$$

Its time derivative and spatial gradient are

$$\begin{aligned}
\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) &= \frac{d\mathbf{P}}{dt} + q\partial_t \mathbf{A} + q(\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}, \\
\partial_{\mathbf{r}} \mathcal{L} &= -q\partial_{\mathbf{r}} V + q\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}).
\end{aligned}$$

The directly verified vector identity

$$(\mathbf{v}(t) \cdot \partial_{\mathbf{r}}) \mathbf{A}(t, \mathbf{r}) = \partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A}).$$

holds because $\mathbf{v}(t)$ has no spatial dependence when $\partial_{\mathbf{r}}$ is applied. It is used in the Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \mathbf{v}} \right) = \partial_{\mathbf{r}} \mathcal{L}.$$

The result is

$$\begin{aligned}
\frac{d\mathbf{P}}{dt} &= -q\partial_{\mathbf{r}} V - q\partial_t \mathbf{A}, \\
&\quad + q(\partial_{\mathbf{r}}(\mathbf{v} \cdot \mathbf{A}) - (\mathbf{v} \cdot \partial_{\mathbf{r}}) \mathbf{A}), \\
&= q(-\partial_{\mathbf{r}} V - \partial_t \mathbf{A} + \mathbf{v} \times (\partial_{\mathbf{r}} \times \mathbf{A})), \\
&= q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \\
&= \frac{m}{\gamma} \text{vector} \left(\frac{dU}{ds} \right).
\end{aligned}$$

The Hamiltonian defined in Eq. (127) is

$$\begin{aligned}
\mathcal{H} &= \frac{\partial \mathcal{L}}{\partial \mathbf{v}} \cdot \mathbf{v} - \mathcal{L} \\
&= (m\gamma \mathbf{v} + q\mathbf{A}) \cdot \mathbf{v} + \frac{m}{\gamma} + qV - q\mathbf{v} \cdot \mathbf{A} \\
&= m\gamma \left(\mathbf{v}^2 + \frac{1}{\gamma^2} \right) + qV \\
&= m\gamma + qV.
\end{aligned}$$

Thus $\mathcal{H}$ is given by the mechanical relativistic energy plus the scalar-potential energy. Along a solution,

$$\begin{aligned}
\frac{d\mathcal{H}}{dt} &= -\partial_t \mathcal{L} = q\partial_t V - q\mathbf{v} \cdot \partial_t \mathbf{A}, \\
\frac{d\mathcal{H}}{dt} &= \frac{d(m\gamma)}{dt} + q\partial_t V + q\mathbf{v} \cdot \partial_{\mathbf{r}} V.
\end{aligned}$$

The following relation is obtained by equating these expressions:

$$\begin{aligned}\frac{d(m\gamma)}{dt} &= q\mathbf{v} \cdot (-\partial_{\mathbf{r}}V - \partial_t\mathbf{A}) \\ &= q\mathbf{v} \cdot \mathbf{E},\end{aligned}$$

in agreement with the scalar component of the Lorentz-force equation.

5.5.9 Energy-Density and Poynting Paravector

The scalar energy per unit volume and the vector energy flow per unit area per unit time are packaged into one paravector by direct multiplication:

$$\begin{aligned}F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ F^H &= (\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ T &= \frac{1}{2}FF^H, \\ &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}, \\ \text{scalar}(T) &= \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2), \quad \text{vector}(T) = \mathbf{E} \times \mathbf{B}.\end{aligned}$$

The differential operator used below is recalled as

$$\partial = \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}.$$

The paravector $T = \frac{1}{2}FF^H \in \mathbb{P}_{\mathbb{R}}$ is the energy density and flow measured relative to the scalar time direction represented by the identity 1. To obtain the corresponding energy–momentum output for an arbitrary real input direction $a \in \mathbb{P}_{\mathbb{R}}$, the function

$$T(a) := \frac{1}{2}F a F^H. \quad (165)$$

is used. Its scalar and vector output gives the energy and momentum flow associated with the selected direction a . Taken for every input direction, this linear function contains the energy density, momentum density, energy flow, and momentum flow. Momentum flow is also called electromagnetic stress, or force per unit area. This complete collection is conventionally called the electromagnetic stress-energy tensor [12, 15, 16].

The time-direction input reproduces the energy-density and flow paravector:

$$T(1) = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + (\mathbf{E} \times \mathbf{B}) \cdot \boldsymbol{\sigma}.$$

For a purely spatial input $a = \mathbf{a} \cdot \boldsymbol{\sigma}$, direct multiplication gives

$$\begin{aligned}\text{scalar}(T(a)) &= -\mathbf{a} \cdot (\mathbf{E} \times \mathbf{B}), \\ \text{vector}(T(a)) &= \mathbf{E}(\mathbf{E} \cdot \mathbf{a}) + \mathbf{B}(\mathbf{B} \cdot \mathbf{a}) - \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)\mathbf{a}.\end{aligned}$$

The first line is the energy flow through the selected spatial direction, with the sign fixed by the one-positive, three-negative interval convention. The second line supplies the corresponding momentum flow.

The argument-free notation is reserved for its value at the identity:

$$T = T(1) = \frac{1}{2}FF^H.$$

Thus the displayed energy-density and flow paravector is one output of the full function, not the full collection by itself. Indeed, for $a' = L^*aL^*$, $F' = L^*FL$, and $(F')^H = LF^HL^*$,

$$T'(a') := \frac{1}{2}F'a'(F')^H. \quad (166)$$

Substitution of the transformed field and input gives

$$T'(a') = L^* \left(\frac{1}{2}F a F^H \right) L^* = L^*T(a)L^*. \quad (167)$$

Thus the output transforms by the same event-paravector rule as its input. Equivalently, the function has the same relation between physical inputs and outputs in both frames.

The scalar energy balance is obtained immediately from Maxwell's equations:

$$\begin{aligned}\partial_t \text{scalar}(T) + \partial_{\mathbf{r}} \cdot \text{vector}(T) &= \mathbf{E} \cdot \partial_t \mathbf{E} + \mathbf{B} \cdot \partial_t \mathbf{B} \\ &\quad + \partial_{\mathbf{r}} \cdot (\mathbf{E} \times \mathbf{B}) \\ &= -\mathbf{J} \cdot \mathbf{E}.\end{aligned}$$

The equation states that any decrease of field energy in a region is accounted for by energy flowing through its boundary and work done on charge. This local energy balance is conventionally called Poynting's theorem [16]. The two quadratic field products and their components are collected below. The quantities preserved by the proper Lorentz transformations and the energy-density/Poynting split measured in a chosen frame are displayed without adding a field equation for F^2 . The two coefficients preserved by the proper, time-orientation-preserving Lorentz transformations are listed in Table 25. The energy-density/Poynting paravector $T = \frac{1}{2}FF^H$, whose components depend on the chosen inertial frame, is given in Table 26.

Table 25. Components of $\frac{1}{2}F^2$.

real scalar:	$\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\mathbf{E} \cdot \mathbf{B}$
real vector:	$\mathbf{0}$
img vector:	$\mathbf{0}$

Table 26. Components of $\frac{1}{2}FF^H$.

scalar:	$\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$	energy density
vector:	$\mathbf{E} \times \mathbf{B}$	Poynting vector

The differentiated field products and the products containing the source are displayed in the next five tables. They are algebraic identities and consequences of Maxwell's equation $\partial F = D^*$, not additional field equations. The algebra products do not generally commute: AB need not equal BA . The spatial part of the derivative acts from the left, so differentiating a product requires the ordered correction in Eq. (139). For the energy-density/Poynting paravector, the result is

$$\partial T = \frac{1}{2}(\partial F)F^H + \frac{1}{2}F(\partial F^H) + \frac{1}{2} \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F^H.$$

The component split of the left side is therefore the derived identity shown next.

Table 27. Components of ∂T .

real scalar:	$\partial_t \text{scalar}(T) + \partial_{\mathbf{r}} \cdot \text{vector}(T)$
img scalar:	0
real vector:	$\partial_{\mathbf{r}} \text{scalar}(T) + \partial_t \text{vector}(T)$
img vector:	$\partial_{\mathbf{r}} \times \text{vector}(T)$

Because F^2 is a complex scalar, its derivative can also be expanded directly:

Table 28. Components of $\partial(F^2)$.

real scalar:	$\partial_t(\mathbf{E}^2 - \mathbf{B}^2)$
img scalar:	$\partial_t(2\mathbf{E} \cdot \mathbf{B})$
real vector:	$\partial_{\mathbf{r}}(\mathbf{E}^2 - \mathbf{B}^2)$
img vector:	$\partial_{\mathbf{r}}(2\mathbf{E} \cdot \mathbf{B})$

The two ordered source products are given by Maxwell's equation:

Table 29. Components of FD^* .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} + \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} - \mathbf{E} \times \mathbf{J}$

Table 30. Components of D^*F .

real scalar:	$-\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-\mathbf{B} \cdot \mathbf{J}$
real vector:	$\rho\mathbf{E} - \mathbf{B} \times \mathbf{J}$
img vector:	$\rho\mathbf{B} + \mathbf{E} \times \mathbf{J}$

Their sum in the two possible product orders is

Table 31. Components of $D^*F + FD^*$.

real scalar:	$-2\mathbf{E} \cdot \mathbf{J}$
img scalar:	$-2\mathbf{B} \cdot \mathbf{J}$
real vector:	$2\rho\mathbf{E}$
img vector:	$2\rho\mathbf{B}$

Only part of the derivative of F^2 is contained in the last table. The complete identity is

$$\partial(F^2) = D^*F + FD^* + \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F.$$

With $F = \mathbf{F} \cdot \boldsymbol{\sigma}$, the correction term is

$$\begin{aligned} \sum_{n=1}^3 \text{com}(\sigma_n, F) \partial_n F &= -2i\mathbf{F} \cdot (\partial_{\mathbf{r}} \times \mathbf{F}) \\ &\quad - 2 \left((\partial_{\mathbf{r}} \cdot \mathbf{F})\mathbf{F} - \frac{1}{2}\partial_{\mathbf{r}}(\mathbf{F}^2) \right) \cdot \boldsymbol{\sigma}. \end{aligned}$$

It is precisely the correction that would be lost by treating the ordered paravector factors as though they commuted while applying the ordinary two-term derivative product rule.

6 Relativistic Wave Equations and Spin Dynamics

The standard relativistic wave equations are obtained from the same ordered product rules. Three allowed wave-function forms are distinguished: a complex scalar for the Klein–Gordon equation; a pair of two-amplitude right-projected paravectors for the two blocks of a Dirac field; and the two-amplitude field that remains in the slow-motion Pauli equation. With these forms separated, unused algebra coefficients cannot be mistaken for independent physical wave-function components [7–10]. Every spatial integral in this section is evaluated at fixed time. Unless a region is named, the integration is extended over all real values of the three spatial components; the measure is written directly as $dr_1 dr_2 dr_3$.

6.1 Klein–Gordon field and frequency splitting

The relativistic energy–momentum relation $E^2 - \mathbf{p}^2 = m^2$ supplies the physical starting point. Wave mechanics is then introduced through the established replacement of energy and momentum by the displayed differential operators; that replacement is physical input and is not obtained from the determinant identity by algebra alone [7, 8, 17]. For a complex scalar field $\psi_{\text{KG}} \in \mathbb{C}$, the free Klein–Gordon equation is

$$(\partial\partial^* + m^2)\psi_{\text{KG}} = (\square + m^2)\psi_{\text{KG}} = 0. \quad (168)$$

The spatial domain means the chosen set of fields together with the decay, periodic, or zero-boundary conditions imposed on them. On such a domain, the positive frequency operator is defined by

$$\Omega := \sqrt{m^2 - \partial_{\mathbf{r}}^2}. \quad (169)$$

Here a spatial Fourier mode means a plane-wave part with a definite spatial wave vector $\mathbf{k}$. The operator Ω multiplies that part by the positive real number $\sqrt{m^2 + \mathbf{k}^2}$ [17]. When Ω^{-1} exists on the chosen data, the two frequency functions are defined by

$$\psi_{\text{KG_split}}(\pm) := \frac{1}{2} (\psi_{\text{KG}} \pm i\Omega^{-1}\partial_t\psi_{\text{KG}}). \quad (170)$$

They satisfy

$$\begin{aligned} (\partial_t + i\Omega)\psi_{\text{KG_split}}(+) &= 0, & (\partial_t - i\Omega)\psi_{\text{KG_split}}(-) &= 0, \\ \psi_{\text{KG}} &= \psi_{\text{KG_split}}(+) + \psi_{\text{KG_split}}(-). \end{aligned} \quad (171)$$

Thus

$$\psi_{\text{KG_split}}(t, \pm) = \exp(\mp i\Omega t) \psi_{\text{KG_split}}(t, \pm)|_{t=0}.$$

When $m = 0$ and $\mathbf{k} = \mathbf{0}$, the multiplier is zero and its inverse does not exist; this massless zero mode is treated separately. The scalar $|\psi_{\text{KG}}|^2$ is not the relativistic Klein–Gordon probability

density; derivatives are contained in the conserved current, which is not positive on the full positive/negative-frequency solution space. Frequency is labeled in this subsection by the sign input of $\psi_{\text{KG_split}}(\pm)$. That input is unrelated to the plus/minus input of the spin-axis projectors in Eq. (71).

6.2 Potential coupling and ordered products

The established prescription called minimal coupling replaces the free energy and momentum operators by the following potential-dependent operators [16, 17]. They are gauge covariant: when the potential and field are changed together as displayed below, their action acquires only the same phase as the field. With the real potential $\Phi = V + \mathbf{A} \cdot \boldsymbol{\sigma}$, the energy operator is defined by

$$E := i\partial_t - qV. \quad (172)$$

The momentum operator is defined by

$$\mathbf{P} := -i\partial_{\mathbf{r}} - q\mathbf{A}. \quad (173)$$

The scalar equation with the stated electromagnetic potential is

$$(-E^2 + \mathbf{P}^2 + m^2)\psi_{\text{KG_with_potential}} = 0, \quad (174)$$

or, after applying the derivative product rule,

$$\begin{aligned} & \left(\square + iqS - q^2 \det(\Phi) + m^2 \right. \\ & \quad \left. 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \right) \\ & \quad \psi_{\text{KG_with_potential}} = 0. \end{aligned} \quad (175)$$

The unchanged form of the scalar equation after a simultaneous gauge change of the potential and phase change of the field is called gauge covariance. Under Eq. (153), the transformed scalar field is defined by

$$\psi'_{\text{KG_with_potential}} := \exp(-iq\lambda)\psi_{\text{KG_with_potential}}. \quad (176)$$

The gauge-dependent energy and momentum operators are defined by

$$E(\lambda) := i\partial_t - qV(\lambda), \quad (177)$$

$$\mathbf{P}(\lambda) := -i\partial_{\mathbf{r}} - q\mathbf{A}(\lambda). \quad (178)$$

The following relations are obtained by direct application of the product rule:

$$\begin{aligned} E(\lambda)\psi'_{\text{KG_with_potential}} &= \exp(-iq\lambda)E\psi_{\text{KG_with_potential}}, \\ \mathbf{P}(\lambda)\psi'_{\text{KG_with_potential}} &= \exp(-iq\lambda)\mathbf{P}\psi_{\text{KG_with_potential}}. \end{aligned} \quad (179)$$

so the same form of Eq. (174) is retained in every gauge. For smooth fields,

$$\begin{aligned} \text{com}(E, P_m) &= iqE_m, \\ \text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m). \end{aligned} \quad (180)$$

Consequently, the written order must be retained:

$$(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF, \quad (181)$$

$$(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF^*. \quad (182)$$

Here

$$\begin{aligned} F^* &= (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ -iqF &= (-iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}, \\ -iqF^* &= (iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

These are ordered star-H products of differential operators, not applications of the scalar-coefficient determinant formula. In a background field the full products do not reproduce only the scalar Klein-Gordon operator: the displayed F or F^* term remains. The scalar equation is recovered by taking the scalar part or by subtracting that remaining term explicitly. The same remaining terms are converted into the established spin-field terms when the Dirac system is squared; they are not additional scalar-wave components.

6.3 Klein-Gordon current and pairing

For $m > 0$, let $\phi_{\text{KG_with_potential}}$ and $\psi_{\text{KG_with_potential}}$ be two scalar solutions of Eq. (174) in the same real potential. A current constructed from these two solutions is called a mixed current; its inputs need not be the same. Its density function is defined by

$$\begin{aligned} \rho & \left(\begin{array}{c} \phi_{\text{KG_with_potential}}, \\ \psi_{\text{KG_with_potential}} \end{array} \right) \\ &:= \frac{1}{2m} \left(\phi_{\text{KG_with_potential}}^* E \psi_{\text{KG_with_potential}} \right. \\ & \quad \left. + (E \phi_{\text{KG_with_potential}})^* \psi_{\text{KG_with_potential}} \right). \end{aligned} \quad (183)$$

The associated spatial-current function is defined by

$$\begin{aligned} \mathbf{j} & \left(\begin{array}{c} \phi_{\text{KG_with_potential}}, \\ \psi_{\text{KG_with_potential}} \end{array} \right) \\ &:= \frac{1}{2m} \left(\phi_{\text{KG_with_potential}}^* \mathbf{P} \psi_{\text{KG_with_potential}} \right. \\ & \quad \left. + (\mathbf{P} \phi_{\text{KG_with_potential}})^* \psi_{\text{KG_with_potential}} \right). \end{aligned} \quad (184)$$

Star here is ordinary complex conjugation because both fields are scalars. Gauge invariance of this mixed current and its pairing is established immediately by Eq. (179) when the same phase is acquired by both solutions. The local identity is obtained from the product rules:

$$\begin{aligned} & \partial_t \rho \left(\begin{array}{c} \phi_{\text{KG_with_potential}}, \\ \psi_{\text{KG_with_potential}} \end{array} \right) \\ & \quad + \partial_{\mathbf{r}} \cdot \mathbf{j} \left(\begin{array}{c} \phi_{\text{KG_with_potential}}, \\ \psi_{\text{KG_with_potential}} \end{array} \right) \\ &= \frac{i}{2m} \left(((E^2 - \mathbf{P}^2) \phi_{\text{KG_with_potential}})^* \psi_{\text{KG_with_potential}} \right. \\ & \quad \left. - \phi_{\text{KG_with_potential}}^* (E^2 - \mathbf{P}^2) \psi_{\text{KG_with_potential}} \right) \\ &= \frac{i}{2m} \left(m^2 \phi_{\text{KG_with_potential}}^* \psi_{\text{KG_with_potential}} \right) \end{aligned}$$

$$-\phi_{\text{KG_with_potential}}^* m^2 \psi_{\text{KG_with_potential}}) = 0. \quad (185)$$

The ordinary conserved Klein–Gordon current is obtained by using the same field for both inputs; its electric current is obtained by multiplication by q . A pairing assigns one scalar to two solutions. The mixed-solution pairing is defined here by the spatial integral

$$\begin{aligned} \langle \phi_{\text{KG_with_potential}} \mid \psi_{\text{KG_with_potential}} \rangle \\ := \iiint \rho \left(\begin{array}{c} \phi_{\text{KG_with_potential}} \\ \psi_{\text{KG_with_potential}} \end{array} \right) \\ dr_1 dr_2 dr_3. \end{aligned} \quad (186)$$

For free positive-frequency solutions, the pairing is

$$\langle \phi_{\text{KG}} \mid \psi_{\text{KG}} \rangle = \frac{1}{m} \iiint \phi_{\text{KG}}^* \Omega \psi_{\text{KG}} dr_1 dr_2 dr_3,$$

which is positive for equal nonzero fields. The local density can nevertheless be negative at some points in a positive-frequency superposition. Therefore a positive local position probability is not supplied by $\rho(\psi_{\text{KG}}, \psi_{\text{KG}})$ in the scalar theory. On the full positive- and negative-frequency solution space, even the integrated pairing can be positive, zero, or negative and therefore cannot serve as a positive norm.

6.4 Fixed right-projector space for one Dirac field

An unrestricted complex paravector contains four complex coefficients, whereas each of the two parts of one Dirac field contains only two. The required two-coefficient space is selected intrinsically by one fixed right-projector. Its third-axis unit direction and positive projector are defined, in that order, by

$$\mathbf{u} := \mathbf{e}_3 = \{0, 0, 1\}, \quad \mathbf{u}^2 = 1, \quad (187)$$

$$\Pi(+) := \frac{1}{2}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(1 + \sigma_3). \quad (188)$$

Each paravector block of one Dirac field is required to satisfy

$$\Psi_{\text{Dirac},n} = \Psi_{\text{Dirac},n} \Pi(+), \quad n \in \{1, 2\}. \quad (189)$$

Every operator below acts from the left, so associativity preserves this condition:

$$(Z \Psi_{\text{Dirac},n}) \Pi(+) = Z(\Psi_{\text{Dirac},n} \Pi(+)) = Z \Psi_{\text{Dirac},n}. \quad (190)$$

The plus input identifies the positive part of this permanently chosen third axis; this right carrier is not itself a spin measurement. Using the ladder element $a(-)$ from Eq. (75), every block satisfying Eq. (189) has the unique form

$$\psi_{\text{block}} = a \Pi(+) + b a(-), \quad a, b \in \mathbb{C}. \quad (191)$$

Indeed, a paravector $\psi_{\text{block}} = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ satisfies the restriction exactly when $V_3 = S$ and $V_2 = -iV_1$, with $a = 2S$ and $b = 2V_1$. Thus exactly two complex amplitudes are retained. Its optional ordinary-matrix translation is supplied in Section 6.8.5.

For any projected block ψ_{block} , the positive density function is defined by

$$\rho(\psi_{\text{block}}) := 2 \text{ scalar}(\psi_{\text{block}}^{\text{H}} \psi_{\text{block}}). \quad (192)$$

For the expansion in Eq. (191), the projector and ladder products give $\rho(\psi_{\text{block}}) = |a|^2 + |b|^2$. The density is therefore real and nonnegative without use of an auxiliary translation.

The two blocks are treated as the ordered pair $\Psi_{\text{Dirac}} := (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2})$. The free Dirac equation is defined by the two coupled paravector equations

$$i\partial \Psi_{\text{Dirac},2} = m \Psi_{\text{Dirac},1}, \quad i\partial^* \Psi_{\text{Dirac},1} = m \Psi_{\text{Dirac},2}. \quad (193)$$

With the real electromagnetic potential, the coupled Dirac equation is

$$\begin{aligned} (i\partial - q\Phi^*) \Psi_{\text{Dirac_with_potential},2} &= m \Psi_{\text{Dirac_with_potential},1}, \\ (i\partial^* - q\Phi) \Psi_{\text{Dirac_with_potential},1} &= m \Psi_{\text{Dirac_with_potential},2}. \end{aligned} \quad (194)$$

Using the energy and momentum operators from Eqs. (172) and (173), the same system is

$$\begin{aligned} (E - \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi_{\text{Dirac_with_potential},2} &= m \Psi_{\text{Dirac_with_potential},1}, \\ (E + \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi_{\text{Dirac_with_potential},1} &= m \Psi_{\text{Dirac_with_potential},2}. \end{aligned} \quad (195)$$

To display every scalar component, the two blocks are written intrinsically as

$$\begin{aligned} \Psi_{\text{Dirac_with_potential},1} &= a \Pi(+) + b a(-), \\ \Psi_{\text{Dirac_with_potential},2} &= c \Pi(+) + d a(-). \end{aligned} \quad (196)$$

The actions of the three sigma generators on $\Pi(+)$ and $a(-)$, obtained from Eq. (76), reduce Eq. (195) to

$$\begin{aligned} (E - P_3)c - (P_1 - iP_2)d &= ma, \\ -(P_1 + iP_2)c + (E + P_3)d &= mb, \\ (E + P_3)a + (P_1 - iP_2)b &= mc, \\ (P_1 + iP_2)a + (E - P_3)b &= md. \end{aligned} \quad (197)$$

These are the four established first-order Dirac component equations [10, 17]; only the defined sigma products were used. The free system is recovered by setting $q = 0$.

A transformation preserves the form of an equation when the primed quantities satisfy the same relation as the unprimed quantities. For a constant transform T with determinant one, this condition fixes the two block transformations:

$$\Psi'_{\text{Dirac},1}(t', \mathbf{r}') = T^{\text{H}} \Psi_{\text{Dirac},1}(t, \mathbf{r}), \quad (198)$$

$$\Psi'_{\text{Dirac},2}(t', \mathbf{r}') = T^{-1} \Psi_{\text{Dirac},2}(t, \mathbf{r}). \quad (199)$$

As shown by the supporting derivation of Eqs. (198) and (199), the first block is therefore acted upon by L under a boost, whereas the second is acted upon by L^* ; both are acted upon by R^{H} under

a rotation. This is the wave-function law, distinct from both event congruence and field similarity.

The potential-coupled ordered pair is

$$\Psi_{\text{Dirac_with_potential}} := \begin{pmatrix} \Psi_{\text{Dirac_with_potential},1}, \\ \Psi_{\text{Dirac_with_potential},2} \end{pmatrix}.$$

For this ordered pair, the Dirac function is defined by

$$\mathcal{D}(\Psi_{\text{Dirac_with_potential}}) := \begin{pmatrix} (E - \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi_{\text{Dirac_with_potential},2} \\ -m \Psi_{\text{Dirac_with_potential},1}, \\ (E + \boldsymbol{\sigma} \cdot \mathbf{P}) \Psi_{\text{Dirac_with_potential},1} \\ -m \Psi_{\text{Dirac_with_potential},2} \end{pmatrix}. \quad (200)$$

Thus Eq. (194) is equivalently $\mathcal{D}(\Psi_{\text{Dirac_with_potential}}) = (0, 0)$. For a generic ordered pair (Y_1, Y_2) , the companion function with the mass sign reversed is defined by

$$\mathcal{D}'(Y_1, Y_2) := ((E - \boldsymbol{\sigma} \cdot \mathbf{P})Y_2 + mY_1, (E + \boldsymbol{\sigma} \cdot \mathbf{P})Y_1 + mY_2). \quad (201)$$

The prime in $\mathcal{D}'$ labels the explicitly stated alternative mass sign, in accordance with Eq. (2); it is not a derivative or a Hermitian operation. This companion function is applied to $\mathcal{D}$ evaluated on the potential-coupled Dirac field. Use of the ordered factorizations (181)–(182) then gives

$$\begin{cases} (-E^2 + \mathbf{P}^2 + m^2 + iqF^*) \Psi_{\text{Dirac_with_potential},1} = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 + iqF) \Psi_{\text{Dirac_with_potential},2} = 0. \end{cases} \quad (202)$$

With $iqF^* = (-iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma}$ and $iqF = (iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma}$, the same equations are written as

$$\begin{cases} (-E^2 + \mathbf{P}^2 + m^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} - iq\mathbf{E} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_with_potential},1} = 0, \\ (-E^2 + \mathbf{P}^2 + m^2 - q\mathbf{B} \cdot \boldsymbol{\sigma} + iq\mathbf{E} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_with_potential},2} = 0. \end{cases} \quad (203)$$

Thus the expected Klein–Gordon operator, the same magnetic spin coupling in both blocks, and the opposite electric spin couplings of the two blocks are obtained component by component by squaring the first-order system.

Under the gauge law (153),

$$\Psi'_{\text{Dirac_with_potential}} := \exp(-iq\lambda) \Psi_{\text{Dirac_with_potential}}. \quad (204)$$

The form of Eq. (194) is then unchanged.

The Hermitian product supplies a nonnegative density whose spatial integral is the squared norm at a fixed time. Its vector part

is not a probability current. With the fixed-projector convention above, the Dirac density function is defined by

$$\rho(\Psi_{\text{Dirac_with_potential}}) := 2 \sum_{n=1}^2 \text{scalar} \left(\Psi_{\text{Dirac_with_potential},n}^H \Psi_{\text{Dirac_with_potential},n} \right). \quad (205)$$

The spatial-current function is defined by

$$\mathbf{j}(\Psi_{\text{Dirac_with_potential}}) := 2 \text{vector} \left(\Psi_{\text{Dirac_with_potential},2} \Psi_{\text{Dirac_with_potential},2}^H - \Psi_{\text{Dirac_with_potential},1} \Psi_{\text{Dirac_with_potential},1}^H \right). \quad (206)$$

The full current paravector is defined by

$$J(\Psi_{\text{Dirac_with_potential}}) := \rho(\Psi_{\text{Dirac_with_potential}}) + \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) \cdot \boldsymbol{\sigma}. \quad (207)$$

Its density and spatial-current components are real, so the current in Eq. (207) belongs to $\mathbb{P}_{\mathbb{R}}$. The two definitions satisfy the continuity equation

$$\partial_t \rho(\Psi_{\text{Dirac_with_potential}}) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) = 0. \quad (208)$$

The normalization condition is the spatial integral

$$\iiint \rho(\Psi_{\text{Dirac_with_potential}}) d\mathbf{r}_1 d\mathbf{r}_2 d\mathbf{r}_3 = 1,$$

not an equation imposed separately at every position. By Eq. (191), $\rho(\Psi_{\text{Dirac_with_potential}})$ is the sum of four absolute squares. The established Dirac spatial current is supplied by the displayed opposite-order products, rather than by the vector part of a single Hermitian norm product [10, 17]. The real current paravector from Eq. (207) obeys the event congruence, as shown in the supporting derivation. The component balances are obtained directly from the two time-evolution equations and the amplitude expansion (196):

$$\begin{aligned} \partial_t(|a|^2 + |b|^2) &= 2\partial_{\mathbf{r}} \cdot \text{vector} \left(\Psi_{\text{Dirac_with_potential},1} \Psi_{\text{Dirac_with_potential},1}^H + im(c^*a + d^*b - a^*c - b^*d), \right. \\ \partial_t(|c|^2 + |d|^2) &= -2\partial_{\mathbf{r}} \cdot \text{vector} \left(\Psi_{\text{Dirac_with_potential},2} \Psi_{\text{Dirac_with_potential},2}^H + im(a^*c + b^*d - c^*a - d^*b). \right. \end{aligned} \quad (209)$$

By addition of the two lines, the mass exchange cancels and the exact component density is

$$\rho(\Psi_{\text{Dirac_with_potential}}) = |a|^2 + |b|^2 + |c|^2 + |d|^2. \quad (210)$$

The exact spatial-current components are

$$\mathbf{j}(\Psi_{\text{Dirac_with_potential}}) = \left\{ \begin{array}{l} c^*d + d^*c - a^*b - b^*a, \\ i(d^*c - c^*d - b^*a + a^*b), \\ |c|^2 - |d|^2 - |a|^2 + |b|^2 \end{array} \right\}. \quad (211)$$

Equations (210) and (211) give the continuity equation (208). Neither block norm is conserved separately: norm is exchanged between them by the mass terms.

6.5 Probabilities and measurement averages

A pairing assigns one complex scalar to two fields. For projected one-block fields, the pairing is defined intrinsically by

$$\langle \phi_{\text{block}} | \psi_{\text{block}} \rangle := 2 \iiint \text{scalar}(\phi_{\text{block}}^H \psi_{\text{block}}) \, dr_1 \, dr_2 \, dr_3. \quad (212)$$

For the amplitude forms $\psi_{\text{block}} = a\Pi(+) + ba(-)$ and $\phi_{\text{block}} = c\Pi(+) + da(-)$, the projector and ladder products give

$$\begin{aligned} \phi_{\text{block}}^H \psi_{\text{block}} &= (c^*a + d^*b)\Pi(+), \\ 2 \text{ scalar}(\phi_{\text{block}}^H \psi_{\text{block}}) &= c^*a + d^*b. \end{aligned} \quad (213)$$

Thus Eq. (212) is exactly the established two-amplitude state pairing [19]. Its optional ordinary-matrix translation is supplied in Section 6.8.5. For two projected two-block fields, the complete Dirac pairing is defined by

$$\begin{aligned} \langle \Psi_{\text{Dirac_left}} | \Psi_{\text{Dirac_right}} \rangle \\ := 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_{\text{Dirac_left},n}^H \Psi_{\text{Dirac_right},n}) \, dr_1 \, dr_2 \, dr_3. \end{aligned} \quad (214)$$

The two inputs are identified by their Dirac_left and Dirac_right role labels; no new kind of field is introduced. When both inputs are the same, the shorter notation $\langle \Psi_{\text{Dirac}} | \Psi_{\text{Dirac}} \rangle$ is used. The factor 2 converts the scalar part of $\Pi(+)$, which is 1/2, into the ordinary two-amplitude pairing. For a specified operator $\mathcal{O}$ acting on ordered field pairs, the operator pairing is defined by

$$\begin{aligned} \langle \Psi_{\text{Dirac_left}} | \mathcal{O}(\Psi_{\text{Dirac_right}}) \rangle \\ := 2 \iiint \sum_{n=1}^2 \text{scalar}(\Psi_{\text{Dirac_left},n}^H (\mathcal{O}(\Psi_{\text{Dirac_right}}))_n) \, dr_1 \, dr_2 \, dr_3. \end{aligned} \quad (215)$$

The resulting scalar need not be real. Its normalized value is

$$\frac{\langle \Psi_{\text{Dirac}} | \mathcal{O}(\Psi_{\text{Dirac}}) \rangle}{\langle \Psi_{\text{Dirac}} | \Psi_{\text{Dirac}} \rangle}. \quad (216)$$

When the same state is used on both sides and $\mathcal{O}$ is Hermitian under this spatial pairing, the real normalized value is called the expectation value of the operator [19]. Hermiticity under the pairing means

$$\langle \Psi_{\text{Dirac_left}} | \mathcal{O}(\Psi_{\text{Dirac_right}}) \rangle = \langle \mathcal{O}(\Psi_{\text{Dirac_left}}) | \Psi_{\text{Dirac_right}} \rangle. \quad (217)$$

For derivatives, this equality also requires the stated integration-by-parts boundary conditions and a common set of allowed fields. If a Hermitian operator is also unchanged when applied twice, the probability of the outcome selected by that projector is given by the same ratio.

For a normalized Dirac field, the probability of finding the system in a specified spatial region at the chosen time is defined by

$$p(\text{region}) := \iiint \rho(\Psi_{\text{Dirac}}) \, dr_1 \, dr_2 \, dr_3, \quad (218)$$

where the integration is restricted to the specified region. The density is nonnegative, but it is the time component of the real current paravector: its scalar and vector components mix under a Lorentz transform in the same way as those of an event. The density alone is therefore not unchanged by a Lorentz transform. The normalization is kept time-independent by conservation when the boundary flux vanishes.

For measurement along a nonzero real axis vector $\mathbf{V}$, the local unit direction is $\mathbf{u} := \mathbf{V}/\|\mathbf{V}\|$, as in Eq. (70). For a projected Pauli block $\psi_{\text{axis_state}}$, the factor 2 is canceled from normalized ratios. The probability function supplied by $\Pi(\pm)$ for this direction is therefore defined by

$$p(\pm) := \frac{\langle \psi_{\text{axis_state}}(\pm) | \psi_{\text{axis_state}}(\pm) \rangle}{\langle \psi_{\text{axis_state}} | \psi_{\text{axis_state}} \rangle}. \quad (219)$$

By Eqs. (71) and (72), each projected field is obtained by left multiplication:

$$\psi_{\text{axis_state}}(\pm) = \Pi(\pm)\psi_{\text{axis_state}}.$$

Consequently,

$$p(+) + p(-) = 1. \quad (220)$$

The expectation value of $\mathbf{u} \cdot \boldsymbol{\sigma}$ is $p(+) - p(-)$; the physical spin component along $\mathbf{u}$ is one half of this value in the natural units used here. The corresponding free two-block energy and momentum-axis probability is defined after those fields have been constructed below.

The Klein–Gordon field is instead paired by Eq. (186). On the set of positive-frequency solutions, the normalized operator pairing is

$$\frac{\langle \psi_{\text{KG}} | \mathcal{O}(\psi_{\text{KG}}) \rangle}{\langle \psi_{\text{KG}} | \psi_{\text{KG}} \rangle}$$

only when that set is preserved and the operator is Hermitian under the Klein–Gordon pairing. The density $\rho(\psi_{\text{KG}}, \psi_{\text{KG}})$ is not thereby converted into a position-probability density at each position. Likewise, the nonscalar parts of a generic two-sided paravector product are only algebraic results. They do not represent additional measurements unless a field equation and a measurement rule assign them that meaning.

6.6 Slow-Motion Pauli Limit

The reduction of the Dirac equation at energies small compared with m is conventionally called the Pauli limit [9, 19]. A single block by itself is not the slow-motion large component. The rapid

factor $\exp(-imt)$ produced by the rest energy is called the rest phase. The construction in this subsection assumes $m > 0$. First, the slowly varying blocks are defined by removing that factor:

$$\Psi_{\text{Dirac_slow},n} := \exp(imt) \Psi_{\text{Dirac_with_potential},n}. \quad (221)$$

The large component is then defined by

$$\psi_{\text{Pauli_large}} := \frac{\Psi_{\text{Dirac_slow},1} + \Psi_{\text{Dirac_slow},2}}{\sqrt{2}}. \quad (222)$$

The small component is defined independently by

$$\psi_{\text{Pauli_small}} := \frac{\Psi_{\text{Dirac_slow},2} - \Psi_{\text{Dirac_slow},1}}{\sqrt{2}}. \quad (223)$$

The words “large” and “small” describe their relative magnitudes only for positive-energy states at energies small compared with m ; they do not define new algebraic types. Equation (195) is then transformed exactly into

$$\begin{aligned} E\psi_{\text{Pauli_large}} &= (\mathbf{P} \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_small}}, \\ (E + 2m)\psi_{\text{Pauli_small}} &= (\mathbf{P} \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_large}}. \end{aligned} \quad (224)$$

At energies small compared with m ,

$$\psi_{\text{Pauli_small}} = \frac{\mathbf{P} \cdot \boldsymbol{\sigma}}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}).$$

By use of

$$(\mathbf{P} \cdot \boldsymbol{\sigma})^2 = \mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma},$$

the following result is obtained:

$$E\psi_{\text{Pauli_large}} = \frac{\mathbf{P}^2 - q\mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}). \quad (225)$$

Equivalently,

$$\begin{aligned} i\partial_t \psi_{\text{Pauli_large}} &= \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi_{\text{Pauli_large}} \\ &\quad - \frac{q}{2m} (\mathbf{B} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}} + O(m^{-2}). \end{aligned} \quad (226)$$

At the next order, the electric field produces a spin–orbit term, which couples spin to motion through that field, and a Darwin term, which is proportional to the local divergence of the electric field [19]. Time-dependent corrections also occur. No term from that order is retained selectively here. The exact change of variables in Eqs. (222) and (223) gives

$$\begin{aligned} \rho(\Psi_{\text{Dirac_with_potential}}) &= \rho(\psi_{\text{Pauli_large}}) + \rho(\psi_{\text{Pauli_small}}), \\ \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) &= 2 \text{ vector} \left(\begin{aligned} &\psi_{\text{Pauli_large}} \psi_{\text{Pauli_small}}^H \\ &+ \psi_{\text{Pauli_small}} \psi_{\text{Pauli_large}}^H \end{aligned} \right). \end{aligned} \quad (227)$$

The projected-block density from Eq. (192) is therefore

$$\rho(\psi_{\text{Pauli_large}}) = 2 \text{ scalar}(\psi_{\text{Pauli_large}}^H \psi_{\text{Pauli_large}}). \quad (228)$$

Writing P_1, P_2, P_3 for the three components of the momentum operator in Eq. (173), the Pauli spatial-current function is defined intrinsically by

$$\begin{aligned} \mathbf{j}(\psi_{\text{Pauli_large}}) &:= \frac{1}{m} \left\{ \begin{aligned} &\text{scalar}(\psi_{\text{Pauli_large}}^H P_1 \psi_{\text{Pauli_large}} \\ &+ (P_1 \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}}) \\ &\text{scalar}(\psi_{\text{Pauli_large}}^H P_2 \psi_{\text{Pauli_large}} \\ &+ (P_2 \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}}) \\ &\text{scalar}(\psi_{\text{Pauli_large}}^H P_3 \psi_{\text{Pauli_large}} \\ &+ (P_3 \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}}) \end{aligned} \right\} \\ &\quad + \frac{1}{m} \partial_{\mathbf{r}} \times \text{vector}(\psi_{\text{Pauli_large}} \psi_{\text{Pauli_large}}^H). \end{aligned} \quad (229)$$

Because $\psi_{\text{Pauli_small}} = O(m^{-1})$, the original Dirac density and current are reduced to $\rho(\Psi_{\text{Dirac_with_potential}}) = \rho(\psi_{\text{Pauli_large}}) + O(m^{-2})$ and $\mathbf{j}(\Psi_{\text{Dirac_with_potential}}) = \mathbf{j}(\psi_{\text{Pauli_large}}) + O(m^{-2})$. The two Hermitian-conjugate contributions inside the first vector summand of Eq. (229) form the convection current associated with motion through the potential-dependent momentum. The curl summand is the magnetization current associated with spin; its divergence is zero. Their standard sum is thereby obtained. The equation $\partial_t \rho(\psi_{\text{Pauli_large}}) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\psi_{\text{Pauli_large}}) = 0$ is obeyed through the retained order.

Where $\mathbf{B} \neq 0$, the local magnetic-field axis is defined by

$$\mathbf{u} := \frac{\mathbf{B}}{\|\mathbf{B}\|}. \quad (230)$$

Equation (71) is then specialized as $\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma})$. The two spin-axis parts are defined by

$$\psi_{\text{Pauli_split}}(\pm) := \Pi(\pm) \psi_{\text{Pauli_large}}. \quad (231)$$

They reconstruct the large component:

$$\psi_{\text{Pauli_large}} = \psi_{\text{Pauli_split}}(+) + \psi_{\text{Pauli_split}}(-). \quad (232)$$

If $\mathbf{B}$ is also uniform and constant, then

$$\begin{aligned} i\partial_t \psi_{\text{Pauli_split}}(\pm) &= \left(\frac{\mathbf{P}^2}{2m} + qV \right) \psi_{\text{Pauli_split}}(\pm) \\ &\quad \mp \frac{q\|\mathbf{B}\|}{2m} \psi_{\text{Pauli_split}}(\pm) + O(m^{-2}). \end{aligned} \quad (233)$$

For a varying axis, the two projected fields are coupled by derivatives of $\Pi(\pm)$ and its commutator with the time-evolution operator on the right of Eq. (226), so they no longer evolve independently.

6.7 Free energy and momentum-axis split

A free momentum mode is a plane-wave solution with a definite momentum $\mathbf{p}$. For such a mode, the positive energy function is defined by

$$E(\mathbf{p}) := \sqrt{m^2 + \mathbf{p}^2}. \quad (234)$$

The split below is defined when $E(\mathbf{p}) > 0$; the massless zero-momentum case has no positive-versus-negative energy split of

this form. The Hamiltonian function on an ordered field pair is defined by

$$\mathcal{H}(\Psi_{\text{Dirac}}) := \begin{pmatrix} -(\boldsymbol{\sigma} \cdot \mathbf{p})\Psi_{\text{Dirac},1} + m\Psi_{\text{Dirac},2}, \\ m\Psi_{\text{Dirac},1} + (\boldsymbol{\sigma} \cdot \mathbf{p})\Psi_{\text{Dirac},2} \end{pmatrix}. \quad (235)$$

Equation (195) then gives

$$\begin{aligned} i\partial_t \Psi_{\text{Dirac}} &= \mathcal{H}(\Psi_{\text{Dirac}}), \\ \mathcal{H}(\mathcal{H}(\Psi_{\text{Dirac}})) &= E(\mathbf{p})^2 \Psi_{\text{Dirac}}. \end{aligned} \quad (236)$$

The function $\mathcal{H}$ is called the Hamiltonian because it supplies $i\partial_t \Psi_{\text{Dirac}}$. It is not a new field: its two components are exactly the two free Dirac evolution equations. The calligraphic symbol $\mathcal{H}$ is distinct from the superscript H , which is reserved for the Hermitian operation [10, 17].

The two energy parts of the field are defined directly, without naming another operator:

$$\Psi_{\text{Dirac_energy}}(\pm) := \frac{1}{2} \left(\Psi_{\text{Dirac}} \pm \frac{\mathcal{H}(\Psi_{\text{Dirac}})}{E(\mathbf{p})} \right). \quad (237)$$

Its two component blocks are defined explicitly by

$$\begin{aligned} \Psi_{\text{Dirac_energy},1}(\pm) &:= \frac{1}{2} \left(\Psi_{\text{Dirac},1} \right. \\ &\quad \left. \pm \frac{-(\boldsymbol{\sigma} \cdot \mathbf{p})\Psi_{\text{Dirac},1} + m\Psi_{\text{Dirac},2}}{E(\mathbf{p})} \right), \\ \Psi_{\text{Dirac_energy},2}(\pm) &:= \frac{1}{2} \left(\Psi_{\text{Dirac},2} \right. \\ &\quad \left. \pm \frac{m\Psi_{\text{Dirac},1} + (\boldsymbol{\sigma} \cdot \mathbf{p})\Psi_{\text{Dirac},2}}{E(\mathbf{p})} \right), \\ \Psi_{\text{Dirac_energy}}(\pm) &:= \begin{pmatrix} \Psi_{\text{Dirac_energy},1}(\pm), \\ \Psi_{\text{Dirac_energy},2}(\pm) \end{pmatrix}. \end{aligned} \quad (238)$$

An eigenvalue is the scalar that multiplies a nonzero state when an operator acts on it [19]. The energy-multiplier and reconstruction identities are

$$\begin{aligned} \mathcal{H}(\Psi_{\text{Dirac_energy}}(\pm)) &= \pm E(\mathbf{p}) \Psi_{\text{Dirac_energy}}(\pm), \\ \Psi_{\text{Dirac}} &= \Psi_{\text{Dirac_energy}}(+) + \Psi_{\text{Dirac_energy}}(-). \end{aligned} \quad (239)$$

The plus and minus inputs select the positive- and negative-energy parts of the same free field.

For $\mathbf{p} \neq 0$, the momentum direction is defined by

$$\mathbf{u} := \frac{\mathbf{p}}{\|\mathbf{p}\|}. \quad (240)$$

For this direction, the existing axis projector from Eq. (71) is

$$\Pi(\pm) = \frac{1}{2}(1 \pm \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2} \left(1 \pm \frac{\mathbf{p} \cdot \boldsymbol{\sigma}}{\|\mathbf{p}\|} \right). \quad (241)$$

The sign input below labels the momentum-axis part. Positive and negative energy are identified by separate field names, so the two independent signs cannot be confused. The simultaneous energy and momentum-axis fields are defined by

$$\begin{aligned} \Psi_{\text{Dirac_positive_energy_axis}}(\pm) &:= \begin{pmatrix} \Pi(\pm) \Psi_{\text{Dirac_energy},1}(+), \\ \Pi(\pm) \Psi_{\text{Dirac_energy},2}(+) \end{pmatrix}, \\ \Psi_{\text{Dirac_negative_energy_axis}}(\pm) &:= \begin{pmatrix} \Pi(\pm) \Psi_{\text{Dirac_energy},1}(-), \\ \Pi(\pm) \Psi_{\text{Dirac_energy},2}(-) \end{pmatrix}. \end{aligned} \quad (242)$$

The two sign choices satisfy

$$\begin{aligned} (\mathbf{p} \cdot \boldsymbol{\sigma}) \Pi(\pm) &= \pm \|\mathbf{p}\| \Pi(\pm), \\ \Psi_{\text{Dirac}} &= \Psi_{\text{Dirac_positive_energy_axis}}(+) \\ &\quad + \Psi_{\text{Dirac_positive_energy_axis}}(-) \\ &\quad + \Psi_{\text{Dirac_negative_energy_axis}}(+) \\ &\quad + \Psi_{\text{Dirac_negative_energy_axis}}(-). \end{aligned} \quad (243)$$

The plus or minus axis input is the scalar by which $\mathbf{u} \cdot \boldsymbol{\sigma}$ multiplies the selected field. In the natural units used here, the corresponding physical spin component along momentum is $+1/2$ or $-1/2$. This spin component along the momentum direction is conventionally called helicity; it is undefined when $\mathbf{p} = \mathbf{0}$ [19]. The Hamiltonian is Hermitian, and the pairing of the two energy parts is zero; they are therefore called orthogonal. The already established $\Pi(\pm)$ projectors are Hermitian, idempotent, and complementary. Because the momentum axis is used, commutation of their left action with $\mathcal{H}$ is obtained, so the four displayed fields are simultaneous energy and momentum-axis parts. At rest any spin axis may be chosen. Negative-frequency classical solutions are not interpreted as antiparticles in this single-particle treatment. That interpretation belongs to quantum field theory, in which particle number may change, and is not developed here [17].

The joint probability function is now defined without referring to a field before its construction:

$$\begin{aligned} p(+, \pm) &:= \frac{\langle \Psi_{\text{Dirac_positive_energy_axis}}(\pm) | \Psi_{\text{Dirac_positive_energy_axis}}(\pm) \rangle}{\langle \Psi_{\text{Dirac}} | \Psi_{\text{Dirac}} \rangle}, \\ p(-, \pm) &:= \frac{\langle \Psi_{\text{Dirac_negative_energy_axis}}(\pm) | \Psi_{\text{Dirac_negative_energy_axis}}(\pm) \rangle}{\langle \Psi_{\text{Dirac}} | \Psi_{\text{Dirac}} \rangle}. \end{aligned} \quad (244)$$

Completeness gives

$$p(+, +) + p(+, -) + p(-, +) + p(-, -) = 1. \quad (245)$$

For a free field with a finite spatial norm, this split is applied to each momentum mode before the spatial pairing is evaluated. An isolated plane wave does not have a finite spatial norm. It must instead be normalized in a finite volume or by the established normalization convention for continuous momentum values [19].

6.8 Supporting Wave-Equation Derivations

The ordered operator expansions, scalar frequency evolution, fixed-projector interpretation, conserved pairings, gauge transformation, squared Dirac system, and controlled Pauli approximation are given in detail in the supporting derivations.

6.8.1 Relativistic Quantum-Mechanical Setting

One fixed operator language is used throughout the wave-equation section. With the differential conventions from Section 4,

$$\begin{aligned}\partial &= \partial_t + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ \partial^* &= \partial_t - \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}}, \\ (i\partial)^* &= -i\partial^*.\end{aligned}$$

The four-dimensional wave operator, conventionally called the d'Alembertian [17], is

$$\begin{aligned}\square &= \partial_t^2 - \partial_{\mathbf{r}}^2 \\ &= \partial\partial^* = \partial^*\partial.\end{aligned}$$

All fields are acted on by the differential operators to their left. In an ordered product, the action of the rightmost factor is applied first, and the written order is retained whenever potentials and derivatives do not commute.

6.8.2 Algebra Blocks and Fixed-Projector Wave Functions

Throughout the spinful part of the wave-equation section, a wave field is not a space-time event but an algebra-valued amplitude field. A generic algebra block may be written with the scalar and vector extraction functions from Eqs. (23) and (24):

$$\begin{aligned}\psi_{\text{block}} : \mathbb{R} \times \mathbb{R}^3 &\rightarrow \mathbb{P}_{\mathbb{C}}, \\ \psi_{\text{block}}(t, \mathbf{r}) &= \text{scalar}(\psi_{\text{block}}(t, \mathbf{r})) \\ &\quad + \text{vector}(\psi_{\text{block}}(t, \mathbf{r})) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The location at which the state is evaluated in space-time is specified by the argument $(t, \mathbf{r})$, while the value $\psi_{\text{block}}(t, \mathbf{r})$ is contained in the internal complex-paravector algebra. The real paravector

$$X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$$

is used to describe an event or interval variable, whereas ψ_{block} is a quantum state amplitude defined *over* such events.

At each fixed point, four complex coefficients are contained in an unrestricted complex paravector. A single two-amplitude block is selected without adding component indices by the fixed right projector $\Pi(+)$ from Eq. (188). A physical block is restricted by

$$\psi_{\text{block}} = \psi_{\text{block}} \Pi(+).$$

Using the ladder element from Eq. (75), every such block has the unique algebraic expansion given in Eq. (191). Every displayed

operator is applied on the left, so the restriction is preserved by associativity: $(Z\psi_{\text{block}})\Pi(+)=Z(\psi_{\text{block}}\Pi(+))=Z\psi_{\text{block}}$. The fixed right factor is only a carrier selector; it is not a measurement of spin along the third axis.

The scalar and vector coefficients of an algebra block are amplitude channels, not separately measurable particles or classical subfields. A bare component count also does not cause physical spin to be assigned: the transformation law, field equation, and fixed-projector restriction must be specified together.

The field $\psi_{\text{KG}} \in \mathbb{C}$ is used in the spin-0 Klein–Gordon subsections. A two-block field is used in the Dirac construction and is treated as the ordered pair

$$\Psi_{\text{Dirac}} = (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2}),$$

with $n \in \{1, 2\}$ and $\Psi_{\text{Dirac},n} = \Psi_{\text{Dirac},n} \Pi(+)$. Two complex amplitudes are then carried by each block, and the four complex amplitudes of one Dirac field are carried by the pair. If the fixed-projector restriction is omitted, two independent Dirac fields are obtained by decomposition with the complementary right projectors, and the same equations remain algebraically valid for both.

6.8.3 Complex-Paravector State Pairings

In the conventional bra-ket terminology, the state on the right of a pairing is called a ket, while its Hermitian counterpart on the left is called a bra [17, 19]. That terminology is made precise in the paravector setting by direct use of the already defined scalar and vector extraction functions:

$$\begin{aligned}\psi_{\text{block}} &= \text{scalar}(\psi_{\text{block}}) + \text{vector}(\psi_{\text{block}}) \cdot \boldsymbol{\sigma}, \\ Z &= \text{scalar}(Z) + \text{vector}(Z) \cdot \boldsymbol{\sigma}, \\ X &= \text{scalar}(X) + \text{vector}(X) \cdot \boldsymbol{\sigma}.\end{aligned}$$

Here $\text{scalar}(\psi_{\text{block}}), \text{scalar}(Z) \in \mathbb{C}$, $\text{vector}(\psi_{\text{block}}), \text{vector}(Z) \in \mathbb{C}^3$, $\text{scalar}(X) \in \mathbb{R}$, and $\text{vector}(X) \in \mathbb{R}^3$. Every differential operator acts on the field to its right. The three involutions on the state are

$$\begin{aligned}\psi_{\text{block}}^* &= \text{scalar}(\psi_{\text{block}})^* - \text{vector}(\psi_{\text{block}})^* \cdot \boldsymbol{\sigma}, \\ \psi_{\text{block}}^{\text{H}} &= \text{scalar}(\psi_{\text{block}})^* + \text{vector}(\psi_{\text{block}})^* \cdot \boldsymbol{\sigma}, \\ \psi_{\text{block}}^{*\text{H}} &= \text{scalar}(\psi_{\text{block}}) - \text{vector}(\psi_{\text{block}}) \cdot \boldsymbol{\sigma}.\end{aligned}$$

The positive pairing follows directly from the fixed-projector expansion. For complex coefficients a, b, c, d , the two fields are written as

$$\psi_{\text{block}} = a\Pi(+) + ba(-), \quad \phi_{\text{block}} = c\Pi(+) + da(-).$$

From Eqs. (71), (75), and (76), the following identities are obtained: $\Pi(+)^{\text{H}} = \Pi(+)$, $a(-)^{\text{H}} = a(+)$, $\Pi(+)\Pi(-) = 0$, $a(+)\Pi(+)=0$, and $a(+)\Pi(-)=\Pi(+)$. Consequently, the product reduces within the algebra as

$$\begin{aligned}\phi_{\text{block}}^{\text{H}} \psi_{\text{block}} &= (c^* \Pi(+) + d^* a(+)) (a \Pi(+) + ba(-)) \\ &= (c^* a + d^* b) \Pi(+), \\ 2 \text{ scalar}(\phi_{\text{block}}^{\text{H}} \psi_{\text{block}}) &= c^* a + d^* b.\end{aligned} \tag{246}$$

Thus ϕ_{block}^H supplies the bra associated with the selected ket, and $2 \text{ scalar}(\psi_{\text{block}}^H \psi_{\text{block}})$ is the sum of the absolute squares of the two amplitudes. This nonnegative sum is the squared norm of the projected block. The factor 2 is required because $\text{scalar}(\Pi(+)) = 1/2$. By summation of this relation over the two projected Dirac blocks, the density is obtained; the spatial current is instead obtained from the opposite-order block products fixed by the Dirac equation. The basic component products are collected below.

The direct action of a complex paravector operator on a complex paravector state is recorded in Table 32.

The three basic ordered two-sided products with a generic complex paravector $Z = \text{scalar}(Z) + \text{vector}(Z) \cdot \sigma$ are given in Tables 33–35.

Tables 36–38 are obtained by specialization to a real paravector $X = \text{scalar}(X) + \text{vector}(X) \cdot \sigma$, with $\text{scalar}(X) \in \mathbb{R}$ and $\text{vector}(X) \in \mathbb{R}^3$.

For a scalar function S , the same three products are reduced to Tables 39–41.

For any real scalar multiplier S , the three-factor Hermitian product is

$$\psi_{\text{block}}^H S \psi_{\text{block}} = S \psi_{\text{block}}^H \psi_{\text{block}}.$$

The case $S = 1$ gives the physical fixed-projector density defined in Eq. (192). Its local form is

$$\begin{aligned} \rho(\psi_{\text{block}}) &= 2 \text{ scalar}(\psi_{\text{block}}^H \psi_{\text{block}}) \\ &= 2 |\text{scalar}(\psi_{\text{block}})|^2 \\ &\quad + 2 \text{ vector}(\psi_{\text{block}})^* \cdot \text{vector}(\psi_{\text{block}}) \\ &= |a|^2 + |b|^2 \geq 0. \end{aligned}$$

The amplitudes from Eq. (191) are used in the last line. The factor is retained explicitly in the scalar-extraction formula. The squared-norm density on one fixed-time spatial slice for a projected Pauli state and, after summation over both blocks, for a Dirac state is supplied by this positive scalar. It is not the conserved Klein–Gordon norm of a scalar relativistic field. The vector part of $\psi_{\text{block}}^H \psi_{\text{block}}$, however, is not automatically a spatial probability current. In fact, under the convention $\psi'_{\text{block}} = R^H \psi_{\text{block}}$, in which the state is rotated while the basis is held fixed, it is unchanged:

$$\psi_{\text{block}}'^H \psi'_{\text{block}} = \psi_{\text{block}}^H R R^H \psi_{\text{block}} = \psi_{\text{block}}^H \psi_{\text{block}}.$$

Spatial spin information instead occurs in products with the opposite order, such as $\psi_{\text{block}} \psi_{\text{block}}^H$, or in three-factor operator products fixed by the field equation. If $\mathcal{O}$ acts from the left and preserves the fixed-projector condition, then

$$\psi_{\text{block}}^H \mathcal{O} \psi_{\text{block}} = 2 \text{ scalar}(\psi_{\text{block}}^H \mathcal{O} \psi_{\text{block}}) \Pi(+).$$

Thus only one complex pairing coefficient is independent in this ordered product; its fixed vector part records the selected fixed-projector space. Four complex coefficient channels can occur only when the fixed-projector restriction is removed or a different product order is used. No generic physical interpretation is assigned to such unrestricted nonscalar channels.

For a nonzero projected block, the scale is defined by

$$N := \rho(\psi_{\text{block}}). \quad (247)$$

The following useful opposite-order identities are obtained from the fixed-projector condition:

$$\begin{aligned} \psi_{\text{block}}^H \psi_{\text{block}} &= N \Pi(+), \\ \left(\frac{\psi_{\text{block}} \psi_{\text{block}}^H}{N} \right)^H &= \frac{\psi_{\text{block}} \psi_{\text{block}}^H}{N}, \\ \left(\frac{\psi_{\text{block}} \psi_{\text{block}}^H}{N} \right)^2 &= \frac{\psi_{\text{block}} \psi_{\text{block}}^H}{N}. \end{aligned} \quad (248)$$

Only the selected fixed-projector space is retained by the first product. The Hermitian idempotent in the second and third lines represents one fully specified two-amplitude state [19]. The conventional name “pure state” means that one definite state, rather than a statistical mixture of several possible states, is represented. Its determinant is zero, and twice the vector part of $\psi_{\text{block}} \psi_{\text{block}}^H / N$ gives the three normalized spin-axis imbalances of that state. Without division by N , the same vector is the norm-weighted local spin density. This is the familiar geometry of a two-amplitude state, not an additional paravector degree of freedom.

At fixed time, the spatially integrated bracket defined in Eq. (212) is

$$\langle \phi_{\text{block}} | \psi_{\text{block}} \rangle = 2 \iiint \text{scalar}(\phi_{\text{block}}^H \psi_{\text{block}}) dr_1 dr_2 dr_3.$$

By Eq. (246), scalar extraction is converted into the ordinary inner product of the two projected amplitudes by the factor 2. It is therefore a normalization convention rather than an additional physical factor. For a projected one-block Pauli state, normalization means that the equal-field specialization is one:

$$\langle \psi_{\text{block}} | \psi_{\text{block}} \rangle = \iiint \rho(\psi_{\text{block}}) dr_1 dr_2 dr_3 = 1.$$

For a complex-paravector operator $\mathcal{O}$,

$$\langle \psi_{\text{block}} | \mathcal{O} \psi_{\text{block}} \rangle = 2 \iiint \text{scalar}(\psi_{\text{block}}^H \mathcal{O} \psi_{\text{block}}) dr_1 dr_2 dr_3.$$

For a state that is not already normalized, the scalar pairing produced when the operator acts between the bra and ket is divided by the squared length:

$$\frac{\langle \psi_{\text{block}} | \mathcal{O} \psi_{\text{block}} \rangle}{\langle \psi_{\text{block}} | \psi_{\text{block}} \rangle}. \quad (249)$$

Suppose the operator output is

$$\mathcal{O} \psi_{\text{block}} = \phi_{\text{operator_output}}.$$

Its scalar–vector decomposition is

$$\begin{aligned} \phi_{\text{operator_output}} &= \text{scalar}(\phi_{\text{operator_output}}) \\ &\quad + \text{vector}(\phi_{\text{operator_output}}) \cdot \sigma. \end{aligned}$$

The local scalar integrand is then

$$\begin{aligned} & \text{scalar}(\psi_{\text{block}}^H \mathcal{O} \psi_{\text{block}}) \\ &= \text{scalar}(\psi_{\text{block}})^* \text{scalar}(\phi_{\text{operator_output}}) \\ &+ \text{vector}(\psi_{\text{block}})^* \cdot \text{vector}(\phi_{\text{operator_output}}). \end{aligned}$$

After spatial integration and normalization by Eq. (249), the result is real and is called the expectation value when $\mathcal{O}$ is Hermitian under that integral and every boundary contribution vanishes. It is the average result associated with repeated measurements of that operator in the specified state [19]. The local integrand need not itself be real. For a general $\mathcal{O}$, the normalized result is only a pairing coefficient and may be complex. For the Dirac field the physical norm is the sum of this same fixed-projector pairing over its two restricted blocks; the associated conserved spatial current is supplied by the field equation. No separately conserved norm is assigned to either massive Dirac block.

By contrast, for an unrestricted algebra-valued paravector field,

$$\begin{aligned} & \text{adj}(\psi_{\text{block}}) \psi_{\text{block}} \\ &= \text{scalar}(\psi_{\text{block}})^2 \\ &- \text{vector}(\psi_{\text{block}}) \cdot \text{vector}(\psi_{\text{block}}) \\ &= \det(\psi_{\text{block}}) \end{aligned}$$

is the determinant quadratic form. It is algebraically useful, but it is not positive-definite: a positive-definite form must be real and strictly positive for every nonzero input, whereas this determinant can be complex, can take either sign when it is real, and can vanish for a nonzero input. It should therefore not be used as the probabilistic normalization. Moreover, after the fixed-projector restriction, this determinant vanishes identically as follows directly from Eq. (191) and the projector–ladder products. In this framework the single bra-ket bracket $\langle \cdot | \cdot \rangle$ is therefore reserved for the scalar pairing built from H .

6.8.4 Wave-Function Spaces and Pairings

In standard physics, spin is not assigned from component count alone. The physically relevant data are

$$(\text{transformation law, field equation, constraints}).$$

Thus the spin-zero Klein–Gordon field is carried by a complex scalar only after its scalar transformation law and field equation are imposed. A one-half-unit Dirac block is carried by a projected two-amplitude block only after the paired first-order equations are imposed. The spin-one electromagnetic field is carried by the pure-vector complex paravector F only with its field transformation law and Maxwell equation. A connection between the present construction and the standard classification is made by these statements in words, without introduction of another symbolic notation.

Scalar pairings. For two complex-paravector fields ϕ_{block} and

ψ_{block} , the three scalar pairings are

$$\begin{aligned} & \text{scalar}(\phi_{\text{block}}^H \psi_{\text{block}}) \\ &= \text{scalar}(\phi_{\text{block}})^* \text{scalar}(\psi_{\text{block}}) \\ &+ \text{vector}(\phi_{\text{block}})^* \cdot \text{vector}(\psi_{\text{block}}), \\ & \text{scalar}(\phi_{\text{block}}^{*H} \psi_{\text{block}}) \\ &= \text{scalar}(\phi_{\text{block}}) \text{scalar}(\psi_{\text{block}}) \\ &- \text{vector}(\phi_{\text{block}}) \cdot \text{vector}(\psi_{\text{block}}), \\ & \text{scalar}(\phi_{\text{block}}^* \psi_{\text{block}}) \\ &= \text{scalar}(\phi_{\text{block}})^* \text{scalar}(\psi_{\text{block}}) \\ &- \text{vector}(\phi_{\text{block}})^* \cdot \text{vector}(\psi_{\text{block}}). \end{aligned}$$

The first and third lines are conjugate-linear in ϕ_{block} and linear in ψ_{block} : multiplication of ϕ_{block} by a complex number conjugates that number in the result, whereas multiplication of ψ_{block} does not. The second line is linear in both fields. When the two fields agree, the first line is the positive Hermitian pairing [17, 19]; the second is the determinant extension; and the third is its indefinite companion, meaning that both positive and negative values can occur. For $\phi_{\text{block}} = \psi_{\text{block}}$, the determinant extension reduces to

$$\text{scalar}(\psi_{\text{block}}^{*H} \psi_{\text{block}}) = \text{adj}(\psi_{\text{block}}) \psi_{\text{block}} = \det(\psi_{\text{block}}).$$

Only the Hermitian pairing is being used here as the quantum bra-ket pairing. The equal-argument determinant is the algebraic quadratic form used in factorization identities, and the companion pairing is algebraically well-defined but does not carry an independent standard quantum interpretation here.

For the spatial bracket already defined in Eq. (212), an operator $\mathcal{O}$ is Hermitian on the chosen fields and boundary conditions when

$$\langle \phi_{\text{block}} | \mathcal{O} \psi_{\text{block}} \rangle = \langle \mathcal{O} \phi_{\text{block}} | \psi_{\text{block}} \rangle.$$

For left multiplication by a fixed paravector Z , moving the operator to the first field replaces it by Z^H , because $(Z^H \phi_{\text{block}})^H = \phi_{\text{block}}^H Z$. For an operator that multiplies at each position, H supplies the Hermitian operation already defined by the algebra. For a differential operator, the allowed set of fields, called its domain, must also be stated. Conditions imposed at the edge of the spatial region or at infinite distance are called boundary conditions [17, 19]. Integration by parts must produce no remaining boundary contribution before the differential operator can be called Hermitian under the spatial bracket. Nonscalar parts of unrestricted or differently ordered products remain available as algebraic results, but an interpretation as a current, transported amount, or spin-axis imbalance must be established separately for the chosen operator and field equation. For a fixed-projector state and a left-acting operator that preserves it, the three-factor product is already restricted to the fixed selector as shown above. **Generalized operator products.** The following identities are algebraic. Operator composition on a paravector-valued state is described by them, but no physical spin is assigned by them alone.

For operator-valued components, reduction of the commutator to the pure cross-product law of Eq. (45) no longer occurs. The already defined extraction functions are retained so that no additional component names are needed. The state is acted on

by the extracted component operators to its left, and those operators commute with the fixed basis elements. With dot and cross products taken componentwise and in the written operator order,

$$\begin{aligned} \text{com}(\mathcal{O}, \mathcal{O}') \psi_{\text{block}} = & \left(\text{com}(\text{scalar}(\mathcal{O}), \text{scalar}(\mathcal{O}')) \right. \\ & + \text{vector}(\mathcal{O}) \cdot \text{vector}(\mathcal{O}') \\ & \left. - \text{vector}(\mathcal{O}') \cdot \text{vector}(\mathcal{O}) \right) \psi_{\text{block}} \\ & + \left(\text{com}(\text{scalar}(\mathcal{O}), \text{vector}(\mathcal{O}')) \right. \\ & + \text{com}(\text{vector}(\mathcal{O}), \text{scalar}(\mathcal{O}')) \\ & + i(\text{vector}(\mathcal{O}) \times \text{vector}(\mathcal{O}')) \\ & \left. - i(\text{vector}(\mathcal{O}') \times \text{vector}(\mathcal{O})) \right) \cdot \sigma \psi_{\text{block}}, \end{aligned} \quad (250)$$

where the scalar-vector commutators are understood componentwise. Unlike the purely algebraic paravector commutator, both scalar and vector parts are generally contained in Eq. (250). Only when all component operators mutually commute is the following simpler law obtained:

$$\text{com}(\mathcal{O}, \mathcal{O}') \psi_{\text{block}} = 2i(\text{vector}(\mathcal{O}) \times \text{vector}(\mathcal{O}')) \cdot \sigma \psi_{\text{block}}.$$

The matching anticommutator is

$$\begin{aligned} \text{acom}(\mathcal{O}, \mathcal{O}') \psi_{\text{block}} = & (\mathcal{O}\mathcal{O}' + \mathcal{O}'\mathcal{O}) \psi_{\text{block}} \\ = & \left(\text{acom}(\text{scalar}(\mathcal{O}), \text{scalar}(\mathcal{O}')) \right. \\ & + \text{vector}(\mathcal{O}) \cdot \text{vector}(\mathcal{O}') \\ & + \text{vector}(\mathcal{O}') \cdot \text{vector}(\mathcal{O}) \left. \right) \psi_{\text{block}} \\ & + \left(\text{acom}(\text{scalar}(\mathcal{O}), \text{vector}(\mathcal{O}')) \right. \\ & + \text{acom}(\text{vector}(\mathcal{O}), \text{scalar}(\mathcal{O}')) \\ & + i(\text{vector}(\mathcal{O}) \times \text{vector}(\mathcal{O}')) \\ & \left. + i(\text{vector}(\mathcal{O}') \times \text{vector}(\mathcal{O})) \right) \cdot \sigma \psi_{\text{block}}, \end{aligned} \quad (251)$$

so that

$$\mathcal{O}\mathcal{O}' = \frac{1}{2}(\text{acom}(\mathcal{O}, \mathcal{O}') + \text{com}(\mathcal{O}, \mathcal{O}')).$$

This is the natural companion to the operator commutator formula. A relation between measurement spreads can be formed from it only after the allowed states, the averaging rule, Hermitian operators, and a shared set of fields on which every operator product exists have been specified. No such physical statement is obtained from this algebraic identity alone.

Wave-function spaces under the operators. A space is said to be closed under an operator when applying that operator to any permitted member produces another member of the same space. The action on the candidate spaces is displayed by the

same algebra:

$$\begin{aligned} \partial : \mathbb{C} &\rightarrow \mathbb{P}_{\mathbb{C}}, \\ \partial : \{\mathbf{V} \cdot \sigma \mid \mathbf{V} \in \mathbb{C}^3\} &\rightarrow \mathbb{P}_{\mathbb{C}}, \\ \partial : \mathbb{P}_{\mathbb{C}} &\rightarrow \mathbb{P}_{\mathbb{C}}, \\ \partial : \mathbb{P}_{\mathbb{C}}\Pi(+) &\rightarrow \mathbb{P}_{\mathbb{C}}\Pi(+). \end{aligned} \quad (252)$$

Indeed,

$$\begin{aligned} \partial \text{ scalar}(\psi_{\text{block}}) &= \partial_t \text{ scalar}(\psi_{\text{block}}) + (\partial_{\mathbf{r}} \text{ scalar}(\psi_{\text{block}})) \cdot \sigma, \\ \partial(\text{vector}(\psi_{\text{block}}) \cdot \sigma) &= \partial_{\mathbf{r}} \cdot \text{vector}(\psi_{\text{block}}) \\ &\quad + \partial_t \text{ vector}(\psi_{\text{block}}) \cdot \sigma \\ &\quad + i(\partial_{\mathbf{r}} \times \text{vector}(\psi_{\text{block}})) \cdot \sigma. \end{aligned}$$

The full complex-paravector space $\mathbb{P}_{\mathbb{C}}$ is the smallest algebra containing the scalar and vector channels, but the projected fixed-projector space is smaller. Two complex amplitudes are contained in $\mathbb{P}_{\mathbb{C}}\Pi(+)$, selected by a fixed right factor, and this space is preserved because every differential and potential operator is applied on the left. The full first-order paravector derivative does not preserve the scalar subspace, whereas scalar second-order combinations such as $\square + m^2$ do. A pure-vector specialization is algebraically meaningful, but it is not preserved by generic first-order paravector action. A field equation that preserves the pure-vector form is therefore required before that form is assigned a physical use. A real-paravector specialization is not a general quantum wave-function space because it is not preserved by multiplication by the unit-magnitude complex phase factor $\exp(i\theta)$.

With those algebraic preliminaries fixed, a wave-function space is chosen first, followed by the equation and operators under which it is preserved, and then by the products from which measured quantities are defined. Those products need not take values in the original wave-function space; scalar, vector, or paravector quantities may be obtained.

Scalar wave-function space. The Klein–Gordon field is the complex scalar

$$\psi_{\text{KG_with_potential}} \in \mathbb{C}.$$

For a chosen nonzero real axis vector $\mathbf{V}$, the unit direction is $\mathbf{u} := \mathbf{V}/\|\mathbf{V}\|$ and the corresponding projector is $\Pi(\pm)$ from Eq. (71). Multiplication by this projector would take $\psi_{\text{KG_with_potential}}$ out of the scalar space, so the axis projectors are not scalar-wave spin projectors. The positive quantity $|\psi_{\text{KG_with_potential}}|^2$ is not the time component of the relativistic conserved current. For $m > 0$, the mixed current is defined in Eq. (183). The following relations are obtained by direct differentiation:

$$\begin{aligned} 2m \partial_t \rho(\phi_{\text{KG_with_potential}}, \psi_{\text{KG_with_potential}}) &= i(E^2 \phi_{\text{KG_with_potential}})^* \psi_{\text{KG_with_potential}} \\ &\quad - i\phi_{\text{KG_with_potential}}^* E^2 \psi_{\text{KG_with_potential}}, \\ 2m \partial_{\mathbf{r}} \cdot \mathbf{j}(\phi_{\text{KG_with_potential}}, \psi_{\text{KG_with_potential}}) &= i\phi_{\text{KG_with_potential}}^* \mathbf{P}^2 \psi_{\text{KG_with_potential}} \\ &\quad - i(\mathbf{P}^2 \phi_{\text{KG_with_potential}})^* \psi_{\text{KG_with_potential}}. \end{aligned}$$

Equation (185) is established by addition of the lines and application of the two Klein–Gordon equations. For one constant-potential plane wave, meaning one term with a single frequency and wave vector,

$$\psi_{\text{KG_with_potential}} = c \exp(-i\omega t + i\mathbf{k} \cdot \mathbf{r}),$$

where c is a constant complex amplitude, the component check is

$$\begin{aligned} E\psi_{\text{KG_with_potential}} &= (\omega - qV)\psi_{\text{KG_with_potential}}, \\ \mathbf{P}\psi_{\text{KG_with_potential}} &= (\mathbf{k} - q\mathbf{A})\psi_{\text{KG_with_potential}}, \\ (\omega - qV)^2 &= (\mathbf{k} - q\mathbf{A})^2 + m^2, \\ \rho(\psi_{\text{KG_with_potential}}, \psi_{\text{KG_with_potential}}) &= \frac{\omega - qV}{m} |c|^2, \\ \mathbf{j}(\psi_{\text{KG_with_potential}}, \psi_{\text{KG_with_potential}}) &= \frac{\mathbf{k} - q\mathbf{A}}{m} |c|^2. \end{aligned}$$

Every potential and current sign is thereby verified.

The free field is obtained by setting $q = 0$. The integrated pairing in Eq. (186) is positive after restriction to the positive-frequency part, meaning the solutions whose individual time factors are $\exp(-i\omega t)$ with $\omega > 0$, although its local density need not be positive at every position for a sum of such solutions. That part of the solution space must be preserved by a scalar measurement operator. Such an operator represents a measured quantity and must be Hermitian under this pairing so that its averages are real [19]. The split is preserved for the free field, or for a background that does not change with time and for which a separate positive-energy space has been established. A single frequency split need not be preserved by a general time-dependent electromagnetic background. Accordingly, no local position-probability rule is claimed for the one-particle Klein–Gordon field.

Projected two-block wave-function space. For the ordered pair of projected blocks, the Hermitian pairing is Eq. (214). A general linear operator on such pairs is defined by its action

$$\begin{aligned} \mathcal{O}(\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2}) \\ := \begin{pmatrix} \mathcal{O}(1,1)\Psi_{\text{Dirac},1} + \mathcal{O}(1,2)\Psi_{\text{Dirac},2}, \\ \mathcal{O}(2,1)\Psi_{\text{Dirac},1} + \mathcal{O}(2,2)\Psi_{\text{Dirac},2} \end{pmatrix}. \end{aligned}$$

where the four component operators act from the left and preserve the fixed right projector. The operator pairing defined in Eq. (215) is then expanded explicitly as

$$\begin{aligned} \langle \Psi_{\text{Dirac_left}} | \mathcal{O}(\Psi_{\text{Dirac_right}}) \rangle \\ = 2 \iiint \text{scalar} \left(\Psi_{\text{Dirac_left},1}^H \mathcal{O}(1,1)\Psi_{\text{Dirac_right},1} \right. \\ + \Psi_{\text{Dirac_left},1}^H \mathcal{O}(1,2)\Psi_{\text{Dirac_right},2} \\ + \Psi_{\text{Dirac_left},2}^H \mathcal{O}(2,1)\Psi_{\text{Dirac_right},1} \\ \left. + \Psi_{\text{Dirac_left},2}^H \mathcal{O}(2,2)\Psi_{\text{Dirac_right},2} \right) dr_1 dr_2 dr_3. \end{aligned}$$

By Eq. (246), the ordinary normalization of each two-amplitude block is provided by the factor 2. Without the fixed right projector,

two independent copies of every displayed Dirac and Pauli equation are produced by decomposition into the two complementary fixed-projector spaces. The conserved density and current in Eq. (208) are supplied by the coupled first-order equation rather than by the vector part of a generic Hermitian product. Equation (216) is the corresponding complete two-block measurement rule.

Pure-vector field space. The electromagnetic field F is a pure-vector complex paravector. That pure-vector space is not preserved under left multiplication by the projectors $\Pi(\pm)$ for the same chosen unit direction, so this two-projector split does not separate the independent electromagnetic field orientations. A free massless electromagnetic wave has two independent field orientations perpendicular to its direction of travel. The field square that is unchanged by the stated transforms and the observer-dependent energy and energy-flow paravector are the products already derived in the electromagnetic section. A generic Hermitian-product bracket is not substituted for those field quantities.

Thus spin is never fixed by component count alone. The wave-function space, transformation law, field equation, restrictions, and products used to calculate measured quantities must agree. For projected Pauli and Dirac fields, the positive wave-function pairing is supplied by H . The scalar Klein–Gordon field instead uses the conserved pairing defined above. The determinant used in factorization is supplied by the star– H composition, and the pure conjugate pairing remains an algebraic companion without an additional physical interpretation.

6.8.5 Matrix Translation of Projected-State Pairings

The intrinsic fixed-projector formulas in Eqs. (191)–(248) have the following ordinary-matrix translation. The operation H is represented by the ordinary Hermitian operation, as established in Eq. (296).

The fixed projector and lowering element are represented by

$$\Pi(+) \longleftrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad a(-) \longleftrightarrow \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}.$$

Consequently, the two intrinsic expansions used in Eq. (246) are represented by

$$\begin{aligned} \psi_{\text{block}} &= a\Pi(+) + ba(-) \longleftrightarrow \text{mat}(\psi_{\text{block}}) = \begin{bmatrix} a & 0 \\ b & 0 \end{bmatrix}, \\ \phi_{\text{block}} &= c\Pi(+) + da(-) \longleftrightarrow \text{mat}(\phi_{\text{block}}) = \begin{bmatrix} c & 0 \\ d & 0 \end{bmatrix}. \end{aligned}$$

The intrinsic pairing is therefore translated into the familiar row–column product by

$$\begin{aligned} 2 \text{ scalar}(\phi_{\text{block}}^H \psi_{\text{block}}) &= \text{tr}(\text{mat}(\phi_{\text{block}})^H \text{mat}(\psi_{\text{block}})) \\ &= \begin{bmatrix} c & d \end{bmatrix}^H \begin{bmatrix} a \\ b \end{bmatrix} = c^*a + d^*b. \end{aligned}$$

This is the ordinary-matrix realization of Eq. (246). Algebraic scalar extraction is thereby identified as one half of the ordinary trace in this realization.

For the nonzero state scale N from Eq. (247), the opposite-order product is represented by

$$\text{mat}\left(\frac{\psi_{\text{block}}\psi_{\text{block}}^H}{N}\right) = \frac{1}{N} \begin{bmatrix} a \\ b \end{bmatrix} \begin{bmatrix} a^* & b^* \end{bmatrix} = \frac{1}{N} \begin{bmatrix} |a|^2 & ab^* \\ ba^* & |b|^2 \end{bmatrix}.$$

The displayed matrix is Hermitian, is unchanged when squared, has determinant zero, and has exactly one independent column direction. These are the ordinary-matrix statements corresponding to the intrinsic identities in Eq. (248). The name “pure state” refers to this single normalized state rather than to a statistical mixture [19].

An unrestricted complex paravector is represented by an arbitrary two-by-two complex matrix. Its two columns are acted upon independently by left multiplication. Under right multiplication by $\Pi(+)$, the first column is retained and the second is set to zero, producing the matrices displayed above. If that right-projector restriction is omitted, two independent copies of each left-acting Pauli or Dirac equation are carried by the two columns.

The ordered pair $\Psi_{\text{Dirac}} = (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2})$ defined intrinsically above can also be displayed as the block column

$$\Psi_{\text{Dirac}} \longleftrightarrow \begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix}.$$

Likewise, the component action of the general pair operator may be packaged as the block matrix

$$\mathcal{O} := \begin{bmatrix} \mathcal{O}(1,1) & \mathcal{O}(1,2) \\ \mathcal{O}(2,1) & \mathcal{O}(2,2) \end{bmatrix}.$$

With this optional packaging, the intrinsic operator pairing from Eq. (215) becomes

$$\begin{aligned} & \langle \Psi_{\text{Dirac_left}} | \mathcal{O} \Psi_{\text{Dirac_right}} \rangle \\ &= 2 \iiint \text{scalar} \left(\Psi_{\text{Dirac_left},1}^H \mathcal{O}(1,1) \Psi_{\text{Dirac_right},1} \right. \\ & \quad + \Psi_{\text{Dirac_left},1}^H \mathcal{O}(1,2) \Psi_{\text{Dirac_right},2} \\ & \quad + \Psi_{\text{Dirac_left},2}^H \mathcal{O}(2,1) \Psi_{\text{Dirac_right},1} \\ & \quad \left. + \Psi_{\text{Dirac_left},2}^H \mathcal{O}(2,2) \Psi_{\text{Dirac_right},2} \right) dr_1 dr_2 dr_3. \end{aligned}$$

No additional state components or physical assumptions are introduced by either matrix display; both are translations of the ordered-pair and fixed-projector algebra defined above.

6.8.6 Spin-0 Wave Mechanics

The scalar relativistic wave operator is motivated by the classical energy–momentum identity developed in the kinematic section. The conventional name “mass shell” refers here only to the relation $E^2 - \mathbf{P}^2 = m^2$ obeyed by a free particle of mass m . Replacement of classical energy and momentum by differential operators is the standard quantization prescription [17] and is an additional physical input, not a consequence of the determinant identity alone. Here $\psi_{\text{KG}} \in \mathbb{C}$ is a complex scalar field, $m \in \mathbb{R}$, and the differential conventions from Section 4 are used for the free-particle operators. The mass-shell identity is

$$\det(mU) = m^2.$$

Its scalar form is

$$-E^2 + \mathbf{P}^2 + m^2 = 0.$$

For a free particle,

$$E = i\partial_t, \quad \mathbf{P} = -i\partial_{\mathbf{r}},$$

and equivalently

$$E + \mathbf{P} \cdot \boldsymbol{\sigma} = i\partial^*.$$

Acting on a scalar field ψ_{KG} , the operator chain is

$$\begin{aligned} (-E^2 + \mathbf{P}^2 + m^2)\psi_{\text{KG}} &= (-(i\partial_t)^2 - \partial_{\mathbf{r}}^2 + m^2)\psi_{\text{KG}} \\ &= (\partial_t^2 - \partial_{\mathbf{r}}^2 + m^2)\psi_{\text{KG}} \\ &= (\partial\partial^* + m^2)\psi_{\text{KG}} \\ &= (\square + m^2)\psi_{\text{KG}}. \end{aligned}$$

Thus the free spin-0 relativistic wave equation is

$$(\square + m^2)\psi_{\text{KG}} = 0 \quad \text{Klein–Gordon equation.}$$

This is the scalar wave equation recorded in Eq. (168). For one plane-wave factor, meaning one complex exponential with one frequency and one spatial wave vector, the relation $\omega^2 = \mathbf{k}^2 + m^2$ fixes the allowed frequency ω for each spatial wave vector $\mathbf{k}$. The conventional name for this frequency–wave-vector relation is the dispersion relation. The later positive/negative-frequency split follows from the same operator identity.

6.8.7 Positive and Negative Frequency Parts

The separation by the sign of the time frequency is one of the central interpretive steps in the free scalar theory. On a stated spatial region and for fields on which the positive square-root operator is defined, the frequency function from Eq. (169) is given by

$$\begin{aligned} \Omega &= \sqrt{m^2 - \partial_{\mathbf{r}}^2} \\ &= \sqrt{\det(m + \boldsymbol{\sigma} \cdot \partial_{\mathbf{r}})}. \end{aligned}$$

Accordingly,

$$\begin{aligned} \square + m^2 &= \partial_t^2 + \Omega^2 \\ &= \partial_t^2 + (m^2 - \partial_{\mathbf{r}}^2) \\ &= (\partial_t - i\Omega)(\partial_t + i\Omega) \\ &= (\partial_t + i\Omega)(\partial_t - i\Omega). \end{aligned}$$

The two frequency-part definitions in Eq. (171) are then restated as

$$\begin{aligned} (\partial_t \pm i\Omega)\psi_{\text{KG_split}}(\pm) &= 0, \\ \psi_{\text{KG_split}}(t, \pm) &= \exp(\mp i\Omega t) \psi_{\text{KG_split}}(t, \pm)|_{t=0}, \\ \psi_{\text{KG}} &= \psi_{\text{KG_split}}(+) + \psi_{\text{KG_split}}(-). \end{aligned}$$

Here the parenthesized sign labels frequency and is not the sign of a spin-axis part. The latter sign is introduced only after its unit axis has been specified in Eqs. (230) and (231). When Ω^{-1}

exists, the two parts of a Klein–Gordon solution at any time can equivalently be extracted as

$$\psi_{\text{KG_split}}(\pm) = \frac{1}{2} (\psi_{\text{KG}} \pm i\Omega^{-1} \partial_t \psi_{\text{KG}}).$$

For a massless field, the spatially constant term with $\mathbf{k} = \mathbf{0}$ is treated separately. The Klein–Gordon pairing has opposite sign on a single negative-frequency wave and is not a positive norm on the combined solution space. In a quantized relativistic field the particle–antiparticle interpretation is supplied by the field-operator construction, not by reinterpreting this one-particle density alone [13, 17].

6.8.8 Fourier Meaning of the Frequency Operator

The square root Ω is most transparent after the field has been written as a sum of plane waves. This is the spatial Fourier decomposition: each term is called a mode and is labeled by one wave vector $\mathbf{k}(n)$ and one frequency $\omega(n)$ [17]. The operator is called nonlocal because its value at one position generally depends on field values at other positions, even though its action on each plane wave is simple. With the stated phase convention, the field and its modes are written schematically as

$$\begin{aligned} \psi_{\text{KG}} &= \sum_n c(n) f(n), \\ f(n) &= \exp(-i\omega(n)t + i\mathbf{k}(n) \cdot \mathbf{r}). \end{aligned}$$

Then

$$\begin{aligned} \square \psi_{\text{KG}} &= \sum_n (\mathbf{k}(n)^2 - \omega(n)^2) c(n) f(n), \\ (\square + m^2) \psi_{\text{KG}} &= \sum_n (\mathbf{k}(n)^2 + m^2 - \omega(n)^2) c(n) f(n), \\ \Omega \psi_{\text{KG}} &= \sum_n \sqrt{m^2 + \mathbf{k}(n)^2} c(n) f(n). \end{aligned}$$

For modes satisfying the Klein–Gordon frequency–wave-vector relation,

$$\mathbf{k}(n)^2 + m^2 = \omega(n)^2, \quad \omega(n) = \pm \sqrt{\mathbf{k}(n)^2 + m^2},$$

and each spatial mode is multiplied by its positive energy magnitude by Ω . The set of frequencies used in the sum is therefore restricted to the two displayed values for every $\mathbf{k}(n)$. The only square-root operator needed in the frequency split is thereby defined. No choice between the possible square-root values of $\square$ is needed.

6.8.9 Klein–Gordon with Potential Fields

The charged-particle Lagrangian, a function from which the classical motion is obtained by the stationary-action rule, is taken as the starting point. Under that rule, a small variation of the physical path produces no first-order change in the time integral of the Lagrangian [16]. The scalar theory with electromagnetic potential is thereby kept in one self-contained place:

$$\mathcal{L} = -\frac{m}{\gamma} - q(V - \mathbf{v} \cdot \mathbf{A}).$$

Here E denotes the energy operator, not the electric field $\mathbf{E}$. The classical Lagrangian first supplies total energy and canonical momentum, where “canonical” names the momentum that includes the potential term. After the standard differential replacement, subtraction of the potential terms gives the kinetic, or gauge-covariant, energy and momentum operators

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}.$$

Here “kinetic” means that the potential contribution has been removed so that these quantities correspond to the particle’s mechanical energy and momentum. “Gauge-covariant” means that their action on the phase-shifted field changes by the same overall phase, as verified below. This differential replacement, together with the stated potential substitution, is the standard quantization input [17].

Acting on a field $\psi_{\text{KG_with_potential}}$ whose displayed derivatives exist, the ordinary derivative product rules needed below are

$$\begin{aligned} \partial_t(V\psi_{\text{KG_with_potential}}) &= (\partial_t V)\psi_{\text{KG_with_potential}} \\ &\quad + V\partial_t\psi_{\text{KG_with_potential}}, \\ \partial_m(A_n\psi_{\text{KG_with_potential}}) &= (\partial_m A_n)\psi_{\text{KG_with_potential}} \\ &\quad + A_n\partial_m\psi_{\text{KG_with_potential}}. \end{aligned}$$

The following expansions are obtained:

$$\begin{aligned} E^2\psi_{\text{KG_with_potential}} &= E(i\partial_t\psi_{\text{KG_with_potential}} - qV\psi_{\text{KG_with_potential}}) \\ &= \begin{pmatrix} -\partial_t^2 - iq\partial_t V \\ -2iqV\partial_t + q^2V^2 \end{pmatrix} \psi_{\text{KG_with_potential}}, \\ \mathbf{P}^2\psi_{\text{KG_with_potential}} &= \mathbf{P} \cdot (-i\partial_{\mathbf{r}}\psi_{\text{KG_with_potential}} - q\mathbf{A}\psi_{\text{KG_with_potential}}) \\ &= \begin{pmatrix} -\partial_{\mathbf{r}}^2 + iq\partial_{\mathbf{r}} \cdot \mathbf{A} \\ + 2iq\mathbf{A} \cdot \partial_{\mathbf{r}} + q^2\mathbf{A}^2 \end{pmatrix} \psi_{\text{KG_with_potential}}. \end{aligned}$$

The required identities are

$$\begin{aligned} \square &= \partial_t^2 - \partial_{\mathbf{r}}^2, \\ \Phi &= V + \mathbf{A} \cdot \boldsymbol{\sigma}, \\ S &= \partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}, \\ \det(\partial) &= \square, \\ \det(\Phi) &= V^2 - \mathbf{A}^2. \end{aligned}$$

Here $\det(\partial) = \partial\partial^*$ is valid because the coordinate derivatives commute. The coupled energy and momentum operators below do not commute and therefore remain explicit ordered products rather than determinants. The Klein–Gordon equation with the stated electromagnetic potential is obtained in explicit form by

substitution of the operator expansions:

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2) \psi_{\text{KG_with_potential}} \\
&= \left(\begin{aligned} &\square + iq(\partial_t V + \partial_{\mathbf{r}} \cdot \mathbf{A}) \\ &+ 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \\ &- q^2(V^2 - \mathbf{A}^2) + m^2 \end{aligned} \right) \psi_{\text{KG_with_potential}} \\
&= \left(\begin{aligned} &\square + iqS - q^2 \det(\Phi) + m^2 \\ &+ 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}}) \end{aligned} \right) \psi_{\text{KG_with_potential}}.
\end{aligned}$$

For the gauge-shifted potential and components defined in Eqs. (153), (154), and (155),

$$\begin{aligned}
V' &= V + \partial_t \lambda, \\
\mathbf{A}' &= \mathbf{A} - \partial_{\mathbf{r}} \lambda, \\
\psi'_{\text{KG_with_potential}} &= \exp(-iq\lambda) \psi_{\text{KG_with_potential}}.
\end{aligned}$$

The real function $\lambda(t, \mathbf{r})$ is the gauge parameter: it labels different potential functions that produce the same electric and magnetic fields. The accompanying phase change of $\psi_{\text{KG_with_potential}}$ ensures that these different descriptions give the same coupled equation [16]. The ordinary product rule is obeyed by the central scalar phase. Consequently,

$$\begin{aligned}
&(i\partial_t - qV') \psi'_{\text{KG_with_potential}} \\
&= \exp(-iq\lambda) (i\partial_t - qV) \psi_{\text{KG_with_potential}}, \\
&(-i\partial_{\mathbf{r}} - q\mathbf{A}') \psi'_{\text{KG_with_potential}} \\
&= \exp(-iq\lambda) (-i\partial_{\mathbf{r}} - q\mathbf{A}) \psi_{\text{KG_with_potential}}.
\end{aligned}$$

Equation (179) is established by a second application of either relation. Thus the coupled Klein–Gordon equation retains the same form after the linked potential and phase changes. Substitution in Eq. (183) also cancels the phase from both current components.

The electromagnetic fields and commutators are incorporated through

$$\begin{aligned}
\mathbf{E} &= -\partial_t \mathbf{A} - \partial_{\mathbf{r}} V, \\
\mathbf{B} &= \partial_{\mathbf{r}} \times \mathbf{A}, \\
F &= (\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad F^* = (-\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma},
\end{aligned}$$

and

$$\begin{aligned}
&\text{com}(E, P_m) \psi_{\text{KG_with_potential}} \\
&= (i\partial_t - qV) \begin{pmatrix} -i\partial_m \psi_{\text{KG_with_potential}} \\ -qA_m \psi_{\text{KG_with_potential}} \end{pmatrix} \\
&\quad - (-i\partial_m - qA_m) \begin{pmatrix} i\partial_t \psi_{\text{KG_with_potential}} \\ -qV \psi_{\text{KG_with_potential}} \end{pmatrix} \\
&= -iq((\partial_t A_m) \psi_{\text{KG_with_potential}} + A_m \partial_t \psi_{\text{KG_with_potential}}) \\
&\quad - iq((\partial_m V) \psi_{\text{KG_with_potential}} + V \partial_m \psi_{\text{KG_with_potential}}) \\
&\quad + iqV \partial_m \psi_{\text{KG_with_potential}} + iqA_m \partial_t \psi_{\text{KG_with_potential}} \\
&= -iq(\partial_t A_m + \partial_m V) \psi_{\text{KG_with_potential}} \\
&= iqE_m \psi_{\text{KG_with_potential}}.
\end{aligned}$$

$$\begin{aligned}
&\text{com}(P_m, P_n) \psi_{\text{KG_with_potential}} \\
&= (-i\partial_m - qA_m) \begin{pmatrix} -i\partial_n \psi_{\text{KG_with_potential}} \\ -qA_n \psi_{\text{KG_with_potential}} \end{pmatrix} \\
&\quad - (-i\partial_n - qA_n) \begin{pmatrix} -i\partial_m \psi_{\text{KG_with_potential}} \\ -qA_m \psi_{\text{KG_with_potential}} \end{pmatrix} \\
&= iq(\partial_m A_n - \partial_n A_m) \psi_{\text{KG_with_potential}}.
\end{aligned}$$

Hence

$$\text{com}(E, \mathbf{P}) = iq\mathbf{E}, \quad \mathbf{P} \times \mathbf{P} = iq\mathbf{B}.$$

The following ordered products are obtained by substitution of those commutators:

$$\begin{aligned}
(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\
&\quad + \boldsymbol{\sigma} \cdot (-\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iqF, \\
(E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - \mathbf{P}^2 \\
&\quad + \boldsymbol{\sigma} \cdot (\text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iqF^*.
\end{aligned}$$

Since

$$\begin{aligned}
E + \boldsymbol{\sigma} \cdot \mathbf{P} &= i\partial^* - q\Phi, \\
E - \boldsymbol{\sigma} \cdot \mathbf{P} &= i\partial - q\Phi^* \\
&= (E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}},
\end{aligned}$$

the two ordered star–H products are

$$\begin{aligned}
&(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}} \\
&= E^2 - \mathbf{P}^2 \\
&\quad + \boldsymbol{\sigma} \cdot (\mp \text{com}(E, \mathbf{P}) - i\mathbf{P} \times \mathbf{P}).
\end{aligned}$$

Unlike the determinant function defined for commuting complex coefficients, these operator products retain a vector part. Operator ordering therefore matters, and the two first-order products remain distinct. Consequently,

$$\begin{aligned}
-(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}} &= (\partial^* + iq\Phi)(\partial + iq\Phi^*), \\
-(E + \boldsymbol{\sigma} \cdot \mathbf{P})^{*\text{H}}(E + \boldsymbol{\sigma} \cdot \mathbf{P}) &= (\partial + iq\Phi^*)(\partial^* + iq\Phi).
\end{aligned}$$

The same factorizations can be rewritten directly in derivative

form:

$$\begin{aligned}
& (E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P})\psi_{\text{KG_with_potential}} \\
&= -(\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi_{\text{KG_with_potential}} \\
&= -\square\psi_{\text{KG_with_potential}} - iq\partial^*(\Phi^*\psi_{\text{KG_with_potential}}) \\
&\quad - iq\Phi\partial\psi_{\text{KG_with_potential}} + q^2\det(\Phi)\psi_{\text{KG_with_potential}} \\
&= -\square\psi_{\text{KG_with_potential}} - iqS\psi_{\text{KG_with_potential}} \\
&\quad + q^2\det(\Phi)\psi_{\text{KG_with_potential}} \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi_{\text{KG_with_potential}} \\
&\quad - iqF\psi_{\text{KG_with_potential}}, \\
& (E - \boldsymbol{\sigma} \cdot \mathbf{P})(E + \boldsymbol{\sigma} \cdot \mathbf{P})\psi_{\text{KG_with_potential}} \\
&= -(\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi_{\text{KG_with_potential}} \\
&= -\square\psi_{\text{KG_with_potential}} - iq\partial(\Phi\psi_{\text{KG_with_potential}}) \\
&\quad - iq\Phi^*\partial^*\psi_{\text{KG_with_potential}} + q^2\det(\Phi)\psi_{\text{KG_with_potential}} \\
&= -\square\psi_{\text{KG_with_potential}} - iqS\psi_{\text{KG_with_potential}} \\
&\quad + q^2\det(\Phi)\psi_{\text{KG_with_potential}} \\
&\quad - 2iq(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi_{\text{KG_with_potential}} \\
&\quad - iqF^*\psi_{\text{KG_with_potential}}.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
& \partial^*(\Phi^*\psi_{\text{KG_with_potential}}) + \Phi\partial\psi_{\text{KG_with_potential}} \\
&= S\psi_{\text{KG_with_potential}} + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi_{\text{KG_with_potential}} \\
&\quad + F\psi_{\text{KG_with_potential}}, \\
& \partial(\Phi\psi_{\text{KG_with_potential}}) + \Phi^*\partial^*\psi_{\text{KG_with_potential}} \\
&= S\psi_{\text{KG_with_potential}} + 2(V\partial_t + \mathbf{A} \cdot \partial_{\mathbf{r}})\psi_{\text{KG_with_potential}} \\
&\quad + F^*\psi_{\text{KG_with_potential}}.
\end{aligned}$$

and their difference isolates the electric contribution:

$$\begin{aligned}
& \left[\partial^*(\Phi^*\psi_{\text{KG_with_potential}}) \right. \\
& \quad \left. + \Phi\partial\psi_{\text{KG_with_potential}} \right] \\
& \quad - \left[\partial(\Phi\psi_{\text{KG_with_potential}}) \right. \\
& \quad \left. + \Phi^*\partial^*\psi_{\text{KG_with_potential}} \right] \\
&= (\partial\Phi^* - \partial\Phi)\psi_{\text{KG_with_potential}} \\
&= (F - F^*)\psi_{\text{KG_with_potential}} \\
&= 2(\mathbf{E} \cdot \boldsymbol{\sigma})\psi_{\text{KG_with_potential}}.
\end{aligned}$$

From the definitions in Eqs. (172) and (173), (135), (136), and (141), the scalar Klein–Gordon operator can therefore be recovered from either ordered first-order product:

$$\begin{aligned}
0 &= (-E^2 + \mathbf{P}^2 + m^2)\psi_{\text{KG_with_potential}} \\
&= \left(\square + iqS - q^2\det(\Phi) + m^2 \right) \psi_{\text{KG_with_potential}} \\
&= (\partial^* + iq\Phi)(\partial + iq\Phi^*)\psi_{\text{KG_with_potential}} \\
&\quad - iqF\psi_{\text{KG_with_potential}} + m^2\psi_{\text{KG_with_potential}} \\
&= (\partial + iq\Phi^*)(\partial^* + iq\Phi)\psi_{\text{KG_with_potential}} \\
&\quad - iqF^*\psi_{\text{KG_with_potential}} + m^2\psi_{\text{KG_with_potential}}.
\end{aligned}$$

The following identities are satisfied by the ordered first-order products:

$$\begin{aligned}
(E \pm \boldsymbol{\sigma} \cdot \mathbf{P})(E \mp \boldsymbol{\sigma} \cdot \mathbf{P}) &= E^2 - (\boldsymbol{\sigma} \cdot \mathbf{P})^2 \mp \text{com}(E, \boldsymbol{\sigma} \cdot \mathbf{P}) \\
&= E^2 - \mathbf{P}^2 - iq(\pm\mathbf{E} + i\mathbf{B}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

The following equation is therefore recovered by taking the scalar part:

$$\begin{aligned}
0 &= \text{scalar}((\partial + iq\Phi^*)(\partial^* + iq\Phi))\psi_{\text{KG_with_potential}} \\
&\quad + m^2\psi_{\text{KG_with_potential}}.
\end{aligned}$$

The Klein–Gordon operator itself is reproduced only by the scalar part; the full ordered products retain the non-scalar field terms $-iqF$ and $-iqF^*$ that distinguish the two factorizations. Explicitly,

$$-iqF = (-iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad -iqF^* = (iq\mathbf{E} + q\mathbf{B}) \cdot \boldsymbol{\sigma}.$$

These remaining terms are not new components of the scalar equation. They become the standard spin–field terms in the two ordered Dirac squares below [17].

6.8.10 Spin- $\frac{1}{2}$ Wave Mechanics

The Dirac field is treated intrinsically as the ordered pair

$$\Psi_{\text{Dirac}} := (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2}).$$

For clarity, the fixed axis and projector from Eqs. (187) and (188) are restated in the same order:

$$\begin{aligned}
\mathbf{u} &= \mathbf{e}_3 = \{0, 0, 1\}, \\
\Pi(+) &= \frac{1}{2}(1 + \mathbf{u} \cdot \boldsymbol{\sigma}) = \frac{1}{2}(1 + \sigma_3).
\end{aligned}$$

One Dirac field is selected by requiring both paravector blocks to satisfy

$$\Psi_{\text{Dirac},n} = \Psi_{\text{Dirac},n}\Pi(+), \quad n \in \{1, 2\}.$$

All differential and potential factors are applied on the left, so this restriction is preserved by associativity.

The projected algebra is spanned by the fixed projector and the previously defined ladder element $a(-)$. Consequently, the pair is specified by four complex scalar amplitudes a, b, c, d :

$$\begin{aligned}
\Psi_{\text{Dirac},1} &= a\Pi(+) + ba(-), \\
\Psi_{\text{Dirac},2} &= c\Pi(+) + da(-).
\end{aligned}$$

This form is obtained from the following right-action identities:

$$\Pi(+)^2 = \Pi(+), \quad a(-)\Pi(+) = a(-).$$

Thus two complex amplitudes are carried intrinsically by each projected paravector.

Before the electromagnetic field is introduced, the first-order system is written directly as the two coupled paravector equations

$$\begin{aligned}
i\partial\Psi_{\text{Dirac},2} - m\Psi_{\text{Dirac},1} &= 0, \\
i\partial^*\Psi_{\text{Dirac},1} - m\Psi_{\text{Dirac},2} &= 0.
\end{aligned}$$

The following reductions are obtained by application of the opposite first-order factor:

$$\begin{aligned} i\partial^*(i\partial\Psi_{\text{Dirac},2} - m\Psi_{\text{Dirac},1}) &= -(\square + m^2)\Psi_{\text{Dirac},2} = 0, \\ i\partial(i\partial^*\Psi_{\text{Dirac},1} - m\Psi_{\text{Dirac},2}) &= -(\square + m^2)\Psi_{\text{Dirac},1} = 0. \end{aligned}$$

Thus the Klein–Gordon equation is required for both blocks.

For a free plane wave proportional to $\exp(-iEt + i\mathbf{p} \cdot \mathbf{r})$, the following scalar component equations are obtained by substitution of the amplitude decomposition:

$$\begin{aligned} (E - p_3)c - (p_1 - ip_2)d &= ma, \\ -(p_1 + ip_2)c + (E + p_3)d &= mb, \\ (E + p_3)a + (p_1 - ip_2)b &= mc, \\ (p_1 + ip_2)a + (E - p_3)b &= md. \end{aligned}$$

These are the standard four first-order Dirac component equations [10, 17]. The relation $E^2 = \mathbf{p}^2 + m^2$ is obtained from their ordered products. With a potential present, the operator-valued component equations in Eq. (197) are obtained from the same algebraic decomposition.

Their transformation law is determined by requiring the two equations to retain the same form rather than by counting components. For a constant determinant-one transform T , with X' given by Eq. (97), the transformed blocks are assigned the laws from Eqs. (198) and (199):

$$\begin{aligned} \Psi'_{\text{Dirac_with_potential},1}(t', \mathbf{r}') &= T^H \Psi_{\text{Dirac_with_potential},1}(t, \mathbf{r}), \\ \Psi'_{\text{Dirac_with_potential},2}(t', \mathbf{r}') &= T^{-1} \Psi_{\text{Dirac_with_potential},2}(t, \mathbf{r}). \end{aligned}$$

The corresponding derivative and potential laws are

$$\partial' = T^H \partial T, \quad (253)$$

$$\partial^* = T^{-1} \partial^* T^*, \quad (254)$$

$$\Phi' = T^{-1} \Phi T^*, \quad (255)$$

$$\Phi'^* = T^H \Phi^* T. \quad (256)$$

For a spatial rotor, the same left factor R^H is assigned to both blocks. For a boost, L is assigned to the first block and L^* to the second. Since $\det(T) = 1$, the identities $TT^{-1} = 1$ and $T^*T^H = 1$ are used in the following explicit substitutions:

$$\begin{aligned} (i\partial' - q\Phi'^*)\Psi'_{\text{Dirac_with_potential},2} &= T^H(i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2} \\ &= m\Psi'_{\text{Dirac_with_potential},1}, \\ (i\partial'^* - q\Phi')\Psi'_{\text{Dirac_with_potential},1} &= T^{-1}(i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} \\ &= m\Psi'_{\text{Dirac_with_potential},2}. \end{aligned}$$

Thus both coupled equations retain the same form after the transformation. Because all transform factors are applied from the left, the fixed right projector is preserved.

The real current paravector defined in Eq. (207) is expanded as

$$\begin{aligned} J(\Psi_{\text{Dirac_with_potential}}) &= 2(\Psi_{\text{Dirac_with_potential},2}\Psi_{\text{Dirac_with_potential},2}^H \\ &\quad + (\Psi_{\text{Dirac_with_potential},1}\Psi_{\text{Dirac_with_potential},1}^H)^*) \\ &= \rho(\Psi_{\text{Dirac_with_potential}}) + \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) \cdot \boldsymbol{\sigma}. \end{aligned}$$

The following transformation is obtained from Eqs. (198) and (199) in each product:

$$\begin{aligned} \Psi'_{\text{Dirac_with_potential},2}\Psi_{\text{Dirac_with_potential},2}^H &= T^{-1}\Psi_{\text{Dirac_with_potential},2}\Psi_{\text{Dirac_with_potential},2}^H(T^{-1})^H, \\ (\Psi'_{\text{Dirac_with_potential},1}\Psi_{\text{Dirac_with_potential},1}^H)^* &= T^{-1}(\Psi_{\text{Dirac_with_potential},1}\Psi_{\text{Dirac_with_potential},1}^H)^*(T^{-1})^H. \end{aligned}$$

Here $(T^H)^* = T^{-1}$ and $T^* = (T^{-1})^H$ are obtained from $\det(T) = 1$. Consequently,

$$\begin{aligned} J(\Psi'_{\text{Dirac_with_potential}}(t', \mathbf{r}')) &= T^{-1}J(\Psi_{\text{Dirac_with_potential}}(t, \mathbf{r}))(T^{-1})^H. \end{aligned}$$

so the current is transformed by the same two-sided rule as a real event paravector. For the amplitude decomposition in the fixed projected algebra, the component form is

$$\begin{aligned} \rho(\Psi_{\text{Dirac_with_potential}}) &= |a|^2 + |b|^2 + |c|^2 + |d|^2, \\ \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) &= \left\{ \begin{aligned} &2 \operatorname{real}(c^*d - a^*b), \\ &2 \operatorname{img}(c^*d - a^*b), \\ &|c|^2 - |d|^2 - |a|^2 + |b|^2 \end{aligned} \right\}. \end{aligned}$$

These are the conventional Dirac density and current components. The two uncanceled block balances are Eq. (209); by their addition, $\partial_t \rho(\Psi_{\text{Dirac_with_potential}}) + \partial_{\mathbf{r}} \cdot \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) = 0$ is established. In particular,

$$\begin{aligned} \mathbf{u} \cdot \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) &= 2 \operatorname{scalar}(\Psi_{\text{Dirac_with_potential},2}^H(\mathbf{u} \cdot \boldsymbol{\sigma})\Psi_{\text{Dirac_with_potential},2}) \\ &\quad - 2 \operatorname{scalar}(\Psi_{\text{Dirac_with_potential},1}^H(\mathbf{u} \cdot \boldsymbol{\sigma})\Psi_{\text{Dirac_with_potential},1}), \end{aligned}$$

so this is the local Hermitian-pairing density for the two-block velocity operator along the unit direction $\mathbf{u}$, where $\mathbf{u}^2 = 1$. The action of this directional operator is $-\mathbf{u} \cdot \boldsymbol{\sigma}$ on $\Psi_{\text{Dirac_with_potential},1}$ and $+\mathbf{u} \cdot \boldsymbol{\sigma}$ on $\Psi_{\text{Dirac_with_potential},2}$. Its spatial integral divided by $\langle \Psi_{\text{Dirac_with_potential}} | \Psi_{\text{Dirac_with_potential}} \rangle$ is the corresponding average velocity component along $\mathbf{u}$. The constant-time density is the time entry measured in one chosen reference frame. It is not a Lorentz scalar, meaning a number unchanged by a Lorentz transformation.

For the free system with $q = 0$ and $m > 0$, the following result is obtained by solving the first equation for the first block

and substituting into the second:

$$\begin{aligned}\Psi_{\text{Dirac},1} &= \frac{i\partial\Psi_{\text{Dirac},2}}{m}, \\ 0 &= i\partial^*\Psi_{\text{Dirac},1} - m\Psi_{\text{Dirac},2} \\ &= -\frac{\partial^*\partial\Psi_{\text{Dirac},2}}{m} - m\Psi_{\text{Dirac},2} \\ &= -\frac{(\square + m^2)\Psi_{\text{Dirac},2}}{m}.\end{aligned}$$

Hence the second block is a Klein–Gordon solution, and the first block is obtained from it by one first-order derivative.

Conversely, a scalar Klein–Gordon solution ψ_{KG} and constant projected amplitude elements $a, b \in \mathbb{P}_{\mathbb{C}}\Pi(+)$ are taken. The corresponding ordered pair is defined intrinsically by

$$\begin{aligned}\Psi_{\text{Dirac},1} &:= \psi_{\text{KG}}a + \frac{i}{m}(\partial\psi_{\text{KG}})b, \\ \Psi_{\text{Dirac},2} &:= \frac{i}{m}(\partial^*\psi_{\text{KG}})a + \psi_{\text{KG}}b, \\ \Psi_{\text{Dirac}} &:= (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2}).\end{aligned}\tag{257}$$

Because the constants occur on the right, only ψ_{KG} is acted on by the derivatives. The following reductions are obtained by direct substitution into the two free Dirac equations:

$$\begin{aligned}i\partial\Psi_{\text{Dirac},2} - m\Psi_{\text{Dirac},1} &= -\frac{1}{m}(\square + m^2)\psi_{\text{KG}}a = 0, \\ i\partial^*\Psi_{\text{Dirac},1} - m\Psi_{\text{Dirac},2} &= -\frac{1}{m}(\square + m^2)\psi_{\text{KG}}b = 0.\end{aligned}$$

The spin-axis choice is carried by the constants a and b ; it is not carried by a bare scalar solution alone. For example, the following ordered pair is obtained with $a = \Pi(+)$ and $b = 0$:

$$\Psi_{\text{Dirac}} = \left(\psi_{\text{KG}}\Pi(+), \frac{i}{m}(\partial^*\psi_{\text{KG}})\Pi(+), 0 \right).$$

A Klein–Gordon solution satisfying $E^2 - \mathbf{p}^2 = m^2$ is thereby converted into a field in the fixed projected Dirac space. The two-input presentation is symmetric but need not be one-to-one. For each single plane wave satisfying the displayed energy–momentum relation, a nonzero b together with $a = -(E - \mathbf{p} \cdot \boldsymbol{\sigma})b/m$ is sent to the zero pair. The same Dirac plane wave can therefore be produced by different input pairs. For a fixed plane wave with $m > 0$, the two spin-axis possibilities are already produced from one projected input block. Four new spin possibilities are not introduced by the symbols a and b .

6.8.11 Dirac with Potential Field

The field is again treated as the ordered pair

$$\Psi_{\text{Dirac_with_potential}} := \begin{pmatrix} \Psi_{\text{Dirac_with_potential},1} \\ \Psi_{\text{Dirac_with_potential},2} \end{pmatrix}.$$

One Dirac field remains restricted by

$$\Psi_{\text{Dirac_with_potential},n} = \Psi_{\text{Dirac_with_potential},n}\Pi(+).$$

Here $n \in \{1, 2\}$. All potential and derivative factors are applied from the left, and the fixed projected space is therefore preserved.

The pair-valued Dirac function is defined by

$$\begin{aligned}\mathcal{D}(\Phi, \Psi_{\text{Dirac_with_potential}}) &:= \begin{pmatrix} (i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2} \\ -m\Psi_{\text{Dirac_with_potential},1} \\ (i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} \\ -m\Psi_{\text{Dirac_with_potential},2} \end{pmatrix}.\end{aligned}\tag{258}$$

The partner function is defined by reversing only the mass sign:

$$\begin{aligned}\mathcal{D}'(\Phi, \Psi_{\text{Dirac_with_potential}}) &:= \begin{pmatrix} (i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2} \\ +m\Psi_{\text{Dirac_with_potential},1} \\ (i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} \\ +m\Psi_{\text{Dirac_with_potential},2} \end{pmatrix}.\end{aligned}\tag{259}$$

This distinct mass-sign choice is labeled by the prime; no derivative is denoted. In terms of Eq. (258), the coupled Dirac equation is

$$\mathcal{D}(\Phi, \Psi_{\text{Dirac_with_potential}}) = (0, 0).$$

The following paravector equations are obtained from its two entries:

$$\begin{aligned}-m\Psi_{\text{Dirac_with_potential},1} + (i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2} &= 0, \\ (i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} - m\Psi_{\text{Dirac_with_potential},2} &= 0.\end{aligned}$$

Equivalently, their time evolution is

$$\begin{aligned}i\partial_t\Psi_{\text{Dirac_with_potential},1} &= (qV - \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_{\text{Dirac_with_potential},1} \\ &\quad + m\Psi_{\text{Dirac_with_potential},2}, \\ i\partial_t\Psi_{\text{Dirac_with_potential},2} &= (qV + \boldsymbol{\sigma} \cdot \mathbf{P})\Psi_{\text{Dirac_with_potential},2} \\ &\quad + m\Psi_{\text{Dirac_with_potential},1}.\end{aligned}$$

The $\mathbf{H}$ operation is applied to these equations, multiplication is performed on the opposite side, and scalar parts are taken. In the fixed projected form, the two separate balance equations are written out in Eq. (209). The real potential terms are canceled within each block, while the mass terms are equal and opposite between the blocks. The following relation is obtained by addition of those two equations:

$$\begin{aligned}0 &= \partial_t [\text{scalar}(\Psi_{\text{Dirac_with_potential},1}^{\mathbf{H}} \Psi_{\text{Dirac_with_potential},1}) \\ &\quad + \text{scalar}(\Psi_{\text{Dirac_with_potential},2}^{\mathbf{H}} \Psi_{\text{Dirac_with_potential},2})] \\ &\quad + \partial_{\mathbf{r}} \cdot [\text{vector}(\Psi_{\text{Dirac_with_potential},2} \Psi_{\text{Dirac_with_potential},2}^{\mathbf{H}}) \\ &\quad - \text{vector}(\Psi_{\text{Dirac_with_potential},1} \Psi_{\text{Dirac_with_potential},1}^{\mathbf{H}})].\end{aligned}$$

The stated density and current are thereby verified, and the distinction between the vector part of a single $\psi_{\text{block}}^{\mathbf{H}} \psi_{\text{block}}$ and the Dirac current is established. The following result is obtained by solving the first equation for the first block and substituting into

the second:

$$\begin{aligned}
& \Psi_{\text{Dirac_with_potential},1} \\
&= \frac{1}{m}(i\partial - q\Phi^*)\Psi_{\text{Dirac_with_potential},2}, \\
0 &= (i\partial^* - q\Phi)\Psi_{\text{Dirac_with_potential},1} - m\Psi_{\text{Dirac_with_potential},2} \\
&= -\frac{1}{m}\left((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2\right) \\
&\quad \Psi_{\text{Dirac_with_potential},2}.
\end{aligned}$$

Thus a Klein–Gordon-type operator with the appropriate ordered field coupling is satisfied by each block.

The following result is obtained by application of the mass-sign partner to the pair returned by $\mathcal{D}$:

$$\begin{aligned}
0 &= \mathcal{D}'(\Phi, \mathcal{D}(\Phi, \Psi_{\text{Dirac_with_potential}})) \\
&= -\left((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2\right)\Psi_{\text{Dirac_with_potential},1}, \\
&\quad \left((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2\right)\Psi_{\text{Dirac_with_potential},2}.
\end{aligned}$$

Equivalently,

$$\begin{aligned}
0 &= ((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2)\Psi_{\text{Dirac_with_potential},1} \\
&= (-E^2 + \mathbf{P}^2 + m^2 + iqF^*)\Psi_{\text{Dirac_with_potential},1}, \\
0 &= ((\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2)\Psi_{\text{Dirac_with_potential},2} \\
&= (-E^2 + \mathbf{P}^2 + m^2 + iqF)\Psi_{\text{Dirac_with_potential},2}.
\end{aligned}$$

Since

$$iqF^* = (-iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma}, \quad iqF = (iq\mathbf{E} - q\mathbf{B}) \cdot \boldsymbol{\sigma},$$

the two equations are exactly the component forms in Eq. (203). Once the square is taken, the coupled Dirac system is separated into two Klein–Gordon-type equations with the same magnetic term and opposite electric terms, as required by the two first-order blocks.

6.8.12 Gauge Transform of the Dirac System

For the real gauge parameter $\lambda(t, \mathbf{r})$ defined above, the potential in Eq. (153) is

$$\Phi' = \Phi + \partial^*\lambda.$$

Because λ is real,

$$\partial\lambda = \Phi' - \Phi^*.$$

The field transformation defined in Eq. (204) is equivalently written as

$$\Psi_{\text{Dirac_with_potential}} = \exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}}.$$

The phase factor is a central scalar, so every commutator correction in Eq. (139) is zero. The ordinary two-term product rule is therefore obeyed by both derivative factors:

$$\begin{aligned}
\partial(\exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}}) &= \exp(iq\lambda)(iq\partial\lambda + \partial) \\
&\quad \Psi'_{\text{Dirac_with_potential}}, \\
\partial^*(\exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}}) &= \exp(iq\lambda)(iq\partial^*\lambda + \partial^*) \\
&\quad \Psi'_{\text{Dirac_with_potential}}.
\end{aligned}$$

The two first-order factors are consequently transformed as

$$\begin{aligned}
& (i\partial - q\Phi^*) \exp(iq\lambda) \\
& \Psi'_{\text{Dirac_with_potential},2} = \exp(iq\lambda)(i\partial - q\Phi'^*) \\
& \quad \Psi'_{\text{Dirac_with_potential},2}, \\
& (i\partial^* - q\Phi) \exp(iq\lambda) \\
& \Psi'_{\text{Dirac_with_potential},1} = \exp(iq\lambda)(i\partial^* - q\Phi') \\
& \quad \Psi'_{\text{Dirac_with_potential},1}.
\end{aligned}$$

The following relation is therefore satisfied by the pair-valued function:

$$\begin{aligned}
& \mathcal{D}(\Phi, \exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}}) \\
&= \exp(iq\lambda) \mathcal{D}(\Phi', \Psi'_{\text{Dirac_with_potential}}).
\end{aligned}$$

Since the left side vanishes for the original field, the following equation is satisfied by the transformed field:

$$\mathcal{D}(\Phi', \Psi'_{\text{Dirac_with_potential}}) = (0, 0).$$

The phase change of the Dirac field is exactly compensated by the shift of the potential. The same form of the coupled Dirac equation is retained for every value of the gauge parameter.

6.8.13 Matrix Packaging of the Dirac Pair

The intrinsic ordered-pair equations in the wave-mechanics section can be packaged by an auxiliary matrix function. This packaging translates those equations without introducing a new physical assumption.

Auxiliary block map. The block identity needed below is defined by

$$\mathbf{I} := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (260)$$

The two-input block-diagonal function needed below is defined by

$$\text{diag}(A, B) := \begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}. \quad (261)$$

A paravector $Z \in \mathbb{P}_{\mathbb{C}}$, including any member of $\mathbb{P}_{\mathbb{R}}$, is packaged into the two-by-two block map

$$\mathbf{W}(Z) := \begin{bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{bmatrix}. \quad (262)$$

For $a, b \in \mathbb{R}$, its complex linearity is

$$\mathbf{W}((a + ib)Z) = (a + ib) \mathbf{W}(Z). \quad (263)$$

Its square is

$$\mathbf{W}(Z)^2 = \text{diag}(\det(Z), \det(Z)) = \det(Z) \mathbf{I}. \quad (264)$$

The block map $\mathbf{W}(Z)$ is a two-by-two matrix over the paravector algebra. After each block is replaced by its Pauli matrix, it is a four-by-four complex matrix. Addition and multiplication by ordinary scalars are preserved, so the packaging map is linear.

General products are not preserved, so it is not an additional representation of the algebra. Equation (264) is obtained because its diagonal block products are reduced to the determinant in Eq. (58); the off-diagonal blocks are zero [12, 15].

If $\det(Z) \neq 0$, Eq. (264) shows that the powers of the block matrix are governed by the scalar equation $x^2 - \det(Z) = 0$. Its two roots are distinct; selecting one square root fixes their labels. An invertible change of basis can then put the block matrix into diagonal form; this property is called diagonalizability. If $\det(Z) = 0$ and $Z \neq 0$, then $\mathbf{W}(Z)^2 = 0$ while $\mathbf{W}(Z) \neq 0$. A nonzero matrix whose positive power is zero is called nilpotent; such a matrix cannot be put into diagonal form. For blocks containing differential operations or other quantities whose order cannot be exchanged, the unreduced statement is

$$\mathbf{W}(Z)^2 = \text{diag}(ZZ^{*\text{H}}, Z^{*\text{H}}Z).$$

A separate operator calculation is required before either ordered product can be called a determinant or both can be reduced to one scalar multiple of the identity.

For a real normalized paravector U with $\det(U) = 1$, Eq. (264) gives $\mathbf{W}(U)^2 = \mathbf{I}$. The two auxiliary projector matrices are therefore

$$\frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)). \quad (265)$$

No additional projector function is introduced by this optional block construction. Direct multiplication gives the complete algebraic properties:

$$\begin{aligned} \left(\frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U))\right)^2 &= \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)), \\ \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\frac{1}{2}(\mathbf{I} - \mathbf{W}(U)) &= 0, \\ \frac{1}{2}(\mathbf{I} + \mathbf{W}(U)) + \frac{1}{2}(\mathbf{I} - \mathbf{W}(U)) &= \mathbf{I}, \\ \mathbf{W}(U)\frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)) &= \pm \frac{1}{2}(\mathbf{I} \pm \mathbf{W}(U)). \end{aligned} \quad (266)$$

For any conformable two-entry column $\Psi_{\text{auxiliary}}$ with paravector entries, the normalized auxiliary condition is equivalently written as

$$\begin{aligned} (\mathbf{W}(U) - \mathbf{I})\Psi_{\text{auxiliary}} &= 0, \\ \Psi_{\text{auxiliary}} &= \frac{1}{2}(\mathbf{I} + \mathbf{W}(U))\Psi_{\text{auxiliary}}. \end{aligned} \quad (267)$$

This is only the +1 eigenvalue restriction of the auxiliary map. No particle-versus-antiparticle meaning follows from the two signs without additional physical structure.

Full-matrix and entrywise Hermitian operations. Two different levels of operation must be distinguished for a matrix whose entries are themselves paravectors. By separate application of the intrinsic Hermitian operation to each entry of $\mathbf{W}(Z)$, the

following matrix is obtained:

$$\begin{aligned} \text{entrywise H : } & \begin{bmatrix} 0 & Z \\ \text{adj}(Z) & 0 \end{bmatrix} \\ \mapsto & \begin{bmatrix} 0 & Z^{\text{H}} \\ Z^* & 0 \end{bmatrix} = \mathbf{W}(Z^{\text{H}}). \end{aligned}$$

The Hermitian operation on the complete matrix also exchanges the two off-diagonal positions. The following result is therefore obtained:

$$\mathbf{W}(Z)^{\text{H}} = \begin{bmatrix} 0 & Z^* \\ Z^{\text{H}} & 0 \end{bmatrix} = \mathbf{W}(Z^*).$$

Thus entrywise application of H is not the Hermitian operation of the complete block matrix. The same final matrix as the full-matrix Hermitian operation is obtained by entrywise intrinsic star:

$$\mathbf{W}(Z)^* = \mathbf{W}(Z^*).$$

In the entrywise operation, however, the block positions are not exchanged.

Free Dirac packaging. The following relations are obtained from complex linearity of the auxiliary function:

$$\begin{aligned} \mathbf{W}(i\partial) &= i \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} = i \mathbf{W}(\partial), \\ \mathbf{W}(i\partial)^2 &= -\text{diag}(\partial\partial^*, \partial^*\partial) \\ &= -\square \mathbf{I}. \end{aligned}$$

The two mass-sign factors consequently satisfy

$$\begin{aligned} (\mathbf{W}(i\partial) + m\mathbf{I})(\mathbf{W}(i\partial) - m\mathbf{I}) &= -(\square + m^2)\mathbf{I}, \\ (\mathbf{W}(i\partial) - m\mathbf{I})(\mathbf{W}(i\partial) + m\mathbf{I}) &= -(\square + m^2)\mathbf{I}. \end{aligned}$$

The free pair $\Psi_{\text{Dirac}} := (\Psi_{\text{Dirac},1}, \Psi_{\text{Dirac},2})$ can therefore be packaged as

$$(\mathbf{W}(i\partial) - m\mathbf{I}) \begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix} = 0.$$

Expansion of the two matrix rows recovers the intrinsic equations

$$\begin{aligned} i\partial\Psi_{\text{Dirac},2} &= m\Psi_{\text{Dirac},1}, \\ i\partial^*\Psi_{\text{Dirac},1} &= m\Psi_{\text{Dirac},2}. \end{aligned}$$

For the fixed projector used in the intrinsic derivation, the two paravectors are represented by the columns

$$\begin{aligned} \Psi_{\text{Dirac},1} & \text{ is represented by } \begin{bmatrix} a \\ b \end{bmatrix}, \\ \Psi_{\text{Dirac},2} & \text{ is represented by } \begin{bmatrix} c \\ d \end{bmatrix}. \end{aligned}$$

For a free plane wave, the coupled matrix equations are

$$(E - \mathbf{p} \cdot \boldsymbol{\sigma}) \begin{bmatrix} c \\ d \end{bmatrix} = m \begin{bmatrix} a \\ b \end{bmatrix}, \quad (E + \mathbf{p} \cdot \boldsymbol{\sigma}) \begin{bmatrix} a \\ b \end{bmatrix} = m \begin{bmatrix} c \\ d \end{bmatrix}.$$

Their four scalar entries are exactly the component equations displayed in the intrinsic derivation. If both columns of each unrestricted Pauli matrix are retained, two independent copies of this Dirac system are obtained.

Matrix form of the Klein–Gordon lift. The intrinsic lift in Eq. (257) is packaged by

$$\begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix} = \frac{1}{m} (\mathbf{W}(i\partial) + m\mathbf{I}) \begin{bmatrix} \psi_{\text{KG}} a \\ \psi_{\text{KG}} b \end{bmatrix}.$$

The following result is then obtained from the free factorization:

$$(\mathbf{W}(i\partial) - m\mathbf{I}) \begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix} = -\frac{1}{m} (\square + m^2) \psi_{\text{KG}} \begin{bmatrix} a \\ b \end{bmatrix} = 0.$$

For $a = \Pi(+)$ and $b = 0$, the result is

$$\begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix} = \begin{bmatrix} \psi_{\text{KG}} \Pi(+) \\ \frac{i}{m} (\partial^* \psi_{\text{KG}}) \Pi(+) \end{bmatrix}.$$

Potential-coupled packaging. The potential-coupled pair is denoted by

$$\Psi_{\text{Dirac_with_potential}} := (\Psi_{\text{Dirac_with_potential},1}, \Psi_{\text{Dirac_with_potential},2}).$$

Its field equation is represented by the block operator

$$\mathbf{D} := \mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I} = i\mathbf{W}(\partial) - q\mathbf{W}(\Phi^*) - m\mathbf{I}. \quad (268)$$

The following elementary block products are obtained:

$$\begin{aligned} \mathbf{W}(\partial)^2 &= \square \mathbf{I}, \\ \mathbf{W}(\partial) \mathbf{W}(\Phi^*) &= \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} \\ &= \text{diag}(\partial\Phi, \partial^*\Phi^*), \\ \mathbf{W}(\Phi^*) \mathbf{W}(\partial) &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix} \begin{bmatrix} 0 & \partial \\ \partial^* & 0 \end{bmatrix} \\ &= \text{diag}(\Phi^*\partial^*, \Phi\partial), \\ \mathbf{W}(\Phi^*)^2 &= \begin{bmatrix} 0 & \Phi^* \\ \Phi & 0 \end{bmatrix}^2 \\ &= \text{diag}(\Phi^*\Phi, \Phi\Phi^*) \\ &= \det(\Phi) \mathbf{I}. \end{aligned}$$

The complete square is expanded as

$$\begin{aligned} \mathbf{W}(i\partial - q\Phi^*)^2 &= -\mathbf{W}(\partial)^2 \\ &\quad - iq(\mathbf{W}(\partial) \mathbf{W}(\Phi^*) + \mathbf{W}(\Phi^*) \mathbf{W}(\partial)) \\ &\quad + q^2 \mathbf{W}(\Phi^*)^2 \\ &= -\text{diag}((\partial + iq\Phi^*)(\partial^* + iq\Phi), \\ &\quad (\partial^* + iq\Phi)(\partial + iq\Phi^*)). \end{aligned}$$

The coupled field equation is therefore packaged as

$$\mathbf{D} \begin{bmatrix} \Psi_{\text{Dirac_with_potential},1} \\ \Psi_{\text{Dirac_with_potential},2} \end{bmatrix} = 0.$$

The reversed-mass partner is

$$\mathbf{D}' := \mathbf{W}(i\partial - q\Phi^*) + m\mathbf{I}. \quad (269)$$

Its product with $\mathbf{D}$ is

$$\begin{aligned} \mathbf{D}'\mathbf{D} &= \mathbf{W}(i\partial - q\Phi^*)^2 - m^2\mathbf{I} \\ &= -\text{diag}((\partial + iq\Phi^*)(\partial^* + iq\Phi) + m^2, \\ &\quad (\partial^* + iq\Phi)(\partial + iq\Phi^*) + m^2). \end{aligned}$$

The two diagonal entries are precisely the two ordered second-order paravector equations obtained from

$$\mathcal{D}'(\Phi, \mathcal{D}(\Phi, \Psi_{\text{Dirac_with_potential}})).$$

Gauge transform in the matrix packaging. With

$$\begin{aligned} \Psi_{\text{Dirac_with_potential}} &= \exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}}, \\ \Phi' &= \Phi + \partial^* \lambda, \end{aligned}$$

the packaged gauge calculation is

$$\begin{aligned} 0 &= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}) \Psi_{\text{Dirac_with_potential}} \\ &= (\mathbf{W}(i\partial - q\Phi^*) - m\mathbf{I}) \exp(iq\lambda) \Psi'_{\text{Dirac_with_potential}} \\ &= \exp(iq\lambda) (\mathbf{W}(i\partial - q\Phi'^*) - m\mathbf{I}) \Psi'_{\text{Dirac_with_potential}}. \end{aligned}$$

The transformed matrix equation is therefore

$$(\mathbf{W}(i\partial - q\Phi'^*) - m\mathbf{I}) \Psi'_{\text{Dirac_with_potential}} = 0.$$

6.8.14 Dirac Equation Components

The previously stated definitions of ∂ , Φ , and F from Eqs. (135), (136), and (141) are retained. A generic complex-paravector block is written intrinsically as

$$\psi_{\text{block}} = \text{scalar}(\psi_{\text{block}}) + \text{vector}(\psi_{\text{block}}) \cdot \sigma.$$

A physical projected block is selected by the previously defined right factor and has the amplitude expansion

$$\psi_{\text{block}} = a\Pi(+) + ba(-), \quad \psi_{\text{block}} = \psi_{\text{block}} \Pi(+).$$

The corresponding scalar and vector extractions are

$$\text{scalar}(\psi_{\text{block}}) = \frac{a}{2}, \quad \text{vector}(\psi_{\text{block}}) = \frac{1}{2} \{b, -ib, a\}.$$

Thus the full-algebra component identities below can be restricted by direct substitution without changing any operator product.

The direct component action of ∂ is

Table 42. Components of $\partial\psi_{\text{block}}$.

scalar:	$\partial_t \text{scalar}(\psi_{\text{block}}) + \partial_{\mathbf{r}} \cdot \text{vector}(\psi_{\text{block}})$
vector:	$\partial_t \text{vector}(\psi_{\text{block}}) + \partial_{\mathbf{r}} \text{scalar}(\psi_{\text{block}}) + i(\partial_{\mathbf{r}} \times \text{vector}(\psi_{\text{block}}))$

The corresponding action of ∂^* is

Table 43. Components of $\partial^* \psi_{\text{block}}$.

scalar:	$\partial_t \text{scalar}(\psi_{\text{block}}) - \partial_{\mathbf{r}} \cdot \text{vector}(\psi_{\text{block}})$
vector:	$\partial_t \text{vector}(\psi_{\text{block}}) - \partial_{\mathbf{r}} \text{scalar}(\psi_{\text{block}}) - i(\partial_{\mathbf{r}} \times \text{vector}(\psi_{\text{block}}))$

The component action of the potential is

Table 44. Components of $\Phi \psi_{\text{block}}$.

scalar:	$V \text{scalar}(\psi_{\text{block}}) + \mathbf{A} \cdot \text{vector}(\psi_{\text{block}})$
vector:	$V \text{vector}(\psi_{\text{block}}) + \text{scalar}(\psi_{\text{block}}) \mathbf{A} + i(\mathbf{A} \times \text{vector}(\psi_{\text{block}}))$

The sigma-conjugate potential acts as

Table 45. Components of $\Phi^* \psi_{\text{block}}$.

scalar:	$V \text{scalar}(\psi_{\text{block}}) - \mathbf{A} \cdot \text{vector}(\psi_{\text{block}})$
vector:	$V \text{vector}(\psi_{\text{block}}) - \text{scalar}(\psi_{\text{block}}) \mathbf{A} - i(\mathbf{A} \times \text{vector}(\psi_{\text{block}}))$

The field-coupling action of iqF is

Table 46. Components of $iqF \psi_{\text{block}}$.

scalar:	$iq(\mathbf{F} \cdot \text{vector}(\psi_{\text{block}}))$
vector:	$iq \text{scalar}(\psi_{\text{block}}) \mathbf{F} - q(\mathbf{F} \times \text{vector}(\psi_{\text{block}}))$

Finally, since $F^* = -\mathbf{F}^* \cdot \boldsymbol{\sigma} = -(\mathbf{E} - i\mathbf{B}) \cdot \boldsymbol{\sigma}$, its field-coupling action is

Table 47. Components of $iqF^* \psi_{\text{block}}$.

scalar:	$-iq(\mathbf{F}^* \cdot \text{vector}(\psi_{\text{block}}))$
vector:	$-iq \text{scalar}(\psi_{\text{block}}) \mathbf{F}^* + q(\mathbf{F}^* \times \text{vector}(\psi_{\text{block}}))$

For the ordered Dirac pair, the two projected blocks are

$$\begin{aligned} \Psi_{\text{Dirac_with_potential},1} &= a\Pi(+) + ba(-), \\ \Psi_{\text{Dirac_with_potential},2} &= c\Pi(+) + da(-). \end{aligned}$$

Substitution of these expansions into the two coupled paravector equations gives the four scalar relations in Eq. (197). The scalar and vector tables above also give the two squared component equations in Eq. (203). Every component result is therefore obtained inside the projected algebra.

6.8.15 Low-Energy Spin-0 Limit

For the scalar theory with electromagnetic potential, the kinetic operators are

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A}.$$

The fast rest-mass oscillation is removed by writing

$$\psi_{\text{KG_with_potential}} = \exp(-imt) \psi_{\text{KG_slow}}.$$

Then

$$\begin{aligned} E^2 \psi_{\text{KG_with_potential}} &= \exp(-imt) (E + m)^2 \psi_{\text{KG_slow}}, \\ \mathbf{P}^2 \psi_{\text{KG_with_potential}} &= \exp(-imt) \mathbf{P}^2 \psi_{\text{KG_slow}}. \end{aligned}$$

The following result is obtained by substitution into the Klein-Gordon equation with the stated electromagnetic potential:

$$\begin{aligned} 0 &= (-E^2 + \mathbf{P}^2 + m^2) \psi_{\text{KG_with_potential}} \\ &= \exp(-imt) (-(E + m)^2 + \mathbf{P}^2 + m^2) \psi_{\text{KG_slow}}, \end{aligned}$$

hence

$$0 = (-E^2 - 2mE + \mathbf{P}^2) \psi_{\text{KG_slow}}.$$

For the free theory, or more generally when E and $\mathbf{P}^2$ commute on the fields being considered, the following two solutions are admitted by the quadratic equation:

$$E = -m \pm \sqrt{m^2 + \mathbf{P}^2}.$$

The root near zero is selected by the slow positive-energy assumption:

$$\begin{aligned} E \psi_{\text{KG_slow}} &= (-m + \sqrt{m^2 + \mathbf{P}^2}) \psi_{\text{KG_slow}} \\ &= \left(\frac{\mathbf{P}^2}{2m} - \frac{\mathbf{P}^4}{8m^3} + O(\mathbf{P}^6/m^5) \right) \psi_{\text{KG_slow}}. \end{aligned}$$

The other root begins at $-2m$ after the rest phase is removed and is incompatible with the assumed slowly varying positive-energy solution. For noncommuting or time-dependent potentials, commutators are also contained in the higher-order terms; the leading low-energy relation remains

$$E \psi_{\text{KG_slow}} \approx \frac{\mathbf{P}^2}{2m} \psi_{\text{KG_slow}}.$$

The Schrödinger equation is obtained by solving for $i\partial_t \psi_{\text{KG_slow}}$ [19]:

$$i\partial_t \psi_{\text{KG_slow}} \approx \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi_{\text{KG_slow}}.$$

If $\mathbf{A} = 0$, this reduces to

$$i\partial_t \psi_{\text{KG_slow}} \approx \left(-\frac{\partial_{\mathbf{r}}^2}{2m} + qV \right) \psi_{\text{KG_slow}}.$$

The scalar low-energy limit is therefore already encoded in the relativistic Klein-Gordon theory.

6.8.16 Low-Energy Spin- $\frac{1}{2}$ and Pauli Limit

The projected two-block field from Eq. (194) is used. Its two block equations are developed here under the assumption $m > 0$:

$$\begin{aligned} (i\partial - q\Phi^*) \Psi_{\text{Dirac_with_potential},2} - m \Psi_{\text{Dirac_with_potential},1} &= 0, \\ (i\partial^* - q\Phi) \Psi_{\text{Dirac_with_potential},1} - m \Psi_{\text{Dirac_with_potential},2} &= 0. \end{aligned}$$

With the kinetic operators from Eqs. (172) and (173),

$$E = i\partial_t - qV, \quad \mathbf{P} = -i\partial_{\mathbf{r}} - q\mathbf{A},$$

the system is

$$\begin{aligned} (E - \mathbf{P} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_with_potential},2} - m \Psi_{\text{Dirac_with_potential},1} &= 0, \\ (E + \mathbf{P} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_with_potential},1} - m \Psi_{\text{Dirac_with_potential},2} &= 0. \end{aligned}$$

The slow fields defined in Eq. (221) are related by

$$\Psi_{\text{Dirac_with_potential},n} = \exp(-imt) \Psi_{\text{Dirac_slow},n}.$$

Since

$$E \Psi_{\text{Dirac_with_potential},n} = \exp(-imt)(E + m) \Psi_{\text{Dirac_slow},n},$$

the system is reduced to

$$\begin{aligned} (E + m - \mathbf{P} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_slow},2} - m \Psi_{\text{Dirac_slow},1} &= 0, \\ (E + m + \mathbf{P} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_slow},1} - m \Psi_{\text{Dirac_slow},2} &= 0. \end{aligned}$$

Neither original block is small at rest. With the energy-basis combinations from Eq. (222),

$$\begin{aligned} \psi_{\text{Pauli_large}} &= \frac{\Psi_{\text{Dirac_slow},1} + \Psi_{\text{Dirac_slow},2}}{\sqrt{2}}, \\ \psi_{\text{Pauli_small}} &= \frac{\Psi_{\text{Dirac_slow},2} - \Psi_{\text{Dirac_slow},1}}{\sqrt{2}}. \end{aligned}$$

The exact system is obtained by addition and subtraction of the reduced equations:

$$\begin{aligned} E \psi_{\text{Pauli_large}} &= (\mathbf{P} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_small}}, \\ (E + 2m) \psi_{\text{Pauli_small}} &= (\mathbf{P} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}}. \end{aligned}$$

Thus $\psi_{\text{Pauli_large}}$ is the large positive-energy field, and $\psi_{\text{Pauli_small}}$ is suppressed by one power of $1/m$. Formally,

$$\begin{aligned} \psi_{\text{Pauli_small}} &= (E + 2m)^{-1} (\mathbf{P} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}}, \\ E \psi_{\text{Pauli_large}} &= (\mathbf{P} \cdot \boldsymbol{\sigma}) (E + 2m)^{-1} (\mathbf{P} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}} \\ &= \frac{(\mathbf{P} \cdot \boldsymbol{\sigma})^2}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}), \end{aligned}$$

where momenta small compared with m , energies small compared with $2m$, and external fields changing little across the length $1/m$ are assumed. In the units used here, $1/m$ is the reduced Compton length associated with the particle [17].

The magnetic term is obtained directly from the kinetic-momentum commutator:

$$\begin{aligned} \text{com}(P_m, P_n) &= iq(\partial_m A_n - \partial_n A_m), \\ \mathbf{P} \times \mathbf{P} &= iq \mathbf{B}, \\ (\mathbf{P} \cdot \boldsymbol{\sigma})^2 &= \mathbf{P}^2 + i(\mathbf{P} \times \mathbf{P}) \cdot \boldsymbol{\sigma} \\ &= \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}. \end{aligned}$$

Consequently,

$$E \psi_{\text{Pauli_large}} = \frac{\mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}),$$

and the Pauli equation is

$$\begin{aligned} i\partial_t \psi_{\text{Pauli_large}} &= \left(\frac{(-i\partial_{\mathbf{r}} - q\mathbf{A})^2}{2m} + qV \right) \psi_{\text{Pauli_large}} \\ &\quad - \frac{q}{2m} (\mathbf{B} \cdot \boldsymbol{\sigma}) \psi_{\text{Pauli_large}} + O(m^{-2}). \end{aligned}$$

At the next order, electric-field commutators generate a coupling between spin and motion in an electric field, conventionally called the spin-orbit term, and a local correction proportional to $\partial_{\mathbf{r}} \cdot \mathbf{E}$, conventionally called the Darwin term [17, 19]. Time-dependent corrections are also generated because E and $\mathbf{P}$ do not commute. All such terms must be retained consistently rather than discarded from the second-order equation for one original block. The leading spin coupling is therefore a controlled consequence of the Dirac system [9, 19].

The Pauli current in Eq. (229) is obtained from the same controlled reduction. Equation (227) gives the intrinsic current relation

$$\begin{aligned} \mathbf{j}(\Psi_{\text{Dirac_with_potential}}) &= 2 \text{vector} \left(\psi_{\text{Pauli_large}} \psi_{\text{Pauli_small}}^H \right. \\ &\quad \left. + \psi_{\text{Pauli_small}} \psi_{\text{Pauli_large}}^H \right), \\ \psi_{\text{Pauli_small}} &= \frac{\mathbf{P} \cdot \boldsymbol{\sigma}}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}). \end{aligned}$$

The sigma product and the differential product rule give each of the three current components without leaving the projected algebra:

$$\begin{aligned} j_n(\psi_{\text{Pauli_large}}) &= \frac{1}{m} \text{scalar} \left(\psi_{\text{Pauli_large}}^H P_n \psi_{\text{Pauli_large}} \right. \\ &\quad \left. + (P_n \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}} \right) \\ &\quad + \frac{1}{m} [\partial_{\mathbf{r}} \times \text{vector}(\psi_{\text{Pauli_large}} \psi_{\text{Pauli_large}}^H)]_n + O(m^{-2}), \end{aligned}$$

where $n \in \{1, 2, 3\}$. The amplitude content of this result is exposed by writing

$$\begin{aligned} \psi_{\text{Pauli_large}} &= a\Pi(+) + b a(-), \\ P_n \psi_{\text{Pauli_large}} &= (P_n a)\Pi(+) + (P_n b)a(-). \end{aligned}$$

The projector-ladder products then give

$$\begin{aligned} 2 \text{scalar} \left(\psi_{\text{Pauli_large}}^H P_n \psi_{\text{Pauli_large}} \right. \\ \left. + (P_n \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}} \right) \\ &= a^* P_n a + b^* P_n b \\ &\quad + (P_n a)^* a + (P_n b)^* b, \\ 2 \text{vector}(\psi_{\text{Pauli_large}} \psi_{\text{Pauli_large}}^H) \\ &= \left\{ \begin{aligned} &a^* b + b^* a, \\ &i(b^* a - a^* b), \\ &|a|^2 - |b|^2 \end{aligned} \right\}, \end{aligned}$$

The first relation supplies the three kinetic-momentum-flow components, and the second supplies the spin-density vector. The

first term is the kinetic-momentum flow. The second is the curl of the spin-density vector and is conventionally called the magnetization current [19]. This curl has zero divergence, so agreement of both the current and the continuity equation with the standard Pauli limit is obtained.

6.8.17 Spin Up and Spin Down

The fixed right carrier already uses the unprimed third-axis projector $\Pi(+)$. Where $\mathbf{B} \neq \mathbf{0}$, its distinct direction is therefore labeled with a prime. Its direction and projector are defined in immediate succession by

$$\mathbf{u}' := \frac{\mathbf{B}}{\|\mathbf{B}\|}, \quad (\mathbf{u}')^2 = 1, \quad (270)$$

$$\Pi'(\pm) := \frac{1}{2}(1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}). \quad (271)$$

The prime has the value-label meaning defined in Eq. (2). The signed action obtained from these definitions is

$$(\mathbf{u}' \cdot \boldsymbol{\sigma})\Pi'(\pm) = \pm\Pi'(\pm).$$

The Pauli state is decomposed by

$$\begin{aligned} \psi_{\text{Pauli_split}}(\pm) &= \Pi'(\pm)\psi_{\text{Pauli_large}}, \\ (\mathbf{u}' \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_split}}(\pm) &= \pm\psi_{\text{Pauli_split}}(\pm), \\ \psi_{\text{Pauli_large}} &= \psi_{\text{Pauli_split}}(+) + \psi_{\text{Pauli_split}}(-), \end{aligned}$$

Thus $\psi_{\text{Pauli_split}}(+)$ and $\psi_{\text{Pauli_split}}(-)$ are the spin-up and spin-down parts along the field axis. The fixed-projector condition is preserved under both left projectors:

$$\begin{aligned} \psi_{\text{Pauli_split}}(\pm)\Pi(+) &= \Pi'(\pm)(\psi_{\text{Pauli_large}}\Pi(+)) \\ &= \psi_{\text{Pauli_split}}(\pm). \end{aligned}$$

At leading order, the Pauli equation is written as

$$\begin{aligned} \mathbf{P} &= -i\partial_{\mathbf{r}} - q\mathbf{A}, \\ i\partial_t\psi_{\text{Pauli_large}} &= \left(\frac{\mathbf{P}^2}{2m} + qV\right)\psi_{\text{Pauli_large}} \\ &\quad - \frac{q}{2m}\|\mathbf{B}\|(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_large}} + O(m^{-2}). \end{aligned}$$

The following result is obtained by applying $\Pi'(\pm)$ to the time evolution:

$$\begin{aligned} i\partial_t\psi_{\text{Pauli_split}}(\pm) &= \left(\frac{\mathbf{P}^2}{2m} + qV\right)\psi_{\text{Pauli_split}}(\pm) \\ &\quad - \frac{q\|\mathbf{B}\|}{2m}(\mathbf{u}' \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_split}}(\pm) \\ &\quad + i(\partial_t\Pi'(\pm))\psi_{\text{Pauli_large}} \\ &\quad + \frac{1}{2m}\text{com}(\Pi'(\pm), \mathbf{P}^2) \\ &\quad \psi_{\text{Pauli_large}} + O(m^{-2}). \end{aligned}$$

Here qV and the local field-axis term commute at each position with $\Pi'(\pm)$; spatial variation is incorporated through the displayed kinetic commutator. The two local spin parts are coupled

by this term together with $\partial_t\Pi'(\pm)$. If $\mathbf{B}$ is uniform and static, then $\Pi'(\pm)$ is constant and the coupling term is zero, so the two spin parts evolve independently. In that case

$$(-\mathbf{B} \cdot \boldsymbol{\sigma})\psi_{\text{Pauli_split}}(\pm) = \mp\|\mathbf{B}\|\psi_{\text{Pauli_split}}(\pm),$$

and the resulting magnetic energy separation is

$$\begin{aligned} i\partial_t\psi_{\text{Pauli_split}}(\pm) &= \left(\frac{\mathbf{P}^2 \mp q\|\mathbf{B}\|}{2m} + qV\right)\psi_{\text{Pauli_split}}(\pm) \\ &\quad + O(m^{-2}). \end{aligned}$$

A preferred axis is therefore selected by the field direction, and opposite energy shifts are acquired by the two projector parts. This magnetic separation of spin energies is conventionally called the Zeeman effect [9, 19].

6.8.18 Free Energy and Momentum-Axis Checks

The pair-valued function $\mathcal{H}$ defined in Eq. (235) generates the free time evolution and is conventionally called the Hamiltonian, or energy operator [17, 19]. Every division by $E(\mathbf{p})$ in this subsection assumes $E(\mathbf{p}) > 0$; the massless zero-momentum case has no energy-sign split of this form. For a pair-valued left action with algebra-valued coefficients, the $\mathcal{H}$ operation is defined intrinsically by

$$\begin{aligned} \mathcal{O}(\Psi_{\text{Dirac}}) &:= (\mathcal{O}(1, 1)\Psi_{\text{Dirac}, 1} + \mathcal{O}(1, 2)\Psi_{\text{Dirac}, 2}, \\ &\quad \mathcal{O}(2, 1)\Psi_{\text{Dirac}, 1} + \mathcal{O}(2, 2)\Psi_{\text{Dirac}, 2}), \\ \mathcal{O}^H(\Psi_{\text{Dirac}}) &:= (\mathcal{O}(1, 1)^H\Psi_{\text{Dirac}, 1} + \mathcal{O}(2, 1)^H\Psi_{\text{Dirac}, 2}, \\ &\quad \mathcal{O}(1, 2)^H\Psi_{\text{Dirac}, 1} + \mathcal{O}(2, 2)^H\Psi_{\text{Dirac}, 2}). \end{aligned}$$

The exchanged input positions follow directly from reversal of product order inside the two-block pairing. For the free Hamiltonian, this operation gives

$$\begin{aligned} \mathcal{H}^H(\Psi_{\text{Dirac}}) &= (-\mathbf{p} \cdot \boldsymbol{\sigma})^H\Psi_{\text{Dirac}, 1} + m\Psi_{\text{Dirac}, 2}, \\ &\quad m\Psi_{\text{Dirac}, 1} + (\mathbf{p} \cdot \boldsymbol{\sigma})^H\Psi_{\text{Dirac}, 2} \\ &= \mathcal{H}(\Psi_{\text{Dirac}}). \end{aligned}$$

Here $(\mathbf{p} \cdot \boldsymbol{\sigma})^H = \mathbf{p} \cdot \boldsymbol{\sigma}$, and the real mass terms are interchanged without change. For any two projected pairs $\Psi_{\text{Dirac_left}}$ and $\Psi_{\text{Dirac_right}}$, the pairing identity is

$$\langle \Psi_{\text{Dirac_left}} | \mathcal{H}(\Psi_{\text{Dirac_right}}) \rangle = \langle \mathcal{H}(\Psi_{\text{Dirac_left}}) | \Psi_{\text{Dirac_right}} \rangle.$$

The Hamiltonian is therefore Hermitian under the pair-valued spatial pairing.

Direct composition of the ordered-pair action gives

$$\begin{aligned} \mathcal{H}(\mathcal{H}(\Psi_{\text{Dirac}})) &= \left(((\mathbf{p} \cdot \boldsymbol{\sigma})^2 + m^2)\Psi_{\text{Dirac}, 1} \right. \\ &\quad + (-m(\mathbf{p} \cdot \boldsymbol{\sigma}) + m(\mathbf{p} \cdot \boldsymbol{\sigma}))\Psi_{\text{Dirac}, 2}, \\ &\quad ((\mathbf{p} \cdot \boldsymbol{\sigma})^2 + m^2)\Psi_{\text{Dirac}, 2} \\ &\quad \left. + (-m(\mathbf{p} \cdot \boldsymbol{\sigma}) + m(\mathbf{p} \cdot \boldsymbol{\sigma}))\Psi_{\text{Dirac}, 1} \right) \\ &= ((m^2 + \mathbf{p}^2)\Psi_{\text{Dirac}, 1}, (m^2 + \mathbf{p}^2)\Psi_{\text{Dirac}, 2}) \\ &= E(\mathbf{p})^2\Psi_{\text{Dirac}}. \end{aligned}$$

The identity $(\mathbf{p} \cdot \boldsymbol{\sigma})^2 = \mathbf{p}^2$ follows directly from the sigma product, so every mixed term cancels inside the ordered pair.

The two energy-sign parts are defined by

$$\Psi_{\text{Dirac_energy}}(\pm) := \frac{1}{2} \left(\Psi_{\text{Dirac}} \pm \frac{\mathcal{H}(\Psi_{\text{Dirac}})}{E(\mathbf{p})} \right).$$

The preceding square then gives

$$\begin{aligned} \mathcal{H}(\Psi_{\text{Dirac_energy}}(\pm)) &= \frac{1}{2} \mathcal{H}(\Psi_{\text{Dirac}}) \\ &\quad \pm \frac{\mathcal{H}(\mathcal{H}(\Psi_{\text{Dirac}}))}{2E(\mathbf{p})} \\ &= \frac{1}{2} \mathcal{H}(\Psi_{\text{Dirac}}) \pm \frac{1}{2} E(\mathbf{p}) \Psi_{\text{Dirac}} \\ &= \pm E(\mathbf{p}) \Psi_{\text{Dirac_energy}}(\pm), \\ \Psi_{\text{Dirac_energy}}(+) + \Psi_{\text{Dirac_energy}}(-) &= \Psi_{\text{Dirac}}. \end{aligned}$$

Their orthogonality for $E(\mathbf{p}) > 0$ follows from Hermiticity by applying the Hamiltonian on either side of the pairing:

$$\begin{aligned} \langle \mathcal{H}(\Psi_{\text{Dirac_energy}}(+)) | \Psi_{\text{Dirac_energy}}(-) \rangle &= \langle \Psi_{\text{Dirac_energy}}(+) | \mathcal{H}(\Psi_{\text{Dirac_energy}}(-)) \rangle, \\ E(\mathbf{p}) \langle \Psi_{\text{Dirac_energy}}(+) | \Psi_{\text{Dirac_energy}}(-) \rangle &= -E(\mathbf{p}) \langle \Psi_{\text{Dirac_energy}}(+) | \Psi_{\text{Dirac_energy}}(-) \rangle, \\ 2E(\mathbf{p}) \langle \Psi_{\text{Dirac_energy}}(+) | \Psi_{\text{Dirac_energy}}(-) \rangle &= 0, \\ \langle \Psi_{\text{Dirac_energy}}(+) | \Psi_{\text{Dirac_energy}}(-) \rangle &= 0. \end{aligned}$$

A number that multiplies a nonzero state under an operator action is called an eigenvalue. The two real energy eigenvalues are $\pm E(\mathbf{p})$ [19].

Because the fixed right carrier remains $\Pi(+)$, the distinct momentum-axis direction and projector pair are primed. For $\mathbf{p} \neq \mathbf{0}$, they are defined in the required order by

$$\mathbf{u}' := \frac{\mathbf{p}}{\|\mathbf{p}\|}, \quad (272)$$

$$\begin{aligned} \Pi'(\pm) &:= \frac{1}{2} (1 \pm \mathbf{u}' \cdot \boldsymbol{\sigma}), \quad (273) \\ (\mathbf{p} \cdot \boldsymbol{\sigma}) \Pi'(\pm) &= \pm \|\mathbf{p}\| \Pi'(\pm). \end{aligned}$$

This is the construction in Eqs. (240) and (241), with the prime carrying the value-label meaning from Eq. (2). The projector commutes with $\mathbf{p} \cdot \boldsymbol{\sigma}$, and its left action commutes with the real mass factors in $\mathcal{H}$. The positive- and negative-energy axis parts are therefore defined separately:

$$\begin{aligned} \Psi_{\text{Dirac_positive_energy_axis}}(\pm) &:= \begin{pmatrix} \Pi'(\pm) \Psi_{\text{Dirac_energy},1}(+) \\ \Pi'(\pm) \Psi_{\text{Dirac_energy},2}(+) \end{pmatrix}, \\ \Psi_{\text{Dirac_negative_energy_axis}}(\pm) &:= \begin{pmatrix} \Pi'(\pm) \Psi_{\text{Dirac_energy},1}(-) \\ \Pi'(\pm) \Psi_{\text{Dirac_energy},2}(-) \end{pmatrix}. \end{aligned}$$

Their energy and momentum-axis equations are

$$\begin{aligned} \mathcal{H}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm)) &= E(\mathbf{p}) \Psi_{\text{Dirac_positive_energy_axis}}(\pm), \\ \mathcal{H}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm)) &= -E(\mathbf{p}) \Psi_{\text{Dirac_negative_energy_axis}}(\pm), \\ (\mathbf{p} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_positive_energy_axis},n}(\pm) &= \pm \|\mathbf{p}\| \Psi_{\text{Dirac_positive_energy_axis},n}(\pm), \\ (\mathbf{p} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_negative_energy_axis},n}(\pm) &= \pm \|\mathbf{p}\| \Psi_{\text{Dirac_negative_energy_axis},n}(\pm). \end{aligned}$$

Writing the pair-valued energy equations one block at a time gives

$$\begin{aligned} m \Psi_{\text{Dirac_positive_energy_axis},2}(\pm) &= (E(\mathbf{p}) \pm \|\mathbf{p}\|) \Psi_{\text{Dirac_positive_energy_axis},1}(\pm), \\ m \Psi_{\text{Dirac_positive_energy_axis},1}(\pm) &= (E(\mathbf{p}) \mp \|\mathbf{p}\|) \Psi_{\text{Dirac_positive_energy_axis},2}(\pm), \\ m \Psi_{\text{Dirac_negative_energy_axis},2}(\pm) &= (-E(\mathbf{p}) \pm \|\mathbf{p}\|) \Psi_{\text{Dirac_negative_energy_axis},1}(\pm), \\ m \Psi_{\text{Dirac_negative_energy_axis},1}(\pm) &= (-E(\mathbf{p}) \mp \|\mathbf{p}\|) \Psi_{\text{Dirac_negative_energy_axis},2}(\pm). \end{aligned} \quad (274)$$

Multiplication of either matching pair of scalar factors recovers

$$\begin{aligned} m^2 &= (E(\mathbf{p}) \pm \|\mathbf{p}\|)(E(\mathbf{p}) \mp \|\mathbf{p}\|) \\ &= (-E(\mathbf{p}) \pm \|\mathbf{p}\|)(-E(\mathbf{p}) \mp \|\mathbf{p}\|) \\ &= E(\mathbf{p})^2 - \mathbf{p}^2. \end{aligned}$$

For the positive-energy family, the two projected blocks have the intrinsic amplitude expansions

$$\begin{aligned} \Psi_{\text{Dirac_positive_energy_axis},1}(\pm) &= a\Pi(+) + b a(-), \\ \Psi_{\text{Dirac_positive_energy_axis},2}(\pm) &= c\Pi(+) + d a(-). \end{aligned}$$

Their scalar and vector extractions are

$$\begin{aligned} \text{scalar}(\Psi_{\text{Dirac_positive_energy_axis},1}(\pm)) &= \frac{a}{2}, \\ \text{vector}(\Psi_{\text{Dirac_positive_energy_axis},1}(\pm)) &= \frac{1}{2} \{b, -ib, a\}, \\ \text{scalar}(\Psi_{\text{Dirac_positive_energy_axis},2}(\pm)) &= \frac{c}{2}, \\ \text{vector}(\Psi_{\text{Dirac_positive_energy_axis},2}(\pm)) &= \frac{1}{2} \{d, -id, c\}. \end{aligned}$$

The negative-energy family has the same expansions with the distinct coefficients a', b', c', d' :

$$\begin{aligned} \Psi_{\text{Dirac_negative_energy_axis},1}(\pm) &= a'\Pi(+) + b' a(-), \\ \Psi_{\text{Dirac_negative_energy_axis},2}(\pm) &= c'\Pi(+) + d' a(-). \end{aligned}$$

The required left actions on the intrinsic carrier basis are

$$\begin{aligned} \sigma_1 \Pi(+) &= a(-), & \sigma_1 a(-) &= \Pi(+), \\ \sigma_2 \Pi(+) &= i a(-), & \sigma_2 a(-) &= -i \Pi(+), \\ \sigma_3 \Pi(+) &= \Pi(+), & \sigma_3 a(-) &= -a(-). \end{aligned}$$

Thus the momentum action on each block is obtained as

$$\begin{aligned}
& (\mathbf{p} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_positive_energy_axis},1}(\pm) \\
&= (p_3 a + (p_1 - ip_2)b) \Pi(+) \\
&\quad + ((p_1 + ip_2)a - p_3 b) a(-), \\
& (\mathbf{p} \cdot \boldsymbol{\sigma}) \Psi_{\text{Dirac_positive_energy_axis},2}(\pm) \\
&= (p_3 c + (p_1 - ip_2)d) \Pi(+) \\
&\quad + ((p_1 + ip_2)c - p_3 d) a(-).
\end{aligned}$$

Equality of the two independent carrier amplitudes gives the four positive-energy momentum-axis relations

$$\begin{aligned}
p_3 a + (p_1 - ip_2)b &= \pm \|\mathbf{p}\| a, & (p_1 + ip_2)a - p_3 b &= \pm \|\mathbf{p}\| b, \\
p_3 c + (p_1 - ip_2)d &= \pm \|\mathbf{p}\| c, & (p_1 + ip_2)c - p_3 d &= \pm \|\mathbf{p}\| d.
\end{aligned}$$

The negative-energy family satisfies the same four relations with a, b, c, d replaced by a', b', c', d' . The energy-axis equations in Eq. (274) give

$$\begin{aligned}
mc &= (E(\mathbf{p}) \pm \|\mathbf{p}\|) a, & md &= (E(\mathbf{p}) \pm \|\mathbf{p}\|) b, \\
ma &= (E(\mathbf{p}) \mp \|\mathbf{p}\|) c, & mb &= (E(\mathbf{p}) \mp \|\mathbf{p}\|) d, \\
mc' &= (-E(\mathbf{p}) \pm \|\mathbf{p}\|) a', & md' &= (-E(\mathbf{p}) \pm \|\mathbf{p}\|) b', \\
ma' &= (-E(\mathbf{p}) \mp \|\mathbf{p}\|) c', & mb' &= (-E(\mathbf{p}) \mp \|\mathbf{p}\|) d'.
\end{aligned}$$

These are the established free Dirac energy and momentum-axis spin relations, obtained entirely from the projector–ladder basis.

The directional current follows directly from the intrinsic current definition. The momentum-axis equation replaces each left action of $\mathbf{u}' \cdot \boldsymbol{\sigma}$ by its sign, and the projector–ladder expansion gives the two block densities. Consequently,

$$\begin{aligned}
& \mathbf{u}' \cdot \mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm)) \\
&= \pm (|c|^2 + |d|^2 - |a|^2 - |b|^2), \\
& \rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm)) \\
&= |a|^2 + |b|^2 + |c|^2 + |d|^2.
\end{aligned}$$

For the negative-energy family, the same component calculation gives

$$\begin{aligned}
& \mathbf{u}' \cdot \mathbf{j}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm)) \\
&= \pm (|c'|^2 + |d'|^2 - |a'|^2 - |b'|^2), \\
& \rho(\Psi_{\text{Dirac_negative_energy_axis}}(\pm)) \\
&= |a'|^2 + |b'|^2 + |c'|^2 + |d'|^2.
\end{aligned}$$

Use of the energy-axis amplitude relations then gives

$$\begin{aligned}
& \frac{\mathbf{u}' \cdot \mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))} \\
&= \pm \frac{|c|^2 + |d|^2 - |a|^2 - |b|^2}{|a|^2 + |b|^2 + |c|^2 + |d|^2} \\
&= \frac{\|\mathbf{p}\|}{E(\mathbf{p})}, \\
& \frac{\mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))} \\
&= \frac{\mathbf{p}}{E(\mathbf{p})}, \\
& \frac{\mathbf{j}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))} \\
&= -\frac{\mathbf{p}}{E(\mathbf{p})}.
\end{aligned}$$

Thus the current divided by the density is the expected relativistic velocity for each energy-sign mode.

Every displayed operation is applied on the left. The fixed-projector condition is therefore preserved directly:

$$\begin{aligned}
& \Psi_{\text{Dirac_positive_energy_axis},n}(\pm) \Pi(+) \\
&= \Psi_{\text{Dirac_positive_energy_axis},n}(\pm), \\
& \Psi_{\text{Dirac_negative_energy_axis},n}(\pm) \Pi(+) \\
&= \Psi_{\text{Dirac_negative_energy_axis},n}(\pm).
\end{aligned}$$

No second projected carrier is introduced. The spin-axis sign measured along the momentum direction is conventionally called helicity. At $\mathbf{p} = 0$, no momentum direction exists and helicity is therefore undefined; any separately chosen spin axis may still be used [17].

6.8.19 Matrix Translation of Dirac and Pauli Component Checks

The intrinsic fixed-projector amplitude expansion from Eq. (191) is translated into one selected two-entry column by

$$\begin{aligned}
\psi_{\text{block}} &= a \Pi(+) + b a(-) \quad \longleftrightarrow \quad \psi_{\text{block}} = \begin{bmatrix} a \\ b \end{bmatrix}, \\
\psi_{\text{block}}^H &\quad \longleftrightarrow \quad \psi_{\text{block}}^H.
\end{aligned}$$

The projector–ladder expansion remains the intrinsic definition.

Pauli current. The large and small projected paravectors $\psi_{\text{Pauli_large}}$ and $\psi_{\text{Pauli_small}}$ are represented by the matrix columns $\psi_{\text{Pauli_large}}$ and $\psi_{\text{Pauli_small}}$, respectively. The row–column form of the current reduction is

$$\begin{aligned}
\mathbf{j}(\Psi_{\text{Dirac_with_potential}}) &= \psi_{\text{Pauli_large}}^H \boldsymbol{\sigma} \psi_{\text{Pauli_small}} \\
&\quad + \psi_{\text{Pauli_small}}^H \boldsymbol{\sigma} \psi_{\text{Pauli_large}}, \\
\psi_{\text{Pauli_small}} &= \frac{\boldsymbol{\sigma} \cdot \mathbf{P}}{2m} \psi_{\text{Pauli_large}} + O(m^{-2}).
\end{aligned}$$

Application of the Pauli product in the written order gives

$$\begin{aligned} \mathbf{j}(\psi_{\text{Pauli_large}}) &= \frac{1}{2m} \left(\psi_{\text{Pauli_large}}^H (\mathbf{P} \psi_{\text{Pauli_large}}) \right. \\ &\quad \left. + (\mathbf{P} \psi_{\text{Pauli_large}})^H \psi_{\text{Pauli_large}} \right) \\ &\quad + \frac{\partial_{\mathbf{r}} \times (\psi_{\text{Pauli_large}}^H \boldsymbol{\sigma} \psi_{\text{Pauli_large}})}{2m} + O(m^{-2}). \end{aligned}$$

In the first numerator, the three entries are obtained by replacing $\mathbf{P}$ in turn by P_1, P_2, P_3 . The first term is the kinetic-momentum flow, and the second is the magnetization current. This is the ordinary row-column translation of the intrinsic formula in Eq. (229).

Free block Hamiltonian. The ordered-pair Hamiltonian from Eq. (235) is packaged as

$$\mathcal{H} := \begin{bmatrix} -\boldsymbol{\sigma} \cdot \mathbf{p} & m \\ m & \boldsymbol{\sigma} \cdot \mathbf{p} \end{bmatrix}, \quad \Psi_{\text{Dirac}} \longleftrightarrow \begin{bmatrix} \Psi_{\text{Dirac},1} \\ \Psi_{\text{Dirac},2} \end{bmatrix}.$$

Direct block multiplication gives

$$\begin{aligned} \mathcal{H}^2 &= \begin{bmatrix} (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 & -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} \\ -m\boldsymbol{\sigma} \cdot \mathbf{p} + m\boldsymbol{\sigma} \cdot \mathbf{p} & (\boldsymbol{\sigma} \cdot \mathbf{p})^2 + m^2 \end{bmatrix} \\ &= (m^2 + \mathbf{p}^2)\mathbf{I} = E(\mathbf{p})^2\mathbf{I}. \end{aligned}$$

Because the momentum components and mass are real and each sigma generator is Hermitian, the block operation satisfies

$$\mathcal{H}^H = \mathcal{H}.$$

The energy-sign projection is consequently checked by

$$\begin{aligned} \mathcal{H}\Psi_{\text{Dirac_energy}}(\pm) &= \frac{1}{2} \left(\mathcal{H}\Psi_{\text{Dirac}} \pm \frac{\mathcal{H}^2\Psi_{\text{Dirac}}}{E(\mathbf{p})} \right) \\ &= \frac{1}{2} (\mathcal{H}\Psi_{\text{Dirac}} \pm E(\mathbf{p})\Psi_{\text{Dirac}}) \\ &= \pm E(\mathbf{p})\Psi_{\text{Dirac_energy}}(\pm), \\ \Psi_{\text{Dirac}} &= \Psi_{\text{Dirac_energy}}(+) \\ &\quad + \Psi_{\text{Dirac_energy}}(-). \end{aligned}$$

These lines are the block-array translation of Eqs. (236) and (239).

Energy and momentum-axis columns. The positive- and negative-energy families are kept distinct. Within either family, the two momentum-axis columns are written as

$$\begin{aligned} \Psi_{\text{Dirac_positive_energy_axis},1}(\pm) &\longleftrightarrow \begin{bmatrix} a \\ b \end{bmatrix}, \\ \Psi_{\text{Dirac_positive_energy_axis},2}(\pm) &\longleftrightarrow \begin{bmatrix} c \\ d \end{bmatrix}. \end{aligned}$$

The negative-energy family is represented in a separate calculation by

$$\begin{aligned} \Psi_{\text{Dirac_negative_energy_axis},1}(\pm) &\longleftrightarrow \begin{bmatrix} a \\ b \end{bmatrix}, \\ \Psi_{\text{Dirac_negative_energy_axis},2}(\pm) &\longleftrightarrow \begin{bmatrix} c \\ d \end{bmatrix}. \end{aligned}$$

For the formulas below, $s = +1$ when the literal axis argument is $+$, and $s = -1$ when that argument is $-$. The momentum-axis equation is therefore

$$\begin{aligned} p_3a + (p_1 - ip_2)b &= s\|\mathbf{p}\|a, & (p_1 + ip_2)a - p_3b &= s\|\mathbf{p}\|b, \\ p_3c + (p_1 - ip_2)d &= s\|\mathbf{p}\|c, & (p_1 + ip_2)c - p_3d &= s\|\mathbf{p}\|d. \end{aligned}$$

The energy relations obtained from Eq. (274) are, for the positive-energy family,

$$\begin{aligned} mc &= (E(\mathbf{p}) + s\|\mathbf{p}\|)a, & md &= (E(\mathbf{p}) + s\|\mathbf{p}\|)b, \\ ma &= (E(\mathbf{p}) - s\|\mathbf{p}\|)c, & mb &= (E(\mathbf{p}) - s\|\mathbf{p}\|)d, \end{aligned}$$

and, for the negative-energy family,

$$\begin{aligned} mc &= (-E(\mathbf{p}) + s\|\mathbf{p}\|)a, & md &= (-E(\mathbf{p}) + s\|\mathbf{p}\|)b, \\ ma &= (-E(\mathbf{p}) - s\|\mathbf{p}\|)c, & mb &= (-E(\mathbf{p}) - s\|\mathbf{p}\|)d. \end{aligned}$$

For each family separately, a, b, c, d denote that family's amplitudes. The positive-energy density and momentum-axis current obey

$$\begin{aligned} \rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm)) &= |a|^2 + |b|^2 + |c|^2 + |d|^2, \\ \mathbf{u} \cdot \mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm)) &= s(|c|^2 + |d|^2 - |a|^2 - |b|^2). \end{aligned}$$

The negative-energy family has the same component form with its own amplitudes:

$$\begin{aligned} \rho(\Psi_{\text{Dirac_negative_energy_axis}}(\pm)) &= |a|^2 + |b|^2 + |c|^2 + |d|^2, \\ \mathbf{u} \cdot \mathbf{j}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm)) &= s(|c|^2 + |d|^2 - |a|^2 - |b|^2). \end{aligned}$$

Their normalized currents distinguish the two energy families:

$$\begin{aligned} \frac{\mathbf{u} \cdot \mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))} &= \frac{\|\mathbf{p}\|}{E(\mathbf{p})}, \\ \frac{\mathbf{u} \cdot \mathbf{j}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))} &= -\frac{\|\mathbf{p}\|}{E(\mathbf{p})}, \\ \frac{\mathbf{j}(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_positive_energy_axis}}(\pm))} &= \frac{\mathbf{p}}{E(\mathbf{p})}, \\ \frac{\mathbf{j}(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))}{\rho(\Psi_{\text{Dirac_negative_energy_axis}}(\pm))} &= -\frac{\mathbf{p}}{E(\mathbf{p})}. \end{aligned}$$

Thus the component equations and current agree exactly with the intrinsic projector-ladder derivations above.

7 Conclusions

The algebra used here is the real Pauli algebra, written as one complex scalar plus one complex three-vector. Its economy is supplied by a structural fact rather than by a new physical postulate: the complex unit is supplied by the central pseudoscalar, dot and cross products are combined by the sigma product, and the determinant is obtained by multiplication by the adjugate. On real event paravectors, that determinant is the Minkowski interval.

Within this framework, spatial rotations, Lorentz boosts, four-velocity and four-momentum, Maxwell's equations, the Lorentz

force, scalar relativistic waves, the two first-order Dirac equations whose product gives the second-order equation, and the leading Pauli equation are derived in one notation. The distinctions between object types are essential. Events are transformed by the determinant-preserving two-sided event law, the electromagnetic field is transformed by an inverse-pair similarity, and one Dirac field is restricted to a fixed right-projector space. Once those distinctions are enforced, agreement between the compact notation and the established component equations is obtained.

Several complementary structures are also exposed by the reformulation. The four Maxwell equations are obtained from the real and imaginary parts of ∂F ; the two established spin-field terms F and F^* are obtained from the two ordered momentum products; and elementary functions, Pauli spin parts, and free spin-along-momentum parts are organized by the same axis projectors. These are connections among established equations, not evidence for an additional interaction.

In particular, no independent measured components are added by the paravector wave-function forms. Each Dirac or Pauli block is reduced by the fixed-projector condition to two complex amplitudes. The scalar $2 \text{ scalar}(\psi_{\text{block}}^H \psi_{\text{block}})$ is their nonnegative squared magnitude, while the opposite-order product $\psi_{\text{block}} \psi_{\text{block}}^H$ retains their spin-direction information. The two squared field terms give the two established Dirac spin couplings. Other non-scalar parts of two-sided products remain algebraic until a separately derived measurement rule assigns them a physical meaning.

7.1 Scope of the Established Results

The results in this paper are established under the assumptions of flat Minkowski space, smooth fields, and sets of fields on which the displayed differential operators and square roots are defined. The electromagnetic discussion is classical. The wave equations describe one particle and do not include particle creation or annihilation. An antiparticle interpretation of a negative-frequency solution requires the separate quantum-field construction and is not made here. Likewise, $FF^H/2$ is the energy-density and flux paravector seen in one frame; the complete electromagnetic energy-momentum description is the linear function $T(a) = FaF^H/2$ on real paravectors a .

The displayed components may be changed by a nonlinear coordinate map or an algebraic change of basis in flat space, but physical curvature is not thereby created. In a curved extension, a local basis would depend on position, a rule would be needed to compare bases at neighboring events, and the failure of that comparison to return unchanged around a small loop would measure curvature. Such a comparison rule is conventionally called a connection. Field equations would also have to retain their form under the corresponding local basis changes [15]. None of that additional structure is derived here. Complex space-time, unseen-matter proposals, and claims about the universe as a whole are also outside the established results.

These and other uncompleted directions are recorded in the following section without being treated as results.

8 Future Work

The developed results constitute a reformulation of established physics in space-time without gravitational curvature. Extensions not yet constructed are separated from points requiring greater mathematical precision. A proposed extension is marked **TODO**. A gap requiring resolution before physical claims can be supported is marked **FIXME**. No premise for the established results is taken from this section.

8.1 TODO: Local Frames and Gravitation

One fixed sigma basis is used throughout space and time in the present calculations. A gravitational extension would instead need a position-dependent local basis and a rule for comparing algebra elements defined at neighboring positions. That comparison rule would have to distinguish an ordinary change of description from a physical gravitational field. In particular, a basis written everywhere as a single transform of a fixed basis is locally only a change of description. A failure to return an algebra element unchanged after comparison around a sufficiently small closed path is called curvature [15]; it is not produced by that basis change alone.

A TODO question is therefore whether one comparison rule can be defined and used consistently in every equation. Such a rule is conventionally called a connection [15]. Curvature would then have to be calculated by the closed-path comparison just defined, rather than by a change of basis alone. If two successive comparisons depend on their order, they are said not to commute. The interval, freely falling motion, Maxwell equation, Dirac system, and energy-momentum source rule would all have to be rebuilt with that same comparison rule. Agreement with the constant-basis formulas would be a required check.

It remains to be determined whether translational motion can be combined with rotation-rate and boost-rate cross-position terms and consistently interpreted as a velocity. Multiplication by mass and construction of a transformed interval would follow only if that interpretation were proved. Those physical identifications are not justified by the moving-basis generator derived in Sec. 1.11 alone. In particular, surrounding each basis element by a transform and its inverse is not the same operation as the two-sided event transformation used for a Lorentz boost, and a complex velocity-like expression is generally produced by the proposed boost-rate term. It must therefore be determined whether a real event law and an observer's measured time and distance can be defined, whether an ordinary derivative can be corrected for the changing local basis so that the corrected derivative transforms with the field it acts on, and whether real measured velocity, energy, momentum, and interval can then be derived. The corrected derivative is conventionally called a covariant derivative [15]. Recovery of the inertial formulas when the basis-change rate is constant or zero would be required.

Space-time curvature is also not yet supplied by a plane-curve analogy. A numerator obtained by differentiating a coordinate quotient is only one piece of a curve formula. The remaining TODO questions are whether the curve parameter has a fixed physical meaning, whether the vector pointing along the curve,

conventionally called its tangent, has a fixed length, how directions are compared at neighboring positions, and whether the proposed result works for motion in more than one direction. Curvature cannot be inferred from that numerator alone.

8.2 TODO: Additional Field-Value Transformations

One phase freedom and one potential paravector are used in the electromagnetic coupling. Here an internal field value means a label carried by the field that is changed without changing the space-time event at which the field is evaluated. Other established field theories use several coupled internal values and transformations for which applying the first and then the second can differ from applying them in the opposite order [15]. The TODO question is whether such transformations can be added while keeping the space-time product and the internal action separate.

The allowed internal field values, their transformation rule, the corresponding potentials, and the quantities preserved by the resulting equations would have to be defined before any field equation is written. The products applied to space-time paravectors and those applied only to the internal field values would also have to be shown explicitly. It must be determined whether the existing algebra can express both actions without ambiguity. Merely replacing the electromagnetic potential by a larger array would not answer that question.

8.3 TODO: Many-Particle and Particle-Number-Changing Fields

Classical fields and single-particle wave functions are described by the wave equations in this manuscript. The term “first quantized” refers to this use of an ordinary wave function while the number of particles remains fixed. In particular, a negative-frequency solution is not by itself an antiparticle [17]. A future theory would require field operators whose action can change the number of particles, operations that add or remove one particle, a transformation that reverses charge, and a no-particle reference state called the vacuum. The TODO question is whether all of these objects can be represented without obscuring the present algebra, and whether the particle–antiparticle interpretation can then be derived rather than assumed [13].

Further TODO questions concern how the state is changed when two identical particles are exchanged, how a state is represented when it cannot be written as one independent state for each particle, and how the remaining state is described after some particles are left unmeasured. The condition that the state cannot be separated is conventionally called entanglement, and the last partial description is conventionally called a reduced state. The rule for extracting quantities measured near one position would have to be specified in the same development [13]. The fixed projected two-amplitude space used for one Dirac field is a useful starting point, but these many-particle constructions are not supplied by it alone. Likewise, the two auxiliary projectors in Eq. (265) only select the two eigenvalues of the block map defined in Section 6.8.13. Their algebraic signs cannot be identified with particle and antiparticle sectors until charge reversal,

particle-number change, and the physical field space have been constructed.

The indefinite Klein–Gordon pairing and the absence of a positive local one-particle position density are also not repaired by a change of algebraic packaging. In a theory with particle-number-changing field operators, the origin of a positive squared state length, an operation that measures charge, and the distinction between particles and antiparticles must be shown. The one-particle formulas must then be recovered when particle-number changes can be neglected.

8.4 TODO: Electromagnetic Response of Materials

Empty-space fields and prescribed sources are treated in the electromagnetic section. A material requires an additional rule that converts the electromagnetic field into the response of that material [16]. Such a rule is conventionally called a constitutive map. Its outputs can include electric dipole moment per unit volume, called polarization; magnetic dipole moment per unit volume, called magnetization; and electric current. An electric dipole moment records the amount and direction of separated positive and negative charge. A magnetic dipole moment records the strength and direction of a small magnet or current loop. Direction dependence is called anisotropy, frequency dependence is called dispersion, current produced by an electric field is described by conductivity, and conversion of field energy into heat or other unavailable forms is called dissipation.

The first TODO question is whether simple media whose response respects addition and multiplication by an ordinary number can be written clearly. Such a response rule is called linear. Direction-dependent and time-dependent responses can then be tested. The distinction between a field paravector and an operation acting on that field must be preserved, and the energy balance must be derived rather than assumed. It must then be determined which material laws fit into one paravector product and which require a separately defined map.

8.5 FIXME: Operator Foundations

Differential, inverse, and square-root operators are used in several factorizations in the wave-mechanics sections. Their algebraic forms are correct on the stated test fields, for which all displayed derivatives exist, but the allowed set of fields must be specified completely. That allowed set is the operator’s domain. Conditions imposed at the edge of the spatial region or at infinite distance are its boundary conditions [17, 19]. The conditions under which integration by parts—moving a derivative from one factor in an integral to the other factor—adds no boundary term, an inverse exists, and an energy operator has real measured values are determined by those choices.

A consistent definition of the square-root frequency operator is also required for every plane-wave term. The nonnegative square root must be selected explicitly, and the spatially constant term with $\mathbf{k} = \mathbf{0}$ must be treated separately when the inverse is used. Enough derivatives and integrals must exist for both written operation orders in every use of the commutator function.

These conditions should eventually be stated once in a common operator-domain section and then cited wherever they are used in a factorization or low-energy expansion.

Likewise, the determinant of a differential operator minus a proposed measured value is not defined merely because the paravector determinant is defined. A finite determinant whose zeros select possible values may be used only after the equation has been reduced to finitely many components, for example one plane wave at a time, a finite set of basis functions, or a finite numerical grid. Merely restricting the spatial region does not make the operator finite-dimensional. At the operator level, only the ordered factorizations and explicitly proved polynomial identities are presently justified. It remains to be determined under what stated conditions an infinite-dimensional determinant can be defined, if one is needed at all.

For scalar relativistic waves, the positive-energy solution space for each permitted background that does not change with time must be constructed, and the measurement operators that preserve it must be determined. General time-dependent backgrounds can mix the positive- and negative-frequency parts, and a position probability cannot be obtained merely by declaring the local Klein–Gordon density to be a position-probability density. For Dirac and Pauli wave-function spaces, the shared operator domains must be specified. It must then be proved that moving an operator from one side of the bracket to the other gives the displayed H operation, that the energy-selecting operators preserve their selected states, and that the calculated measurement averages are real.

8.6 FIXME: Bounds on Measurement Spread

A spin-measurement uncertainty bound cannot be inferred directly from the axis commutator function and a determinant inequality. For a measured quantity, its centered operator is obtained by subtracting its average from the operator. Its variance is the average square of that difference, and its uncertainty is the non-negative square root of the variance [11, 19]. The normalized projected-wave bracket in Eq. (212) must be used to define those averages. A shared set of fields on which both operator orders and their H operations exist must also be required. The standard lower bound built from the commutator function, and the stronger bound that also retains the anticommutator function, should then be proved in the manuscript’s notation. Spin-axis measurement operators should be checked as a worked example rather than treated as the proof.

8.7 TODO: Higher-Order Pauli Reduction

The leading kinetic and magnetic spin terms are proved by the present Dirac reduction. The next TODO question is whether the next-order terms can be derived with the same sign and charge conventions. These include the coupling between spin and motion in an electric field, called the spin–orbit term, and the local correction produced by spatial variation of the electric field, often called the contact or Darwin term [17, 19]. Corrections required for time-dependent potentials must also be derived. The density and current must be expanded to the same order as the energy

operator; retaining higher-order energy terms while leaving the measurement rule at leading order would be inconsistent. These terms are established consequences of the Dirac equation, but their derivation in this notation remains future work.

8.8 TODO: Classify Companion Components

Paired real and imaginary components and paired scalar and vector components are repeatedly produced by the algebra. Some of these are familiar physical laws: the magnetic-divergence and induction equations, for example, are encoded by the imaginary parts of the Maxwell equation. Other components, such as the imaginary part of the force product, are valid algebraic companions without a separate physical interpretation established here.

Two cases are already identified with established physics. A projected block’s normalized opposite-order product is Hermitian and unchanged when squared, so it represents one fully specified two-amplitude state [19]. The remaining F and F^* terms in the two ordered products are the standard two Dirac spin couplings [17]. For the remaining nonscalar channels of unrestricted or differently ordered products, a separate derivation as a current, spin-axis quantity, or measurement result is still required before physical meaning is assigned. For a fixed-projector state and a left-acting operator that preserves that space, the product $\psi_{\text{block}}^H \mathcal{O} \psi_{\text{block}}$ is already fixed by its one complex scalar coefficient and contains no additional independent channels.

The following exact identity may also be studied:

$$\square(F^2) = \partial^* \partial(F^2).$$

Because F^2 is a complex scalar, this equality is an exact use of the defined wave operator, but it is not an independent field equation. Any useful source expansion must be derived from Maxwell’s equation with the ordered product rule of Eq. (139). A TODO question is whether the resulting scalar identity describes a balance unchanged by the stated transformations or merely repeats information already contained in Maxwell’s equation. It must not be presented as an independent law of field evolution without such a derivation.

Each companion formula should be placed by a systematic classification into one of three categories: an alternate form of an established law, redundant information implied by another displayed equation, or an algebraic channel whose physical meaning is presently unassigned. The operations that return the original input when applied twice, the order functions, $\Pi(\pm)$ selection functions for a stated unit direction $\mathbf{u}$, and field products should all be included in that classification. No unassigned component should be advertised as a new interaction merely because it is made available by the algebra. Maps that preserve the defined addition and multiplication, together with the selection, factorization, exponential, and logarithmic constructions, should be separated into those proved directly from the algebra and those that depend on an optional translation collected in an appendix.

8.9 TODO: Complex Space-Time and Imaginary Coordinates

A complex paravector $Z \in \mathbb{P}_{\mathbb{C}}$ is already permitted by the generic algebra: $Z = X + iY$, with $X, Y \in \mathbb{P}_{\mathbb{R}}$. Treating such an element as a physical event, rather than as an algebraic field value, would be a new hypothesis. The following self-contained consistency identity is obtained from the existing determinant and adjugate definitions:

$$\det(Z) = \det(X) - \det(Y) + i(X \operatorname{adj}(Y) + Y \operatorname{adj}(X)).$$

The quantity in parentheses is a scalar, but the determinant of Z is generally complex. No interval interpretation is fixed by this identity. In a proposed complex-event theory, it would have to be stated which, if any, real quantity is measured as an interval and why that choice is preserved by the allowed transformations.

The scalar part of Y could be described as an imaginary-time coordinate, and its vector part could be described as imaginary space. Those descriptions alone carry no physical content. A metric is a stated rule that assigns an interval to a coordinate difference [6]. The TODO question is whether a real interval rule can be selected for a complex paravector and preserved by the allowed transformations. Taking the real part, absolute magnitude, or squared absolute magnitude of a complex determinant would give different interval rules, so none can be selected by terminology alone. The ordinary real interval would have to be recovered when the imaginary part vanishes.

Only after that choice is justified could the time measured along an observer's path, the possible time order of events connected by a signal, velocity, momentum, or force be defined for a complex event. For real space-time, the zero-interval directions form the light cone and separate event differences that can be joined by a signal no faster than light from those that cannot [5, 6]. A TODO question is whether that same measured boundary is recovered to experimental accuracy. Complex mass, unobserved paths, or new force channels should not be introduced before these more basic questions are resolved.

Faster-than-light propagation is not implied by complex coordinates. A propagation speed can be assigned only after field equations, the surfaces along which disturbances propagate, observers, and measured time and distance have been defined. If the measured projection remains the real paravector X , the ordinary light cone may remain unchanged. If Y changes how disturbances propagate, the new boundary between permitted and forbidden signal directions and the possibility of controllable faster-than-light signaling would have to be derived explicitly.

An algebraic relationship between the real and imaginary parts is shown by the mixed terms in $\det(Z)$, but a weak physical interaction is not established. Such an interaction would require an equation of evolution and a source law. Only then could constants to be measured experimentally be identified. TODO questions could then be asked about a real-imaginary mixing strength, a mass or length scale that reduces the imaginary part, or a relative propagation or decay scale. None of these constants is predicted by the algebra alone; each would have to be connected to a definite measured quantity and constrained by experiment.

Allowing complex coefficients in the algebra cannot by itself be tested experimentally because it is a mathematical extension. A proposed physical interpretation could be rejected if measured intervals or energies became unavoidably complex, probabilities failed to remain real and nonnegative, time evolution failed to preserve normalization, source-free solutions grew without bound, or predicted departures from the observed light cone and ordinary Lorentz behavior exceeded experimental limits. The final TODO question is whether any nontrivial complex-event evolution law can pass these tests or whether every consistently measured result reduces to the real-paravector theory already developed.

8.10 FIXME: Proposed Connections to Large-Scale Observations

No derivation is established here for energy assigned to otherwise empty space, for the observed speeding-up of large-scale expansion, or for the additional gravitational effects commonly attributed to unseen matter [18]. These topics are collectively described here as dark-sector questions. A contribution to gravity, a tendency to gather with matter, or a measured energy is not implied by the existence of imaginary algebra components.

A consistent local gravitational equation and a conserved physical source must first be supplied by any such model. Ordinary local gravity must then be recovered. The TODO question is whether quantitative predictions can be made for the changing distance between far-separated galaxies, the bending of light by gravity, the formation of large matter concentrations, and the motion of matter [18]. A nearly constant contribution spread over large distances and a contribution that gathers into dense regions are physically different requirements and must not be inferred from the same algebraic term without separate derivations.

8.11 TODO: Limits of the Notation-Light Form

The compact notation is most effective when the relevant scalar and vector data are carried by one algebra element. In some extensions, maps with several independent inputs, information that is valid only on overlapping local regions, or conditions on the whole spatial region that cannot be recovered from one local formula are instead required. That mathematical structure is not removed by avoidance of indices; only its recorded form is changed.

The conditions under which a paravector is sufficient, a map that preserves addition and multiplication by ordinary numbers is required, or several-input or whole-region data must be introduced should be stated explicitly. Any new notation should be justified by a construction that cannot be expressed clearly with the existing functions and operators. The raw-text character of the framework is thereby preserved while notational economy is prevented from being mistaken for a reduction in physical content.

Table 38. Components of $\psi_{\text{block}}^{*\text{H}} X \psi_{\text{block}}$.

scalar:	$\text{scalar}(X) (\text{scalar}(\psi_{\text{block}})^2 - \text{vector}(\psi_{\text{block}}) \cdot \text{vector}(\psi_{\text{block}}))$
vector:	$(\text{scalar}(\psi_{\text{block}})^2 + \text{vector}(\psi_{\text{block}}) \cdot \text{vector}(\psi_{\text{block}})) \text{vector}(X)$ $- 2(\text{vector}(X) \cdot \text{vector}(\psi_{\text{block}})) \text{vector}(\psi_{\text{block}}) + 2i \text{scalar}(\psi_{\text{block}}) \text{vector}(X) \times \text{vector}(\psi_{\text{block}})$

Table 39. Components of $\psi_{\text{block}}^* S \psi_{\text{block}}$.

scalar:	$S (\text{scalar}(\psi_{\text{block}}) ^2$ $- \text{vector}(\psi_{\text{block}})^* \cdot \text{vector}(\psi_{\text{block}}))$
vector:	$S (\text{scalar}(\psi_{\text{block}})^* \text{vector}(\psi_{\text{block}})$ $- \text{scalar}(\psi_{\text{block}}) \text{vector}(\psi_{\text{block}})^*$ $- i \text{vector}(\psi_{\text{block}})^* \times \text{vector}(\psi_{\text{block}}))$

Table 40. Components of $\psi_{\text{block}}^{\text{H}} S \psi_{\text{block}}$.

scalar:	$S (\text{scalar}(\psi_{\text{block}}) ^2$ $+ \text{vector}(\psi_{\text{block}})^* \cdot \text{vector}(\psi_{\text{block}}))$
vector:	$S (\text{scalar}(\psi_{\text{block}})^* \text{vector}(\psi_{\text{block}}) + \text{scalar}(\psi_{\text{block}}) \text{vector}(\psi_{\text{block}})^*$ $+ i \text{vector}(\psi_{\text{block}})^* \times \text{vector}(\psi_{\text{block}}))$

Table 41. Components of $\psi_{\text{block}}^{*\text{H}} S \psi_{\text{block}}$.

scalar:	$S (\text{scalar}(\psi_{\text{block}})^2 - \text{vector}(\psi_{\text{block}}) \cdot \text{vector}(\psi_{\text{block}}))$
vector:	0

9 Verification and Formula Status

Exact algebraic identities are separated in Table 48 from physical statements for which a stated domain or approximation is required. In this section, “verified” means that the intrinsic algebra was checked symbolically under the stated assumptions; the optional bridge in Appendix A was checked independently. It does not mean empirical confirmation and does not settle questions about infinite-domain differential operators. The accompanying symbolic report is generated by `tools/verify_sta_results.py`.

Table 48. Status of the principal constructions.

Statement	Standard correspondence	Status	Qualification
Eqs. (7) and (42)	Real Pauli algebra and three-dimensional real Clifford algebra [12, 15]	verified	Associative scalar–vector form with eight real degrees of freedom.
Eqs. (93)–(97)	Similarity transport and the standard double-sided Lorentz action	verified	Generic algebra elements are transported by similarity; real events are transported by congruence. The adjugate equals the inverse only after determinant-one normalization.
Eqs. (52) and (58)	Positive coefficient norm and inverse determinant	verified	The first is nonnegative and real; the second is generally complex and is the exact inverse divisor.
Eqs. (101) and (102)	Spatial rotations and Lorentz boosts [4, 5]	verified	For spatial rotors, the event congruence is reduced to similarity; for event boosts, the congruence form is retained.
Eq. (116)	Minkowski interval [6]	verified	Valid on real event paravectors with signature $(+, -, -, -)$.
Eqs. (146) and (149)	Maxwell electrodynamics [2, 16]	verified	Signs are fixed by $F = (\mathbf{E} + i\mathbf{B}) \cdot \sigma$ and the passive-boost convention.
Eq. (159)	Relativistic Lorentz force [16]	verified	The coordinate-time power and force equations are obtained by division by γ .
Eq. (161)	Electromagnetic energy density and flux	verified	T is the chosen-observer paravector; the full stress-energy object is the linear map on real paravectors.
Eqs. (168) and (174)	Klein–Gordon equation [7, 8]	verified	The frequency split is defined on the stated set of fields and boundary conditions.
Eqs. (183)–(186)	Conserved Klein–Gordon solution pairing	verified	Its integral is positive on the set of free positive-frequency solutions; its local density is not a general position probability.
Eqs. (194) and (188)	Standard Dirac equation in two coupled two-entry blocks [10, 17]	verified	One Dirac field is represented by two fixed projected algebra blocks, each containing two complex amplitudes.
Eqs. (197)–(216)	Four Dirac components, conserved current, and normalized Hermitian measurement average [19]	verified	The positive norm is conserved only for the complete two-block field; normalized outcome probabilities are supplied by Hermitian projectors.
Eq. (202)	Squared Dirac system with the stated electromagnetic potential	verified	The terms $+iqF^*$ and $+iqF$ are carried by the upper and lower blocks, respectively.
Eqs. (224)–(229)	Leading Pauli reduction [9, 19]	verified	The expansion is in inverse mass after the large/small energy-basis split; convection and magnetization terms are contained in the inherited current.

Statement	Standard correspondence	Status	Qualification
Eq. (242)	Free energy and spin-along-momentum projectors	verified	The energy and spin projectors commute when the spin axis follows momentum (or when any axis is chosen at rest), but not for an arbitrary fixed axis.

Automated Checks of Main Algebraic and Operator Identities

The identities below were subjected to automated algebraic consistency checks in Python using SymPy. The sigma-basis product was implemented directly, and differential identities were evaluated on generic test functions $\psi_{\text{KG}}(t, \mathbf{r})$ with symbolic potentials and field components. The optional matrix bridge was checked separately against the definitions in Appendix A. The verification script is `tools/verify_sta_results.py`.

Check	Status	Detail
Eqs. (3)–(4)	verified	The identity $\sigma_n \sigma_m = \delta_{nm} + \epsilon_{nmk} i \sigma_k$ is satisfied by all nine basis products in the Pauli realization.
Eq. (41)	verified	The product $(\mathbf{A} \cdot \boldsymbol{\sigma})(\mathbf{B} \cdot \boldsymbol{\sigma})$ is reduced to $\mathbf{A} \cdot \mathbf{B} + i(\mathbf{A} \times \mathbf{B}) \cdot \boldsymbol{\sigma}$ for symbolic components A_n, B_n . The order-sensitive vector identities $(\mathbf{A} \cdot \boldsymbol{\sigma})\boldsymbol{\sigma} = \mathbf{A} - i\mathbf{A} \times \boldsymbol{\sigma}$ and $\boldsymbol{\sigma}(\mathbf{A} \cdot \boldsymbol{\sigma}) = \mathbf{A} + i\mathbf{A} \times \boldsymbol{\sigma}$ are checked component by component.
Eqs. (42)–(58)	verified	Correct reduction is obtained for the complex paravector product, the adjugate identity $Z \text{adj}(Z) = \det(Z) = S^2 - \mathbf{V}^2$, and the Euclidean scalar part $\text{scalar}(ZZ^\text{H}) = S ^2 + \ \mathbf{V}\ ^2$.
Eqs. (55) and (296)	verified	Every bridge identity defined in Appendix A is verified entry by entry. The two-way <code>mat(·)</code> translation, Hermitian operation, star operation, adjugate, determinant, and their stated distinctions all agree.
Eqs. (262) and (264)	verified	The auxiliary construction in Section 6.8.13 squares to the determinant times its identity element. The check has the same scalar-entry restriction stated there.
Eqs. (93)–(95)	verified	For generic complex $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$ and $T = f + \mathbf{g} \cdot \boldsymbol{\sigma}$, the stated scalar-preserving vector formula for $T^{-1}ZT$ is reproduced after multiplication through by $\det(T) = f^2 - \mathbf{g}^2$. The full complex-paravector similarity calculation is thereby checked.
Eqs. (102) and (149)	verified	For a boost along the first axis, standard time/longitudinal mixing is produced by the event congruence $X' = L^{-1}X(L^{-1})^\text{H} = L^*XL^*$; the transverse coordinates are left real and unchanged, and $\det(X)$ is preserved. The event intertwiner $LX'L^\text{H} = X$, equivalently $LX' = XL^*$, is also satisfied. That event boost is not produced by either similarity ordering. Standard electric–magnetic mixing is produced by the field similarity $F' = L^{-1}FL$, whereas the alternative two-sided field product with two star-related factors introduces an unwanted scalar component.
Eqs. (92) and (101)	verified	For a positive right-angle rotation about the third axis, $R^\text{H}XR$ sends $\{r_1, r_2, r_3\}$ to $\{-r_2, r_1, r_3\}$. This fixes the displayed positive Rodrigues sign and preserves the event determinant.
Eqs. (71), (117), (123), and (121)	verified	The projector identities are satisfied by the idempotents $\Pi(\pm)$, the normalization $\det(U) = 1$ is obtained, and $\text{scalar}(U^*A) = 0$ follows for symbolic three-velocity and three-acceleration components. $E^2 - \mathbf{P}^2 = m^2$ is implied by $mU = E + \mathbf{P} \cdot \boldsymbol{\sigma}$.
Eq. (139)	verified	The ordered product rule is satisfied for two generic space-time-dependent paravectors. The extra term $\sum_n \text{com}(\sigma_n, Z) \partial_n Z'$ is exactly the correction produced by moving the left field through the sigma basis.

Eqs. (144), (145), (146)	verified	The relations $S = \partial_t V + \partial_r \cdot \mathbf{A}$, $\mathbf{E} = -\partial_t \mathbf{A} - \partial_r V$, and $\mathbf{B} = \partial_r \times \mathbf{A}$ are obtained from the potential identity, while Gauss, Faraday, and Ampère–Maxwell component equations with the manuscript’s signs are obtained from ∂F .
Eqs. (153) and (159)	verified	The component laws $V(\lambda) = V + \partial_t \lambda$ and $\mathbf{A}(\lambda) = \mathbf{A} - \partial_r \lambda$ are obtained from $\Phi(\lambda) = \Phi + \partial^* \lambda$, and the resulting electric and magnetic fields are unchanged. The expected power term $\gamma \mathbf{v} \cdot \mathbf{E}$ and force term $\gamma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ are obtained from the product FU ; their real paravector is orthogonal to U in the stated determinant pairing.
Eqs. (161) and (165)	verified	The product $FF^H/2$ has scalar part $(\mathbf{E}^2 + \mathbf{B}^2)/2$ and vector part $\mathbf{E} \times \mathbf{B}$. For a generic real input a , $FaF^H/2$ is again Hermitian, is linear in that input, and reduces to the displayed energy-density/flux paravector when $a = 1$. Its scalar and vector parts for a purely spatial input reproduce the displayed Poynting and momentum-flow components.
Eqs. (175), (185), and (181)	verified	The expanded Klein–Gordon operator and the commutators $\text{com}(E, P_m) = iqE_m$ and $\text{com}(P_m, P_n) = iq(\partial_m A_n - \partial_n A_m)$ are reproduced on a generic test function. The simultaneous potential change and field phase give $E'\psi'_{\text{KG_with_potential}} = \exp(-iq\lambda)E\psi_{\text{KG_with_potential}}$ and the corresponding three momentum relations. The mixed-current divergence is reduced to the difference of the two Klein–Gordon equations, and the ordered factorization $(E + \boldsymbol{\sigma} \cdot \mathbf{P})(E - \boldsymbol{\sigma} \cdot \mathbf{P}) = E^2 - \mathbf{P}^2 - iqF$ is obtained.
Eqs. (170) and (171)	verified	For a spatial mode on which Ω has a nonzero value, the two defined parts reconstruct the original Klein–Gordon solution and obey their respective first-order equations. The stated exclusion of a zero eigenvalue is necessary because the definition contains Ω^{-1} .
Eqs. (202), (225), and (233)	verified	The two squared equations are obtained by elimination of one Dirac component. After selection of the common overall sign in Eq. (202), $+iqF^*$ is carried by the upper component and $+iqF$ by the lower component. The exact Pauli identity $(\boldsymbol{\sigma} \cdot \mathbf{P})^2 = \mathbf{P}^2 - q \mathbf{B} \cdot \boldsymbol{\sigma}$ is satisfied on a test function. For uniform $\mathbf{B} = \ \mathbf{B}\ \mathbf{u}$, the relation $(\mathbf{B} \cdot \boldsymbol{\sigma})\Pi(\pm) = \pm\ \mathbf{B}\ \Pi(\pm)$ is satisfied by the projectors.
Eqs. (236)–(243)	verified	The free pair-valued Hamiltonian satisfies $\mathcal{H}(\mathcal{H}(\Psi_{\text{Dirac}})) = E(\mathbf{p})^2 \Psi_{\text{Dirac}}$. Its two displayed energy functions are complementary, mutually annihilating projectors with multipliers $\pm E(\mathbf{p})$. The componentwise momentum-axis action commutes with this Hamiltonian, which justifies the simultaneous free energy and momentum-axis split.
Eqs. (188), (197), and (208)	verified	Exactly two complex amplitudes are retained by the fixed right projector. All four displayed Dirac component equations are reproduced by direct sigma-basis multiplication. The intrinsic products give $\rho(\Psi_{\text{Dirac_with_potential}}) = a ^2 + b ^2 + c ^2 + d ^2$ and the three displayed components of $\mathbf{j}(\Psi_{\text{Dirac_with_potential}})$, including every relative sign. The projected norm and one-sided carrier restriction are also verified.
Eq. (109)	verified	The identities $QQ^H = a^2 + \mathbf{B}^2$ and $Q^2 = a^2 - \mathbf{B}^2 - 2ia \mathbf{B} \cdot \boldsymbol{\sigma}$ are satisfied by the quaternion correspondence.

The literature comparison in Section 9 is complemented by these automated checks. Purely algebraic and differential-operator identities are thereby separated from statements whose interpretation is still determined by physical conventions.

A Standard Two-by-Two Matrix Realizations

The intrinsic algebra is translated here into ordinary two-by-two complex matrices. The functions $\text{mat}(Z)$ and $\text{mat}^{-1}(\mathbf{M})$ give the translation in both directions. Addition and multiplication are preserved, and the intrinsic operations are compared directly with matrix C, T, and H. The matrix origins of $\text{tr}(\cdot)$, $\text{adj}(\cdot)$, and $\text{det}(\cdot)$ are made explicit. The quaternion matrix is then obtained only as a specialization of this same map [9].

A.1 Standard Two-by-Two Complex Realization

The sigma algebra is represented faithfully by two-by-two matrices with complex scalar entries [9, 12, 14]. A translation is called faithful when it is one-to-one and addition, complex scalar multiplication, and the ordered product are preserved. The ordinary matrix operations needed for that statement are defined first.

Two generic matrices are written as

$$\mathbf{M} := \begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix}, \quad \mathbf{N} := \begin{bmatrix} n_{11} & n_{12} \\ n_{21} & n_{22} \end{bmatrix}, \quad m_{jk}, n_{jk} \in \mathbb{C}, \quad j, k \in \{1, 2\}. \quad (275)$$

Their ordered product is defined by

$$\mathbf{MN} := \begin{bmatrix} m_{11}n_{11} + m_{12}n_{21} & m_{11}n_{12} + m_{12}n_{22} \\ m_{21}n_{11} + m_{22}n_{21} & m_{21}n_{12} + m_{22}n_{22} \end{bmatrix}. \quad (276)$$

The multiplicative identity and trace are defined by

$$\mathbf{I} := \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad (277)$$

$$\text{tr}(\mathbf{M}) := m_{11} + m_{22}. \quad (278)$$

Thus $\mathbf{IM} = \mathbf{MI} = \mathbf{M}$, and the trace is the sum of the diagonal entries.

The transpose is defined by

$$\mathbf{M}^T := \begin{bmatrix} m_{11} & m_{21} \\ m_{12} & m_{22} \end{bmatrix}. \quad (279)$$

For a complex scalar entry $z = x + iy$, entrywise complex conjugation is defined first by

$$z^C := x - iy, \quad x, y \in \mathbb{R}, \quad (280)$$

and then on a matrix by

$$\mathbf{M}^C := \begin{bmatrix} m_{11}^C & m_{12}^C \\ m_{21}^C & m_{22}^C \end{bmatrix}. \quad (281)$$

The superscript C is used only for matrices and their scalar entries. The matrix Hermitian operation is defined by

$$\mathbf{M}^H := \mathbf{M}^{CT} = \mathbf{M}^{TC} = \begin{bmatrix} m_{11}^C & m_{21}^C \\ m_{12}^C & m_{22}^C \end{bmatrix}. \quad (282)$$

Thus $\mathbf{H} = \mathbf{CT} = \mathbf{TC}$ in ordinary matrix notation.

The matrix adjugate and determinant are defined by

$$\text{adj}(\mathbf{M}) := \begin{bmatrix} m_{22} & -m_{12} \\ -m_{21} & m_{11} \end{bmatrix}, \quad (283)$$

$$\text{det}(\mathbf{M}) := m_{11}m_{22} - m_{12}m_{21}. \quad (284)$$

The following identity is obtained directly from the product in Eq. (276):

$$\mathbf{M} \text{adj}(\mathbf{M}) = \text{adj}(\mathbf{M}) \mathbf{M} = \text{det}(\mathbf{M}) \mathbf{I}. \quad (285)$$

When $\text{det}(\mathbf{M}) \neq 0$, the multiplicative inverse is therefore defined by

$$\mathbf{M}^{-1} := \frac{\text{adj}(\mathbf{M})}{\text{det}(\mathbf{M})}, \quad \mathbf{MM}^{-1} = \mathbf{M}^{-1}\mathbf{M} = \mathbf{I}. \quad (286)$$

The three sigma generators are represented by the Pauli matrices

$$\sigma_1 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_2 = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \quad (287)$$

The same defining rule as for the intrinsic generators is satisfied by their matrix products:

$$\sigma_n \sigma_m = \delta_{nm} \mathbf{I} + \sum_{k=1}^3 \epsilon_{nmk} i \sigma_k. \quad (288)$$

Consequently,

$$\sigma_n^2 = \mathbf{I}, \quad \sigma_m \sigma_n = -\sigma_n \sigma_m \quad (m \neq n), \quad \sigma_1 \sigma_2 \sigma_3 = i \mathbf{I}. \quad (289)$$

The matrix entry i is the same central unit defined intrinsically by Eq. (3); no second imaginary unit is introduced.

For $Z = S + \mathbf{V} \cdot \boldsymbol{\sigma}$, the forward translation is defined by

$$\begin{aligned} \text{mat}(Z) &:= S \mathbf{I} + \sum_{n=1}^3 V_n \sigma_n \\ &= \begin{bmatrix} S + V_3 & V_1 - i V_2 \\ V_1 + i V_2 & S - V_3 \end{bmatrix}. \end{aligned} \quad (290)$$

For the generic matrix in Eq. (275), the inverse translation is defined by

$$\text{mat}^{-1}(\mathbf{M}) := \frac{m_{11} + m_{22}}{2} + \frac{m_{12} + m_{21}}{2} \sigma_1 + \frac{m_{21} - m_{12}}{2i} \sigma_2 + \frac{m_{11} - m_{22}}{2} \sigma_3. \quad (291)$$

The superscript -1 on the name of the inverse translation belongs to that function name. It is distinct from the multiplicative matrix inverse in Eq. (286). For $Z \in \mathbb{P}_{\mathbb{C}}$ and $\mathbf{M} \in \mathbb{C}^{2 \times 2}$, the two descriptions are related simply by

$$\text{mat}(Z) = \mathbf{M}, \quad Z = \text{mat}^{-1}(\mathbf{M}). \quad (292)$$

Direct substitution in Eqs. (290) and (291) verifies both directions. The following identities are then obtained from the Pauli products in Eq. (288):

$$\begin{aligned} \text{mat}(Z + Z') &= \text{mat}(Z) + \text{mat}(Z'), \\ \text{mat}(cZ) &= c \text{mat}(Z), \quad c \in \mathbb{C}, \\ \text{mat}(ZZ') &= \text{mat}(Z) \text{mat}(Z'). \end{aligned} \quad (293)$$

Faithfulness of the translation and reproduction of the ordered paravector product by matrix multiplication are established by these identities.

A fixed matrix is required for the matrix action representing intrinsic star:

$$\mathbf{T} := \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \quad \mathbf{T}^{-1} = -\mathbf{T}. \quad (294)$$

For an arbitrary complex matrix, transported star is defined directly by

$$\mathbf{M}^* := \mathbf{T} \mathbf{M}^{\mathbb{C}} \mathbf{T}^{-1} = \begin{bmatrix} m_{22}^{\mathbb{C}} & -m_{21}^{\mathbb{C}} \\ -m_{12}^{\mathbb{C}} & m_{11}^{\mathbb{C}} \end{bmatrix}. \quad (295)$$

This transported star is intrinsic star written in matrix coordinates. It is not entrywise matrix conjugation $\mathbb{C}$.

Under the correspondence in Eq. (292), the complete operation dictionary is

$$\begin{aligned} \text{mat}(Z^*) &= \mathbf{M}^* = \mathbf{T} \mathbf{M}^{\mathbb{C}} \mathbf{T}^{-1}, \\ \text{mat}(Z^{\mathbb{H}}) &= \mathbf{M}^{\mathbb{H}} = \mathbf{M}^{\mathbb{C}^{\mathbb{T}}}, \\ \text{mat}(\text{adj}(Z)) &= \text{adj}(\mathbf{M}) = \mathbf{M}^{*\mathbb{H}} = \mathbf{T} \mathbf{M}^{\mathbb{T}} \mathbf{T}^{-1}, \\ \det(Z) &= \det(\mathbf{M}), \\ \text{mat}(0) &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \text{mat}(1) = \mathbf{I}, \\ \text{mat}(Z^{-1}) &= \mathbf{M}^{-1} = \frac{\text{adj}(\mathbf{M})}{\det(\mathbf{M})}, \quad \det(Z) \neq 0. \end{aligned} \quad (296)$$

Thus intrinsic H is represented exactly by the familiar matrix Hermitian operation, while intrinsic $*H$ is represented by the matrix adjugate. In contrast, matrix C is fixed by the chosen complex entries and has no independent intrinsic-algebra definition. In particular,

$$\mathbf{M}^* = \mathbf{T} \mathbf{M}^C \mathbf{T}^{-1} \neq \mathbf{M}^C \quad \text{in general.} \quad (297)$$

Under the same correspondence, their distinct effects on the intrinsic coordinates are

$$\begin{aligned} \text{mat}^{-1}(\mathbf{M}^C) &= S^* + V_1^* \sigma_1 - V_2^* \sigma_2 + V_3^* \sigma_3, \\ \text{mat}^{-1}(\mathbf{M}^T) &= S + V_1 \sigma_1 - V_2 \sigma_2 + V_3 \sigma_3, \\ \text{mat}^{-1}(\mathbf{M}^H) &= S^* + V_1^* \sigma_1 + V_2^* \sigma_2 + V_3^* \sigma_3 = Z^H, \\ \text{mat}^{-1}(\mathbf{M}^*) &= S^* - V_1^* \sigma_1 - V_2^* \sigma_2 - V_3^* \sigma_3 = Z^*. \end{aligned} \quad (298)$$

Matrix C and T are basis-dependent: their displayed coordinate sign patterns are fixed by the particular Pauli matrices in Eq. (287) and can change when different representing matrices are selected. Their composition H , however, represents the intrinsic Hermitian operation, and transported star represents the intrinsic star operation.

The extraction functions are translated without additional definitions:

$$\begin{aligned} \text{scalar}(Z) &= \frac{1}{2} \text{tr}(\mathbf{M}), \\ \text{mat}(\text{vector}(Z) \cdot \boldsymbol{\sigma}) &= \mathbf{M} - \frac{1}{2} \text{tr}(\mathbf{M}) \mathbf{I}, \\ \text{mat}(\text{real}(Z)) &= \frac{1}{2} (\mathbf{M} + \mathbf{M}^H), \\ \text{mat}(\text{img}(Z)) &= \frac{\mathbf{M} - \mathbf{M}^H}{2i}. \end{aligned} \quad (299)$$

The product-order distinctions are displayed directly by

$$\begin{aligned} \mathbf{M}\mathbf{N}^C &= \mathbf{M}^C \mathbf{N}^C, \\ \mathbf{M}\mathbf{N}^T &= \mathbf{N}^T \mathbf{M}^T, \\ \mathbf{M}\mathbf{N}^H &= \mathbf{N}^H \mathbf{M}^H, \\ \mathbf{M}\mathbf{N}^* &= \mathbf{M}^* \mathbf{N}^*, \\ \text{adj}(\mathbf{M}\mathbf{N}) &= \text{adj}(\mathbf{N}) \text{adj}(\mathbf{M}). \end{aligned} \quad (300)$$

Product order is preserved by matrix C and transported star, whereas it is reversed by matrix T , matrix H , and the adjugate. The relation

$$\mathbf{M}^H = \text{adj}(\mathbf{M})^* = \text{adj}(\mathbf{M}^*) \quad (301)$$

is the matrix translation of Eq. (55).

The distinction between C and star is made especially visible by the individual basis matrices:

$$\begin{aligned} \sigma_1^C &= \sigma_1, & \sigma_2^C &= -\sigma_2, & \sigma_3^C &= \sigma_3, \\ \sigma_1^T &= \sigma_1, & \sigma_2^T &= -\sigma_2, & \sigma_3^T &= \sigma_3, \\ \sigma_n^H &= \sigma_n, & \sigma_n^* &= -\sigma_n, & \text{adj}(\sigma_n) &= -\sigma_n. \end{aligned} \quad (302)$$

The same signs happen to be produced on the three individual basis matrices by entrywise C and transpose, but different operations remain: product order is preserved by C and reversed by T .

For the real event $X = t + \mathbf{r} \cdot \boldsymbol{\sigma}$, the translation reduces to

$$\begin{aligned} \text{mat}(X) &= \begin{bmatrix} t + r_3 & r_1 - ir_2 \\ r_1 + ir_2 & t - r_3 \end{bmatrix}, \\ \text{tr}(\text{mat}(X)) &= 2t, \\ \det(\text{mat}(X)) &= t^2 - \mathbf{r}^2 = \det(X). \end{aligned} \quad (303)$$

Its two eigenvalues are the two roots of

$$x^2 - \text{tr}(\text{mat}(X))x + \det(\text{mat}(X)) = 0, \quad x = \lambda(\pm) = t \pm \|\mathbf{r}\|. \quad (304)$$

Thus twice the scalar part is recovered by the matrix trace, and the real space-time quadratic form is recovered by the ordinary matrix determinant.

A.2 Quaternion Specialization of the Two-by-Two Map

The intrinsic quaternion basis from Eq. (90) is obtained by restricting the same Pauli matrix map already defined in Eq. (290). Its three basis values are

$$\begin{aligned}\text{mat}(q_1) &= -i\sigma_1 = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix}, \quad \text{mat}(q_2) = -i\sigma_2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \\ \text{mat}(q_3) &= -i\sigma_3 = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}.\end{aligned}\tag{305}$$

For the quaternion in Eq. (108), the two-by-two value of the map is

$$\text{mat}(Q) = \begin{bmatrix} a - iB_3 & -B_2 - iB_1 \\ B_2 - iB_1 & a + iB_3 \end{bmatrix}.\tag{306}$$

Because intrinsic star fixes Q , its induced matrix operation also fixes $\text{mat}(Q)$:

$$\text{mat}(Q)^* = \text{mat}(Q^*) = \text{mat}(Q).$$

This star is the induced operation from Eq. (296); it is not entrywise matrix C, which does not fix the displayed matrix in general. The Hermitian and adjugate operations coincide on this specialization:

$$\text{mat}(Q)^H = \text{mat}(Q^H) = \text{adj}(\text{mat}(Q)).$$

Consequently,

$$\begin{aligned}\text{mat}(Q) \text{mat}(Q)^H &= \text{mat}(Q)^H \text{mat}(Q) = (a^2 + \mathbf{B}^2)\mathbf{I}, \\ \det(\text{mat}(Q)) &= a^2 + \mathbf{B}^2, \quad \text{tr}(\text{mat}(Q)) = 2a.\end{aligned}\tag{307}$$

Thus a unit quaternion is represented by a matrix whose Hermitian product is $\mathbf{I}$ and whose determinant is 1. This is a specialization of the single Pauli matrix realization, not a second representation of the paravector algebra.

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